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Smart Energy Management for a Hybrid Solar–Diesel–Battery Microgrid in Rural Yemen

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A graduation project document submitted to the department of Mechatronics in partial fulfillment to the requirements for Bachelor Degree in Mechatronics Engineering

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DECLARATION

We hereby certify that the thesis entitled: " **Smart Energy Management for a Hybrid Solar–Diesel–Battery Microgrid in Rural Yemen** " submitted in partial fulfillment of the requirements for the Bachelor's Degree of Mechanical Engineering, represents an authentic record of our original work conducted in the Department of Mechatronics Engineering (Faculty of Engineering) at Emirates international University. This work was carried out from August 2025 to May 2026 under the supervision of:

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Abstract

This graduation project proposes a hybrid photovoltaic–battery–diesel microgrid for a rural coastal community in Yemen with limited and unreliable electricity access. The main objective is to provide a practical electrification solution that improves supply continuity, reduces diesel dependence, and increases the use of renewable energy.

The proposed system combines a 150 kW photovoltaic array, a battery energy storage system of about 100–123 kWh, and a 100 kW diesel generator. The community load is divided into three feeders with different priority levels so that essential loads are supplied first, followed by less critical residential and community loads. This prioritization helps the microgrid use available energy more efficiently during periods of limited generation.

To manage the system, a rule-based Energy Management System is developed. Solar power is always given priority when available, the battery supports short-term balancing and backup operation, and the diesel generator is used only when renewable and stored energy are not sufficient. This control strategy is simple, transparent, and suitable for industrial implementation using platforms such as the Siemens S7-1200 PLC and the DSE8610 MKII controller.

The project also combines design and simulation tools to evaluate the proposed system. MATLAB/Simulink and Stateflow are used to test the energy management logic, while EPLAN, PLC control mapping, and Node-RED/HTML monitoring interfaces are used to show how the system can be implemented in practice. The results indicate that the hybrid microgrid can improve critical-load continuity, increase renewable energy utilization, and reduce diesel runtime compared with a diesel-only supply.

Overall, the project presents a realistic and technically practical solution for rural electrification in Yemen. It links system sizing, load prioritization, control logic, and supervisory monitoring into one integrated framework that bridges academic analysis and real-world application.

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Chapter 1

Introduction

1. Chapter 1

1.1 Introduction

Modern power systems are undergoing a rapid transition from centralized fossil-fuel generation toward decentralized architectures that integrate renewable energy sources, energy storage, and advanced digital control. Microgrids have emerged as a key enabling technology in this transition because they can operate either grid-connected or islanded, provide high reliability for critical loads, and enable efficient local use of distributed energy resources. Hybrid microgrids that combine photovoltaic (PV) generation, battery energy storage systems (BESS), and diesel generators are particularly attractive in remote areas or weak grids where fuel costs are high and power quality is poor. [1]

In parallel with these developments, advances in power electronics, programmable logic controllers (PLCs), digital genset controllers, and industrial Internet of Things (IoT) platforms make it possible to implement sophisticated energy management strategies on relatively low-cost hardware. By combining local measurements, supervisory control algorithms, and remote monitoring over secure communication links, [2] these smart microgrids can dynamically adapt to changing irradiance, load demand, and system contingencies while minimizing operating cost and fuel consumption. This project is positioned within that context and focuses on the design and implementation of a PV–Diesel–Battery hybrid microgrid with a smart Energy Management System (EMS) and IoT-based monitoring and control infrastructure. [3]



Figure 1-1 Conceptual illustration of the proposed intelligent hybrid microgrid for prioritized load supply in a rural Yemeni community.

As shown in figure 1.1, the work presented in this thesis targets an academic yet practically grounded implementation suitable for a bachelor-level engineering capstone project. The system is sized around a 146 kWp PV array, a high-voltage lithium-iron-phosphate battery bank of approximately 123 kWh, a 100 kW diesel generator, and three prioritized load feeders, all interconnected through a 400 V three-phase alternative current (AC) bus. On top of this physical layer, the project develops and validates an EMS implemented in MATLAB/Simulink and Python, integrated with Node-RED, an HTML dashboard, a PLC, and Deep-Sea Electronics (DSE) controller modules to form a complete prototype of a smart hybrid microgrid.

1.2 Background

A microgrid can be defined as a group of interconnected loads and distributed energy resources within clearly defined electrical boundaries that acts as a single controllable entity with respect to the grid [4]. It can operate in grid-connected mode—supporting the main grid—or in islanded mode, supplying local loads when the grid is unavailable or unreliable. Hybrid microgrids that integrate PV, batteries, and diesel generators leverage the complementary characteristics of these resources: PV provides low-cost, clean energy but is intermittent; batteries smooth short-term variability and shift energy over time; diesel generators provide dispatchable power and inertia but incur fuel and maintenance costs and produce emissions. [5]

Several studies have shown that hybrid PV–battery–diesel microgrids can significantly reduce fuel consumption and operating costs compared to diesel-only solutions, while maintaining or even enhancing reliability. For example, optimization-based studies using tools such as renewables energy optimization (REopt) or hybrid optimization model for electric renewables (HOMER) demonstrate that appropriately sized PV and storage can reduce diesel runtime and fuel use by 20–60% in typical off-grid applications, depending on solar resource and load profile. Experimental work on islanded microgrids further highlights the importance of coordinated control strategies: an EMS must determine when to curtail PV, when to charge or discharge the battery, and when to start or stop diesel units to satisfy constraints on frequency, voltage, state of charge (SOC), and generator loading. [6]

Recent literature also emphasizes the role of IoT-based monitoring and smart controllers in microgrids. By using PLCs, embedded controllers, and cloud-connected gateways, operators can collect real-time data, implement remote control actions, and integrate predictive algorithms based on weather and load forecasting [7]. Several industrial case studies demonstrate how DSE

controller families and supervisory modules can coordinate gensets with renewable sources and storage to realize multi-energy microgrids with automated start-stop sequences and load sharing logic. These developments motivate the educational and practical value of building a complete, small-scale hybrid microgrid demonstrator that combines hardware-oriented design with advanced control and IoT integration. [8]

1.3 Problem Statement

Despite the extensive body of literature on hybrid microgrids, many academic projects treat the PV array, battery system, diesel generator, and EMS only at a conceptual or simulation level. Students often investigate isolated aspects—such as PV sizing, battery modeling, or PI control of a converter—without integrating them into a coherent end-to-end system that includes detailed electrical design, protection coordination, EMS logic, and real-time monitoring. This leads to a gap between theoretical analyses and the practical engineering tasks required to implement a functioning microgrid in the field. [9]

Furthermore, many published EMS algorithms assume idealized conditions and abstract away the constraints of commercially available components. In practice, inverter maximum direct current (DC) voltages, maximum power point tracking (MPPT) current limits, string fuse ratings, cable ampacities, battery SOC windows, and DSE controller capabilities strongly influence feasible operating strategies. Without capturing these constraints, simulation results may overestimate achievable performance or propose control actions that cannot be implemented safely. [10]

There is therefore a need for a reference-quality educational project that:

- Designs the PV array, battery bank, and diesel generator integration based explicitly on real datasheets and protection requirements. [11]
- Develops an EMS that respects these physical limits and load priorities, and that is actually executed on real-time platforms (MATLAB/Simulink, PLC, DSE, and Node-RED) rather than remaining purely theoretical. [12]
- Demonstrates the dynamic behavior of the system under different operating scenarios, including variations in PV availability and load demands, with clear visualization and data logging. [13]

This project addresses that need by delivering a complete design, implementation, and evaluation of a PV–Diesel–Battery microgrid, emphasizing both rigor in the electrical and control design and clarity in the educational documentation.

1.4 Motivation

The primary motivation for this work is to contribute a comprehensive, academically rigorous yet practically oriented case study of a hybrid microgrid suitable for bachelor-level education and future extension by other students and researchers. Remote communities, industrial plants, and mission-critical facilities such as hospitals and data centers all face challenges related to unreliable grid supply, high diesel fuel costs, and increasing pressure to decarbonize. Hybrid microgrids offer a pathway to mitigate these issues, but their design and operation are non-trivial, particularly when incorporating multiple resources, load priorities, and communication layers.

From a learning perspective, the project integrates several domains that are usually treated separately in coursework:

- Power system design (sizing, cabling, and protection of PV arrays, AC distribution, and diesel systems).
- Power electronics and inverter behavior, including maximum DC voltage and MPPT characteristics.
- Energy storage modeling, SOC management, and battery lifetime considerations.
- Control theory and EMS algorithms implemented in MATLAB/Simulink and Python.
- Industrial automation via PLCs and DSE controllers.
- IoT, Node-RED dashboards, and web-based HMIs for monitoring and supervisory control.

By bringing these aspects together in one coherent framework, the project not only delivers a functioning prototype but also serves as a rich teaching and demonstration platform. The detailed documentation produced in this project aims to be sufficiently complete that another student team could reproduce, modify, or extend the system with minimal ambiguity.

In addition to the global drivers for hybrid microgrids, this project is specifically motivated by the situation of rural coastal communities in Yemen, where access to reliable and affordable electricity remains severely limited. Many villages and small towns either lack grid connection altogether or depend on small diesel generators with high fuel costs, frequent outages, and significant environmental impacts. Recent initiatives deploying solar microgrids in Yemen have demonstrated that decentralized PV-based systems can substantially improve quality of life and economic opportunities for local residents, but these pilots also highlight the need for robust energy management and prioritization of critical loads in order to cope with variable solar resources and constrained battery capacities. Against this backdrop, the hybrid PV–Diesel–Battery microgrid and EMS proposed in this thesis are conceived as a scalable reference architecture that could be adapted to supply prioritized loads—such as health centers, water pumping, and essential household consumption—in Yemeni coastal villages over extended periods with reduced fuel dependence and improved reliability. [14]

1.5 Project Overview

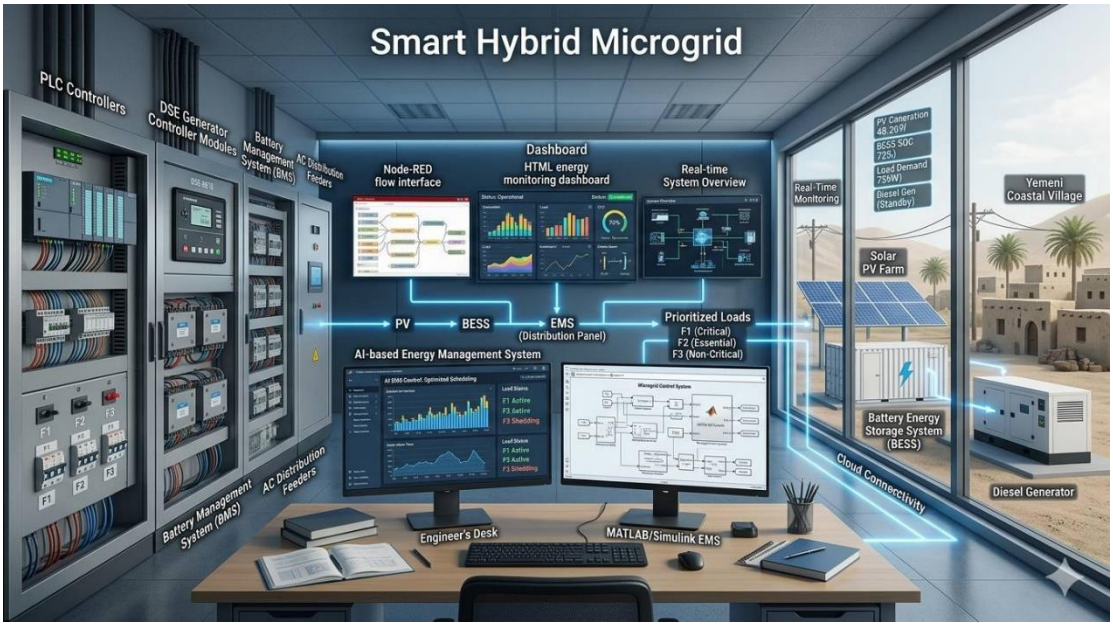


Figure 1-2 Smart hybrid microgrid control room showing monitoring, EMS, and load allocation interfaces

As shown in figure 1-2, this project implements a hybrid microgrid composed of a PV generation subsystem, a battery energy storage subsystem, a diesel generator, an AC distribution board with prioritized feeders, and a multi-layer control and monitoring architecture. The core idea is to use PV as the primary energy source whenever available, use the battery to buffer

shortfalls and charge from surplus energy within an SOC window of 30–90%, and finally rely on the diesel generator as a backup resource when PV and battery cannot meet the demand of critical loads.

1.5.1 Power Hardware

On the generation side, the PV plant consists of 266 Jinko 550 W modules of the Tiger Neo 72HL4-BDV family, arranged as 14 strings of 19 modules each, yielding an installed DC capacity of approximately 146 kWp. Each string operates around a maximum power point voltage of about 789 V and an open-circuit voltage that remains safely below the 1100 V DC limit of the Huawei SUN2000-100KTL-M2 inverter under the minimum design temperature. String fuses rated 20 A gPV, 6 mm^2 copper PV cables, and appropriately rated DC isolators and surge protection devices (SPDs) provide electrical protection and safety.

The AC side operates at 400 V three-phase, with the PV inverter feeding a main distribution board through a 200 A, four-pole molded-case circuit breaker. From this board, three feeders supply prioritized loads: F1 (critical, up to 57 kW), F2 (important, up to 26 kW), and F3 (non-critical, up to 15 kW), each with its own breaker and cable sizing derived from the expected currents and installation conditions. A 100kW diesel generator, controlled by a DSE genset controller, can connect to the same AC bus to support the system when required.

The battery subsystem is based on high-voltage $LiFePO_4$ technology. For the design adopted here, 24 modules rated around 51.2 V and 100 Ah are arranged as two parallel strings of 12 series-connected batteries, producing an approximate nominal DC voltage of 614 V and a usable energy content of roughly 73.7 kWh when operated between 90% and 30% SOC. A 100 kW bidirectional PCS interfaces this battery bank with the 400 V AC bus, respecting current and voltage limits derived from manufacturer specifications and cable ampacities.

1.5.2 Control and EMS Architecture

The control architecture is layered. At the lowest level, the PV inverter, PCS, and DSE genset controllers perform local regulation of currents, voltages, and protection functions according to their internal firmware and parameterization. At the supervisory level, a custom EMS algorithm developed in MATLAB/Simulink and implemented as stateflow charts evaluates PV availability, battery SOC, and load demand on a discrete-time basis to generate setpoints and

switching decisions. These include deciding when to enable or disable F2 and F3, when to charge or discharge the battery, and when to start or stop the diesel generator. [15]

The EMS logic is mirrored in a Python application that provides a scenario-based simulation interface. This tool allows users to specify peak values for PV power and each feeder, run the EMS algorithm, and visualize results such as power balance, feeder contributions, generator output, and SOC trajectories under various scenarios. Four representative scenarios are defined in this work: a default case with maximum PV and full load, a reduced-PV case with lower feeder set-points, a case in which F3 is shed due to PV deficit, and a severe deficit case in which F2 is also shed and F1 is maintained using battery and diesel support. [16]

An IoT layer implemented in Node-RED connects to the PLC and controllers via industrial communication protocols, exposing process variables and control commands to an HTML-based dashboard. This dashboard allows operators to monitor PV power, feeder loading, battery SOC, and generator status, and to manually enable or disable non-critical feeders if needed. The bidirectional link ensures that changes in the field (e.g., automatic EMS decisions) are reflected in the dashboard, and commands issued from the dashboard propagate back to the PLC and control devices. [17]

1.6 Research Aim and Objectives

1.6.1 Aim

The overall aim of this thesis is to design, implement, and evaluate a smart EMS-driven PV–Diesel–Battery microgrid with prioritized loads and IoT-based monitoring, using real component specifications and industrial-grade controllers, and to document the entire process at a level suitable for replication and educational use. [18]

1.6.2 Specific Objectives

To achieve this aim, the research is structured around the following objectives:

1. **To perform a comprehensive literature review** on hybrid microgrids, EMS strategies, and IoT-enabled monitoring architectures, with emphasis on PV–battery–diesel systems and PLC-based load management.

2. **To define detailed system requirements and models** for the PV array, battery bank, diesel generator, AC distribution, and prioritized loads, based on realistic operating conditions and manufacturer datasheets.

3. **To design the electrical architecture** of the microgrid, including string configuration, DC and AC cable sizing, protection coordination (fuses, breakers, SPDs, isolators), grounding, and main distribution board layout.

4. **To develop an EMS algorithm** that schedules PV utilization, battery charging/discharging, and diesel operation while enforcing SOC limits and load priorities among F1, F2, and F3.

5. **To implement and integrate the EMS** using MATLAB/Simulink, State flow, PLC logic, DSE controllers, Node-RED, and an HTML dashboard, ensuring reliable communication and consistent behavior across platforms.

6. **To construct a simulation and testing framework** (MATLAB/Python tools and scenario-based studies) to evaluate system performance in terms of energy balance, fuel savings, load shedding behavior, and resilience under varying PV and load conditions.

7. **To analyze and discuss the results** in an academically rigorous manner, highlighting the benefits, limitations, and potential improvements of the proposed microgrid design and EMS.

1.7 Scope and Limitations

The scope of this project is deliberately defined to balance practical relevance with feasibility for a bachelor-level thesis. The system considered is a single-bus AC microgrid operating at 400 V three-phase, integrating one PV plant, one battery bank, and one diesel generator. The focus is on **power management and control**, not on detailed converter design or low-level control of power electronic hardware. Manufacturer-provided inverters and PCS units are treated as black-box components with configurable parameters and operational limits. [19]

The EMS operates at a supervisory level with discrete-time steps on the order of minutes, which is appropriate for scheduling decisions such as feeder shedding, battery dispatch, and generator on/off commands. Fast dynamics such as sub-second voltage transients, harmonic distortion, and detailed grid-forming behaviors are outside the scope of this work. Likewise, the

study focuses on a single-day or multi-day operation with representative load and irradiance profiles, rather than long-term techno-economic optimization over many years. [20]

From a hardware integration perspective, the thesis assumes that the PV inverter, power conversion system (PCS), PLC, DSE controllers, and communication devices conform to their datasheet specifications and are correctly installed according to relevant standards; detailed mechanical design, civil works, and full compliance testing are not covered. Additionally, the battery model used in simulations simplifies some phenomena such as temperature effects, aging, and detailed electrochemical dynamics, focusing instead on SOC tracking and power limits consistent with high-voltage lithium-iron-phosphate systems. [21]

These limitations are intentional and allow the work to concentrate on the key educational contributions: rigorous yet tractable system sizing, clear protection and cabling design, realistic EMS logic with prioritized loads, and a fully integrated monitoring and control stack that can be reproduced in a laboratory or pilot-scale installation.

1.8 Methodology Overview

The methodology adopted in this thesis combines analytical design, model-based simulation, and prototype-oriented implementation. It can be summarized in the following main phases, which correspond closely to the chapter structure:

1. **Requirement definition and modeling:** The first step is to characterize the load feeders F1–F3, define design criteria for reliability and autonomy, and specify target PV and battery sizes. Manufacturer datasheets for PV modules, inverters, batteries, PCS, and DSE controllers are collected and analyzed to extract electrical ratings and constraints.

2. **System design:** Using standard electrical engineering methods and relevant codes, the PV string configuration, DC and AC cabling, and protective devices are computed. This includes verifying that string voltages remain within inverter limits over the expected temperature range, that currents remain within cable ampacity limits, and that breakers and fuses are coordinated.

3. **Control and EMS development:** The EMS is first designed conceptually as a set of rules and state machines that enforce SOC windows and load priorities. The logic is then

implemented and validated in MATLAB/Simulink using State flow charts, enabling rapid prototyping and visualization of system behavior.

4. **Simulation and scenario analysis:** A Python-based tool and MATLAB scripts are used to run multiple daily scenarios featuring different PV availability and load settings. For each scenario, time-series plots of PV power, feeder demands, battery power, generator output, and SOC are generated to assess the effectiveness of the EMS in minimizing load shedding and diesel usage.

5. **Integration with PLC, DSE, and IoT:** The validated EMS logic is mapped to PLC code and DSE controller configurations. Simultaneously, Node-RED flows and HTML dashboards are developed to expose system states and control commands over an Industrial 4G LTE router, enabling remote supervision and manual override functions.

6. **Testing and evaluation:** The integrated system is exercised under the defined scenarios and, where possible, against real or emulated hardware to verify correct operation, communication reliability, and adherence to design constraints. Results are compared to expectations from the design and literature.

Throughout these phases, the work is documented in detail, with diagrams, flowcharts, and screenshots of Simulink models, Node-RED dashboards, and PLC/DSE configurations to support reproducibility and academic clarity. [22]

1.9 Project Structure

To guide the reader through the various aspects of the project, the thesis is organized into seven chapters in addition to references and appendices:

- **Chapter 1 – Introduction:** Presents the background, problem statement, motivation, project overview, objectives, scope, methodology, and thesis outline.

- **Chapter 2 – Background and Literature Review:** Surveys fundamental concepts of microgrids, hybrid PV–battery–diesel systems, EMS strategies, and IoT-based monitoring architectures, drawing from recent journal articles, conference papers, standards, and industrial case studies.

- **Chapter 3 – Requirements Analysis and Modeling:** Defines the functional and non-functional requirements of the system, characterizes load profiles and operating scenarios, and develops mathematical models for the PV array, battery bank, diesel generator, and EMS.
- **Chapter 4 – Project Design:** Details the electrical design of the PV and AC systems, including sizing of strings, cables, and protective devices, as well as the architecture of the control and communication layers (PLC, DSE, Node-RED, and dashboard).
- **Chapter 5 – Implementation and Test:** Describes the practical realization of the Simulink models, PLC and DSE code, Node-RED flows, and HTML interfaces, and documents the integration steps and test setup.
- **Chapter 6 – Results and Discussions:** Presents and analyzes the results obtained from simulations and experiments, including the four main EMS scenarios, highlighting how the system responds to changes in PV availability and load conditions.
- **Chapter 7 – Conclusions and Recommendations:** Summarizes the main findings, discusses limitations, and proposes directions for future work, such as extending the system with additional resources, incorporating predictive control, or deploying it in a real installation.

Chapter 2

Microgrid Hybrid System Background

2. Chapter 2

Background and Literature Review

Hybrid microgrids that combine photovoltaic (PV) generation, battery energy storage systems (BESS), and diesel generators have become an attractive solution for supplying reliable and affordable power to remote and weak-grid communities. This chapter provides an in-depth review of the concepts, technologies, and control strategies that underpin the design of such systems, with particular emphasis on architectures and methodologies relevant to the PV–Diesel–Battery microgrid developed in this thesis.

The chapter is organized into ten main sections. Section 2.1 introduces general microgrid concepts and architectures. Section 2.2 focuses on hybrid PV–Diesel–Battery systems in rural electrification. Section 2.3 discusses load characterization and priority-based management. Sections 2.4–2.6 review the main hardware technologies: PV systems, battery storage, and diesel generators with genset controllers. Sections 2.7–2.9 address energy management strategies, monitoring/IoT architectures, and communication and cyber-security aspects. Section 2.10 presents relevant case studies for rural and coastal communities, including projects in Yemen and similar contexts. [23]

2.1 Microgrid Concepts and Architectures

2.1.1 Definition and Operating Modes

A microgrid is generally defined as a localized group of electricity sources and loads that can disconnect from the traditional grid to operate autonomously and often includes distributed generation, storage, and controllable loads. The IEEE and various research institutions describe microgrids as systems with clearly defined electrical boundaries and a single point of common coupling (PCC) to the upstream network, enabling both grid-connected and islanded.

In grid-connected mode, the microgrid exchanges power with the main grid, drawing energy when local resources are insufficient and exporting surplus when local generation exceeds demand. In islanded mode, the microgrid relies entirely on its internal resources to maintain voltage and frequency within acceptable limits. Transitions between these modes must be carefully controlled to avoid instability, especially when high levels of inverter-based generation are present. [24]

2.1.2 AC, DC, and Hybrid Configurations

Microgrids can be broadly classified into AC, DC, and hybrid topologies.

- **AC microgrids** use an AC bus (typically low- or medium-voltage) as the main backbone. Synchronous machines and grid-tied inverters operate in parallel, and most loads are AC. This configuration is common in retrofits of existing distribution networks and rural mini-grids.
- **DC microgrids** employ a DC bus to interconnect PV arrays, DC loads, and DC/AC converters feeding local AC loads. They reduce conversion stages for DC sources and loads but require careful protection design.
- **Hybrid AC/DC microgrids** combine both AC and DC buses linked through bidirectional converters, enabling flexible integration of different types of resources and loads.

The system developed in this thesis follows the widely used **AC microgrid** configuration with a 400 V three-phase AC bus. PV and battery resources are connected via grid-tied inverters and PCS units, while the diesel generator connects directly to the AC bus through a circuit breaker and DSE controller. This choice simplifies compatibility with standard industrial loads and distribution equipment.

2.1.3 Core Components of a Microgrid

Typical components of a microgrid include:

- **Distributed energy resources (DERs):** PV arrays, wind turbines, diesel gensets, small hydro, etc.
- **Energy storage:** Batteries, flywheels, or other storage systems that provide fast response and energy shifting.
- **Loads:** Residential, commercial, and industrial consumers with varying sensitivity to interruptions.
- **Power electronic interfaces:** Inverters and converters that connect DERs and storage to the AC or DC bus.
- **Protection devices:** Fuses, circuit breakers, relays, and surge protectors that ensure safe operation.

- **Control and communication infrastructure:** EMS, PLCs, controllers, communication networks, and SCADA/IoT platforms.

To better understand the evolution of distributed energy systems, it is important to distinguish between the principal microgrid architectures reported in the literature. These architectures differ in bus type, component integration, control complexity, and suitability for practical applications.

Table 2-1 Comparison of Main Microgrid Architectures

Architecture Type	Main Bus Type	Typical Applications	Main Advantages	Main Limitations
AC Microgrid	AC	Rural electrification, campuses, industrial plants	High compatibility with conventional loads and generators	More conversion stages for DC sources
DC Microgrid	DC	Data centers, telecom systems, DC buildings	Efficient for PV and battery systems with DC loads	Protection and standardization remain more complex
Hybrid AC/DC Microgrid	Both AC and DC	Advanced multi-energy systems, research platforms	High flexibility and efficient integration of mixed loads and sources	Increased control and hardware complexity

As shown in Table 2.1, AC microgrids remain the most practical architecture for conventional rural electrification projects because they are directly compatible with diesel generators, most household and institutional loads, and standard protection devices. Hybrid AC/DC systems provide additional flexibility, but at the expense of greater design and control complexity.

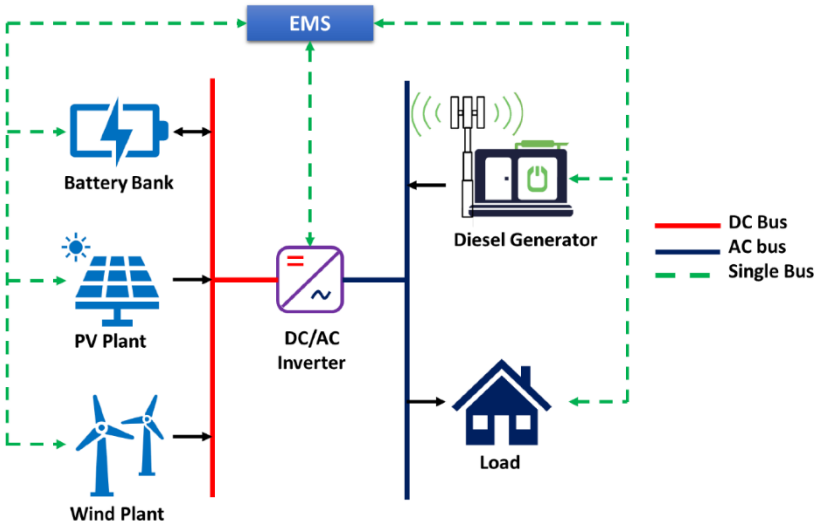


Figure 2-1 A coordinated system in which power electronics, control layers, monitoring systems, and prioritized loads interact continuously to maintain reliable operation.

A conceptual architectural as shown in figure 2-1 is useful here because it gives the reader a visual summary of the microgrid building blocks before the chapter moves into component-specific discussions.

2.2 Hybrid PV–Diesel–Battery Systems for Rural Electrification

2.2.1 Motivation for Hybridization

Historically, many remote communities have relied on diesel-only power plants or small stand-alone generators. While robust and dispatchable, diesel generation is associated with high fuel costs, noisy operation, and significant greenhouse gas and pollutant emissions. The cost and logistical complexity of fuel delivery—especially in conflict-affected or geographically isolated regions—can further undermine the reliability and economic viability of such systems.

Integrating PV and battery storage can significantly reduce diesel fuel consumption and operating costs. Simulation-based studies show that introducing PV and storage into diesel mini-grids can cut fuel use by 20–60%, depending on local solar resources, load profiles, and system sizing. Batteries enable higher PV penetration by absorbing surplus generation during midday and supplying energy during evening peaks or cloudy periods. Hybrid systems therefore

combine the **low operating cost and cleanliness of PV**, the **flexibility and fast response of batteries**, and the **firm capacity of diesel generators**.

2.2.2 Typical Architectures

Hybrid PV–Diesel–Battery systems are usually implemented as AC microgrids with the following:

- PV arrays connected through one or more grid-tied inverters (often with multiple MPPT channels).
- A battery bank interfaced via a bidirectional PCS, capable of charging from the AC bus and discharging to it.
- One or more diesel gensets controlled by genset controllers that can start/stop automatically and share loads.
- An AC distribution board that supplies loads and hosts protection devices and metering.

Different control philosophies exist. In some systems, the diesel generator is always running and PV plus battery serve mainly as fuel-saving resources ("diesel-dominant" operation). In more advanced schemes, the system can operate in **PV-dominant** or **battery-dominant** modes during the day, with diesel used only as backup when renewable output is insufficient or storage is depleted. The latter approach is aligned with the strategy adopted in this thesis, where PV and battery have priority, and diesel is invoked primarily to protect critical loads (F1).

2.2.3 Sizing Considerations

Designing hybrid systems involves jointly sizing PV capacity, battery capacity, and diesel generator rating. Key criteria include:

- **Load coverage and reliability:** ensuring that critical loads can be served across typical and extreme conditions.
- **Fuel savings vs. capital cost:** balancing the extra investment in PV and storage against long-term reductions in fuel consumption and maintenance.
- **Battery cycling and lifetime:** avoiding oversizing that leads to under-utilization, or under sizing that causes excessive cycling and degradation.

- **Diesel loading constraints:** maintaining minimum loading (e.g., ≥30–40% rated power) to avoid inefficient operation and wet-stacking.

Multi-objective optimization studies employ genetic algorithms and other techniques to identify trade-offs between total cost, fuel use, emissions, and reliability. Although such formal optimization is beyond the scope of this bachelor project, the design presented in later chapters qualitatively incorporates these considerations, aiming for a realistic and educational configuration rather than purely theoretical

Rural power systems may be configured in different ways depending on cost, desired reliability, and resource availability. A comparison between common system types helps explain why hybrid PV–Diesel–Battery systems are increasingly preferred. [25]

Table 2-2 Comparison of Common Rural Power Supply Configurations

System Configuration	Renewable Contribution	Fuel Dependence	Reliability	Complexity	Suitability for Rural Communities
Diesel Only	None	Very High	High if fuel is available	Low	Limited due to cost and emissions
PV Only	High	None	Low to medium due to intermittency	Low	Limited unless loads are very small
PV-Battery	High	None	Medium to high	Medium	Good for small autonomous systems
PV-Diesel	Medium	Medium	High	Medium	Good but less flexible than storage-based systems
PV-Diesel-Battery	High	Low to medium	Very high	High	Excellent for resilient rural electrification

Table 2-2 indicates that the PV–Diesel–Battery configuration provides the most balanced solution in terms of renewable penetration, operational flexibility, and continuity of supply. This makes it particularly suitable for remote or weak-grid communities where diesel backup remains necessary but should be minimized.

2.3 Load Characterization and Priority-Based Management

2.3.1 Categories of Loads in Rural Microgrids

In rural mini-grids and microgrids, loads are diverse and often include a mix of residential, commercial, and community services such as schools, health centers, and water pumping. For design and control purposes, loads are typically categorized as:

- **Critical loads:** whose interruption can cause serious safety or economic consequences, e.g., hospital equipment, water supply pumps, communication systems, and essential lighting.
- **Important (or priority) loads:** desirable to supply under normal conditions but tolerable to curtail for limited periods, such as small businesses, refrigeration for shops, or some public services.
- **Non-critical or deferrable loads:** loads that can be shed with minimal impact, including certain household appliances, entertainment loads, or discretionary industrial processes.

The distribution of these categories depends heavily on the socio-economic context and the objectives of the microgrid project. Studies in off-grid electrification programs emphasize the importance of involving communities in defining what is “critical” versus “non-critical,” as this can vary by region and culture.

2.3.2 Load Profiles and Temporal Variability

Load demand in rural communities often exhibits strong daily and seasonal patterns. Typical daily profiles show low consumption during late night, increasing in the morning with domestic activities, peaks around midday in commercial centers or for irrigation pumping, and evening peaks due to residential lighting and entertainment. Seasonal variations arise from agricultural cycles, school terms, and climate-related needs (e.g., fans in hot seasons).

Accurate characterization of these profiles is crucial for sizing PV and storage and for designing EMS logic. Many microgrid studies use measured or synthetic load profiles derived from surveys, smart meter data, or proxies such as similar communities. In the context of this thesis, the three feeders F1–F3 are conceptually mapped to groups of loads with different priority

levels but share a common daily pattern, which is used in EMS scenarios described in later chapters. [26]

2.3.3 Priority-Based Load Shedding

Priority-based load shedding is a widely adopted mechanism in microgrids to maintain system stability and protect critical loads during resource shortages or faults. Loads are grouped into priority classes, and the EMS or microgrid controller disconnects lower-priority groups when available generation plus storage is insufficient.

The resilience-oriented microgrid in Cordova, Alaska, for example, uses smart meters and control logic to classify customers into four groups from critical to flexible. Under extreme conditions, the system automatically disconnects flexible loads while keeping critical services such as hospitals and emergency infrastructure energized. Similar approaches are documented in rural mini-grid programs and in simulation studies that implement hierarchical load shedding schemes. [27]

The present project adopts a three-tier scheme with feeders F1 (critical), F2 (important), and F3 (non-critical). The EMS ensures that F3 is shed first when deficits occur, followed by F2 if necessary, while F1 is preserved as long as PV, battery, or diesel resources can sustain it. This strategy is consistent with best practices observed in resilient microgrids and aligns with the needs of rural coastal communities in Yemen, where health centers and water supply have higher priority than some residential or commercial loads. [28]

One of the defining features of resilient hybrid microgrids is the classification of loads into priority groups. This enables the EMS to preserve essential services when generation resources are insufficient to satisfy all demand simultaneously.

Table 2-3 Classification of Loads in Rural Hybrid Microgrids

Load Class	Typical Examples	Tolerance to Interruption	Operational Priority	Typical EMS Action
Critical	Health center devices, communication systems, essential lighting, water pumping	Very low	Highest	Must remain energized whenever possible
Important	Refrigeration, workshops, community facilities, small businesses	Medium	Intermediate	May be curtailed during severe deficits
Non-Critical	Entertainment loads, comfort appliances, discretionary domestic loads	High	Lowest	Shed first during shortages

Table 2-3 demonstrates that the categorization of loads is not merely an operational convenience, but a central design principle in community-scale microgrids. Such prioritization is particularly important in the present project, where feeders F1, F2, and F3 reflect different levels of service importance.

2.4 Photovoltaic Technologies, Modeling, and Design in Microgrids

2.4.1 PV Technologies and Module Characteristics

PV modules are typically based on crystalline silicon (mono- or multi-crystalline) or thin-film technologies (e.g., CdTe, CIGS). Crystalline silicon dominates the market due to its high efficiency, maturity, and declining cost. Key datasheet parameters include rated power at standard

test conditions (STC), open-circuit voltage VocV_{oc}Voc, short-circuit current IscI_{sc}Isc, maximum power voltage VmpV_{mp}Vmp, maximum power current ImpI_{mp}Imp, and temperature coefficients that describe how voltage and power vary with cell temperature.

The Jinko Tiger Neo 72HL4-BDV family, as an example of modern high-power modules, employs N-type mono-crystalline cells with multi-busbar technology and half-cut cell design to reduce resistive losses and improve shade tolerance. Such modules can achieve efficiencies above 21% and rated powers around 550 W, enabling compact arrays with higher capacity per surface area. These features are particularly useful in space-constrained sites such as rooftops or small ground-mount plots in rural communities.

2.4.2 I–V and P–V Characteristics, Temperature Effects

The electrical behavior of a PV module is often modeled by a single-diode equivalent circuit. The resulting current–voltage (I–V) curve exhibits a characteristic "knee" where maximum power occurs. The maximum power point (MPP) depends on irradiance and cell temperature: increasing irradiance raises both current and power, while higher temperatures mainly reduce voltage and hence power. MPPT algorithms implemented in inverters continuously track the MPP to maximize energy capture. [29]

In microgrid design, temperature coefficients are particularly important for determining worst-case open-circuit voltage at low ambient temperatures. As temperature decreases, VocV_{oc}Voc increases; thus, string sizing must ensure that the sum of module VocV_{oc}Voc at the minimum design temperature remains below the maximum allowed DC input voltage of the inverter (e.g., 1100 V). Conversely, at high temperatures, VmpV_{mp}Vmp decreases, reducing power output and affecting DC bus voltage margins.

2.4.3 String Sizing, Inverter Selection, and Protection

PV arrays are formed by connecting modules in series to form strings and connecting multiple strings in parallel to achieve the desired voltage and power. String design must satisfy several constraints:

- **Voltage constraints:**
    - Voc,string(Tmin)<VDC,max,inverterV_{oc, string}(T_{min}) < V_{DC,max, inverter}Voc,string(Tmin)<VDC,max,inverter.

- Vmp,string(Tmax)V_{mp, string}(T_{max})Vmp,string(Tmax) within the MPPT voltage range of the inverter.

- **Current constraints:**
  - String current not exceeding module and fuse ratings.
  - Aggregated string currents within inverter input and MPPT channel limits.

Protection components include string fuses (typically gPV type), DC disconnects, and surge protection devices. Standards and manufacturer guidelines recommend selecting string fuses rated above the maximum short-circuit current (often with a safety factor) but below the maximum fuse rating specified for the module. Cable sizing must consider current-carrying capacity, voltage drop, installation method, and environmental conditions. [30]

In AC-coupled microgrids, inverters must also comply with grid code requirements such as voltage/frequency ride-through, anti-islanding, and reactive power capabilities. Research on microgrids shows that advanced inverters can support grid-forming or grid-support functions, but in many commercial systems they operate in grid-following mode with external controls handling system-level dynamics. [31]

PV module datasheets provide the basic electrical parameters needed for array configuration, inverter matching, and protection design. Understanding these parameters is essential for translating theoretical solar energy potential into a technically safe and efficient system.

Table 2-4 Key PV Module Parameters Used in Array and Inverter Design

Parameter	Symbol	Engineering Importance	Design Impact
Open-Circuit Voltage	VocV_{oc}Voc	Determines maximum string voltage	Limits number of modules in series
Short-Circuit Current	IscI_{sc}Isc	Required for protection and conductor sizing	Affects fuse and cable selection
Maximum Power Voltage	VmpV_{mp}Vmp	Determines operating voltage of the string	Must match inverter MPPT range
Maximum Power Current	ImpI_{mp}Imp	Used to estimate string current at rated conditions	Helps size inputs and conductors
Power Rating	PmaxP_{max}Pmax	Indicates nominal module output	Used in total array sizing
Temperature Coefficient	—	Reflects sensitivity to ambient conditions	Critical for cold and hot weather design

Table 2-4 identifies the most important PV module parameters typically used in engineering calculations. These variables later become central in determining the number of modules per string.

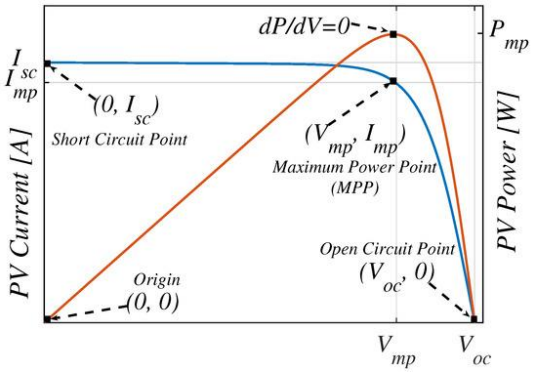


Figure 2-2 Series–parallel arrangement of PV modules into strings and arrays for inverter-based solar generation systems.

As shown in figure 2.2 the allowable PV range, and the current-carrying requirements of cables and protective devices.

2.5 Battery Energy Storage Systems for Microgrids

2.5.1 Battery Technologies and Comparative Overview

Several battery technologies are employed in microgrids, each with distinct characteristics in terms of cost, performance, and environmental impact. [32]

- **Lead-acid batteries** (flooded or valve-regulated) are mature and low-cost but have relatively low energy density, limited cycle life (often 500–2000 cycles at moderate depth of discharge), and require careful temperature management.
- **Lithium-ion batteries** based on NMC or NCA chemistries offer higher energy density and efficiency but can present safety concerns due to thermal runaway if not properly managed.
- **Lithium-iron-phosphate (LiFePO₄ or LFP)** batteries provide excellent thermal stability, long cycle life (often 4000–6000 cycles at 80% DOD), and high round-trip efficiency (~90–96%), at the expense of slightly lower energy density than some NMC systems.

For stationary microgrid applications, safety, cycle life, and total cost of ownership (TCO) are often more important than volumetric energy density. LFP batteries are therefore increasingly favored in BESS deployments due to their robustness and reduced risk of fire, especially in remote or harsh environments where fire-fighting resources may be limited.

2.5.2 High-Voltage Modular Battery Systems

High-voltage modular batteries, such as BYD Battery-Box Premium HVS/HVM, are specifically designed for integration with high-voltage PCS units and hybrid inverters. In these systems, several modules are stacked in series to reach nominal voltages between roughly 150 V and 400 V, while parallel connections of stacks allow capacities ranging from a few kilowatt-hours up to tens of kilowatt-hours or more. [33]

The HVM series, for instance, uses 51.2 V modules of 2.76 kWh each; stacks of 3–8 modules in series provide usable energies between approximately 8.3 kWh and 22.1 kWh at nominal voltages of 153–409 V. Up to three stacks can be paralleled to reach capacities around 66 kWh. This modularity simplifies scaling and allows later expansion without replacing existing batteries.

Although the specific bank used in this thesis is modeled as 2P×12S of 51.2 V, 100 Ah modules (approx. 614 V nominal, ≈123 kWh), the underlying concept is similar: modular high-voltage LFP units interfaced to a 100 kW PCS. The design adopts a SOC window of 30–90% to limit degradation while providing approximately 73.7 kWh of usable energy, sufficient to support the critical feeder for over an hour at maximum loading.

2.5.3 SOC, DOD, and Degradation Considerations

State of charge (SOC) represents the ratio of remaining capacity to nominal capacity. Depth of discharge (DOD) is complementary (DOD = 100% − SOC). Battery lifetime is highly sensitive to DOD, cycling frequency, temperature, and C-rate (charge/discharge rate).

Empirical studies and manufacturer data show that limiting the DOD (e.g., to 60–70%) can significantly increase the number of cycles before end-of-life. LFP batteries commonly offer 4000–6000 cycles to 80% remaining capacity when cycled at moderate DOD and operated within recommended temperature ranges. For this reason, EMS strategies often enforce SOC boundaries such as 20–80% or 30–90% and avoid unnecessary cycling to maintain long-term performance.

In microgrid EMS design, SOC is treated as a key state variable. Controllers may reserve a portion of battery capacity as spinning reserve to handle contingencies, or they may prioritize charging to high SOC levels when weather forecasts predict extended cloudy periods. Optimization-based models sometimes include degradation costs proportional to throughput (kWh cycled) to discourage aggressive cycling.

2.5.4 Role of Batteries in PV–Diesel Microgrids

The role of BESS in PV–Diesel microgrids has been extensively studied. For example, a recent analysis demonstrates that integrating a battery storage system into a PV–Diesel microgrid can enable peak shaving, reduce required diesel generator capacity, and substantially cut fuel consumption and CO_2 emissions compared to diesel-only or PV-Diesel configurations. Other works show that batteries allow higher PV penetration and more flexible diesel scheduling, enabling generators to operate at more efficient loading levels and reducing start-stop cycles.

Because the performance of a hybrid microgrid strongly depends on its storage technology, a comparison between available battery chemistries is necessary to justify the growing preference for lithium-based systems, particularly $LiFePO_4$.

Table 2-5 Comparison of Battery Technologies Used in Microgrids

Battery Technology	Energy Density	Cycle Life	Safety	Typical Cost Trend	Suitability for Stationary Microgrids
Lead-Acid	Low	Low to medium	Moderate	Low	Suitable for low-cost basic systems
NMC Lithium-Ion	High	High	Moderate	Medium to high	Suitable for compact high-performance applications
$LiFePO_4$	Medium	Very high	High	Medium	Highly suitable for safe stationary BESS

Table 2-5 shows that $LiFePO_4$ batteries offer an excellent balance between cycle life, safety, and suitability for stationary applications. These characteristics explain why they are now widely used in high-voltage modular storage systems for microgrids.

2.6 Diesel Generator Technology and Genset Controllers

2.6.1 Diesel Generators in Isolated and Hybrid Systems

Diesel generators provide fast, dispatchable power and have long been the backbone of isolated power systems such as island communities, remote mining sites, and rural mini-grids.

Their advantages include high reliability, maturity, and the ability to operate independently of weather conditions. However, drawbacks include fuel costs (which can dominate operating expenses), emissions, noise, and dependence on fuel logistics—issues that are particularly acute in fragile or conflict-affected regions like Yemen

Operating diesel gensets efficiently requires respecting minimum loading thresholds (often 30–40% of rated power) to avoid incomplete combustion, carbon deposits, and reduced lifetime. Frequent start-stop cycles can also increase wear and maintenance costs. Hybrid microgrids seek to reduce generator runtime and improve loading profiles by coordinating diesel with PV and batteries through EMS strategies.

2.6.2 Genset Controllers and DSE Solutions

Modern gensets are typically equipped with digital controllers that handle engine start/stop sequences, protection (over-current, over-voltage, under-frequency, etc.), synchronization to an AC bus, and load sharing when multiple units operate in parallel. Deep Sea Electronics (DSE) produces a range of such controllers (e.g., DSE7320, DSE8610) widely used in industrial and microgrid applications.

A case study published by DSE illustrates the use of their controllers in a multi-energy microgrid where diesel gensets operate alongside renewables and storage. The controllers communicate with a supervisory system to receive start/stop and power setpoint commands, while providing measurements and status information. This arrangement closely parallels the architecture adopted in this thesis, where a DSE unit controls the 100 kW diesel generator and interfaces with the PLC and EMS logic.

2.6.3 Diesel Scheduling and Optimal Loading

Research on diesel dispatch in hybrid microgrids focuses on minimizing fuel consumption, maintenance cost, and emissions while satisfying reliability constraints. Some studies propose multi-objective optimization frameworks that simultaneously size PV, battery, and diesel

capacity and determine operating policies that maintain diesel loading near an optimal percentage (e.g., 70–80%) where efficiency is highest.

In the present project, the EMS uses simpler rule-based logic but still respects the principle of avoiding very low diesel loading. The generator is primarily used when PV and battery cannot sustain F1, and it is typically dispatched to support a reasonable share of the critical load rather than being kept idling at minimal power. This approach aligns with insights from optimization studies while remaining implementable on PLC and DSE hardware without heavy computational requirements.

2.7 Energy Management Systems in Hybrid Microgrids

2.7.1 Role and Functions of the EMS

The Energy Management System (EMS) is responsible for coordinating the operation of distributed energy resources, storage, and controllable loads to achieve technical and economic objectives. In hybrid PV–Diesel–Battery microgrids, key EMS functions include:digital-library.

- Maintaining real-time power balance between generation, storage, and loads.
- Enforcing limits on battery SOC, inverter and PCS ratings, and generator operating ranges.
- Prioritizing critical loads and executing load shedding or reconnection of lower-priority feeders.
- Minimizing fuel consumption and operating costs by maximizing the use of PV and storage.
- Logging data and providing information to operators and higher-level decision makers.

The EMS typically operates on a timescale of seconds to minutes, sitting above local control loops (e.g., inverter current controllers or engine governors) and issuing setpoints or discrete commands such as start/stop signals and breaker operations.

2.7.2 Rule-Based and Heuristic EMS Strategies

Rule-based EMS strategies are among the most commonly implemented approaches in real microgrids due to their simplicity, transparency, and ease of implementation on PLC platforms.

These strategies encode human-designed rules such as:

- If SOC > 70% and PV power is available, supply all loads and charge the battery.
- If SOC < 30% and PV power is insufficient to support critical loads, start the diesel generator.
- If total available power (PV + battery discharge + diesel) is less than total demand, shed non-critical loads first.

Such rules can be represented as state machines or decision trees. They are straightforward to validate and communicate to operators, who may be reluctant to rely on black-box optimization algorithms in critical infrastructure. Several real-world microgrid demonstrations report successful deployment of rule-based EMSs in remote communities, emphasizing their robustness and maintainability.

The EMS implemented in this project follows a rule-based philosophy. It utilizes measured PV power, battery SOC, and feeder demands to choose between states such as “All feeders ON,” “F3 shed,” “F2 and F3 shed,” or “Diesel ON,” according to predefined thresholds. These rules are encoded in MATLAB/Simulink State flow charts, translated into PLC logic, and mirrored in the Python simulation tool, ensuring consistent behavior across simulation and implementation.

2.7.3 Optimization-Based EMS Approaches

In contrast, optimization-based EMS strategies formulate the dispatch problem as an optimization problem subject to technical constraints. Common objective functions include minimization of fuel cost, maintenance cost, or a combination of cost and emissions. Constraints capture power balance, SOC dynamics, generator minimum up/down times, and operational limits.

Several works employ mixed-integer linear programming (MILP) to co-optimize generator on/off schedules and battery charge/discharge profiles over planning horizons of 24 hours or more. These approaches can theoretically yield globally optimal schedules under given forecasts of PV and load. Other authors use dynamic programming or heuristic optimization (e.g., genetic algorithms, particle swarm optimization) when non-linearities make MILP formulations difficult.

Model predictive control (MPC) extends these ideas by solving a finite-horizon optimization problem at each time step, incorporating updated states and forecasts. For PV–Battery microgrids, MPC has been shown to manage SOC more effectively and reduce curtailment,

particularly when high-quality forecasts are available. However, these methods require more computational resources and reliable communication, which may be challenging to guarantee in rural microgrid deployments.

Although this thesis does not implement an optimization-based EMS, the rule-based strategy is designed in a way that aligns qualitatively with many optimization results: PV utilization is maximized, battery SOC is maintained within a healthy window, and diesel is used primarily as a last resort to protect critical loads. This makes the system a realistic platform for future research where optimization algorithms could be tested against the existing rule-based baseline.

2.7.4 EMS for PV–Diesel–Battery Systems in the Literature

Specific studies on PV–Diesel–Battery microgrids propose EMS algorithms tailored to the characteristics of these systems. One work presents a rule-based controller that minimizes diesel operation by starting the generator only when the battery SOC drops below a threshold and PV cannot cover demand; the diesel then charges the battery to a higher SOC before shutting down to avoid frequent cycling. Another study investigates an MPC-based strategy where generator dispatch, battery power, and load shedding are jointly optimized to maintain frequency and voltage while reducing fuel consumption. [34]

Common features across these works include:

- Use of **SOC thresholds** to prevent deep discharges and ensure reserve capacity.
- Enforcement of **minimum generator loading** and **minimum on/off times**.
- Implementation of **load shedding schemes** that prioritize critical loads.
- Consideration of **forecast information** when available.

The literature on microgrid EMS proposes a wide spectrum of control philosophies, ranging from simple heuristic rules to advanced optimization and artificial intelligence-based methods.

Table 2-6 Comparison of Common EMS Strategies in Hybrid Microgrids

EMS Strategy	Main Principle	Strengths	Weaknesses	Suitability for This Thesis
Rule-Based	Thresholds and logical conditions	Simple, transparent, easy to implement	May be suboptimal	Very high
Optimization-Based	Solves dispatch optimization problem	High theoretical efficiency	Computationally demanding	Medium
Model Predictive Control	Forecast-based rolling optimization	Adaptive and powerful	Requires accurate models and forecasts	Medium
Fuzzy Logic	Linguistic rules and inference	Flexible in uncertain conditions	Harder to tune systematically	Medium
AI / Machine Learning	Data-driven decision making	Potentially highly adaptive	Requires data and training	Low to medium

Table 2-7 indicates that no single EMS strategy is universally superior. Instead, the most appropriate method depends on system size, data availability, hardware constraints, and the need for transparency. For this reason, a rule-based approach is especially suitable for practical educational and field-oriented systems.

2.8 Monitoring, SCADA, and IoT Architectures for Microgrids

2.8.1 Evolution from Traditional SCADA to IoT

Traditional Supervisory Control and Data Acquisition (SCADA) systems have long been used in power networks to collect measurements, display alarms, and enable remote control of substations and plants. These systems are typically built around centralized servers, proprietary protocols, and fixed HMI software. [35]

With the advent of the Industrial Internet of Things (IoT), microgrid monitoring architectures are increasingly adopting more flexible, modular, and web-based technologies. Instead of rigid SCADA platforms, engineers can deploy edge gateways, message brokers, and web dashboards that leverage open standards and cloud services. This shift allows easier integration of heterogeneous devices, remote access via secure VPNs, and rapid development of custom visualizations.

2.8.2 Role of PLCs and Industrial Gateways

Programmable logic controllers (PLCs) remain central in many microgrids because of their robustness, deterministic behavior, and compatibility with industrial fieldbuses and protocols such as Modbus, Profibus, and CAN. PLCs can acquire data from sensors, inverters, PCS units, and genset controllers, execute local protection and control logic, and provide aggregated information to higher-level systems.

Industrial gateways and 4G/LTE routers complement PLCs by offering IP connectivity, VPNs, and sometimes built-in compute resources to run IoT applications. For example, some commercial routers allow hosting Node-RED instances directly on the device, enabling local dashboards and control flows even with intermittent internet connectivity. This architecture is particularly appealing for rural microgrids where connectivity may be unreliable. [36]

2.8.3 Node-RED for Microgrid Dashboards

Node-RED is an open-source, flow-based programming tool built on Node.js that enables users to connect devices, APIs, and online services in a simple graphical way. It has gained popularity for rapid prototyping of IoT applications, including energy monitoring dashboards.

In the context of microgrids, Node-RED can be used to:

- Poll data from PLCs, inverters, and batteries using Modbus/TCP, HTTP, or MQTT.
- Aggregate and log time-series data in databases such as Influx DB.
- Present real-time dashboards in web browsers, including charts of PV output, load, SOC, and generator status.
- Implement simple control logic, such as threshold-based alerts or manual override buttons.

Community examples show Node-RED flows that monitor solar PV systems, calculate surplus energy, and control loads such as water heaters or EV chargers accordingly. These examples provide a strong precedent for using Node-RED in the present project to visualize and interact with the microgrid EMS.

In this thesis, Node-RED is deployed on an industrial 4G/LTE gateway and configured to exchange data with the PLC and EMS via Modbus or HTTP interfaces. The HTML dashboard displays feeder powers, PV output, battery SOC, and generator operation, and allows manual enabling or disabling of F2 and F3. The Python scenario tool and MATLAB simulations reflect the same logic, providing a consistent bridge between design, simulation, and real-time monitoring. [37]

2.8.4 Web-Based HMIs and User Interaction

Web-based HMIs offer several advantages over traditional desktop-bound HMIs: platform independence, ease of deployment, and accessibility from multiple devices including smartphones and tablets. For rural microgrids, this can simplify operation and maintenance by allowing technicians or project managers to access the system from different locations without installing specialized software.

However, web-based HMIs must be carefully designed to present information clearly and avoid overwhelming users with excessive detail. Common design principles include:

- Prioritizing key performance indicators (KPIs) such as total load, PV output, SOC, and generator status on the main screen.
- Using color coding to indicate normal, warning, and alarm states.
- Providing drill-down views for advanced users to inspect detailed logs or component statuses.

IoT-enabled microgrids are typically structured as layered architectures in which local devices, control hardware, communication gateways, and user interfaces work together to provide supervisory awareness and control.

Table 2-7 Layers of an IoT-Based Monitoring Architecture for Hybrid Microgrids

Architecture Layer	Typical Components	Main Function
Field Layer	Inverters, PCS, generators, sensors, meters	Data acquisition and actuation
Control Layer	PLCs, local controllers, relays	Local control and coordination
Gateway Layer	Protocol converters, industrial routers	Communication bridging and remote access
Application Layer	Node-RED, databases, dashboards	Visualization, data processing, alarms
User Layer	Browser, tablet, operator PC	Human interaction and supervision

Table 2-8 clarifies that IoT monitoring is not limited to a dashboard. It is a multi-layer architecture that starts from field measurements and extends all the way to user interfaces and remote access services.

2.9 Communication Protocols and Cyber-Security Considerations

2.9.1 Communication Protocols in Microgrids

Microgrids rely on a variety of communication protocols to connect field devices with controllers and monitoring systems. Commonly used protocols include:

- **Modbus (RTU/TCP):** A widely adopted open protocol for industrial automation; simple and easy to implement on PLCs, inverters, and BESS. Modbus/TCP is frequently used over Ethernet for microgrid monitoring.
- **CAN/RS-485 based protocols:** Used by many battery systems and genset controllers; provide robust communication over short to medium distances.
- **Proprietary protocols:** Vendor-specific interfaces for advanced features or configuration.
- **Higher-level protocols:** Such as MQTT or HTTP/REST for integration with IoT platforms and web dashboards.

Real-time simulation studies using platforms like RTDS highlight how Modbus communication can be used to link microgrid controllers with hardware-in-the-loop setups, providing realistic testing of control strategies.

In the microgrid of this thesis, Modbus/TCP is the primary protocol between the PLC, DSE controller, Node-RED, and potentially the EMS running on a supervisory PC. This choice reflects a balance between ubiquity, simplicity, and support from commercial devices.

2.9.2 Cyber-Security and Reliability

As microgrids become more connected, they also become more exposed to cyber-security threats and communication failures. Potential risks include unauthorized control actions, data tampering, denial-of-service attacks, or simple misconfigurations that could compromise system stability or safety.

Best practices for cyber-secure microgrids include:

- Using **VPN tunnels** and firewalls for remote access to PLCs and gateways.
- Applying **authentication and authorization** mechanisms for web dashboards and APIs.
- Segregating control networks from public internet access through proper network zoning.
- Implementing **fail-safe defaults**, such that loss of communication leads to safe operation (e.g., generators fall back to local control; feeders default to a conservative state).

Although a complete cyber-security assessment is beyond the scope of this thesis, acknowledging these concerns is essential, particularly if the design is to be replicated in real installations.

Communication is a critical enabler of smart microgrids, especially when multiple controllers and software layers must exchange measurements, alarms, and commands.

Table 2-8 Communication Protocols Commonly Used in Hybrid Microgrids

Protocol	Communication Medium	Typical Application	Advantages	Limitations
Modbus RTU	RS-485 serial	Meters, controllers, field devices	Simple and widely supported	Lower speed
Modbus TCP	Ethernet	PLCs, gateways, dashboards	Easy IP integration	Requires network infrastructure
CAN	Bus network	Batteries and embedded controllers	Robust and fast	More specialized implementation
MQTT	IP / broker-based	IoT and cloud messaging	Lightweight and scalable	Depends on broker services
HTTP/REST	IP / web	APIs and dashboards	Universal support	Less deterministic for control

Table 2-9 highlights that no single communication protocol is sufficient for all tasks. Instead, practical microgrids often combine industrial protocols such as Modbus with internet-oriented protocols for dashboard integration and remote access.

2.10 Case Studies of Hybrid Microgrids for Rural and Coastal Communities

2.10.1 International Case Studies

Numerous case studies illustrate the deployment of hybrid microgrids in remote and coastal communities around the world. The Rocky Mountain Institute (RMI) has compiled profiles of renewable microgrids on islands and in remote communities, showing how PV, wind, batteries, and diesel generators can be combined to reduce fuel consumption and improve reliability. These projects highlight key design choices such as resource assessment, community engagement, tariff structures, and technical architectures.

In some Pacific Island microgrids, PV and BESS have been introduced to displace a large portion of diesel consumption, with EMS strategies that prioritize renewable utilization while maintaining system frequency and voltage. Similar projects in coastal communities in Southeast Asia and the Caribbean adopt modular microgrid architectures that can be scaled as demand grows.

These international case studies provide validation that hybrid PV–Diesel–Battery systems, combined with load prioritization and modern control, can deliver reliable power in isolated settings similar to rural coastal Yemen.

2.10.2 Rural Electrification in Yemen

As discussed in Chapter 1, Yemen faces severe challenges in its electricity sector, with large portions of the population lacking reliable grid supply and relying instead on small diesel generators or informal networks. UNDP and other organizations have implemented solar microgrid projects that provide clean electricity to small businesses and households, demonstrating significant socio-economic benefits, particularly for women and youth entrepreneurs.

Media reports describe how groups of women known as “microgrid girls” have been trained to install, operate, and maintain solar microgrids in rural villages, empowering them economically and socially while providing essential services to their communities. However, these projects also reveal limitations such as limited battery capacity, the need for effective tariff structures, and challenges in maintenance and spare parts supply.

The design presented in this thesis can be seen as a **technical reference architecture** that could be adapted to such contexts. By explicitly incorporating prioritized feeders F1–F3, the system ensures that critical services—such as health centers, water pumps, and essential household loads—are supplied first, while less critical loads can be curtailed when resources are scarce. The inclusion of a diesel generator as a backup provides resilience against extended periods of low irradiance or unexpected load surges, an important consideration in fragile and uncertain environments like Yemen.

2.10.3 Lessons for the Present Project

From the reviewed case studies, several lessons inform the project design:

- **Community engagement and load prioritization** are crucial. Technical design must reflect local perceptions of what constitutes critical services.
- **Hybridization with diesel** remains necessary in many contexts to ensure reliability, but careful EMS design can dramatically reduce runtime and fuel consumption.
- **Modularity and scalability** in PV and battery systems facilitate staged deployment and later expansion as funding and demand evolve.
- **Robust, simple control and monitoring solutions** (PLCs, DSE controllers, Node-RED dashboards) are favored over highly complex systems that may be difficult to support in remote areas.

These insights reinforce the suitability of the PV–Diesel–Battery microgrid and EMS developed in this thesis as a realistic and educational example of how hybrid microgrids can support rural coastal communities in Yemen and similar regions.

Case studies provide an important bridge between theory and practice because they demonstrate how hybrid microgrid concepts are implemented under real technical, economic, and geographical constraints.

Table 2-9 Summary of Selected Rural and Coastal Hybrid Microgrid Case Studies

Location / Context	Main Resources	Main Objective	Key Technical Lesson	Relevance to This Thesis
Rural Yemen	PV, battery, local backup	Community electrification	Local relevance of decentralized solar systems	High
Remote island community	PV, diesel, battery	Reduce diesel dependence	Hybridization increases resilience	High
Mini-grid development program	Mixed renewables and backup	Expand energy access	Modularity and prioritization are essential	High

Table 2-10 shows that despite differences in scale and location, many rural and coastal microgrid projects converge on the same principles: hybridization of resources, prioritization of critical services, and reliance on simple but reliable control strategies.

Chapter 3

Requirements Analysis and Modeling

3. Chapter 3

3.1 Project Context and Problem Statement

3.1.1 Electricity Access Challenges in Yemen

Yemen faces one of the most severe electricity crises in the world, driven by years of conflict that have damaged generation assets, transmission lines, and distribution networks. Only around 40% of households had access to electricity in 2021, and many rural communities experience daily blackouts that can last 20 hours or more. The national grid now serves only limited urban areas such as Aden and nearby governorates, while large parts of the country, especially in the north and west, rely on fragmented and unreliable local solutions.

In this context, rural households, farmers, and small businesses have historically depended on small diesel generators or private diesel-based mini-grids to meet basic energy needs. However, diesel fuel in Yemen is expensive, subject to supply disruptions, and highly sensitive to geopolitical constraints, making electricity both costly and unreliable for low-income communities. Fuel prices have risen dramatically over the past decade; for example, the price of 20 liters of diesel increased from approximately 4 USD in 2015 to 35 USD in 2022, placing significant financial pressure on rural users. These conditions create a clear need for alternative energy solutions that can improve reliability while reducing dependence on imported fossil fuels.

3.1.2 Role of Decentralized Solar Microgrids

Against this backdrop, decentralized solar microgrids have emerged as a promising pathway to restore and expand electricity access in Yemen's rural and peri-urban communities. Several projects led by development agencies have demonstrated that community-scale solar microgrids can provide reliable power for households, shops, and social services while simultaneously creating local employment opportunities. For example, UNDP-supported solar microgrids in frontline communities have reduced the cost of energy by up to 65% compared with diesel-based supply, making electricity substantially more affordable for local users. [38]

Solar microgrids have also shown strong social impacts. In multiple Yemeni villages, women-led cooperatives operate and maintain village-scale solar plants, using the resulting income to support their families and invest in local enterprises. These initiatives highlight how clean energy projects can simultaneously address energy poverty, gender inclusion, and livelihood creation when they are designed with community participation. Nevertheless, most

existing projects focus on relatively simple PV-battery configurations without sophisticated energy management or integration of dispatchable diesel generation.[39]

3.1.3 Need for Hybrid PV–Diesel–Battery Solutions

While pure PV-battery systems can significantly reduce fuel consumption, they are inherently constrained by solar resource variability and the finite capacity of storage. In many rural communities, critical services such as health centers, water pumping, and communications require continuous, high-reliability power that cannot be fully guaranteed by renewable resources alone. Hybrid systems that combine photovoltaic generation, battery energy storage, and diesel gensets can provide a more balanced trade-off between reliability, operating cost, and environmental impact. [40]

Hybrid PV–diesel–battery microgrids are designed to maximize solar energy utilization during the day while retaining diesel as a firm backup source for extended cloudy periods, evening peaks, or emergencies. Batteries absorb short-term fluctuations and allow higher PV penetration by shifting surplus energy from midday to evening demand hours. Simulation and techno-economic studies in various countries show that such hybrid systems can reduce diesel fuel consumption by 20–60% compared with diesel-only operation, while maintaining or improving supply reliability for rural customers. For Yemen, where fuel logistics are costly and the national grid is unlikely to recover quickly, hybrid microgrids offer a particularly attractive architecture.

3.1.4 Microgrids and Rural Electrification in Yemen

Beyond standalone projects, hybrid microgrids fit into broader national and regional strategies for rural electrification in fragile settings. International studies on rural mini-grids identify hybrid renewable energy systems as a key tool for achieving universal access in off-grid areas, particularly where extending central networks is economically or politically infeasible. In Yemen, techno-economic assessments of mini-grid solutions indicate strong potential for solar-based microgrids to deliver Tier 3–4 access levels at competitive life-cycle cost, especially in villages already hosting diesel generators and partial infrastructure. [41]

Case studies from Yemeni communities demonstrate that carefully designed mini-grids can dramatically improve service quality compared with ad-hoc individual generators and household solar kits. However, these studies also stress the importance of robust technical design, proper tariff structures, and sustainable governance models to avoid underutilization, technical failures,

or unsustainable business models. The present project situates itself within this context by targeting a realistic hybrid microgrid architecture for a rural coastal community, taking into account both technical and socio-economic constraints highlighted in the literature.

3.1.5 Limitations of Conventional Diesel-Only Operation

Traditional diesel-only power plants, which remain common in Yemeni villages, exhibit several structural weaknesses that motivate the transition toward hybrid systems. First, the levelized cost of electricity is dominated by fuel expenditures, which are highly volatile and vulnerable to supply interruptions, especially under conditions of import restrictions or conflict. Second, continuous diesel operation at low load factors leads to inefficient fuel use, increased maintenance requirements, and phenomena such as wet stacking, which further degrade generator performance.

Third, diesel power plants provide no inherent buffering or storage capability, meaning that any mismatch between generation and demand is either curtailed or compensated by running multiple generators at suboptimal loading. Finally, diesel combustion contributes to local air pollution and greenhouse gas emissions, which is increasingly important for development programs that aim to align with global climate objectives. These limitations collectively underscore the need for intelligent control strategies that minimize diesel runtime while preserving system reliability—an objective that lies at the core of hybrid PV–battery–diesel microgrid design.

3.1.6 Rationale for an Intelligent EMS-Based Approach

Hybrid microgrids require an energy management system (EMS) capable of coordinating PV generation, battery charge/discharge, and diesel operation according to predefined objectives and constraints. Literature on hybrid renewable systems emphasizes that naive control strategies, such as simple diesel-following or static scheduling, can lead to suboptimal fuel savings, excessive battery cycling, or poor utilization of renewable resources. Advanced EMS implementations often use rule-based logic, optimization algorithms, or model predictive control to balance competing goals like minimizing fuel consumption, maintaining battery health, and ensuring uninterrupted supply to critical loads. [42]

In fragile contexts like Yemen, however, there is a strong practical argument for control schemes that are transparent, explainable, and implementable on industrial hardware such as PLCs and genset controllers. Rule-based EMS designs, when carefully structured and validated

through simulation, can achieve substantial performance improvements while remaining understandable to local engineers and operators. This project therefore focuses on a rule-based EMS that prioritizes PV and battery usage, enforces SOC limits, and only dispatches the diesel generator when necessary to protect critical loads.

3.1.7 From Simulation to Real Implementation

To ensure that the proposed EMS concept is realistic and transferable, the project combines simulation models with representations of actual industrial components. The MATLAB/Simulink and Stateflow environment is used to implement and test the rule-based logic under daily PV and load profiles, with inputs and outputs corresponding to measurable quantities such as PV power, feeder demands, battery state of charge, and generator status. These simulations support the analysis of operational scenarios, including PV surplus, battery support, diesel start/stop events, and feeder shedding or restoration.

At the same time, EPLAN schematics specify the physical architecture of the microgrid, including PV array configuration, DC and AC buses, inverter/charger connections, battery bank, diesel generator, and protection devices. Industrial hardware such as the Siemens S7-1200 PLC and the DSE8610 MKII genset controller are modeled conceptually, with documented digital I/O mapping and Modbus TCP monitoring channels, to show how the EMS logic could be deployed on real equipment. This dual focus on simulation and real-world implementation forms the basis for analyzing requirements in subsequent sections of Chapter 3.

3.1.8 Problem Statement

Given the above context, the central problem addressed in this project can be stated as follows:

How can a hybrid PV–battery–diesel microgrid for a rural coastal community in Yemen be designed and controlled, using an intelligible rule-based energy management system, to prioritize renewable energy use, protect critical loads, and reduce diesel fuel consumption while remaining compatible with industrial PLC and genset controller hardware?

This problem statement encapsulates several intertwined challenges. The system must reliably supply a mix of critical, important, and non-critical loads under highly variable solar irradiance and uncertain fuel availability. It must do so using a hardware architecture that is realistic for deployment in resource-constrained environments, where maintenance capacity, spare parts, and technical training may be limited. Finally, the control strategies must be

transparent and verifiable, enabling engineers and operators to understand and adapt the EMS logic over time.

3.1.9 Specific Objectives of the Project

In line with the broad problem statement, the project pursues the following specific objectives:

1. **Characterize the load demand and priority structure** of a representative rural coastal community, mapping real-world services and consumers to three feeders (F1–F3) with different criticality levels.
2. **Define and size the main generation and storage resources**—a 150 kW PV array, approximately 100–123 kWh high-voltage battery bank, and 100 kW diesel generator—based on realistic technical parameters and manufacturer data.
3. **Develop a rule-based EMS algorithm** that coordinates PV, battery, and diesel resources to minimize fuel consumption while keeping critical loads supplied under a defined SOC operating window.
4. **Implement and test the EMS in MATLAB/Simulink and Stateflow**, using daily PV and load profiles to simulate a variety of operational scenarios and stress conditions.
5. **Map the EMS logic to industrial control hardware**, including PLC ladder logic for the S7-1200 and I/O interactions with the DSE8610 MKII, demonstrating how the concept could be realized in a real microgrid installation.
6. **Design a monitoring and control interface** using Node-RED and an HTML dashboard, and outline how similar functionality would be achieved in the real system via PLC/SCADA and IoT connectivity.
7. **Assess qualitatively the expected benefits and limitations** of the proposed architecture in terms of reliability, fuel savings, operational complexity, and suitability for rural Yemeni contexts.

These objectives guide the requirements analysis and modeling tasks in the remainder of Chapter 3, and they provide a clear link between literature insights, simulation activities, and practical implementation considerations.

3.1.10 Structure of the Remaining Chapter 3

The remainder of Chapter 3 builds upon this context to derive detailed system requirements and models. Section 3.2 defines the system scope and boundaries, distinguishing between physical, control, and simulation domains. Section 3.3 formalizes the load structure and priority modeling, while Section 3.4 characterizes the generation and storage resources at both conceptual and parameter levels. Section 3.5 consolidates electrical and operational requirements, including ratings, protection needs, and operating constraints derived from standards and manufacturer data.

Section 3.6 then translates the high-level objectives into concrete EMS control requirements and logic, followed by Section 3.7, which develops the mathematical representations used in simulation. Section 3.8 addresses communication and integration requirements across MATLAB, Node-RED, PLC, and DSE layers. Finally, Sections 3.9 and 3.10 define verification scenarios and real-system mapping requirements, preparing the ground for the design, implementation, and results presented in subsequent chapters.

3.2 System Scope and Boundary Definition

3.2.1 Definition and Purpose of System Scope

In engineering projects, especially microgrid projects that combine electrical infrastructure, control logic, and digital monitoring, scope definition is not a formal step added for documentation only; it is a design necessity. Conceptual microgrid design guides emphasize that the project team must define goals, requirements, risks, and microgrid boundaries early so that later sizing, control design, and validation tasks remain consistent with the actual problem being solved. In other words, system scope determines what the project is responsible for, which assets are under coordinated control, which interfaces must be modeled, and which simplifications are acceptable for the academic objectives of the work.

For the present project, the scope is deliberately broader than a pure simulation study and narrower than a full field deployment package. It includes a complete hybrid PV–battery–diesel microgrid concept for a rural village application, a rule-based EMS implemented and tested in MATLAB/Simulink logic, an industrially interpretable control layer centered on a Siemens S7-1200 PLC and a DSE8610 MKII generator controller, and a supervisory software layer built

around Node-RED and an HTML monitoring dashboard. At the same time, it does not claim to include final procurement drawings, relay coordination studies, closed-loop hardware-in-the-loop commissioning, or detailed field installation practices beyond the level necessary to show realism and implementation readiness.

The purpose of defining scope in this chapter is therefore fourfold. First, it separates the physical energy system from the digital supervisory layer. Second, it distinguishes between what is modeled explicitly in MATLAB/Simulink and what is represented as a realistic implementation mapping. Third, it clarifies which components participate directly in EMS decisions and which components act only as monitored or commanded devices. Fourth, it establishes a defensible academic boundary so that the claims made in later chapters remain aligned with the evidence produced by the project.

3.2.2 Physical Boundary of the Proposed Hybrid Microgrid

The physical boundary of the proposed microgrid includes all generation, storage, conversion, distribution, switching, and load-group elements that participate in standalone power delivery to the target rural community. According to the EPLAN package prepared for the project, the core system consists of a 150kW photovoltaic array, a battery bank of approximately 123 kWh nominal energy at 600 V DC, an inverter/charger unit interfacing the DC and AC sides, a 100kW diesel generator, a main AC bus, a DC bus, and three feeders classified by priority as F1, F2, and F3. This boundary represents the minimum physical infrastructure required for local generation, short-term storage, source switching, and prioritized load supply under off-grid operation.

Within this physical boundary, the PV array, battery bank, and diesel generator are not treated as isolated devices. They are coordinated sources connected through shared power conversion and bus structures. The DC side groups the PV array, the MPPT charge controller, the battery interface, and the inverter/charger around a common DC bus, while the AC side groups the inverter output, the diesel generator output, and the feeder breakers around the main AC bus. Such a boundary definition is consistent with standard microgrid descriptions in the literature, where a microgrid is understood as a coordinated cluster of distributed energy resources and loads capable of controlled islanded operation within a well-defined electrical perimeter. [43]

The physical boundary does not extend to an upstream utility grid interconnection because the project is intentionally framed as a rural standalone system. Although the PLC input table

includes an optional GRID_AVAILABLE signal for future expansion, the present design does not rely on an external grid as a normal source of energy. This means that the microgrid boundary is drawn around the local autonomous supply system itself, not around a grid-connected point of common coupling. The resulting scope is therefore well suited to off-grid or weak-grid rural electrification scenarios where community resilience depends on local energy autonomy rather than central utility support. [44]

3.2.3 Electrical Boundary and Interface Nodes

A clear electrical boundary requires identifying the main nodes at which power enters, is converted, is switched, and is consumed. In the present design, the first major node is the **PV generation node**, where the 150 kWp PV field feeds the DC combiner and charge control stage. The EPLAN description specifies 266 modules arranged as 14 strings of 19 modules, with approximate values of ≈ Vmp 789V, ≈ oc,coldV 952V, string current around 13.25 A, and total short-circuit current around 190 A. These values define the electrical envelope of the renewable source and therefore determine the voltage class, insulation requirements, DC protection level, and compatibility with the selected inverter/charger and DC bus. [45]

The second major node is the **DC bus**, which acts as the coupling point between the PV generation branch, the battery branch, and the inverter/charger branch. The DC bus is important not only physically but conceptually, because it is the point at which renewable production, battery charging/discharging, and inverter demand interact. From the standpoint of system scope, the DC bus lies fully inside the project boundary. Its voltage behavior is not modeled in electromagnetic detail in the MATLAB energy management scripts, but it is represented functionally through source availability, charging logic, and battery power limits.

The third major node is the **AC bus**, defined in the EPLAN diagrams as BUS_AC01. This bus receives power from the inverter output and from the diesel generator through their respective breakers and distributes power to the three feeders F1, F2, and F3. At this node, the study is concerned primarily with source selection, feeder availability, load prioritization, and bus-level continuity of supply rather than with transient protection studies or harmonic analysis. This is an intentional boundary decision: the project addresses operational energy management and control, not detailed power-quality certification or grid-code compliance.

The fourth interface category consists of the **feeder nodes**, each representing a grouped load class rather than an individually modeled consumer network. The feeder outputs are therefore treated as controllable distribution interface points. Breakers ACBR03, ACBR04, and ACBR05

define the switching boundaries between the microgrid core and the respective load groups. In practical terms, once power passes through these breaker-defined interfaces, the internal wiring of each feeder is outside the modeling boundary unless it affects priority classification, aggregate load demand, or breaker status feedback.

3.2.4 Source Boundary: PV, Battery, and Diesel Roles

An important part of scope definition is not only stating which sources exist, but also defining the role boundary of each source. In this project, the PV system is the **primary energy source**. It is responsible for supplying the active load whenever solar power is available, beginning with the highest-priority feeder and then extending to the lower-priority feeders as power permits. This source boundary is not arbitrary; it directly reflects the project objective of reducing diesel dependency and maximizing renewable utilization in a rural fuel-constrained context.

The battery system has a narrower but critical role boundary. It is not treated as the main long-duration source, and it is not modeled as an independent market participant or grid-support asset. Instead, it is defined as a **short-term balancing and critical-load support resource**. In the MATLAB EMS, the battery supports F1 when PV is insufficient, subject to a usable SOC window between 90% and 30% and a power limit of about 60 kW in the simplified model. In the EPLAN representation, the battery is described as a 123 kWh, 600 V nominal bank with approximately 125 A maximum charge/discharge at 1C, monitored through a BMS and connected through a DC breaker and contactor. Thus, the battery boundary includes energy shifting, transient support, and operational reserve for critical loads, but excludes high-fidelity electrochemical aging modeling and long-term degradation analysis.[46]

The diesel generator has the most restricted energy role but one of the most important reliability roles. It is defined as a **backup and continuity source**, not as a base-load generator. In the EMS logic, the generator is started only when PV is insufficient for the critical feeder and the battery SOC has reached its lower threshold, or under specific manual or supervisory conditions. The generator therefore sits inside the physical and control scope, but its intended operating philosophy is secondary and conditional. This boundary is academically important because it frames the generator as a resilience asset that protects essential service continuity rather than as the dominant source of daily energy.

3.2.5 Load Boundary and Feeder Segmentation

Load modeling is often where scope becomes ambiguous, especially in rural electrification projects where many small and diverse consumers exist. To avoid this ambiguity, the project defines the load boundary at the level of **three aggregated feeders**, each corresponding to a priority class rather than an individually metered branch. According to the EPLAN overview, F1 contains critical loads such as part of the residential lighting/fan demand, the hospital, and markets; F2 contains medium-priority residential electrical device loads; and F3 contains lower-priority loads such as the water pump, school, and mosque in the project classification. In the MATLAB models, these feeders are similarly represented as grouped load channels with maximum ratings of approximately 57 kW for F1, 26 kW for F2, and 15 kW for F3.

This feeder-based boundary has several advantages. First, it aligns naturally with industrial control practice, because breakers and digital outputs can be assigned at feeder level rather than at appliance level. Second, it matches the logic of load shedding and restoration, which is usually implemented through sectional switching rather than device-by-device scheduling in practical village microgrids. Third, it keeps the problem tractable for both simulation and implementation mapping, while still preserving the fundamental social priority structure of the community demand.

However, feeder aggregation also introduces deliberate exclusions. The study does not model individual household appliance duty cycles, reactive power consumption by specific end users, or detailed intrafeeder voltage drop. The internal composition of each feeder is represented only to the extent needed to justify its priority and nominal demand envelope. This is a valid academic simplification, because the research question focuses on system-level EMS coordination and feeder priority logic, not end-use behavioral modeling. The feeder boundary therefore serves as the interface between community demand complexity and manageable control abstraction.[47]

3.2.6 Protection and Switching Boundary

Protection devices form an essential boundary layer between safe operation and uncontrolled fault propagation. In the present project, the protection and switching boundary includes the PV-side DC breaker, surge protection device, battery breaker, battery contactor, inverter output breaker, diesel generator breaker, feeder breakers, and PLC-side DC protection shown in the EPLAN package. These devices are part of the project scope because they define physical

switching points, control authority points, and fault isolation points that directly affect how the EMS and control architecture can be implemented.

At the same time, the study does not attempt to produce a full relay coordination report, time-current selectivity analysis, arc-flash study, or certified protection settings schedule. Those tasks belong to detailed engineering and commissioning rather than to the conceptual-to-implementation academic scope of this work. The project instead adopts a functional protection boundary: breakers and contactors are included insofar as they define whether sources and feeders can be connected, disconnected, enabled, or isolated in response to EMS commands, protective trips, or emergency conditions. [48]

This distinction is especially relevant for the battery and diesel branches. The battery branch uses both a DC breaker and a main contactor, which implies a clear boundary between energy storage presence and storage availability to the bus. Similarly, the diesel branch includes not only the generator itself but also the DSE-controlled breaker close/trip interface, meaning that "generator running" and "generator connected to the AC bus" are conceptually distinct states, even if the simplified EMS sometimes treats them jointly. Recognizing such distinctions strengthens the realism of the boundary definition and prepares the groundwork for later implementation discussion.

3.2.7 Control Boundary of the EMS

The EMS control boundary specifies what decisions are made by the supervisory logic and what remains the responsibility of local device-level controllers. In the present project, the EMS is responsible for high-level operational decisions such as determining the active source sequence, deciding whether the generator should start or stop, deciding whether battery support is allowed, and determining whether feeders F2 and F3 should remain connected or be shed. These are classic secondary or supervisory decisions in microgrid control architecture, where the controller optimizes resource usage and service continuity according to system priorities.

By contrast, the EMS does **not** directly implement low-level inverter pulse-width modulation, diesel engine fuel injection timing, AVR dynamics, or BMS cell balancing. Those behaviors remain outside the EMS boundary and are assumed to be handled by local embedded controllers inside the inverter, DSE controller, and battery management system. This separation is consistent with hierarchical microgrid control theory, which distinguishes high-level energy management from local primary control loops and equipment-level protection.[49]

In practical terms, this means that the EMS boundary begins at the level of **commands, permissions, setpoints, and status interpretation**. It can command the start/stop of a source, enable or disable a feeder, choose operational mode, and process state information such as SOC, PV power, and generator status. But it does not simulate semiconductor switching, governor tuning, or thermal runaway mitigation. This boundary is critical for academic clarity: it prevents the chapter from claiming device-level controller design when the actual contribution lies in system-level decision logic and integration.

3.2.8 PLC-Centered Operational Boundary

The project places the Siemens S7-1200 PLC at the center of the real implementation boundary. The EPLAN documentation identifies the PLC as the main control unit, equipped with digital and analog I/O, Ethernet communication, and optional serial and CAN interfaces, receiving measurements and statuses from the DSE, inverter, BMS, and feeders while sending control commands back to them. This establishes the PLC not merely as a peripheral component, but as the operational boundary through which system-wide decisions are materialized in industrial form.

The PLC-centered boundary has strong engineering justification. In real microgrid practice, a PLC often acts as the reliable deterministic executor of supervisory logic, especially in remote or harsh environments where transparency, maintainability, and ruggedness are more valuable than algorithmic complexity alone. In this project, the PLC is therefore treated as the physical executor of the EMS philosophy, even though the logic is first developed and analyzed in MATLAB/Simulink.

This boundary also clarifies an important modeling choice: the PLC implementation is represented through logic sketches, tag mappings, and control flow diagrams rather than a fully exported validated TIA Portal project package. The ladder-style screenshots and functional diagrams show realistic command pathways such as Gen_Start_Cmd, Gen_Stop_Cmd, feeder control commands, and manual override conditions. Therefore, the PLC lies inside the implementation scope as a realistic execution platform, but detailed software certification, factory acceptance testing, and full commissioning parameter files remain outside the present work.

3.2.9 DSE8610 MKII Boundary: Control Versus Monitoring

The DSE8610 MKII generator controller occupies a particularly important boundary position because it bridges engine-level control and system-level supervisory coordination. The EPLAN file describes the DSE unit as receiving generator electrical and mechanical sensing signals and providing outputs for starter relay, fuel solenoid, engine stop, breaker close/trip, alarm relays, and status return to the PLC. The additional project images further clarify that the DSE is used for start/stop execution and protective supervision, while Modbus TCP is primarily used for monitoring parameters such as generator status, kW, voltage, frequency, alarms, run hours, and controller settings readback.

This distinction defines the DSE boundary very clearly. **Digital I/O control** remains the pathway for operational commands such as start, stop, enable, and emergency stop, while **Modbus TCP** remains the pathway for read-only or supervisory monitoring in the present architecture. The project therefore does not claim that the generator is remotely dispatched over a generic data network alone. Instead, it adopts a realistic industrial split between command authority and monitoring visibility, which enhances safety and reduces ambiguity in the control architecture. [50]

From a scope perspective, the DSE is fully inside the real-system mapping because generator availability is central to the EMS. However, the internal parameterization of every DSE protection curve, event-log register, or load-sharing feature is not fully explored. The DSE images show features such as protection & trips, auto/manual mode, and commissioning settings, but the study uses them mainly to justify the control philosophy and implementation realism rather than to reproduce the complete manufacturer commissioning manual. Thus, the DSE boundary is included functionally and architecturally, but not exhaustively at vendor-documentation depth.

3.2.10 Inverter, BMS, and Power Conversion Boundary

The inverter/charger and BMS occupy the boundary between energy management logic and energy conversion safety. According to the EPLAN documentation, the inverter/charger is fed from the shared DC bus and supplies the AC bus at 400 V, 50 Hz, three-phase, while also supporting charging mode from the generator back into the battery. This dual-mode role means the inverter is not simply a one-direction converter; it is the physical interface through which

EMS decisions about charging, inverting, and source transfer become actual power exchange behavior. [51]

Nevertheless, the project boundary treats the inverter in a functional rather than electromagnetic sense. The MATLAB model assumes inverter availability and power transfer capability up to designated limits, but it does not include switching-frequency models, converter loss maps, control-loop tuning, or harmonic distortion analysis. This is appropriate because the objective is to analyze whether the EMS allocates energy sources correctly, not to validate converter waveform quality.

The BMS is similarly included only at the system-integration level. It is responsible for SOC estimation, protection alerts, and interface signals such as battery low, battery high, and battery alarm in the I/O mapping. However, cell balancing, pack thermal diffusion, state-of-health estimation, and internal chemistry dynamics remain outside the modeling scope. In boundary terms, the study uses the BMS as the authoritative source of battery permission and status information, not as a subject of electrochemical research. This keeps the focus aligned with hybrid microgrid coordination while preserving enough realism to support an implementation-oriented thesis.

3.2.11 Communication Boundary in the Supervisory Layer

A modern hybrid microgrid is not defined only by wires carrying power; it is also defined by communication channels carrying status, commands, and supervisory context. The literature on microgrid control frequently distinguishes between electrical layers and communication-enabled supervisory layers, emphasizing that higher-level control often depends on reliable data exchange even when lower-level stability does not. In the present project, the communication boundary includes PLC digital I/O, Modbus TCP monitoring from the DSE, optional RS485 links, Ethernet supervisory communication, REST APIs within Node-RED, MQTT publishing for status exchange, and the HTML dashboard polling mechanism. [52]

This communication boundary is intentionally layered. At the lowest supervisory level, the PLC exchanges hard signals with field devices. At an intermediate digital level, supervisory software exchanges structured status and commands via network protocols. At the user-facing level, the dashboard reads and writes to a clean API rather than interacting directly with field hardware. Such layering is important because it localizes responsibility: field safety remains with industrial hardware, while visualization and experimentation remain with software interfaces. [53]

The project does not attempt to implement full cybersecurity architecture, IEC 61850 modeling, redundant industrial networking, or cloud-native fleet orchestration. Those belong to larger utility-scale or enterprise deployments. The present communication boundary is narrower and practical: it is sufficient to demonstrate status acquisition, operator visibility, manual override capability, and integration potential between simulation and supervisory interfaces.

3.2.12 Node-RED Middleware Boundary

Node-RED is included in the project as a **middleware and supervisory integration layer**, not as the primary real-time controller. The project documentation explicitly describes Node-RED as a layer sitting between the HTML dashboard, a microgrid simulator engine, and future external systems such as MATLAB through MQTT. It exposes a clean REST API, including GET /api/status and POST /api/control/... routes, and can also publish microgrid state to an MQTT topic such as microgrid/status.

This places Node-RED in a very specific boundary position. It belongs inside the project because it demonstrates how system states can be visualized, how operator commands can be collected, and how a software integration layer can mediate between control logic and user interaction. However, it is outside the hard real-time protection and primary control boundary. It is not the entity responsible for inverter PWM, breaker trip security, or engine start permissives at millisecond scale. Instead, it is a supervisory orchestration and interface boundary.

This distinction is crucial for avoiding overstatement. In the thesis, Node-RED should be presented as a flexible integration and visualization tool for the simulation-oriented digital layer and for possible IoT/SCADA-style extensions, not as a replacement for industrial PLC determinism. By defining this boundary clearly, the project gains the benefits of modern open supervisory integration without misrepresenting the actual role of industrial control hardware.

3.2.13 HTML Dashboard Boundary and User Interaction Layer

The HTML dashboard defines the outermost human–system boundary of the project. It is the interface through which an operator can observe system mode, battery SOC, generator status, PV output, total load, active source, feeder states, and alarms, while also issuing selected manual commands such as generator start/stop, feeder toggling, parameter saving, and alarm reset. In scope terms, the dashboard lies fully inside the project because it demonstrates how supervisory visibility and user intervention are made accessible and interpretable.

However, the dashboard is not treated as an HMI certified for industrial deployment. It is a browser-based supervisory interface built for project demonstration, system understanding, and interactive scenario control. Its role boundary is therefore similar to that of a lightweight SCADA front end or engineering dashboard rather than a plant-floor safety console. This is a reasonable academic boundary because it supports explanation and validation without requiring proprietary HMI runtime platforms.

The dashboard boundary also clarifies the role of manual override. When the user toggles F2 or F3 or sends generator commands, the action does not bypass the broader system architecture; it travels through the Node-RED API layer and is then translated into system state changes or control requests. Thus, the user interaction boundary remains structured and mediated, which is a key design principle in reliable supervisory systems. It also helps explain how the project supports both monitoring and limited operator intervention without collapsing the distinction between observation and control.

3.2.14 MATLAB/Simulink/Stateflow Modeling Boundary

The MATLAB and Simulink environment defines the principal **simulation boundary** of the study. Within this boundary, the project models daily PV production, aggregated feeder loads, battery SOC evolution, source dispatch priority, load shedding, battery charging, and generator start/stop behavior under rule-based EMS logic. The scripts and models implement discrete-time representations over a 24-hour horizon, using a PV maximum of 150 kW, feeder maxima of approximately 57 kW, 26 kW, and 15 kW, a battery capacity of about 100 kWh in the simplified simulation, and SOC bounds between 30% and 90%.

This simulation boundary is intentionally behavior-oriented. It represents **what the system decides and how energy is allocated**, not every physical transient that would occur in the real plant. Therefore, it includes power balance, feeder activation state, battery support, and source switching logic, but excludes electromagnetic transients, exact diesel startup delays, converter efficiency maps across all operating points, and distribution-network voltage drop at every branch. Such simplifications are standard and acceptable when the main research contribution lies in EMS logic and system operation rather than in detailed hardware dynamics.

Stateflow-related materials further support the simulation boundary by showing how operational modes and transitions can be represented as explicit state logic, which is especially appropriate for feeder priority control, source activation, and manual override behavior. Therefore, the simulation boundary includes rule-based supervisory state evolution and scenario analysis, while

the exact embedded implementation of every device remains mapped, not replicated, in the simulation model. This boundary makes the model academically meaningful and computationally manageable.

3.2.15 Data Boundary: Inputs, Outputs, Setpoints, and Tags

Another critical aspect of scope definition is the **data boundary**—that is, which variables are treated as authoritative inputs, controllable outputs, tunable setpoints, and status tags. In the MATLAB EMS, the primary numerical inputs include PV input power and feeder demand scaling, while the outputs include feeder supplied power, generator power, battery power, SOC trajectory, active load, and total supply. In the Node-RED and dashboard layer, the exchanged data include mode, SOC, generator status, PV power, total load, active source, feeder states, setpoint thresholds, and alarm lists.

The PLC and EPLAN mappings extend this data boundary into industrial tags. Digital inputs include signals such as generator running, generator fault, battery low, battery high, inverter ready, inverter fault, feeder faults, emergency stop, and mode selector. Digital outputs include commands such as generator start, generator stop, inverter mode selection, inverter enable, battery contactor close/open, feeder close/trip, and alarm relay activation. Analog I/O includes PV voltage/current, battery voltage/current/temperature/SOC, inverter AC values, generator AC values, and setpoint-oriented analog outputs. This mapping is extremely important because it links simulation variables to field-significant information structures.

What lies outside the data boundary is equally important. The study does not attempt to collect every raw sensor sample, oscilloscope trace, or manufacturer register block. Nor does it model database historians, long-term archival schemas, or full cybersecurity event logs. Instead, it selects the subset of data necessary to explain source dispatch, feeder prioritization, protection awareness, operator visibility, and implementation realism. This focused data boundary is suitable for a bachelor-level engineering thesis because it preserves completeness of function without overwhelming the study with nonessential telemetry infrastructure.

3.2.16 Real Implementation Boundary Versus Simulation Boundary

A central theme of this project is that the work is neither purely theoretical nor fully deployed in the field. This makes it necessary to define a careful boundary between the **real implementation representation** and the **simulation representation**. The simulation representation includes the MATLAB/Simulink/Stateflow logic used to evaluate daily operation,

feeder priority, SOC management, and generator dispatch under different scenarios. The real implementation representation includes EPLAN single-line and control diagrams, PLC logic sketches, DSE control-and-monitoring architecture, and diagrams of the real microgrid power and communication layout.

The key point is that these two boundaries are connected by **functional equivalence**, not by claiming that the exact real plant was fully commissioned and measured under all simulated scenarios. For example, the simulated decision "shed F3 first" corresponds to a real implementation pathway via feeder breaker control logic and PLC outputs. Likewise, a simulated condition for generator start corresponds to a real PLC-to-DSE start command pathway shown in the project images. This is a strong and defensible academic position: the project demonstrates consistency between the modeled control philosophy and a realistic implementation architecture, even where full field validation is beyond scope.

By explicitly maintaining this distinction, the thesis avoids two common errors. The first would be understating the realism of the project by presenting it as only a software model. The second would be overstating the project by implying that every modeled scenario has already been executed on a fully commissioned rural site. The boundary adopted here avoids both extremes. It presents the work as a hybrid academic-engineering study with simulation-based validation and implementation-oriented architectural grounding.

3.2.17 Included Elements of the Study

To avoid ambiguity, the following elements are explicitly included within the project scope:

- The hybrid microgrid architecture composed of PV generation, battery storage, inverter/charger, diesel generation, AC/DC buses, and three prioritized feeders.
- Rule-based EMS logic for source coordination, feeder supply priority, battery charge/discharge behavior, and generator start/stop decisions.
- A simulation environment based on MATLAB and associated logic materials, including feeder-based operation and daily scenario evaluation.
- Realistic implementation mapping through EPLAN, PLC logic sketches, DSE8610 MKII architecture, and control/monitoring diagrams.
- A supervisory integration layer using Node-RED APIs, MQTT publication concepts, and an HTML dashboard for visibility and selected manual interaction.

- Discussion of communication pathways, digital/analog tags, and the practical distinction between operational control and monitoring.

These included elements collectively define a scope that is sufficiently broad to support a full engineering narrative from concept to implementation mapping, while remaining focused enough to be completed and defended coherently at bachelor level.

3.2.18 Excluded Elements and Deliberate Simplifications

The study also excludes several elements, either because they are beyond the academic goal of the work or because they would require field resources not available within the project boundary. These exclusions include:

- Detailed PV module thermal modeling, shading analysis, and long-term meteorological uncertainty studies.
- Full electrochemical battery aging, degradation, and lifetime optimization modeling.
- Electromagnetic transient analysis of converters, harmonic studies, and protection selectivity calculations.
- Full TIA Portal software package validation, FAT/SAT procedures, and commissioned hardware-in-the-loop testing.
- Detailed civil, mechanical, and installation design such as foundations, enclosure thermal management, and cable tray routing.
- Cybersecurity hardening, redundancy engineering, and enterprise SCADA historian architecture.
- Utility interconnection studies and grid-code compliance, since the project is framed as an off-grid rural system.

These exclusions are not weaknesses if they are stated transparently. On the contrary, they are signs of disciplined scope control. Academic engineering projects are strongest when they solve a clearly bounded problem thoroughly rather than attempting to address every possible subsystem superficially. The present thesis therefore focuses on the integrated problem of hybrid microgrid requirements analysis, supervisory energy management, and implementation-oriented architecture.

3.2.19 Boundary Justification from Literature and Engineering Practice

The boundary adopted in this project is strongly supported by microgrid design literature. Conceptual microgrid design guides emphasize defining microgrid boundaries, identifying critical loads, clarifying goals, and matching design depth to the development stage of the project. Reviews of microgrid architecture similarly show that practical microgrid studies often separate physical source/load architecture, supervisory control hierarchy, and communication structure into distinct but interacting layers. This layered viewpoint aligns closely with the present project's combination of electrical design, EMS logic, PLC/DSE industrial mapping, and dashboard-level supervision.

Furthermore, literature on hierarchical microgrid control distinguishes primary local control, secondary coordination, and tertiary supervisory optimization, often with different communication requirements and implementation platforms. The project's rule-based EMS, PLC execution mapping, and supervisory Node-RED/dashboard layer correspond well to this general structure. The study does not claim to design all three layers in full mathematical depth, but it locates itself clearly in the supervisory and coordination domain while respecting equipment-level local controllers.

From an engineering-practice perspective, the adopted boundary is also realistic because it mirrors how real projects are often developed: concept first, architecture second, simulation and control logic third, and detailed field engineering after feasibility and functional validation. The presence of EPLAN documentation, PLC tag structures, DSE monitoring/control separation, and a supervisory dashboard adds practical credibility without forcing the thesis into the scope of a commercial EPC delivery package.

3.2.20 Implications for Subsequent Chapters

The scope and boundary decisions established here shape the rest of the thesis in a direct way. In Chapter 4, the design chapter will build only on components and interfaces that have been explicitly admitted this boundary: the PV array, battery, diesel generator, inverter/charger, feeders, PLC, DSE, and supervisory software layer. This prevents the design narrative from drifting into areas not modeled or justified.

In Chapter 5, implementation and testing will rely on the same boundary to explain how MATLAB/Simulink logic, PLC-style decision pathways, DSE command/monitoring links, and Node-RED/dashboard interactions represent different views of one coherent system rather than

disconnected artifacts. In Chapter 6, results and discussion will be interpreted only within the validity range of the adopted model and implementation mapping, ensuring that conclusions about reliability improvement, feeder prioritization, and diesel reduction remain credible and properly bounded.

For this reason, the present section is not peripheral background. It is the structural foundation that makes the later chapters analytically consistent. By defining the physical, electrical, functional, control, communication, and simulation boundaries together, the thesis gains a unified framework within which both the real-system representation and the simulation-based investigation can be understood as parts of the same engineering problem

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3.3 Load Structure and Priority Modeling

3.3.1 Load Classification Philosophy

Load classification is one of the most important conceptual steps in hybrid microgrid design because the value of a microgrid does not arise only from generating electricity, but from allocating limited electricity to the most important services at the right time. In rural and standalone microgrid literature, load prioritization is widely recognized as a practical method for maintaining reliability under fluctuating renewable generation, limited storage capacity, and backup generator constraints. Rather than assuming that all electrical demand has identical urgency, engineers classify loads according to criticality, social importance, time sensitivity, and the consequences of interruption.

In the present project, load classification is based on a **service-continuity philosophy** rather than a purely consumption-based philosophy. This means that the ranking of loads is not determined only by power magnitude in kilowatts, but by the consequences of losing supply to that load group. A hospital load with moderate power demand may be more critical than a larger but deferrable community load; likewise, water pumping may be essential at some times but not necessarily require the same uninterrupted continuity as medical or basic residential lighting loads.

The adopted classification strategy therefore reflects three parallel principles. First, loads that directly affect life, safety, and minimal social continuity must remain energized for as long

as technically possible. Second, loads that improve quality of life and productive use of electricity should be supplied whenever power is sufficient, but they may be interrupted temporarily without immediate catastrophic consequences. Third, loads that are useful but deferrable or schedule-flexible should be the first candidates for disconnection during energy shortage. This philosophy naturally leads to a three-tier feeder model composed of F1 critical loads, F2 medium-priority loads, and F3 low-priority loads.

From a modeling perspective, this three-level structure offers an excellent balance between realism and controllability. It is realistic because rural communities do not consume electricity as a single undifferentiated block; they have essential, desirable, and deferrable demands. It is controllable because feeder-level grouping allows the EMS to make decisions using discrete switching priorities instead of needing to solve appliance-level optimization across dozens of end-use categories. As a result, the project uses feeder-class logic as the bridge between real community energy needs and manageable simulation-based supervisory control.

3.3.2 F1 Critical Loads

Feeder F1 is the **critical-load feeder**, and it represents the part of the community demand that must be preserved under almost all operating conditions. According to the EPLAN overview, F1 includes half of the residential houses for basic lighting and fans, the hospital, and market-related loads identified as part of the highest-priority service group. In the MATLAB EMS formulation, F1 is also treated as the first load segment to be supplied from PV, then battery, and finally diesel generation if required. This establishes F1 not merely as another branch of the load model, but as the foundational demand block around which the entire dispatch logic is organized.

The rationale for assigning hospital demand to F1 is straightforward and strongly supported by microgrid resilience literature. Medical loads are among the clearest examples of critical loads because interruptions can directly affect health outcomes, emergency response, and community trust in the system. Even when a hospital is small or rural, its refrigeration, lighting, treatment devices, ventilation, communication, and essential service continuity justify protected feeder status. Therefore, placing the hospital in F1 is not only technically valid but socially and ethically consistent with resilience-oriented microgrid design.

The inclusion of basic residential lighting and fans in F1 also reflects a defensible rural-electrification philosophy. In low-resource or off-grid contexts, not all household consumption is equally critical. Basic lighting, minimum thermal comfort, and phone-charging-equivalent support often form the baseline service level expected from a community microgrid, while

heavier discretionary appliances can be placed in lower classes. By assigning only part of the residential demand to F1, the project avoids the unrealistic assumption that every household load is critical, yet it preserves the minimum household service necessary for dignity, safety, and night-time functionality.

In the simulation scripts, the F1 demand envelope emerges from the combination of half the residential base profile, hospital demand, shops, and selected social-service loads such as mosque events in the scaled formulation. The maximum modeled value of F1 is around 57 kW, making it the largest and most operationally protected feeder in the system. This design choice is important because it ensures that the EMS always evaluates source adequacy first against the most important aggregated demand. In other words, F1 acts as the non-negotiable anchor of the dispatch sequence. If F1 cannot be maintained, the system is already operating in a degraded state regardless of what happens on F2 or F3.

Another important feature of F1 is that it is not simply "always on" in a naive sense. It is **priority-protected**, which is different. The EMS attempts to maintain F1 through PV first, then through battery discharge subject to SOC limits, and finally through generator support. This means that criticality does not eliminate energy constraints; rather, it determines how scarce energy is allocated. Such a distinction is central to microgrid engineering because priority does not create energy—it only decides where limited energy should go first.

3.3.3 F2 Medium-Priority Loads

Feeder F2 is defined as the **medium-priority feeder**, representing loads that are desirable and socially important but not as urgent as F1. According to the EPLAN description, F2 consists mainly of half of the residential houses associated with electrical devices rather than only minimum essential lighting/fans. In the EMS structure, F2 is supplied only after F1 has been secured, and only from remaining available PV in the simplified rule-based logic. This immediately gives F2 a conditional supply status: it is important enough to be intentionally modeled and controllable, but not important enough to consume the last available reserves before F1 is protected.

This is a rational classification for community demand. Medium-priority loads often include appliances and services that support comfort, convenience, productivity, and normal domestic activity, but whose temporary interruption does not create the same immediate risk as loss of medical support or minimum lighting. In real communities, such loads may include televisions,

small domestic devices, fans above minimum essential levels, general sockets, and other end uses that improve living conditions but can tolerate limited curtailment during shortages.

The decision to assign part of the residential demand to F2 rather than keeping all household demand in one feeder is particularly strong from a modeling perspective. It reflects the reality that household demand has its own internal hierarchy. Basic lighting is not equivalent to discretionary plug loads, and an EMS that cannot distinguish between them is less effective in energy-constrained operation. Therefore, splitting residential demand across F1 and F2 allows the project to represent partial service continuity rather than all-or-nothing household electrification.

In the MATLAB EMS, F2 is not supported by generator dispatch in the same direct way as F1 in the simplified logic. Instead, it receives surplus renewable energy after the critical feeder is satisfied. This makes F2 an important indicator of renewable sufficiency. When F2 remains energized for long periods, it signals that PV availability and/or overall demand conditions are favorable. When F2 is dropped, it signals that the system has entered a conservation mode focused on preserving higher-priority service. This is analytically useful because F2 becomes a visible marker of system stress without requiring the thesis to introduce additional resilience indices.

Moreover, medium-priority classification gives the control system operational flexibility. F2 can be disconnected earlier than F1 during shortages, reconnected earlier than F3 during recovery, and manually controlled for testing or scenario management without undermining the basic mission of the microgrid. Thus, F2 functions as a buffer zone between essential survival-oriented supply and nonessential or deferrable service. That middle layer is often what gives a hybrid microgrid practical usefulness in daily life rather than reducing it to a binary emergency-only system.

3.3.4 F3 Low-Priority Loads

Feeder F3 is the **low-priority feeder**, intended for loads that can tolerate interruption most easily when generation is limited. In the EPLAN documentation, F3 includes the water pump, school, and mosque in the project classification, and the MATLAB scripts model F3 as a lower-power feeder with a maximum around 15 kW. In the rule-based dispatch, F3 receives supply only after F1 and F2 have been considered, making it the first load group to lose supply and the last to regain it under constrained operation.

At first glance, some of the F3 components, especially water pumping, may appear important enough to deserve higher priority. This is where load classification philosophy requires nuance. Low priority does not mean “unimportant in absolute terms”; it means “more schedulable or deferrable relative to the other loads in the system.” Water pumping, for example, may be very important to village life, but if water can be pumped during solar hours or stored in tanks, then short interruptions may be more tolerable than interruptions to hospital service or minimum residential lighting. Similarly, school loads are socially valuable, but they are usually time-bound and not continuously life-critical across all hours.

The same logic applies to community or religious loads that are periodic rather than continuous. In the MATLAB profile, mosque demand appears at specific times rather than as a constant base load, illustrating that some community loads are event-driven rather than always active. This temporal nature supports lower priority classification in an EMS that must maintain service continuity under renewable variability. A load that is flexible in timing or limited in duration is a stronger candidate for controlled interruption than a load that must remain available continuously.

F3 therefore plays a major role in operational resilience even though it is the lowest-priority feeder. It is the feeder that allows the system to adapt gracefully rather than collapsing into full shortage. By shedding F3 first, the EMS gains a controlled way to reduce total demand quickly while preserving the social core of service. In this sense, low-priority loads are not marginal to the design; they are essential to the design because they provide the controllable margin through which the microgrid can remain stable and socially useful under scarcity.

It is also important to note that F3 is not defined as permanently optional. During strong solar availability, low demand periods, or times of adequate battery condition, F3 can and should be restored to support broader community benefit. The low-priority feeder is therefore part of a **dynamic service envelope** rather than a permanently sacrificed branch. This dynamic role strengthens the realism of the project because rural electrification success depends not only on critical load survival but also on the ability to deliver broader community value whenever resources allow.

3.3.5 Feeder Mapping to Real Community Loads

A major strength of the project is that the feeder model is not abstractly invented; it is mapped to recognizable real community services. The EPLAN system overview explicitly associates the feeders with residential houses, hospital, markets, water pump, school, and mosque, thereby

grounding the load model in a plausible rural village context rather than in generic anonymous demand categories. This mapping gives the simulation social meaning. When the EMS protects F1, it is not merely protecting a number in kilowatts; it is protecting medical service and minimum residential continuity. When it sheds F3, it is not merely reducing load mathematically; it is postponing lower-priority community services.

The use of feeder mapping is especially valuable in rural microgrid projects because community acceptance depends heavily on whether the allocation of energy appears fair, transparent, and aligned with local needs. A microgrid that treats all loads identically may be simple mathematically, but it often fails socially because users experience outages as arbitrary rather than purposeful. By mapping feeders to meaningful community functions, the project creates a basis for explaining energy management decisions to operators, stakeholders, and end users.

The simulation data also reinforce this mapping. The load construction in the MATLAB code explicitly includes residential demand, hospital demand, school demand, water pumping, mosque timing, and shops, which are then aggregated into feeder groups. This means that feeder labels are not cosmetic; they emerge from an underlying decomposition of community load categories. As a result, the project can legitimately compare feeder-level logic with community service categories instead of presenting a purely symbolic feeder hierarchy.

Still, feeder mapping in the project remains an **aggregation**, not a full distribution study. The model does not show every house, every branch circuit, or every appliance-level schedule within the village. This is a necessary simplification. The purpose of the mapping is to retain social and operational interpretability while keeping the control problem tractable. Therefore, the feeder-to-community relationship should be understood as a structured representation of clustered services rather than a literal one-breaker-per-building deployment blueprint.

3.3.6 Load Shedding and Restoration Priority Rules

Once loads are classified, the next engineering step is to define how they are shed and restored. In microgrid practice, emergency or constrained-operation load shedding is typically based on criticality differences between load classes, since not all loads contribute equally to safety, resilience, or frequency recovery. The present project follows a priority-based shedding framework in which F3 is shed first, F2 is shed second, and F1 is preserved as long as any technically feasible source combination remains available.

In the simplified EMS logic, source allocation already embodies this hierarchy. Available PV power is directed first toward F1, then toward F2, then toward F3. If PV is insufficient for F1, the battery is used within its allowable SOC and power constraints. If the battery cannot maintain F1 alone, the generator is brought in to preserve the critical feeder. This means that "load shedding" in the project is not only an explicit breaker action; it is also the natural consequence of a sequential source-priority structure in which lower-priority feeders simply do not receive power under shortage.

Restoration follows the opposite direction. Once sufficient renewable power becomes available, or once system conditions improve through battery recovery or generator support, the EMS can restore service in ascending order of priority sacrifice: F2 is restored before F3, and F3 is restored only after higher-priority supply is secure. This mirrored structure is important because restoration without rule consistency can produce oscillation, unfairness, or poor user confidence. If lower-priority feeders were reconnected too aggressively, the system could jeopardize the stability of F1 or repeatedly force the generator into avoidable cycling.

The project's rule set is therefore defensible on both technical and social grounds. Technically, it helps preserve power balance and avoid wasting scarce energy on noncritical demand during shortage. Socially, it communicates a consistent logic: essential services first, useful but less critical services second, and deferrable services last. This consistency is a major advantage in rural microgrids, where service predictability can be almost as important as average energy quantity.

Another benefit of explicit shedding and restoration rules is analytical clarity. Because the feeder classes are ordered, results can be interpreted directly. If F3 is lost, the system is under mild or moderate stress. If F2 is also lost, the system is in a severe conservation mode. If F1 is threatened, the system is in critical deficiency. Such qualitative interpretation makes the results chapter stronger because system behavior can be discussed in terms of service levels, not just power curves.

3.3.7 Manual Override of Feeder Behavior

Although automatic EMS logic is central to the project, the architecture also recognizes the need for **manual override**. In practical microgrids, operators may need to force a feeder on or off for maintenance, testing, fault isolation, community events, or abnormal operating conditions that are not fully represented in the default control logic. The Node-RED dashboard and API

structure in this project already support feeder control commands through routes such as /api/control/feeders, where the operator can send enable or disable instructions for F2 and F3.

This manual override capability is significant for two reasons. First, it reflects the reality that supervisory systems are rarely left entirely to autonomous control without any operator intervention path. Second, it shows that the feeder model is not merely passive; it is operationally actionable within the control architecture. In other words, feeder priority is encoded not only in simulation mathematics but also in the digital control and monitoring concept built around Node-RED and the dashboard.

The project's current supervisory logic focuses especially on F2 and F3 for manual override, which is reasonable because these feeders are the ones most likely to be intentionally shed or re-enabled without compromising critical service continuity. F1, being the critical feeder, should generally remain protected from casual manual disconnection except under maintenance or safety procedures. This selective override philosophy is good engineering practice because it preserves the integrity of the priority hierarchy while still giving operators enough flexibility to manage noncritical or medium-priority service dynamically.

Manual override also raises an important conceptual point: automatic priority logic and operator authority must coexist without becoming contradictory. In a well-structured system, manual override should either temporarily supersede automatic logic with clear status indication or operate within bounded permissions so that unsafe or irrational states cannot be created. The present project supports this principle by routing user actions through a mediated API and control layer rather than allowing direct uncontrolled hardware manipulation from the dashboard. Thus, manual override is included as a structured supervisory function, not as an uncontrolled bypass of the EMS architecture.

3.3.8 Comparison Between Real Feeder Allocation and Simulation Feeder Logic

A strong thesis must show not only that the simulation works, but that it corresponds meaningfully to the real system architecture being proposed. In this project, the **real feeder allocation** is expressed through EPLAN documentation and implementation-oriented control diagrams, while the **simulation feeder logic** is expressed through aggregated MATLAB load channels and EMS dispatch rules. The two are not identical in every engineering detail, but they are clearly aligned at the functional level.

In the real allocation, feeders correspond to physically distinct load groups connected through dedicated breakers at the AC distribution level: F1 critical, F2 medium priority, and F3 low priority. In the simulation, these same feeder classes are represented as aggregated time-varying power demands with explicit priority sequencing in the dispatch algorithm. This means that the simulation preserves the most important real-world feature of feeder allocation: differential service continuity based on priority. What it does not preserve in full detail are branch-level wiring characteristics, fault localization granularity, and exact downstream consumer diversity.

This kind of correspondence is appropriate and academically strong. A simulation does not need to reproduce every cable and junction box to be valid for EMS analysis. What it must reproduce is the **decision-relevant structure** of the real system. In this case, the decision-relevant structure includes which loads are grouped together, which feeder is protected first, which feeder is shed first, and how sources are allocated across feeder priority levels. The project achieves exactly that alignment.

There are also useful differences between the two representations. The real feeder allocation is naturally hardware-oriented, emphasizing breakers, PLC I/O, DSE interactions, and power distribution equipment. The simulation feeder logic is naturally behavior-oriented, emphasizing time profiles, source adequacy, active load after EMS, and SOC-dependent dispatch outcomes. Rather than being a weakness, this difference is a strength because it allows each representation to do what it does best: the real allocation explains implement ability, and the simulation explains system behavior.

Therefore, the relationship between the two can be summarized as one of **functional abstraction with architectural consistency**. The simulation simplifies the physical feeder structure into aggregated demand channels, but it preserves the same hierarchy, same service logic, and same operational intention found in the real design package. This makes the results defensible. When the simulation shows F3 curtailment or F2 restoration, those events can be interpreted as realistic feeder-level actions rather than as abstract numerical artifacts. That alignment significantly strengthens the technical credibility of the overall thesis.

3.4 Generation and Storage Resources

3.4.1 PV Array as Primary Renewable Source

The photovoltaic array is the principal renewable generation resource in the proposed hybrid microgrid and serves as the first energy source in the operational hierarchy of the system. This role is consistent with modern hybrid rural microgrid practice, where solar generation is commonly positioned as the primary daytime source because of its low operating cost, modularity, and suitability for off-grid communities with limited fuel logistics. In the present project, the EPLAN documentation identifies the PV field as a 150 kW array, while the simulation model uses a maximum PV capacity of 150 kW with a daytime sinusoidal generation profile between sunrise and sunset.

The physical characteristics of the PV subsystem reinforce its central role. The EPLAN package specifies a 150 kWp arrangement composed of 266 modules configured in 14 strings of 19 modules, with approximate operating voltage Vmp≈789V_{mp} \approx 789Vmp≈789 V, cold open-circuit voltage Voc≈952V_{oc} \approx 952Voc≈952 V, string current of about 13.25 A, and total short-circuit current around 190 A. These values place the PV plant firmly within the high-power DC generation segment of the microgrid and justify the presence of a dedicated combiner, surge protection, MPPT stage, and DC bus interface. The array is therefore not represented as a generic renewable source but as a defined electrical subsystem with realistic voltage and current characteristics.

From an operational viewpoint, the PV array is treated as the **preferred source of active power** because it can directly offset diesel runtime and reduce battery cycling when irradiance is available. In the EMS logic, available PV is dispatched first to the critical feeder F1, then to F2, and finally to F3, before any excess is considered for battery charging. This priority assignment is not arbitrary; it reflects the general microgrid principle that zero-fuel renewable generation should be exploited first, provided that source variability is managed through storage and backup support.

The simulation scripts further show that PV is the dominant daytime energy-shaping resource. In both one-2.m and run_ems_matlab.m, the PV curve defines the primary time window during which feeder restoration becomes possible and battery charging can occur. When PV output rises above critical demand, medium- and low-priority loads can re-enter service. When it exceeds total active demand, the residual energy becomes an opportunity for charging

rather than curtailment. In this sense, the PV array does more than produce electricity; it sets the daily rhythm of system abundance and scarcity.

However, being the primary source does not mean being a sufficient source at all times. Solar production is intermittent, weather-dependent, and time-limited by daylight. For this reason, the PV subsystem is positioned as the **first** source in the hierarchy but not the **only** one. Its strength lies in energy economy and sustainability, while its weakness lies in temporal variability. The architecture of the whole microgrid is therefore built around compensating for that variability through storage, controllable backup, and prioritized load allocation.

Another important point is that the PV resource is linked to both the electrical and control layers. Electrically, it feeds the DC bus through the combiner and MPPT stage. Functionally, it provides a real-time input to the EMS as a measure of renewable availability. Supervisory and monitoring layers can expose PV power to operators through the dashboard and PLC tags, turning solar availability into an observable decision variable rather than a hidden background process. This is important in implementation terms because PV power is one of the core signals around which all higher-level dispatch decisions are made.

Thus, in the present project, the PV array is best understood as the **economic and ecological foundation** of the microgrid. It is the cheapest source per operating hour, the first source used by the EMS, the source that enables load restoration during the day, and the source that creates charging opportunities for the battery. Without the PV subsystem, the architecture would lose its hybrid character and regress toward a diesel-dominated backup system. With it, the microgrid becomes a renewable-led autonomous energy platform.

3.4.2 Battery Bank as Short-Term Balancing Source

The battery bank is the principal energy storage resource in the system and is designed to function as a **short-term balancing and continuity support source** rather than as a long-duration primary generator. In hybrid microgrids, batteries are commonly used to absorb renewable variability, support critical loads during temporary generation deficits, reduce generator cycling, and preserve service continuity during transitions between source states. That exact philosophy is visible in the present project, where the battery intervenes mainly when PV is insufficient but the system still aims to avoid or delay generator startup.

The electrical specification of the battery in the design package gives this role clear physical grounding. The EPLAN file describes the battery bank as approximately 123 kWh nominal

energy, 600 V nominal voltage, 100 Ah capacity, and maximum charge/discharge current around 125 A at approximately 1C, connected to the DC bus through a battery DC breaker, a main contactor, and a BMS interface. In the simulation layer, the simplified battery model uses 100 kWh capacity, SOC limits between 30% and 90%, and a power limit of 60 kW. Although these values are simplified relative to detailed field behavior, they are fully adequate for modeling energy support, SOC evolution, and dispatch constraints.

The battery's core mission in the EMS is to provide **short-duration energy continuity for critical service**. In the rule-based algorithm, when PV cannot fully satisfy F1, the battery is discharged before the diesel generator is started, as long as SOC remains above the minimum threshold. This shows that the battery is not treated as a generic reserve but as a selective bridging source that helps preserve the renewable-first philosophy of the system. It buys time, smooths deficits, and prevents unnecessary generator starts when the shortage is temporary or moderate.

This balancing role is particularly important in rural microgrids because many source deficits are not sustained over long periods. They may arise during cloud passages, morning/evening transitions, short-lived demand peaks, or transient mismatches between load and solar output. A properly managed battery can absorb these variations much more efficiently than a diesel generator that would otherwise need to start frequently, operate at suboptimal loading, and consume fuel for events that do not require long-duration thermal generation. The battery therefore acts as the **flexibility engine** of the hybrid architecture.

The charge behavior of the battery is equally important. In the EMS code, remaining PV power after serving the active loads is directed toward charging the battery, subject to battery power limits and SOC ceiling. This means the battery is replenished opportunistically from surplus daytime renewable energy rather than being charged arbitrarily. In implementation terms, this transforms the battery into a renewable capture device as well as a shortage support device. The battery does not merely discharge to protect the loads; it also stores renewable excess so that solar energy can influence system operation beyond the daylight window.

Still, the battery is intentionally not oversized to cover all nighttime demand independently. This is a crucial design message. The battery exists to provide **short-term autonomy and balancing**, not to eliminate the diesel generator entirely under all circumstances. Such a boundary is realistic and economically sensible. Long-duration storage is expensive, and many rural hybrid systems achieve better techno-economic balance by combining moderate storage

with renewable generation and backup thermal support rather than forcing the battery to carry the entire system for extended periods.

In system behavior terms, the battery also serves as an indicator of operating resilience. A healthy SOC means the system still has maneuvering space before thermal backup becomes necessary. A declining SOC means resilience margin is shrinking. A battery at the lower SOC threshold means the system must move from renewable-balancing mode into backup-preservation mode. Therefore, the battery is simultaneously an energy device, a flexibility device, and a state-of-system indicator. This triple role makes it one of the most strategically important resources in the whole microgrid.

3.4.3 Diesel Generator as Backup and Reliability Source

The diesel generator is included in the proposed microgrid as the **backup and reliability source**, not as the preferred daily source of energy. This distinction is essential. In a hybrid renewable microgrid, the diesel generator should preserve continuity when renewable production is inadequate and battery reserve becomes insufficient, but it should not dominate normal operation unless the system has already entered a constrained state. The present project reflects that philosophy very clearly by assigning the generator a secondary and conditional role in the source hierarchy.

According to the EPLAN design, the generator is rated at 100 kW, 400 V, 50 Hz, three-phase, and is connected to the main AC bus through a dedicated generator breaker controlled by the DSE8610 MKII controller. This rating is highly significant because it is large enough to sustain a substantial share of the village load and, most importantly, large enough to preserve F1 under severe renewable deficiency. The generator therefore provides not only energy backup but also capacity security.

In the EMS logic, the generator is used only after two conditions have effectively been met: PV is insufficient for the critical demand, and the battery cannot continue to cover the deficit within its permitted SOC operating window. This makes the generator the **last-resort energy guarantor** for continuity of essential supply. Such a role is common in islanded hybrid systems, where thermal generation remains the most dependable dispatchable source despite its fuel cost and environmental disadvantage.

The reliability value of the generator extends beyond simple kW supply. Diesel generation provides firmness, controllability, and predictable availability during low-irradiance periods,

high-load periods, and long-duration renewable deficits. In practical rural systems, it also serves as psychological and operational assurance: the presence of a dispatchable engine-backed source means that community-critical services do not depend entirely on uncertain weather or battery reserve. This is especially important when the protected loads include medical and basic social continuity functions.

At the same time, the project deliberately avoids presenting the diesel unit as a base-load source. The very structure of the EMS, which allocates PV first and battery second, demonstrates that the system is designed to **minimize generator runtime**, not normalize it. This design choice aligns with fuel-economy goals, emissions reduction, maintenance reduction, and the broader strategic objective of moving rural electrification away from diesel dependence without sacrificing reliability.

The generator also plays an indirect role in supporting the battery, because the inverter/charger architecture allows AC energy from the diesel source to be converted into DC charging energy under appropriate operating mode. This means the diesel unit is not just a load-serving asset; it can also help restore reserve margin after deep battery discharge conditions. In resilience terms, this is highly valuable. A system that can only serve load from diesel but cannot rebuild storage remains vulnerable after the immediate shortage has passed. By contrast, a generator that can support both load continuity and battery recovery contributes to **recovery resilience**, not only survival.

Therefore, the diesel generator in this project should be understood as the **reliability backbone of last resort**. It is expensive relative to solar, less desirable environmentally, and intentionally subordinated in the dispatch order. Yet it remains indispensable because it provides the controllable firmness that renewable-led systems need in order to guarantee protected service under extended scarcity.

3.4.4 An Inverter Role in AC/DC Conversion

The inverter/charger is the central power-conversion interface of the microgrid and is one of the most strategically important devices in the architecture because it physically links the DC resource side to the AC distribution side. Without it, the PV array and battery would remain electrically separated from the main AC bus and the community feeders. In the EPLAN package, the inverter/charger receives energy from the DC bus, which is shared by the PV subsystem and battery subsystem, and feeds the AC bus at 400 V, 50 Hz, three-phase through its dedicated output breaker.

Its role is best understood as **bidirectional energy conversion**. In one direction, it converts DC power from PV and/or battery into AC power for the feeders. In the opposite direction, it can convert AC power from the diesel generator into DC charging power for the battery. This duality is explicitly described in the EPLAN document through three operating modes: inverting, charging, and standby/idle. As a result, the inverter/charger is not a passive interface but an active operating bridge that supports both supply continuity and storage recovery.

From a resource-management standpoint, the inverter is what makes the hybrid architecture coherent. The PV system produces on the DC side. The battery stores on the DC side. The village loads exist on the AC side. The diesel generator is also on the AC side. The inverter/charger therefore becomes the conversion core through which renewable generation becomes usable load power and through which AC backup energy can become stored DC reserve. This makes it one of the main enablers of source cooperation across the whole microgrid.

In the project's control philosophy, the inverter is also a commandable and monitorable device. The PLC mapping indicates that the inverter can receive start/stop or mode-related commands and can return measurements such as voltage, current, and frequency feedback. This means the inverter lies not only in the power path but also in the control path. It is therefore both an energy interface and an information interface. The EMS does not need to solve internal converter dynamics in full detail, but it must assume and supervise the correct macro-level mode of the inverter for the dispatch philosophy to work.

The simulation abstracts away switching-level converter behavior, which is appropriate for the thesis objective. However, the functional presence of the inverter remains evident everywhere in the results because every flow between DC resources and AC loads presupposes successful power conversion. When PV supplies F1, the inverter is implicitly active. When the battery supports the AC feeders, the inverter is implicitly active. When the generator recharges the battery through AC-to-DC conversion, the inverter/charger is again the enabling device. In this sense, the inverter is the silent mediator behind almost every major source interaction.

Another reason the inverter deserves special emphasis is that it shapes the architectural distinction between a loosely connected hybrid system and a truly integrated hybrid microgrid. If PV, battery, and diesel simply existed side by side without controlled bidirectional conversion, the system would behave more like a set of parallel subsystems than a coordinated microgrid. The inverter/charger provides the common conversion layer that turns multiple resources into

one operationally unified energy system. That is why it should be seen as a **system-integration component**, not merely as a hardware accessory.

3.4.5 Generator Controller Role (DSE8610 MKII)

The DSE8610 MKII generator controller is the device that turns the diesel generator from a simple mechanical source into a controllable and monitorable microgrid participant. In the project architecture, the diesel engine itself produces energy, but the DSE provides the logic and interfacing needed to start it, stop it, monitor its health, supervise its operating conditions, and coordinate its connection to the rest of the system. This makes the DSE one of the most important control-interface components in the entire resource layer.

The EPLAN design specifies that the DSE receives sensing inputs such as generator voltage, generator frequency, engine speed, oil pressure, coolant temperature, fuel level, remote start input from the PLC, and alarm-related signals. It also provides outputs for starter relay control, fuel solenoid actuation, engine stop, breaker close/trip commands, alarm relays, and running/fault status signals returned to the PLC. These input–output relationships clearly show that the DSE is not only a monitoring device. It is the operational brain of the generator subsystem at equipment level.

Project screenshots further indicate that the DSE is used with a separation between **digital control authority** and **network monitoring visibility**. Start/stop and discrete operational commands are associated with hardwired or PLC-mediated control pathways, while Modbus TCP is used to monitor generator electrical values, alarms, run status, and related operational information. This is a very realistic industrial design decision because it preserves deterministic and secure command execution while still allowing rich supervisory visibility through networked monitoring.

From the perspective of the EMS, the DSE is the execution gateway for all generator-related decisions. The EMS may decide that backup generation is needed, but that decision becomes physically meaningful only when translated through the PLC and then executed by the DSE controller. Likewise, a generator fault condition becomes relevant to supervisory energy management only when the DSE reports it back as a usable status signal. In this way, the DSE is a **boundary device** between system-level reasoning and engine-level reality.

The DSE also contributes to protection integrity. Because it monitors engine condition and generator electrical status, it helps ensure that the diesel source is not treated as an infinitely

available ideal source in the system. This is important for thesis realism. A generator can fail to start, may enter alarm states, may require protective trip, or may need operator attention. Including the DSE in the architecture acknowledges these operational realities and prevents the diesel source from being represented too simplistically.

Therefore, the DSE8610 MKII should be described in the thesis as the **local supervisory controller of the generator subsystem**. It executes start/stop logic, monitors health and electrical state, provides protective response pathways, and exports usable generator status to the higher-level microgrid controller. Its presence strengthens the implementation credibility of the whole project by showing that the diesel generator is integrated as an industrial asset rather than as an abstract on/off block.

3.4.6 BMS Role and Battery Protection Logic

The Battery Management System, or BMS, is the device that makes battery participation in the microgrid safe, observable, and controllable. In energy management discussions, batteries are sometimes treated merely as energy reservoirs with SOC values attached to them. In real systems, however, a battery cannot be responsibly dispatched unless a BMS supervises its operating limits, safety condition, and permission status. The present project reflects this reality by explicitly placing the BMS within the battery branch and connecting it to the PLC-centered control architecture.

According to the EPLAN documentation, the battery bank is associated with a BMS and linked to the DC bus through a battery breaker and main contactor, while the PLC receives battery-related status and can send commands that affect contactor behavior and integration with the wider system. This means the BMS is not merely a passive monitor. It is part of the protection logic that determines whether the battery may remain connected, whether charging/discharging is permitted, and whether alarm conditions must be communicated upward.

In the simulation environment, battery protection is represented in simplified but meaningful form through SOC minimum and maximum thresholds and a battery power limit. The EMS only allows discharge while SOC is above the minimum threshold and only allows charging while SOC is below the maximum threshold. Although this is not a full battery safety model, it is the logical macro-level expression of what a BMS enforces in practice: avoid overdischarge, avoid overcharge, and keep the battery within its safe usable operating region.

This protection role matters greatly because the battery is both valuable and vulnerable. Unlike the diesel generator, which can in principle continue operating as long as fuel and mechanical health are adequate, the battery has electrochemical constraints that can be damaged by abuse, excessive current, temperature excursions, or operation outside safe SOC windows. Therefore, the BMS acts as the **guardian of storage integrity**. It ensures that the battery is not sacrificed in the name of short-term continuity in a way that would reduce its long-term usefulness or compromise safety.

The BMS also has an important informational role. The PLC and supervisory layers need battery-related signals such as SOC, battery alarms, and availability states in order to make correct dispatch decisions. A battery that exists physically but is unavailable due to protective lockout is functionally different from a battery that is healthy and dispatchable. Thus, the BMS transforms raw battery presence into qualified battery availability. That distinction is central to accurate EMS design.

In a broader architectural sense, the BMS is what allows the battery to participate in the source hierarchy as a managed resource instead of an uncontrolled buffer. It provides the permission structure through which charging and discharging can occur within acceptable limits, and it supports contactor-level isolation when those limits are violated. Therefore, the BMS should be presented as both a **protection subsystem** and a **resource-enablement subsystem**. Without it, the battery could not be responsibly integrated into an autonomous village microgrid.

3.4.7 Resource Interaction Hierarchy in the System

The individual descriptions of PV, battery, diesel generator, inverter/charger, DSE, and BMS become fully meaningful only when they are viewed as part of a single **resource interaction hierarchy**. The microgrid is not designed as a flat set of equal sources; it is structured as a layered interaction system in which each resource has a preferred operational position and a specific response role under changing conditions. This hierarchy is one of the strongest conceptual features of the whole project.

At the top of the energy preference hierarchy is the **PV array**, which acts as the first-choice source whenever solar production is available. Its priority comes from its zero-fuel nature and renewable character. Beneath it sits the **battery**, which provides short-term balancing and preserves critical service when PV is temporarily insufficient. At the next level sits the **diesel generator**, which is activated when renewable-led operation can no longer maintain protected

supply within battery safety limits. This three-step sequence—renewable first, storage second, thermal backup third—is the core energy interaction logic of the system.

Supporting this energy hierarchy is a parallel **conversion and control hierarchy**. The inverter/charger is the conversion bridge that enables energy to move between DC and AC domains. The BMS determines whether storage participation is safe and permitted. The DSE determines whether generator participation is healthy and executable. The PLC supervises commands and state integration across the overall architecture. Node-RED and the dashboard then expose selected operational states and override functions to the supervisory level. Together, these components create a layered system in which energy flow, device safety, equipment control, and operator visibility reinforce one another rather than acting independently.

The MATLAB EMS makes this hierarchy visible in simplified algorithmic form. The code first allocates PV to F1, then uses the battery if F1 remains under-supplied, then invokes the generator for the residual critical demand, and only afterward considers lower-priority feeders and battery charging with remaining renewable surplus. This sequence effectively encodes the resource philosophy of the whole design. It is not simply a control routine; it is the operational constitution of the microgrid.

This hierarchy also provides interpretive strength in later chapters. For example, if the generator runtime decreases while F1 continuity remains protected, the reader can understand that the renewable–storage interaction is functioning well. If battery SOC repeatedly reaches its lower threshold before sunrise, the design may require either higher PV capture, revised load management, or different reserve strategy. If F2 and F3 are frequently curtailed despite high daytime PV, this may indicate a mismatch between resource sizing and feeder demand assumptions. Thus, the hierarchy is not only architectural; it is diagnostic.

In final analytical terms, the resource interaction hierarchy can be summarized as follows:

- **PV provides the preferred primary energy path**.
- **Battery provides short-term flexibility and continuity reserve**.
- **Diesel provides controllable backup when renewable-led operation reaches its safe limit**.
- **Inverter/charger provides the bidirectional conversion platform that makes hybrid interaction physically possible**.

- **BMS and DSE provide local protection and executable resource readiness**.
- **PLC and supervisory layers provide coordination, visibility, and operational control**.

This makes the proposed system more than a simple collection of components. It is a structured energy ecosystem in which each resource has a defined priority, a defined boundary, and a defined interaction rule. That is precisely what gives the project its engineering coherence.

3.5 Electrical and Operational Requirements

3.5.1 PV Array Rating and String-Level Requirements

The photovoltaic subsystem must satisfy both energy-production requirements and inverter/DC-bus compatibility requirements. In the project design, the PV field is specified as a 150 kWp array composed of 266 modules arranged as 14 strings of 19 modules each, with approximate operating voltage 789 V, cold open-circuit voltage Voc,cold≈952*Voc*,*cold*≈952 V, string current Imp,string≈13.25*Imp*,*string*≈13.25 A, and total short-circuit current Isc,array≈190*Isc*,*array*≈190 A These values are not just nameplate data; they establish the electrical envelope within which the combiner, DC breaker, surge protection, MPPT stage, and inverter input must all safely operate.

A correctly designed PV string arrangement must satisfy two simultaneous voltage conditions. First, the maximum cold-weather open-circuit string voltage must remain below the maximum DC input limit of the inverter or charge-control interface. Second, the hot-weather MPP voltage must remain high enough to stay within the operating MPPT range of the converter. This principle is directly relevant to the present design because the inverter/charger section indicates a DC bus capability up to about 1100 V, while the EPLAN PV field reports a cold VocV_{oc}Voc near 952 V, leaving a usable engineering margin below the converter ceiling.

The project therefore appears electrically consistent at the string-voltage level, but the thesis should explicitly state that this consistency is conditional on temperature-corrected verification, not only nominal STC module values. In real PV engineering, string count is never accepted merely from nominal module voltage; it must be checked against site minimum temperature for VocV_{oc}Voc and site maximum module temperature for VmpV_{mp}Vmp. Since the

EPLAN values already present a cold-voltage estimate, this strengthens the design and suggests that the array sizing has moved beyond a purely nominal layout.

At the current level, parallel stringing increases total array current while string voltage remains essentially unchanged. This means the combiner box, DC cabling, DC breaker, MPPT input, and surge devices must all be selected with sufficient current rating and short-circuit withstand capability for the full array contribution, not merely one string. The reported array short-circuit current of about 190 A is especially important because it informs the interrupting and thermal requirements of upstream protection and conductor sizing.

The PV array also has installation-quality requirements that should be acknowledged in the thesis because they affect actual performance even when the electrical ratings are acceptable. Good PV string design requires uniform string voltage, minimized module mismatch, avoidance of partial shading across mixed modules in the same string, and routing that limits avoidable voltage drop in DC runs. In practical terms, these are not optional refinements; they are necessary to ensure that the 150 kWp installed field behaves close to its intended production profile rather than suffering hidden energy losses or unequal string loading.

Another requirement is coordination with the inverter loading ratio and the DC/AC conversion stage. General PV design guidance commonly accepts a PV-to-inverter power ratio around 90–110% depending on site conditions and desired energy yield strategy. In the present project, the PV field is 150 kWp while the inverter/charger identified in the EPLAN excerpt is 100 kW AC output equipment, which implies that the DC side is intentionally larger than the single inverter AC nameplate. This does not automatically mean the design is wrong; rather, it suggests one of two engineering interpretations: either controlled clipping at very high irradiance is acceptable, or the conversion stage is being treated functionally in the control model rather than as a strict one-to-one PV nameplate bottleneck. The thesis should state this clearly so the reader understands whether the oversizing is deliberate for improved energy harvest or whether additional conversion staging is assumed elsewhere in the full hardware concept.

Table 3-1 PV electrical design summary

Parameter	Project value	Engineering requirement
Array rated power	**150 kWp**	**Must cover primary daytime renewable contribution and match EMS source assumptions**
Module configuration	**266 modules, 14 strings × 19 modules**	**Strings must be electrically uniform and temperature-verified**
String MPP voltage	**About 789 V**	**Must stay within inverter MPPT operating range at high temperature**
Cold open-circuit voltage	**About 952 V**	**Must remain below inverter/DC bus maximum voltage**
String current	**About 13.25 A**	**Used for string conductor and combiner design**
Array short-circuit current	**About 190 A**	**Used for protective-device and bus current withstand selection**

As shown in table 3-1, the operational requirement tied to the PV array is equally important: PV is the first source to be used by the EMS. In the simulation, the PV output is generated between sunrise and sunset with a peak of 150 kW and is dispatched first to the feeders before battery charging is considered. Therefore, the PV subsystem must not only be electrically adequate but also operationally observable, because the controller needs real-time power availability feedback to determine feeder restoration, charging opportunities, and backup generator necessity.

3.5.2 Battery Voltage, Capacity, and SOC Window Requirements

The battery subsystem must satisfy three categories of requirements simultaneously: voltage compatibility with the DC bus and converter, energy adequacy for short-term balancing, and

protection-oriented operating limits that preserve battery life and safety. In the project's electrical documentation, the battery bank is specified as approximately 123 kWh, 600 V nominal, 100 Ah, with maximum charge/discharge current around 125 A and an intended SOC operating window of 90% to 30%. In the simulation layer, the battery is represented in simplified form as a 100 kWh storage block with the same 90% initial SOC, minimum SOC of 30%, maximum SOC of 90%, and a power limit of 60 kW.

The nominal 600 V battery requirement is important because it places the battery squarely in the high-voltage DC domain of the microgrid rather than in a low-voltage auxiliary storage architecture. This allows efficient power transfer to the inverter/charger and reduces current for a given power level compared with lower-voltage battery systems. At the same time, a high-voltage battery imposes stricter insulation, contactor, breaker, isolation, and BMS coordination requirements. Therefore, the battery voltage requirement is not only about efficient energy transfer; it is also a driver of the safety architecture.

Capacity requirements must be interpreted correctly. The battery is not sized to supply the whole village through long renewable deficits; instead, it is intended to support **short-term bridging and balancing**. This is reflected in the EMS, where the battery is used mainly to sustain F1 when PV is temporarily inadequate and before generator support becomes necessary. Thus, the battery's required energy capacity is tied less to full autonomy duration and more to the need for absorbing renewable fluctuations, covering transition periods, and delaying generator starts during moderate shortages.

The SOC window of 90% to 30% is a particularly important operational requirement. This range defines the usable energy band rather than the theoretical full electrochemical capacity of the battery. In the simulation, the controller never allows charging above 90% or discharging below 30%. Such a policy is consistent with battery management practice because limiting the active SOC window helps reduce stress, preserve cycle life, and avoid unsafe or highly degradative operating regions. The thesis should explicitly emphasize that the available dispatchable battery energy is therefore only a fraction of nominal nameplate energy, not the full stored energy rating.

The battery power limit is also as important as energy capacity. In the simplified model, the battery maximum power is 60 kW, which means that even if sufficient energy remains in the battery, the converter and battery interface together are not expected to supply unlimited instantaneous power. This is fundamental to system realism. Batteries are constrained by both

kWh and kW, and microgrid reliability depends on respecting both. A battery with adequate energy but inadequate discharge power may still fail to support a rapid feeder demand increase. Conversely, a battery with high power capability but low remaining SOC may fail to sustain support for the required duration.

The current rating around 125 A at approximately 1C in the EPLAN document reinforces this point. It signals that the battery must be operated within bounded charging and discharging current, and that BMS-enforced current protection is part of the design intent. This matters because excessive current can create thermal stress, accelerate degradation, and trigger protective disconnection. Therefore, battery operation must always be treated as a **permitted dispatch resource**, not an unrestricted power reservoir.

Table 3-2 Battery requirement summary

Parameter	Project value	Operational meaning
Nominal voltage	**600 V**	**Must match high-voltage DC bus architecture and converter interface**
Nominal energy	**123 kWh in design, 100 kWh in simulation**	**Supports short-term balancing, not full long-duration autonomy**
Capacity	**100 Ah**	**Defines stored charge base with nominal DC voltage**
SOC window	**90% to 30%**	**Limits dispatch to a protected usable band**
Maximum current	**About 125 A**	**Bounds safe charge/discharge operation and informs protection/BMS logic**
Simulation power limit	**60 kW**	**Caps instantaneous battery contribution even when energy is available**

3.5.3 Generator Rated Power and Operating Constraints

The diesel generator must be specified not only by its nameplate rating, but also by the conditions under which it may be started, loaded, and integrated with the AC system. In the present design, the generator is rated at 100 kW, 400 V, 50 Hz, three-phase, and is connected to the main AC bus through its own breaker and DSE8610 MKII controller. In the EMS model, generator output is limited to a rated value of 100 kW and is invoked only after PV and battery support are insufficient to preserve the critical load feeder.

The most basic generator requirement is adequacy of rated power relative to protected load. Because F1 is the highest-priority feeder and may reach roughly 57 kW in the simulation formulation, a 100 kW generator provides enough capacity to preserve critical demand with useful operating margin under backup conditions. This margin is important because diesel sets should not be sized only to exactly match average critical load; they must be capable of handling realistic fluctuations, motor starts, voltage regulation needs, and nonideal transient conditions.

Operating constraints are equally important. A diesel generator is not an ideal source that can be switched from zero to any load forever without consequence. It has warm-up behavior, AVR-controlled voltage response, fuel and cooling dependencies, and mechanical health limitations reflected in monitored variables such as oil pressure, coolant temperature, and engine speed. These sensed variables are explicitly listed in the DSE page of the project and therefore should be incorporated conceptually into the operating requirements section, even if the mathematical model abstracts them.

The generator also has a fundamental dispatch constraint: it is intended as backup, not as the primary daily energy source. In the EMS, the generator is only used when PV and battery can no longer maintain critical service. This means the generator requirement is not only "can it produce 100 kW?" but also "can it start reliably when commanded, synchronize with the protected AC bus condition, and sustain stable operation once accepted into service?" Reliability of start and sustained run are especially important in microgrid backup planning, because the practical value of a generator depends as much on dependable availability as on rated capacity.

A further operational constraint is generator loading quality, which becomes especially relevant in small microgrids with varying demand. Generators that operate continuously at excessively low load are known to suffer from poor combustion behavior and long-term damage mechanisms such as wet stacking, making low-load operation undesirable over extended periods. For this reason, the generator must be controlled with both upper and lower loading considerations in mind. The thesis should therefore distinguish between generator **capacity rating** and generator **healthy operating region**, since the latter determines whether dispatch is sustainable in real practice.

3.5.4 AC Bus Voltage and Frequency Requirements

The AC bus is the common electrical platform on which source integration and feeder distribution occur, so its voltage and frequency requirements are central to both power quality and equipment compatibility. In the project design, the inverter output and the diesel generator

are both defined to operate at 400 V, 50 Hz, three-phase, and they connect into the main AC bus identified as BUS_AC01. This common rating is essential because no hybrid microgrid can function coherently if major AC sources and feeders are not aligned to the same bus voltage and frequency framework.

The 400 V, 50 Hz requirement has several implications. First, all downstream feeders, breakers, protection settings, and connected three-phase equipment must be compatible with that bus level. Second, the inverter/charger must regulate its AC output so that the bus remains within acceptable voltage and frequency limits during renewable-led operation. Third, the generator AVR and speed governor must support the same nominal bus conditions during diesel-backed operation. Thus, bus requirements are not passive specifications; they are active obligations imposed on both major sources.

Voltage stability at the AC bus matters because undervoltage can reduce motor performance, increase currents, and disrupt equipment, while overvoltage can damage insulation and electronic devices. Frequency stability matters because it reflects power balance and source control quality. In islanded microgrids especially, poor frequency control can signal inadequate source dispatch, improper transitions, or unstable sharing between sources. Even though the project does not model full transient frequency dynamics in detail, the bus frequency requirement remains fundamental because the generator controller, inverter monitoring, and PLC measurements all depend on valid electrical reference values.

Table 3-3 AC platform summary

Element	Required value	Why it matters
Main AC bus	**400 V, 50 Hz, 3-phase**	**Common operating platform for inverter, generator, and feeders**
Inverter AC output	**400 V, 50 Hz, 3-phase**	**Must match bus for stable renewable-led operation**
Generator output	**400 V, 50 Hz, 3-phase**	**Must match bus for backup service and safe breaker closure**
Feeder supply basis	**From BUS_AC01 through ACBR03–05**	**Feeder energization depends on healthy bus state**

As shown in table 3-3, the bus also serves as the point from which feeder switching decisions become meaningful. F1, F2, and F3 breakers branch from the AC bus, and therefore feeder energization is valid only if bus conditions are acceptable. This means feeder supply logic should be considered conditional not only on power sufficiency but also on bus health. In a refined implementation, feeder closure permissives should logically require healthy bus voltage, valid source status, and absence of relevant trip conditions.

3.5.5 Protection Requirements: Breakers, Fuses, Contactors, SPDs

Protection in this microgrid must be understood as a layered requirement rather than a single device requirement. The project already reflects this philosophy in EPLAN by distributing protective elements across the PV branch, battery branch, AC source paths, PLC supply, and feeder outgoing circuits. This is the correct architectural approach because hybrid microgrids contain multiple fault domains: DC generation faults, battery faults, converter interface faults, AC feeder faults, control-power faults, and surge-related disturbances.

On the AC side, the design includes dedicated breakers for the inverter output, generator input, and each feeder: ACBR01 for PV inverter output, ACBR02 for the diesel generator, and ACBR03–05 for feeders F1–F3. These breakers are required not only for fault interruption but

also for operational isolation, selective load control, and maintenance safety. In other words, an AC breaker in this architecture serves a dual function: protection and controllable switching. This dual role is especially important for feeder priority logic because controlled shedding and restoration rely on switchgear that can be commanded safely and repeatedly.

On the DC PV side, the design includes a DC combiner, PV-side SPD, MPPT stage, and DC breaker. This reflects the fact that PV arrays can continue producing under daylight and therefore require DC-capable isolation and protective devices rather than AC substitutes. The SPD on the PV side is particularly important because long outdoor DC cable runs and exposed array structures are especially vulnerable to lightning-induced and switching surges. Good surge protection practice recommends staged protection and short, well-grounded SPD connections to reduce let-through voltage and improve protection performance.

On the battery side, the presence of both a battery DC breaker and a main battery contactor is a strong design feature. The breaker provides fault isolation and manual/service protection, while the contactor enables controlled connection and disconnection under BMS/PLC authority. This distinction matters. A breaker is not the same as a routine operational control element, and a contactor is not a substitute for full fault-clearing protection. Using both is therefore a sound requirement in a high-voltage DC battery subsystem.

Control and auxiliary circuits also require protection. The PLC panel includes a dedicated DC protection element for the 24 V control supply. This is often overlooked in academic descriptions, but it is essential in real systems because loss of the control supply can disable monitoring, breaker commands, remote start logic, alarms, and safe interlocking. A microgrid with well-designed power hardware but poorly protected control power can still fail unsafely or become inoperable.

Table 3-4 Protection device mapping

Protection element	Location	Primary purpose
ACBR01	**Inverter AC output**	**Isolate inverter from AC bus, protect source-side AC connection**
ACBR02	**Generator to AC bus**	**Generator connection protection and controlled coupling**
ACBR03–05	**Feeders F1, F2, F3**	**Feeder fault protection and load-control switching**
DCBR_PV01	**PV DC side**	**PV array isolation and DC fault protection**
SPD_PV01	**PV branch**	**Surge suppression for lightning/switching transients**
DCBR_BATT01	**Battery branch**	**Battery DC fault isolation**
CONT_BATT01	**Battery branch**	**Controlled battery connection/disconnection**
DCBR_PLC01	**PLC 24 V supply**	**Control-power protection**

As shown in table 3-4, the thesis should also mention selectivity and coordination as implicit protection requirements. The goal is not merely that "a breaker trips," but that the **correct** protective element trips with minimal unnecessary outage extension. For example, a feeder fault should ideally trip the corresponding feeder breaker rather than unnecessarily disconnecting the entire AC bus or source path. Likewise, a battery fault should isolate the battery branch without forcing loss of healthy PV conversion if operating conditions permit.

3.5.6 Minimum Generator Loading and Safe Operation Limits

One of the most important practical operating requirements in diesel-backed microgrids is minimum acceptable generator loading. While the generator is rated at 100 kW in the project, that does not mean it should be operated continuously at any very low output level just because it can technically remain online. Industry guidance commonly warns against prolonged operation below about 30% of rated load because persistent light loading can lead to incomplete combustion, carbon buildup, and wet stacking. For a 100 kW generator, this implies that sustained operation much below roughly 30 kW should be treated as undesirable unless there is a specific mitigation strategy.

This requirement has direct implications for the EMS. If the generator is started solely to preserve a small residual load while the battery is available and the generator remains deeply underloaded, then the system may protect service continuity but at the cost of unhealthy generator operation. Therefore, a good operational strategy should ensure that generator starts are justified by meaningful support need, and once the generator is online, dispatch logic should avoid leaving it idling at very low load for extended periods whenever an alternative operating mode is possible.

In practical terms, this means generator control should consider at least four safe-operation limits:

- Minimum healthy loading region, typically avoiding prolonged operation below about 30% of rating.
- Maximum loading region that leaves a margin below full continuous overload exposure and preserves voltage/frequency regulation quality.
- Thermal and engine-condition limits, such as coolant temperature, oil pressure, and speed health, already monitored by the DSE.
- Start/stop cycling limits, because excessively frequent starts can reduce equipment life and undermine reliability.

The Node-RED dashboard architecture already contains a parameter called maxGenLoad, which indicates that supervisory setpoint thinking is present in the control concept. That is useful because it suggests the project can accommodate generator loading rules at the supervisory level rather than treating the generator as a binary start/stop device. Even if the current simulation is

simplified, the thesis can legitimately state that safe generator operation should include not only emergency availability but also acceptable load band management.

The project should also make clear that “minimum loading” does not mean the generator must instantly trip whenever output drops below a threshold. Rather, it means prolonged sub-threshold operation should be avoided as a normal strategy. In some situations, brief low-load operation may be unavoidable during transitions, synchronization, or temporary critical support. The engineering requirement is therefore best stated as a **time-qualified operating rule** rather than an absolute instantaneous prohibition.

3.5.7 Feeder Control and Synchronization Requirements

Feeder control requirements in this microgrid arise from two parallel needs: priority-based service allocation and electrically safe switching. On the first side, feeders F1, F2, and F3 must be controllable independently because the EMS depends on selective energization and load shedding to preserve critical service under constrained resources. On the second side, feeder switching must occur only when the upstream AC bus is healthy and the relevant source path is already established.

The EPLAN single-line diagram shows dedicated feeder breakers ACBR03, ACBR04, and ACBR05 for F1, F2, and F3 respectively. This hardware separation is the electrical basis for feeder-priority implementation. In the control philosophy used in the MATLAB model, F1 is supplied first, F2 next if surplus remains, and F3 last. Therefore, feeder control is not merely a distribution convenience; it is a central operational mechanism through which the resource hierarchy is translated into actual service outcomes.

Synchronization requirements here should be interpreted carefully. The feeders themselves are downstream load branches and do not require source-style synchronization in the same way two active AC generators would. However, feeder closure still requires **switching permissives** linked to the synchronized state of the upstream bus-source combination. In practical terms, this means that a feeder should not be reclosed unless:

- The AC bus is energized and within acceptable voltage/frequency conditions.
- The supplying source path, inverter or generator, is confirmed healthy and connected.
- No feeder trip or fault lockout is active.

- The EMS or operator command authorizes that feeder's restoration according to priority logic.

The supervisory control architecture already supports feeder control through Node-RED API routes such as /api/control/feeders, where feeder enable/disable states are managed and returned in status JSON as f2Status and f3Status. This is valuable because it means the project does not stop at conceptual feeder prioritization; it includes an actual digital pathway by which feeder states can be monitored and commanded. For the thesis, this strengthens the argument that feeder control has both electrical meaning and implementation realism.

It is also advisable to mention anti-oscillation behavior as an operational requirement. If feeder restoration is attempted too aggressively when source margins are marginal, repeated on/off cycling can occur, damaging equipment confidence and creating unstable service perception. Therefore, feeder restoration should ideally include hysteresis, source margin checks, or short persistence timers so that reconnection happens only when the supply condition is genuinely sustainable. Even where not fully coded in the current simplified EMS, this is an important engineering requirement to state in the design chapter.

3.5.8 Alarm, Trip, and Emergency Stop Requirements

A hybrid microgrid is not safe merely because it contains generation and protection hardware; it must also have a clear philosophy for alarming abnormal conditions, tripping unsafe states, and forcing emergency shutdown when required. The present project already contains the architectural elements needed for this philosophy. The DSE controller provides generator alarms and status back to the PLC, the BMS supervises battery condition, the inverter supplies monitored values and operational status, and the Node-RED/dashboard layer exposes alarms to the operator interface. This creates the foundation for a tiered abnormal-condition management strategy.

Alarm requirements should cover any condition that is abnormal but not yet necessarily unsafe enough to justify immediate disconnection. Examples include low SOC warning, generator running state changes, feeder disabled by command, loss of communication, voltage or frequency abnormality warning, battery availability restriction, or inverter mode abnormality. The dashboard already represents alarms as a list in returned JSON and supports reset functionality via /api/control/reset-alarms. That means the project has a practical supervisory path for annunciation, acknowledgment, and operator awareness, which is a major implementation strength.

Trip requirements are more severe than alarms and should correspond to conditions where continued energization becomes unsafe or technically unacceptable. On the generator side, such trips may be associated with serious engine faults or electrical problems reported through the DSE, such as severe oil pressure loss, overtemperature, or unacceptable electrical condition. On the battery side, the BMS may need to force contactor opening when overcurrent, unsafe voltage window, or other battery-protection conditions are violated. On the feeder side, protective devices should isolate faulted branches without propagating unnecessary system-wide outage.

Emergency stop requirements represent the highest level of intervention. An emergency stop should immediately place equipment into a predefined safe state, prioritizing personnel and asset protection over service continuity. Although the available excerpts do not detail a physical E-stop pushbutton layout, the project's control architecture strongly implies that stop commands, breaker trip paths, contactor control, and alarm logic are already organized through the PLC/DSE/control layer. The thesis should therefore define emergency stop functionally as a command pathway that disables hazardous operation, opens or blocks relevant power paths, and annunciates the event clearly to operators.

Table 3-5 Distinguish the three layers

Event layer	Purpose	Typical response
Alarm	Warn operator of abnormal but non-immediate-danger state	Annunciate, log, continue with caution or degraded mode
Trip	Automatically isolate unsafe equipment or faulted branch	Open breaker/contactor, block restart until condition clears
Emergency stop	Force rapid safe shutdown in critical hazard scenario	Immediate stop/trip logic with highest priority

As shown in table 3-5, this distinction is important because not every abnormal condition deserves the same response. If every alarm caused a trip, the system would become unnecessarily unavailable. If serious trip-level conditions were treated as mere warnings, the system would become unsafe. The purpose of this requirement structure is to preserve both reliability and protection integrity.

The project's existing status model in Node-RED, where mode, SOC, generator status, source identity, feeder states, and alarms are assembled into a supervisory JSON object, is especially useful here. It means abnormal conditions can be detected not only locally at equipment level but also represented coherently at system level. That is exactly what a real hybrid microgrid needs: not isolated fault signals alone, but an intelligible operational picture that supports correct operator intervention and automated control behavior.

Integrated Design View

When considered together, the electrical and operational requirements in this section define the allowable operating envelope of the whole microgrid. The PV subsystem must remain within verified string-voltage and current limits, the battery must remain within protected SOC and current boundaries, the generator must operate within a healthy loading region, the AC bus must remain at 400 V and 50 Hz, and the protection/control architecture must be able to isolate faults without destroying service continuity for the most important feeders. In other words, this section defines the difference between a microgrid that merely exists on paper and a microgrid that can be built, operated, protected, and trusted in practice.

A useful engineering interpretation is that the project operates through four nested envelopes:

- **Electrical envelope**: voltage, current, frequency, power, and rating compatibility.
- **Operational envelope**: dispatch order, SOC band, minimum generator loading, feeder restoration logic.
- **Protection envelope**: breakers, contactors, SPDs, trip logic, branch isolation.
- **Supervisory envelope**: alarms, status visibility, remote commands, and operator override pathways through PLC and Node-RED.

That layered interpretation is very powerful for the thesis because it shows the reader that the system is not governed by a single rule or a single component. It is governed by a coordinated set of requirements that must all be satisfied simultaneously for stable, safe, and resilient operation.

3.6 Control Requirements and EMS Logic

3.6.1 EMS Objectives and Priorities

The Energy Management System is the supervisory decision layer that transforms the hybrid microgrid from a collection of electrical assets into a coordinated service-delivery system. Its purpose is not merely to switch sources on and off, but to allocate limited generation and storage resources in a way that preserves critical service continuity, protects equipment, reduces unnecessary diesel operation, and maintains a stable and explainable operating state for the village microgrid. In the present project, these goals are evident in both the MATLAB rule-based logic and the Node-RED/PLC supervisory architecture, where source status, feeder status, SOC, generator state, and operator commands are all tied into one control framework.

The first objective of the EMS is **continuity of critical load supply**. The feeder hierarchy and dispatch code clearly show that F1 has the highest operational priority and is treated as the non-negotiable service block whenever technically feasible. This means that every control decision is implicitly evaluated against one core question: can the system continue supplying F1 safely and reliably? If the answer is uncertain, the EMS must immediately shift from optimization behavior into preservation behavior.

The second objective is **renewable maximization with controlled storage support**. The EMS gives PV the first opportunity to meet demand, then uses the battery within a protected SOC window, and only afterward invokes the diesel generator when the renewable-storage combination cannot preserve critical supply. This objective is important because it aligns operational practice with the technical and economic identity of the system as a renewable-led hybrid microgrid rather than a diesel system with solar accessories.

The third objective is **asset protection through bounded operation**. The controller does not dispatch the battery below the SOC minimum, does not charge above the SOC maximum, and does not assign the generator as the first-choice source when lower-cost resources are available. In implementation terms, this objective extends beyond the MATLAB model into BMS, inverter, PLC, and DSE interactions, where local equipment protection must remain consistent with supervisory decisions. Therefore, the EMS should be understood as a constrained optimization layer in the broad sense, even though its current realization is rule-based rather than mathematically optimized.

The fourth objective is **transparent system operability**. A strong microgrid controller must produce states that operators can interpret: which source is active, why feeders are on or off, whether the system is in Auto or manual mode, whether alarms are present, and which thresholds

are currently applied. The Node-RED dashboard structure, with fields such as mode, soc, genStatus, activeSource, f2Status, f3Status, and EMS setpoints, shows that this interpretability requirement is part of the project itself, not an afterthought. This is an important strength because a controller that cannot explain its own behavior is difficult to trust, test, and maintain.

Table 3-6 These objectives can be summarized as a practical priority ladder

EMS objective	Control meaning	Evidence in project
Preserve F1 continuity	Critical feeder is supplied before all lower priorities r	F1 served first from PV, then battery, then generator
Maximize PV utilization	Use renewable energy before fuel-backed generation	PV dispatched first; surplus PV charges battery
Protect battery and equipment	Respect SOC and power limits; bounded source usage	SOC window 30–90%, battery power limit 60 kW
Minimize unnecessary generator runtime	Generator is backup, not default source	Generator only supports remaining critical deficit
Maintain operator visibility and controllability	State and commands exposed through dashboard/API	Mode, SOC, feeder states, alarms, and setpoints visible

As shown in table 3-6, this layered objective structure gives the EMS a clear identity: it is neither a pure economic dispatcher nor a pure protection system alone. It is a **service-priority supervisory controller** that balances continuity, renewable usage, equipment safety, and operator awareness at the same time.

3.6.2 Rule-Based Control Philosophy

The control logic adopted in the project is fundamentally **rule-based**, meaning that supervisory decisions are produced through ordered logical conditions and priority rules rather than through real-time optimization solvers. This is a very appropriate approach for the present

hybrid microgrid because the system has a clear resource hierarchy, a limited number of major decision variables, and a strong requirement for interpretability and implementability in PLC/Node-RED style platforms. In small and medium standalone microgrids, rule-based EMS structures are often favored when robustness, clarity, and straightforward field deployment are more important than algorithmic complexity.

The specific rule structure in the MATLAB EMS is sequential and deterministic. At each time step, the controller evaluates available PV, feeder demand, battery SOC, and remaining deficit, then allocates sources according to a fixed order: PV first to F1, battery second for remaining F1 deficit, generator third for any residual critical deficit, then leftover PV to F2 and F3, and finally any remaining PV to battery charging. This sequencing is not merely a coding convenience; it is the formal expression of the system's operational philosophy.

A rule-based philosophy offers several important engineering advantages in this project. First, it is easy to verify because each control decision can be traced to a specific condition, threshold, or sequence branch. Second, it is easy to map into PLC logic, state machines, or Node-RED flows because Boolean decisions and ordered permissives are natural forms of implementation in industrial control platforms. Third, it is easy to explain in the thesis, which matters academically because a control strategy that cannot be clearly defended often weakens the design chapter even if it performs well numerically.

Another important point is that rule-based control in this project is **structured**, not arbitrary. The logic is built around explicit bounds: source ratings, SOC minimum and maximum, feeder priorities, and constrained generator use. Thus, the EMS does not behave like a vague heuristic; it behaves like a constrained supervisory policy. This distinction is important because well-designed rule-based EMS architectures can be highly reliable and technically mature even without optimization routines.

The rule-based philosophy also creates a natural pathway to future expansion. Because the present architecture already uses named variables such as socStartThreshold, socStopThreshold, and maxGenLoad in the dashboard and control API, the system is ready for progressive refinement into more advanced supervisory logic without needing to redesign the entire control interface. In other words, the current rule-based EMS is both a valid final controller for a practical microgrid and a sound foundation for later adaptive or predictive enhancement.

Table 3-7 Rule structure overview

Decision step	Rule action	Operational purpose
Step 1	**Serve F1 from PV first**	**Protect critical load with renewable source**
Step 2	**If F1 deficit remains, discharge battery above SOC minimum**	**Preserve continuity without immediate generator start**
Step 3	**If F1 deficit still remains, start/use generator**	**Guarantee critical-service continuity**
Step 4	**Use remaining PV for F2 then F3**	**Restore lower-priority service only after F1 security**
Step 5	**Use remaining PV to charge battery up to SOC maximum**	**Rebuild reserve margin for later deficits**

As shown in table 3-7, this structure shows that the EMS is not merely “PV-first” in a superficial sense. It is **priority-first**, and PV-first is one expression of that broader philosophy.

3.6.3 PV-First Supply Logic

The PV-first principle is one of the defining features of the control strategy. In the EMS code, the first available source at every step is solar generation, and the first allocation of that source is to the critical feeder F1. This means the controller treats PV not as auxiliary energy but as the preferred primary supply whenever irradiance is present. Such a structure is fully consistent with the system architecture, where the PV array is the dominant renewable source and the diesel generator is explicitly designated as backup.

The logic begins by setting pv_available = Ppv(i) at each time step, then assigning as much of that available PV as possible to F1 demand through p_from_pv_to_f1 = min(pv_available,

f1_demand). This simple line captures a very important policy: the renewable source is always given the first opportunity to preserve the most important load class. Only after this allocation is completed does the controller evaluate whether any deficit remains and whether additional resources must intervene.

This PV-first logic has several operational consequences. During strong solar hours, F1 may be fully supplied without battery discharge or generator participation. If PV exceeds F1, the controller can restore F2 and then F3. If PV exceeds all active load, it can then charge the battery. In this sense, PV-first logic does not simply reduce fuel consumption; it shapes the entire service-restoration behavior of the microgrid across the day.

A major strength of the present design is that PV-first logic is not isolated from the feeder-priority model. The controller does not spread PV evenly across all loads or maximize total energized load at the expense of criticality. Instead, it applies renewable energy through the same social-priority lens used everywhere else in the system. This is a strong and defensible decision because it aligns energy economics with service ethics: the cheapest source is used first, but it is also directed first to the most important services.

The daily PV profile in the MATLAB scripts further supports this operating philosophy. PV rises from zero after sunrise, reaches a midday peak up to 150 kW, and falls back to zero at sunset. This naturally creates daytime windows of abundance and evening/night scarcity. The EMS therefore treats daytime PV not only as current power but also as an opportunity for reserve rebuilding through battery charging and feeder restoration. In other words, PV-first logic is both an instantaneous supply rule and a daily operational planning principle embedded in the hour-by-hour dispatch.

Table 3-8 PV-first logic sequence

Condition	EMS action	Result
PV < F1	**All PV goes to F1 first**	**Critical feeder partly supplied, deficit remains**
PV = F1	**F1 fully supplied from PV**	**No critical deficit from solar side**
PV > F1 but < F1+F2	**F1 supplied first, remaining PV to F2**	**Partial restoration of medium-priority service**
PV > F1+F2 but < F1+F2+F3	**F1 and F2 supplied, remaining PV to F3**	**Lower-priority loads partially restored**
PV > all active feeder demand	**Surplus PV used for battery charging**	**Energy shifted into storage reserve**

As shown in table 3-8, this table shows that PV-first supply logic is tightly coupled to both load priority and storage behavior. It is not just a source preference rule; it is the **entry point for the entire daily EMS sequence**.

3.6.4 Battery Discharge and Charge Logic

The battery logic in the project is bidirectional and state-dependent. It includes one set of rules for discharge during shortage and another set for charge during renewable surplus. This is important because the battery is not treated as a generic energy buffer that is always active. Instead, its use is conditional on both system need and battery state of charge, making it a controlled flexibility resource rather than an automatic balancing sink.

During shortage conditions, the battery is used only after PV has already been allocated to F1 and a remaining F1 deficit still exists. The controller then checks whether SOC(i) > SOC_min before allowing discharge. If that condition is true, the battery contributes up to the minimum of three quantities: the remaining F1 deficit, the battery power limit, and the maximum energy that

may be withdrawn during the current time step without crossing the SOC minimum. This is one of the strongest parts of the EMS because it shows that battery dispatch is limited simultaneously by demand need, power capability, and energy availability.

The relevant discharge rule can be understood conceptually as:

- The battery may discharge only if there is still unmet critical demand after PV allocation.
- The battery may discharge only if SOC is above the minimum threshold.
- The battery discharge power may not exceed the battery power limit of 60 kW.
- The battery discharge energy in the current time step may not reduce SOC below 30%.

This makes the battery a **protected F1 support source**, not a universal supply source for all feeders. In the current EMS, battery discharge is dedicated to preserving critical demand, while F2 and F3 are supplied only from remaining PV. This design choice is conservative and technically strong because it prevents the battery from being diluted across lower-priority loads when the system is already under stress.

Battery charging follows a different logic. Once F1, F2, and F3 allocations from available PV have been completed, any remaining PV is examined for possible charging action. Charging is allowed only if SOC(i) < SOC_max, and charging power is limited by remaining PV, battery power capability, and the amount of energy that can be added without exceeding the SOC maximum. The controller records charging as negative battery power, which is then used to update SOC in the next time step. This means charging is surplus-driven rather than schedule-driven.

The SOC update equation is also important conceptually. The controller computes the battery energy change from power over the time step and updates SOC accordingly, then clips the result inside the allowed 30–90% range. This means the battery model directly influences subsequent dispatch decisions because future discharge eligibility depends on the current SOC trajectory. Therefore, battery logic is not only a response mechanism; it is also a memory mechanism inside the EMS.

Table 3-9 Battery control matrix

Battery state	System condition	EMS decision
SOC > 30%, F1 deficit exists after PV	**Shortage mode**	**Discharge battery up to power/energy limits**
SOC = 30%	**Protected minimum reached**	**Block further discharge**
SOC < 90%, surplus PV remains after load allocation	**Surplus mode**	**Charge battery up to allowable limit**
SOC = 90%	**Protected upper bound reached**	**Block further charging**
No F1 deficit and no surplus PV	**Neutral state**	**Battery remains idle**

As shown in table 3-9, this battery logic greatly strengthens the EMS because it introduces temporal intelligence into the system. PV reflects present renewable availability, generator reflects present firm backup, but the battery reflects both **present flexibility and future resilience margin**.

3.6.5 Generator Auto/Manual Logic

The generator logic in the project should be understood in two parallel layers: **automatic generator logic** inside the EMS and **manual generator logic** exposed through the supervisory interface. This distinction is very important because it separates autonomous continuity-preservation behavior from operator authority. A robust microgrid normally requires both.

In automatic mode, the generator is not the first source considered by the EMS. It becomes active only when PV has already been applied to F1 and the battery, within its safe SOC window, cannot eliminate the remaining critical deficit. Thus, automatic generator logic is inherently conditional and backup-oriented. The controller does not start the generator to maximize comfort

loads or to preserve battery at all cost; it starts the generator to prevent unacceptable loss of critical service.

At the hardware level, this automatic logic is implemented through the relationship between the PLC and the DSE8610 MKII controller. The EPLAN documentation shows that the PLC sends start/stop commands to the DSE and receives running/fault feedback, while the DSE itself manages starter relay, fuel solenoid, engine stop, breaker close/trip coils, and generator status signals. This means the EMS decides **when** generator support is required, while the DSE decides **how** that engine-level sequence is executed safely. That is a strong and realistic division of control responsibility.

Manual logic is represented in the Node-RED/dashboard control layer. The dashboard can send POST /api/control/generator with commands such as start or stop, and the system responds through status and alarm updates. This shows that generator authority is not locked entirely inside the automatic algorithm; it can be intentionally overridden by an operator. Such override capability is essential for testing, maintenance, commissioning, fault recovery, and controlled operational experiments.

However, manual control should not be interpreted as uncontrolled freedom. In a well-structured system, manual commands should still be mediated by device permissives, fault checks, and interlocks. For example, a manual start command may be accepted by the supervisory layer, but actual generator start should still depend on DSE health logic and engine availability. Likewise, a manual stop should be blocked or delayed if it would create an unsafe condition for a critical feeder unless the emergency procedure specifically requires shutdown. Even if these nuances are not all explicitly coded in the simplified model, they are fully consistent with the architecture shown in the project and should be stated as control requirements.

Table 3-10 Auto vs manual generator control

Control layer	Role	Project evidence
Automatic EMS logic	**Starts/uses generator only after PV and battery are insufficient for F1**	**Rule-based dispatch in MATLAB**
PLC supervisory control	**Sends generator commands and receives status/faults**	**PLC mapped to DSE and device feedback**
DSE local controller	**Executes engine-level start/stop and breaker actions**	**Starter relay, fuel solenoid, breaker close/trip outputs**
Manual supervisory interface	**Allows operator start/stop commands**	**/api/control/generator in Node-RED/dashboard**

As shown in table 3-10, this layered structure is one of the strongest features of the project because it avoids two extremes: a generator that is purely manual and operationally blind, or a generator that is fully automatic but inaccessible for maintenance and field intervention.

3.6.6 Generator Start/Stop Conditions

The generator start and stop conditions are among the most important parts of the EMS because they define when the system transitions from renewable-led operation into fuel-backed continuity mode and when it can safely return. In the MATLAB EMS, the start condition is implicit in the dispatch sequence: if F1 still has unmet demand after PV allocation and permissible battery discharge, the remaining deficit is assigned to the generator up to its rated power. This means the start condition can be expressed conceptually as:

Start generator if F1remaining after PV and battery>0\text{Start generator if} F1_{\text{remaining after PV and battery}} > 0Start generator if F1remaining after PV and battery>0

This is a very strong rule because it ties generator use directly to critical-service preservation rather than to generic low renewable output alone. The generator is not started just because PV is low; it is started because low PV plus limited battery support creates an actual critical-load deficit. That makes the start logic service-aware rather than source-centric.

In the supervisory layer, the project introduces explicit start/stop thresholds through dashboard setpoints such as socStartThreshold and socStopThreshold, shown in the Node-RED status JSON and update API. This is extremely valuable because it indicates the architecture is prepared for a threshold-based control refinement beyond the simplified MATLAB rule. In a practical implementation, a reasonable interpretation is:

- Start generator automatically when SOC falls below the configured start threshold and available renewable power cannot sustain required service.For-NODE-RED-3.txt+1
- Keep generator online until the stop threshold is reached or until the supply condition is sufficiently restored.
- Use separate start and stop thresholds to avoid rapid cycling and chatter.

The use of separate thresholds, such as start at 30% and stop at 50%, is a classic and sound control technique because it introduces hysteresis. Without hysteresis, a generator could repeatedly start and stop around one boundary, causing wear, poor stability, and poor operator confidence. The project dashboard's explicit inclusion of both start and stop values suggests that the control design already understands this practical requirement.

Generator stop conditions should be defined as carefully as start conditions. A robust stop policy should not simply stop the generator the moment PV reappears or SOC moves slightly upward. Instead, stop should occur when:

- Critical load supply is secure without continued generator dependence.
- Battery reserve has recovered above the stop threshold, if threshold-based logic is applied.
- No relevant generator alarm, cooldown requirement, or transition constraint prevents an orderly stop.
- The system will not immediately re-enter shortage if the generator is removed.

Table 3-11 Generator start/stop logic summary

Condition type	Practical meaning	Project basis
Start by deficit	**F1 still unmet after PV and battery**	**Explicit in rule-based EMS**
Start by SOC policy	**Battery reserve too low for continued autonomous support**	**Dashboard includes socStartThreshold**
Stop by reserve recovery	**Battery or renewable condition sufficiently recovered**	**Dashboard includes socStopThreshold**
Stop by service security	**Generator no longer needed to preserve critical continuity**	**Implied by renewable/battery sufficiency logic**
Manual stop/start	**Operator forces action through dashboard/API**	**/api/control/generator**

As shown in table 3-11, the project writing, this section becomes much stronger if you explicitly distinguish between the **current simulated start rule** and the **practical supervisory threshold rule** supported by the control interface. That shows both what the simulation already does and how the real system logic is intended to evolve.

3.6.7 Feeder Enable/Disable Logic

Feeder control is the part of the EMS where energy priorities become visible as service outcomes. The project's feeder hierarchy consists of F1 as critical, F2 as medium priority, and F3 as low priority, and the EMS allocates power to them in exactly that order. Therefore, feeder enable/disable logic is not peripheral to the controller; it is one of its main mechanisms.

In the MATLAB EMS, F1 is effectively always attempted first, and the controller computes F1_supplied, F2_supplied, and F3_supplied as separate outputs. F2 and F3 are enabled only when enough PV remains after preserving F1. This means that in the current control design, the default source-based feeder logic is:

- F1 is protected and supplied first.
- F2 is supplied only from PV remaining after F1.
- F3 is supplied only from PV remaining after F2.
- Under shortage, lower-priority feeders are disabled by omission of available power allocation rather than by an optimization solver.

This is a very clear and defensible approach because it creates a one-to-one relationship between source scarcity and feeder shedding sequence. If F3 disappears, the system is under moderate stress. If F2 also disappears, the system is under severe conservation. If F1 becomes threatened, the system has reached critical deficiency. That makes feeder state itself a diagnostic output of the EMS.

The Node-RED supervisory layer adds a second dimension: explicit operator feeder control. The dashboard contains f2Status and f3Status indicators and supports /api/control/feeders commands of the form { "feeder": "F2", "enabled": true } or similar. This means feeder logic exists both as automatic service-allocation behavior and as explicit commandable switch-state behavior. That is a major strength because it brings the model closer to real field operation.

The thesis should also clearly state that F1 is normally excluded from casual manual disable actions because it represents protected critical service. In contrast, F2 and F3 are intentionally suitable for selective enable/disable control, since they form the controllable service margin of the microgrid. This distinction reflects good operational discipline: the same hierarchy used by the EMS should also shape operator permissions.

Table 3-12 Feeder logic interpretation

Feeder	Priority	Automatic EMS behavior	Manual supervisory behavior
F1	**Critical**	**Supplied first from PV, then battery, then generator if needed**	**Normally protected from routine disable actions**
F2	**Medium**	**Enabled only after F1 is secure and PV remains**	**Can be enabled/disabled from dashboard/API**
F3	**Low**	**Enabled only after F1 and F2 are satisfied and PV remains**	**Can be enabled/disabled from dashboard/API**

As shown in table 3-12, this feeder logic is one of the clearest places where the EMS demonstrates **socially meaningful control**, because the controller is not optimizing anonymous kilowatts but managing differentiated community service levels.

3.6.8 SOC Thresholds and Setpoints

State of charge thresholds are among the most important setpoints in the entire EMS because they define when the battery is considered available, protected, or depleted from the perspective of dispatch decisions. In the current simulation, SOC limits are fixed at 30% minimum and 90% maximum, and battery charging/discharging is clipped accordingly. This means SOC thresholds are already an active part of the operational logic, not just display values.

The lower threshold, 30%, serves as a **battery preservation boundary** and also as a system-level reserve signal. Once SOC reaches this minimum, further battery discharge is blocked. At that point, if PV is insufficient, the EMS must transition toward generator support to preserve F1. Thus, SOC_min is both a battery-protection threshold and a trigger for change in source hierarchy.

The upper threshold, 90%, serves as a controlled charging limit. Once SOC reaches this level, the battery is considered adequately charged and further charging is blocked even if surplus PV remains. This reflects the practical principle that not all theoretical battery capacity should be used continuously, especially when long-term battery health and operational headroom matter.

In the supervisory layer, the Node-RED dashboard exposes setpoints such as socStartThreshold and socStopThreshold and allows them to be updated through /api/control/setpoints. This is highly significant because it shows the project is designed with configurable EMS behavior in mind. Instead of hardcoding all thresholds forever, the architecture allows operators or higher-level logic to tune the response of the generator and storage interaction according to field conditions, maintenance policy, seasonal demand, or experimental scenario analysis.

The presence of editable setpoints also suggests a mature supervisory philosophy: control logic should be parameterized where reasonable, not frozen unnecessarily. For example, the start threshold could be raised if battery aging or critical-load sensitivity demands earlier generator intervention. The stop threshold could be raised to ensure more reserve after generator operation, or lowered if fuel minimization is prioritized. Such flexibility makes the EMS more adaptable without altering its core hierarchy.

Table 3-13 SOC-related setpoints

Setpoint	Function	Current role in project
SOC_min = 30%	**Prevent overdischarge; define minimum usable reserve**	**Blocks battery discharge below threshold**
SOC_max = 90%	**Prevent overcharge; define usable upper bound**	**Blocks battery charge above threshold**
socStartThreshold	**Supervisory trigger for generator assist/start logic**	**Exposed in dashboard/API**
socStopThreshold	**Supervisory stop or recovery threshold**	**Exposed in dashboard/API**

As shown in table 3-13, this section becomes especially convincing if you describe SOC not only as a battery number but as a **control-state variable**. In this EMS, SOC is one of the main variables that determines whether the system is still in renewable-buffered mode or has crossed into backup-dependence mode.

3.6.9 Manual Override and Restoration Logic

Manual override is essential in any serious microgrid control architecture because no automatic EMS, however well designed, can anticipate every maintenance case, testing requirement, fault recovery procedure, or operator-intent scenario. The project already supports this through dashboard/API control of generator commands, feeder commands, parameter updates, and alarm reset actions. This means manual override is an explicit feature of the architecture rather than an informal assumption.

The key principle in manual override is that it should **temporarily redirect supervisory authority without destroying safety structure**. In practice, this means an operator may start or stop the generator, disable or enable F2/F3, and modify setpoints, but those actions should still pass through the mediated control pathway of Node-RED, PLC logic, and device-level permissives rather than bypassing them directly. This is exactly the architecture suggested by the project, where the dashboard communicates through REST APIs to Node-RED, which in turn mediates state updates and commands.

Restoration logic is the companion to manual override. Once a feeder has been manually disabled or once the system has been operating under constrained conditions, the controller must determine how and when normal automatic sequencing is re-established. A robust restoration policy should generally follow the same priority order used during automatic service recovery:

- Restore F1 first or preserve it continuously.
- Restore F2 only after source sufficiency for F1 is secure.
- Restore F3 only after F1 and F2 are adequately supported.
- Return to normal Auto mode only when manual commands are released or explicitly cleared.

This last point is especially important in thesis writing. Manual override should not be described as a random operator action layered on top of automation. It should be described as a **temporary supervisory mode** with defined entry and exit conditions. The dashboard status

structure already includes a mode field, which appears as “Auto” in the example JSON and user interface. This strongly suggests that mode awareness exists in the design and can be used to distinguish between normal automatic operation and manual-intervention periods.

A good restoration philosophy also avoids immediate uncontrolled rollback. If an operator re-enables a feeder, the controller should ideally still check source sufficiency before accepting sustained restoration. Similarly, after manual generator use, the system should not instantly shut the generator down the moment Auto mode is resumed unless stop conditions are genuinely satisfied. These points make the EMS description look much more mature because they show that mode transfer itself is part of the control design.

Table 3-14 Manual override pathways

Override function	Interface path	Expected EMS role
Generator start/stop	**/api/control/generator**	**Permit operator intervention in source availability**
Feeder enable/disable	**/api/control/feeders**	**Permit operator control over F2/F3 service states**
Setpoint update	**/api/control/setpoints**	**Permit tuning of EMS thresholds and limits**
Alarm reset	**/api/control/reset-alarms**	**Clear annunciated alarms after review or correction**
Mode awareness	**mode in status JSON/dashboard**	**Distinguish Auto operation from manual supervisory control**

As shown in table 3-14, this makes manual override an integrated supervisory feature rather than a contradiction of automation. In a strong design, **manual mode is not the absence of logic; it is a different layer of logic**.

3.6.10 Active Source Determination and State Reporting

A sophisticated EMS is not complete unless it can clearly report which source is actively sustaining the system at a given time and what the system state currently means. The project already contains the foundations for this through both the MATLAB results and the Node-RED dashboard JSON structure. In the dashboard, fields such as activeSource, genStatus, soc, pvPower, totalLoad, mode, and feeder statuses are explicitly reported. This is extremely important because it turns control behavior into a readable operational narrative.

In the MATLAB environment, active source determination can be inferred from the relative values of Ppv, P_batt, and P_gen, as well as the time-series outputs for supplied feeders and total active load. If PV is nonzero and adequate for the active feeders, the system is operating in renewable-led mode. If battery discharge is positive while generator output remains zero, the system is in storage-supported mode. If generator output is nonzero, the system has entered backup-supported mode. Thus, the simulation already contains the raw logic required to classify operational states, even if some of that classification is displayed more explicitly in the dashboard layer.

The Node-RED structure makes this state reporting much more user-facing. Example status objects include mode, soc, genStatus, pvPower, totalLoad, activeSource, f2Status, f3Status, threshold settings, and alarm lists. This means the EMS is expected not only to act but also to explain itself to supervisory users. For practical microgrid control, this is essential. Operators need to know whether the current supply is PV-dominant, battery-backed, or generator-supported, and whether feeder restrictions are due to automatic conservation or manual command.

A strong way to frame this in the thesis is to treat active-source determination as a **state abstraction layer**. The raw electrical variables are many, but the operator often needs a smaller set of meaningful operating states, for example:

- PV Supply State: PV is the dominant active source and no backup source is required.
- Battery Support State: PV is insufficient and battery discharge is currently preserving service.
- Generator Backup State: The generator is running because renewable-storage support is insufficient for required continuity.

- Charge Recovery State: PV surplus is charging the battery while loads are already supplied.
- Manual Intervention State: The system is being influenced by operator override or non-default supervisory commands.

Table 3-15 Example state-reporting model

Reported state	Indicative condition	Meaning
PV	**Ppv > 0, P_gen = 0, battery not discharging significantly**	**Renewable-led supply state**
Battery	**P_batt > 0, P_gen = 0**	**Stored energy actively preserving continuity**
Generator	**P_gen > 0**	**Backup generation is required**
Charging	**P_batt < 0 and PV surplus exists**	**Reserve is being rebuilt**
Manual/Override	**mode or command path indicates supervisory intervention**	**Operator authority affecting normal auto sequence**

As shown in table 3-15, this kind of reporting is not cosmetic. It directly improves diagnosability, operator trust, and control validation. If a feeder goes offline, the operator should be able to tell whether the reason was insufficient PV, depleted battery reserve, generator unavailability, or manual disable command. By exposing source identity and system mode explicitly, the project moves from “control exists” to “control is intelligible,” which is a much stronger engineering position.

Integrated EMS View

Table 3-16 That layered control philosophy can be summarized in one control chain

Layer	Main role	Typical variables/signals
Device layer	Safe execution of local functions	Inverter mode, battery contactor, DSE run/fault
EMS layer	Source allocation and load-priority logic	PV power, SOC, feeder demand, generator dispatch
Supervisory layer	Operator interaction, status display, command entry	Mode, activeSource, alarms, feeder toggles, setpoints

As shown in table 3-16, when all the previous subsections are combined, the control architecture of the project can be described as a **hierarchical rule-based EMS with supervisory configurability**. At the bottom level, local devices such as the inverter, BMS, and DSE enforce equipment-specific capabilities and safety constraints. At the middle level, the PLC and rule-based EMS evaluate available PV, battery SOC, feeder demands, and generator need. At the top level, Node-RED and the dashboard expose commands, states, alarms, and setpoints to the operator.

From a thesis perspective, this is a very strong architecture because it proves that the project does not stop at simulation curves. It includes a coherent decision philosophy, a practical rule base, configurable thresholds, manual override capability, feeder-level service logic, and explicit state reporting. That combination gives the EMS section both academic depth and implementation credibility.

3.7 Modeling Assumptions and Mathematical Representation

3.7.1 Time-Step and Discrete-Time Modeling Assumptions

The mathematical model adopted in this project is a **discrete-time supervisory energy management model** evaluated over a 24-hour horizon using hourly sampling points. In both

MATLAB scripts, the simulation time vector is explicitly defined as t=0:23*t*=0:23, which means the entire microgrid behavior is represented by 24 sequential hourly states rather than by a continuous-time dynamic formulation. This modeling choice is fully appropriate for supervisory EMS studies because the primary focus is not electromagnetic transients or converter switching behavior, but daily source allocation, feeder priority enforcement, battery reserve evolution, and generator intervention logic.

The discrete-time approach assumes that within each one-hour interval, power values can be treated as approximately constant or sufficiently represented by their hourly average equivalent. In other words, the model does not attempt to resolve second-level irradiance fluctuations, sub-minute load spikes, inverter control transients, or breaker actuation dynamics. Instead, each time step represents a supervisory decision window in which the controller observes the available PV, the feeder demands, the current battery SOC, and then applies the dispatch logic for that interval. This is a common and defensible simplification when the purpose of the simulation is operational planning and EMS validation rather than protection coordination or dynamic stability analysis.

A key implication of the one-hour time step is that energy and power become closely linked through the relation E=PΔt*E*=*P*Δ*t*, with Δt=1Δ*t*=1 hour. Because the time step is one hour, a power value in kW numerically corresponds to an ऊर्जा transfer in kWh over that step, which simplifies SOC updating and energy accounting. This is why the battery SOC model in the MATLAB script can convert between battery power and state-of-charge change directly through battery capacity without needing an additional time scaling factor beyond dt=1*dt*=1.

However, this convenience comes with modeling consequences. A one-hour step smooths intra-hour variations and may underrepresent short generator starts, short battery bursts, or rapid PV cloud-transient events. Therefore, the model should be interpreted as a **high-level operational model** rather than a fast dynamic or protection model. It is especially suitable for answering questions such as: Which source supports which load class over the day? How does SOC evolve under the EMS? When is the generator likely to be needed? How much feeder curtailment occurs?

The discrete-time framework can be written generally as:

$$f(x_k, u_k, w_k) = {}_{k+1}x \qquad \textbf{3.1}$$

where *xk* represents the system state at time step k*k*, *uk* represents the EMS control action, and *wk* represents exogenous inputs such as PV availability and load demand. In the present

project, the dominant state variable is battery SOC, the dominant exogenous inputs are PV and load profiles, and the dominant control actions are battery dispatch, generator dispatch, and feeder service allocation.

Table 3-17 Discrete-time modeling summary

Modeling element	Representation in project	Meaning
Time horizon	**24 hours**	**Daily operational cycle**
Time index	**t=0, 1,…,23t = 0,1,\dots,23t=0,1,…,23**	**Hourly supervisory snapshots**
Time step	**Δt=1\Delta t = 1Δt=1 hour**	**Power-to-energy conversion becomes direct**
Main purpose	**Supervisory EMS analysis**	**Not converter transient or protection study**
Dominant state	**SOC**	**Memory of stored energy condition**
Dominant decisions	**PV allocation, battery use, generator support, feeder supply**	**Core EMS outputs**

As shown in table 3-17, this discrete-time foundation is important because every other equation in the chapter must be interpreted through it. The PV equation, load equations, SOC update, and dispatch rules are all **time-indexed supervisory relations**, not continuous differential equations.

3.7.2 PV Power Profile Representation

The PV generation profile is represented in the simulation by a deterministic sinusoidal daytime function bounded by sunrise and sunset. Specifically, both MATLAB scripts define sunrise at 06:00 and sunset at 18:00, and assign PV power as zero outside this interval. During

daylight hours, PV output is computed by a sine-shaped function that rises smoothly from zero after sunrise, reaches a peak around midday, and returns to zero at sunset.

The base PV profile can be written as:

$$P_{PV,base}(k) = \begin{cases} P_{PV,\max} \sin\left(\pi \dfrac{t_k - {}_{sr}t}{{}_{ss}t - {}_{sr}t}\right), & {}_{sr}t \le t_k \le {}_{ss}t \\ 0, & \text{otherwise} \end{cases} \quad \textbf{3.2}$$

where PPV,max=150$P_{PV,max}$=150 kW, tsr=6, and tss=18. This mathematical form reflects the daily shape of solar availability in a simplified but physically intuitive way: low in the early morning, highest around solar noon, and decreasing afterward.

In the EMS script , this base profile is then scaled by a user-defined factor:

$$P_{PV}(k) = \alpha_{PV} P_{PV,base}(k) \quad \textbf{3.3}$$

Where $\alpha_{PV} = \dfrac{pv_input_kw}{PV_MAX}$, constrained within the allowable maximum. This is an important modeling feature because it allows the same normalized PV shape to represent different operating scenarios without redefining the entire daily irradiance pattern. For example, the same daily shape can represent a full-capacity day, a reduced irradiance day, or a scenario input from an external interface simply by adjusting the scaling factor.

The chosen sinusoidal PV representation has several strengths. It is mathematically smooth, easy to interpret, and computationally stable. It also captures the most important supervisory characteristic of solar generation: daytime availability concentrated around midday. For a rule-based EMS whose task is to coordinate dispatch at the hourly level, this is often sufficient to reproduce the essential renewable-driven operating sequence of the microgrid.

At the same time, the thesis should explicitly state what this PV model does **not** capture. It does not include stochastic cloud cover, module temperature effects, inverter clipping details, seasonal variation in day length, dust losses, shading mismatch, MPPT dynamics, or irradiance data measured from a real site. Therefore, the PV equation should be described as a **representative daily supervisory profile**, not a meteorological forecasting model. This distinction strengthens the academic credibility of the chapter because it shows awareness of both the utility and the limits of the simplification.

3.7.3 Load Profile Formulation

The total load profile in the project is built as an aggregation of several community demand categories, each represented by a simple time-dependent rule. This is a strong modeling approach because it avoids treating the village as a single featureless demand block. Instead, the total load emerges from residential demand, hospital demand, school demand, water pumping demand, mosque demand, and shop demand, each with its own temporal behavior.

The overall load can therefore be written as:

$$P_{shop}(k) + P_{mos}(k) + P_{pump}(k) + P_{sch}(k) + P_{hos}(k) + P_{res}(k) = P_{load}(k) \quad \textbf{3.4}$$

where each term is constructed from explicit scheduling assumptions in the MATLAB model. This decomposition is highly valuable because it permits feeder grouping, social interpretation, and priority-based service logic, all of which would be much weaker if the total load were modeled as one undifferentiated curve.

Residential load formulation

The residential base load is modeled using 30 houses with a peak household demand coefficient of 1.2 kW, multiplied by a time-dependent factor that changes across the day. Thus, the residential load is:

$$N_h P_{house,peak} \lambda_{res}(k) = P_{res,base}(k) \quad \textbf{3.5}$$

where Nh=30Nh=30, Phouse,peak=1.2$Phouse,peak$=1.2 kW, and λres(k)$\lambda res(k)$ is a piecewise factor equal to 0.8, 0.9, 1.0, or 1.1 depending on the hour. This is a reasonable way to represent household diversity at a simplified level, since it captures the idea that domestic demand is lower late at night, moderate in the morning, strong in the daytime, and highest in the evening period.

Hospital load formulation

The hospital load is modeled as a nearly continuous high-priority demand with two levels: 28 kW during most hours and 32 kW during the interval from 10:00 to 16:00. This can be expressed as:

$$\left.\begin{matrix} t_k < 16 \geq 10 & ,32 \\ \text{otherwise} & ,28 \end{matrix}\right\} = P_{hos}(k) \quad \textbf{3.6}$$

This representation is particularly important because it reflects the operational idea that medical demand is persistent and only moderately variable, which is exactly the kind of load that justifies placement inside the critical feeder.

School, pump, mosque, and shops

The school load is modeled as 8 kW during school hours from 07:00 to 14:00 and zero otherwise. The water pump is modeled as 15 kW during the interval from 11:00 to 14:00 and zero otherwise. The mosque load is event-based and takes a value of 3 kW at selected hours {5,12,15,18,19}{5,12,15,18,19}. Shop demand is modeled as 4 kW between 08:00 and 22:00 and zero outside that period.

Together, these profiles produce a structured daily demand curve that contains both continuous and intermittent components. This is mathematically useful because it creates realistic periods of strong overlap, such as midday when hospital, school, pump, shops, and residential demand may coexist, and evening periods when residential and shop loads dominate while PV is declining or absent.

Feeder-level formulation

The model then maps these category loads into feeders. In run_ems_matlab.m, the feeder base profiles are constructed as:

$$P_{mos}(k) + P_{shop}(k) + P_{hos}(k) + P_{res,F1}(k) = F1_{base}(k)$$

$$P_{res,F2}(k) = F2_{base}(k) \qquad \textbf{3.7}$$

$$P_{pump}(k) + P_{sch}(k) = F3_{base}(k)$$

with residential demand divided equally between F1 and F2. This formulation is especially strong because it mathematically embeds the social-priority philosophy directly into the load equations. The feeders are not arbitrary labels added later; they are algebraic aggregations of actual community load components.

In the EMS script, these feeder profiles are further scaled by user-defined factors so that scenario-specific feeder magnitudes can be generated while preserving the same normalized temporal shapes. That means the final feeder demands can be written as:

$$\alpha_{F3} F3_{base}(k) = \alpha_{F2} F2_{base}(k), \quad F3(k) = \alpha_{F1} F1_{base}(k), \quad F2(k) = F1(k) \qquad \textbf{3.8}$$

where each $\alpha\alpha$ is obtained from the ratio between user input and the maximum base feeder value.

Table 3-18 Load component summary

Load component	Mathematical style	Time behavior
Residential	Piecewise-scaled household aggregate	Varies across night, morning, day, evening
Hospital	Two-level piecewise constant	Continuous, high-priority load
School	Scheduled block load	Active 07:00–14:00
Pump	Scheduled block load	Active 11:00–14:00
Mosque	Event-based pulse load	Active at selected hours
Shops	Scheduled commercial load	Active 08:00–22:00

As shown in table 3-18, this shows load formulation is one of the strongest parts of the modeling framework because it gives the EMS a demand structure that is both mathematically manageable and socially interpretable.

3.7.4 Battery SOC Update Model

The battery state of charge is the main internal state variable of the supervisory EMS model. Unlike PV and load, which are external or scheduled inputs, SOC carries information from one time step to the next and therefore gives the model memory. This makes SOC the key link between current dispatch decisions and future system resilience.

The simulation initializes the battery state of charge at 90% and constrains it between a minimum of 30% and a maximum of 90%. In run_ems_matlab.m, battery power Pbatt(k)*Pbatt*

(k) is defined with a sign convention in which positive values correspond to discharge and negative values correspond to charging. This sign convention is particularly useful because it allows both charge and discharge behavior to be represented in one variable.

The SOC update equation in the script is:

$$100 \times \frac{P_{batt}(k)\Delta t}{_{batt}E} - {}_{k}SOC = {}_{k+1}SOC \qquad \textbf{3.9}$$

where Ebatt=100$Ebatt$=100 kWh in the simulation model and Δt=1Δt=1 hour. Because discharge is positive, it reduces SOC; because charging is represented by negative power, it increases SOC through the same equation. This is mathematically elegant and compact, and it matches the implemented code exactly.

After the update, SOC is clipped to remain inside the allowable range:

$$\min\left(SOC_{\max}, \max\left(SOC_{\min}, SOC_{k+1}\right)\right) = {}_{k+1}SOC \qquad \textbf{3.10}$$

with SOCminf_0=30%SOCmin=30% and SOCmaxf_0=90%SOCmax=90%. This saturation step is critical because it ensures that the model respects the supervisory operating window even if a dispatch calculation would otherwise push the battery beyond safe or intended limits.

The discharge logic also computes the maximum energy that can be withdrawn during the current step without violating the SOC minimum:

$$_{batt}E\,\frac{^{\min}SOC_k - SOC}{100} = E_{dis,\max}(k) \qquad \textbf{3.11}$$

and therefore the maximum feasible discharge power over one step is:

$$\frac{E_{dis,\max}(k)}{\Delta t} = P_{dis,\max}(k) \qquad \textbf{3.12}$$

which is then compared with the battery power limit and the remaining F1 demand. Similarly, the charging logic computes the remaining available storage margin before hitting SOCmax$f_0$$SOC$max, and uses that to cap charging power.

This means battery dispatch is bounded by three simultaneous constraints:

- battery power capability Pbatt,maxf_0=60$Pbatt$,max=60 kW,
- available energy margin above SOCmin$f_0$$SOC$min for discharge or below SOCmax$f_0$$SOC$max for charge,

- and the actual system need or surplus power at that time step as shows in Table 3.11:

Table 3-19 Battery SOC model summary

Battery quantity	Representation	Meaning
Initial SOC	90%	Battery starts fully within usable upper band
Minimum SOC	30%	Lower protected operating bound
Maximum SOC	90%	Upper protected operating bound
Capacity	100 kWh in simulation	Energy base for SOC updates
Power variable sign	+++ discharge, −-− charge	One variable represents both operating modes
SOC update	Energy-balance discrete equation	Tracks stored energy from one step to next

As shown in table 3-19, one important note is that the simpler script one-2.m introduces an efficiency parameter η=0.95η= 0.95, but the SOC update shown there does not fully propagate an efficiency-adjusted equation through the final logic, while uses a direct idealized power-to-energy update without explicit charge/discharge efficiency losses. This is worth noting in the thesis because it highlights that the adopted SOC model is intentionally simplified for supervisory clarity rather than electrochemical precision.

3.7.5 Power Balance Equations

Power balance is the central physical principle underlying the entire EMS. At every time step, the supplied active load must be supported by an equal combination of available PV power,

generator power, and battery discharge power, subject to the simplifications of the model. In the implemented MATLAB EMS, the total active supplied load after control is defined as:

$$F3_{sup}(k) + F2_{sup}(k) + F1_{sup}(k) = P_{active}(k) \quad \textbf{3.13}$$

where each supplied feeder term is an output of the dispatch logic rather than necessarily equal to its original demanded value. This distinction is crucial. The model does not assume that all demand is always served; it explicitly computes how much of each feeder is actually energized after EMS action.

The source-side supplied power is represented in the script as:

$$P_{batt,dis}(k) + P_{gen}(k) + P_{PV}(k) = P_{supply}(k) \quad \textbf{3.14}$$

where Pbatt,dis(k)=max(Pbatt(k),0)$P_{batt,dis}(k)=\max(P_{batt}(k),0)$, meaning only battery discharge is counted as supply to the AC side in the plotted power balance. Charging power is excluded from this supply expression because it represents absorption of surplus energy rather than delivery to the load.

Under idealized lossless supervisory assumptions, the intended balance relation is:

$$P_{supply}(k) \geq P_{active}(k) \quad \textbf{3.15}$$

with equality in constrained operating periods and inequality possible when surplus PV is available for charging or curtailed after all feasible actions have been taken. Because the controller first allocates PV to loads and only then uses remaining PV for battery charging, the effective power balance is implemented procedurally rather than by solving a simultaneous algebraic dispatch optimization.

Another useful relation is the total original demand before control:

$$F3(k) + F2(k) + F1(k) = P_{load}(k) \quad \textbf{3.16}$$

and the curtailed or unsupplied demand can therefore be interpreted as:

$$P_{active}(k) - P_{load}(k) = P_{shed}(k) \quad \textbf{3.17}$$

This quantity is not explicitly named in the code as a standalone variable, but it is conceptually present in the difference between original feeder demand and feeder supplied values. This is extremely useful in thesis interpretation because it allows the EMS to be discussed not only in terms of generation and battery usage, but also in terms of service curtailment resulting from source limitations.

Table 3-20 Power balance quantities

Quantity	Equation	Role
Original total load	Pload=F1+F2+F3P_{load}= F1+F2+F3Pload=F1+F2+F3	Demand before EMS intervention
Active supplied load	Pactive=F1sup+F2sup+F3supP_{active}= F1_{sup}+F2_{sup}+F3_{sup}Pactive=F1sup +F2sup+F3sup	Demand actually served
Source supply	Psupply=PPV+Pgen+Pbatt,disP_{supply}= P_{PV}+P_{gen}+P_{batt,dis}Psupply=PPV +Pgen+Pbatt,dis	Power delivered to active load
Unsupplied demand	Pshed=Pload−PactiveP_{shed}= P_{load}-P_{active}Pshed=Pload−Pactive	Implied curtailed demand

As shown in table 3-20, this power-balance formulation is fundamental because it turns the EMS from a simple switching sequence into a mathematically traceable dispatch system. Every reported result—served load, generator energy, battery use, and feeder curtailment—rests on these relationships.

3.7.6 Source Dispatch Equations

The source dispatch equations in this project are not derived from global optimization, but from an ordered rule sequence. That means the mathematical representation is best written as a set of conditional allocation equations rather than a single objective-minimization problem. This is fully consistent with the rule-based EMS philosophy described earlier.

Step 1: PV allocation to F1

The first allocation is PV to the critical feeder:

$$\min\left(P_{PV}(k), F1(k)\right) = P_{PV\rightarrow F1}(k) \qquad \textbf{3.18}$$

and the remaining critical deficit becomes:

$$P_{PV\to F1}(k) - F1(k) = F1_{rem}(k) \tag{3.19}$$

with remaining PV after critical allocation:

$$P_{PV\to F1}(k) - P_{PV}(k) = P_{PV,rem1}(k) \tag{3.20}$$

This step mathematically encodes the PV-first and F1-first philosophy at the same time.

Step 2: Battery discharge to remaining F1

If critical demand remains and SOC is above the minimum, the battery may discharge:

$$\min\left(F1_{rem}(k), P_{batt,\max}, \frac{{}_{\min}SOC_k - SOC}{100}\frac{{}_{batt}E}{\Delta t}\right) = P_{batt,dis}(k) \tag{3.21}$$

provided SOCk>SOCmin*SOCk*>*SOC*min; otherwise Pbatt,dis(k)=0*Pbatt*,*dis*(*k*)=0. After battery support, the residual critical deficit becomes:

$$P_{batt,dis}(k) - F1_{rem}(k) = F1_{rem2}(k) \tag{3.22}$$

This equation is especially strong because it combines demand need, battery power limit, and available energy margin inside one dispatch relation.

Step 3: Generator support to remaining F1

If a critical deficit still remains, the generator supplies up to its rating:

$$\min\left(F1_{rem2}(k), P_{gen,rated}\right) = P_{gen}(k) \tag{3.23}$$

with Pgen,rated=100*Pgen*,*rated*=100 kW. The actually supplied critical feeder is therefore:

$$(P_{gen}(k), 0 - \max(F1_{rem2}(k) - F1(k) = F1_{sup}(k) \tag{3.24}$$

which in practice becomes full F1 support when the combined PV, battery, and generator capacity is sufficient.

Step 4: Remaining PV to F2 and F3

After F1 support is secured, the remaining PV is distributed sequentially to F2 and then F3:

$$\begin{aligned}
&\min\left(P_{PV,rem1}(k), F2(k)\right) = P_{PV\to F2}(k)\\
&P_{PV\to F2}(k) - P_{PV,rem1}(k) = P_{PV,rem2}(k)\\
&\min\left(P_{PV,rem2}(k), F3(k)\right) = P_{PV\to F3}(k)
\end{aligned} \tag{3.26}$$

Therefore,

$$P_{PV\to F3}(k) = P_{PV\to F2}(k), \qquad F3_{sup}(k) = F2_{sup}(k) \tag{3.27}$$

which reveals a key property of the current model: lower-priority feeders are supplied only by remaining PV, not by battery or generator support. This is one of the most important mathematical characteristics of the EMS because it encodes a very conservative service hierarchy.

Step 5: Battery charging from residual PV

If PV still remains after F1, F2, and F3 allocations, the battery may charge:

$$\min\left(P_{PV,rem3}(k), P_{batt,\max}, \frac{{}_kSOC_{\max} - SOC}{100}\frac{{}_{batt}E}{\Delta t}\right) = P_{batt,ch}(k) \tag{3.28}$$

where PPV,rem3(k)$P_{PV,rem3}(k)$ is the PV remaining after feeder supply. In the code, charging is recorded as negative battery power, so the net battery dispatch variable can be written as:

$$P_{batt,ch}(k) - P_{batt,dis}(k) = P_{batt}(k) \tag{3.29}$$

This compact formulation explains why positive values reduce SOC and negative values increase SOC in the update equation.

Table 3-21 Dispatch hierarchy summary

Dispatch stage	Equation type	Priority meaning
PV to F1	Minimum of PV and F1 demand	Critical renewable-first allocation
Battery to remaining F1	Minimum of deficit, power limit, energy margin	Protected storage support
Generator to remaining F1	Minimum of deficit and generator rating	Backup continuity support
Remaining PV to F2	Minimum of remaining PV and F2 demand	Medium-priority renewable-only supply
Remaining PV to F3	Minimum of remaining PV and F3 demand	Low-priority renewable-only supply
Residual PV to battery	Minimum of surplus PV, power limit, SOC headroom	Reserve recovery

As shown in table 3-21, this sequential mathematical representation is one of the strongest parts of the model because it directly mirrors the implemented algorithm and avoids any gap between theory and code.

3.7.7 Simplifications Adopted in Simulation

The simulation deliberately adopts several simplifications so that the model remains transparent, computationally light, and strongly aligned with supervisory EMS analysis rather than full electro-mechanical emulation. These simplifications should not be hidden; they should be stated clearly as design choices that define the scope of the mathematical model.

The first simplification is the use of **hourly time steps**. This suppresses fast transient behavior, intra-hour PV variability, and sudden load spikes. It is appropriate for daily dispatch studies, but it does not represent second-level control phenomena.

The second simplification is the use of a **deterministic sinusoidal PV profile** rather than measured irradiance data or weather-driven forecasting. This provides a clean and repeatable renewable input but does not capture cloud intermittency, seasonal variation, soiling, or temperature derating.

The third simplification is the use of **scheduled and piecewise-constant load components**. While these loads are socially meaningful and better than a single lumped curve, they still do not capture random customer behavior, appliance diversity at high resolution, or stochastic event uncertainty.

The fourth simplification is the **idealized battery model**. The battery is represented mainly by energy capacity, power limit, and SOC bounds. The model does not fully include charge/discharge efficiency asymmetry, temperature effects, voltage dependence, aging, internal resistance, or BMS-triggered dynamic restrictions.

The fifth simplification is the **rule-based dispatch without dynamic source transition modeling**. The model determines generator and battery contributions algebraically per step, but it does not simulate engine cranking time, inverter synchronization transients, breaker closing delays, ramp-rate limits, or detailed generator minimum load control behavior.

The sixth simplification is the assumption of **lossless or near-lossless internal conversion at the EMS level**. DC/DC, DC/AC, AC/DC, cabling, transformer, and auxiliary losses are not explicitly propagated through the balance equations in the presented supervisory model. Thus, the power balance is operationally useful but not a full efficiency-accounting network model.

Table 3-22 Major simulation simplifications

Simplification	What is kept	What is neglected
Hourly time step	**Daily dispatch pattern**	**Fast dynamics and transients**
Sinusoidal PV	**Daytime shape and peak level**	**Weather variability and temperature effects**
Scheduled loads	**Community demand structure**	**Stochastic end-user variation**
SOC-based battery model	**Storage energy and bounds**	**Electrochemical detail and aging**
Rule-based dispatch	**Clear operational logic**	**Global optimization and predictive control**
Idealized power balance	**Source-load allocation**	**Detailed losses and converter dynamics**

As shown in table 3-22, these simplifications shows are entirely acceptable for the purpose of the study because the goal is to validate control logic, feeder prioritization, and daily source coordination. In fact, overcomplicating the model at this stage could make the EMS behavior harder to interpret without proportionally improving the central thesis argument.

3.7.8 Limitations of the Mathematical Model

Every mathematical model is useful precisely because it is selective, and the present EMS model is no exception. Its strength lies in operational clarity, but that same clarity depends on assumptions that limit how far the results can be generalized. A strong thesis should therefore state these limitations explicitly rather than leaving them implicit.

The first limitation is temporal resolution. Because the simulation is hourly, it cannot evaluate sub-hour disturbances such as sudden cloud cover, short motor starting transients,

momentary overloads, or switching oscillations. As a result, the model is not suitable for detailed voltage/frequency transient analysis or protection-trip timing evaluation.

The second limitation is the simplified PV representation. The sinusoidal profile is useful for supervisory testing, but it cannot predict actual PV output under site-specific weather, dust, module temperature, seasonal solar geometry, or partial shading conditions. Therefore, the renewable results should be interpreted as representative operational behavior rather than exact field yield prediction.

The third limitation concerns load realism. Although the load model is richer than a single aggregate curve, it still uses deterministic schedules for each demand type. Real rural communities exhibit random appliance use, behavioral diversity, interruptions, and special-event deviations that are not captured by the present formulation.

The fourth limitation is that battery behavior is represented only at the supervisory energy level. The model does not include nonlinear voltage-SOC dependence, charge/discharge efficiency losses in full detail, degradation over cycles, thermal constraints, or high-fidelity BMS intervention logic. Therefore, the simulated battery trajectories should be interpreted as EMS-level reserve behavior rather than exact electrochemical performance predictions.

The fifth limitation is source transition realism. Generator start is represented as an immediate available support when required by the rule-based logic, but the model does not incorporate actual crank delay, warm-up, synchronization delay, ramp constraints, or forced outage probability. Similarly, inverter mode changes and feeder switching are not modeled with physical actuation times. This means the mathematical model is strong at representing **who should supply what**, but weaker at representing **exactly how fast the physical transition occurs**.

The sixth limitation is that the dispatch structure is conservative and intentionally restricted. In the current model, battery and generator support are effectively reserved for F1, while F2 and F3 are supplied only from remaining PV. This simplifies the hierarchy and strengthens critical-load protection, but it may underrepresent operational strategies in which the generator or battery also temporarily supports medium-priority loads for economic or social reasons. Thus, the present EMS is valid as a priority-protection strategy, but it does not span the full design space of possible hybrid dispatch policies.

Table 3-23 Model limitations summary

Limitation	Effect on interpretation
Hourly resolution	Good for daily EMS behavior, weak for fast dynamics
Simplified PV profile	Good for representative solar availability, weak for weather realism
Deterministic loads	Good for structured scheduling, weak for random demand behavior
Simplified battery model	Good for SOC-based supervision, weak for electrochemical realism
Immediate source transitions	Good for dispatch logic, weak for startup/synchronization timing
Restricted support of F2/F3	Strong critical-load focus, conservative service recovery behavior

As shown in table 3-23, these limitations do not invalidate the model. On the contrary, they clarify its intended role. The model is best understood as a **supervisory EMS representation for daily operational analysis**, capable of evaluating feeder priority behavior, battery reserve evolution, generator dependence, and active load after control. It is not intended to replace detailed power system transient simulation, protection studies, weather-driven forecasting, or electrochemical battery modeling.

Mathematical framing

Taken together, the modeling assumptions and equations define a coherent mathematical abstraction of the hybrid microgrid. The model combines a deterministic daytime PV source, structured community load equations, a bounded SOC-based battery state model, and sequential source dispatch rules operating in hourly discrete time. That combination is especially well matched to the objectives of the study because it allows the thesis to analyze not only whether

power is available, but also **how power is prioritized, when reserves are consumed, and under what conditions the system transitions into generator-backed operation**.

3.8 Communication and Integration Requirements

3.8.1 MATLAB/Simulink to Node-RED Interface

The interface between MATLAB/Simulink and Node-RED is designed conceptually as a **simulation-to-supervisory integration layer**, where MATLAB/Simulink provides the underlying EMS and microgrid state variables while Node-RED acts as the middleware that normalizes, routes, and exposes these variables to operator-facing and future external interfaces. The project description explicitly states that Node-RED was structured as a middle layer between the dashboard, the microgrid simulator engine, and future external systems such as MATLAB via MQTT. This makes Node-RED not a simple visualization tool, but a communication broker and control orchestrator between analytical logic and field-style supervisory control.

From an architectural perspective, the MATLAB/Simulink side is expected to generate or update the principal operational states of the microgrid, such as mode, SOC, generator status, PV power, total load, feeder states, and active source identity. These are precisely the same variables that appear in the Node-RED status object returned to the dashboard, which suggests that the interface requirement is primarily a **structured data exchange contract** rather than an ad hoc signal transfer. In other words, successful integration depends less on the specific transport mechanism and more on having a consistent semantic model of the system state.

The project already identifies MQTT as the future-ready communication bridge for MATLAB integration. This is a strong design choice because it decouples the simulation engine from the UI and from other consumers of system state. Instead of tightly coupling MATLAB directly to the dashboard, MATLAB can publish and/or subscribe through Node-RED using a brokered message layer, allowing multiple clients to monitor or interact with the same state model without rewriting the supervisory stack. That architecture is much more scalable than point-to-point integration.

A robust MATLAB/Simulink-to-Node-RED interface therefore requires four core properties:

- A common state schema, including variables such as mode, soc, genStatus, pvPower, totalLoad, activeSource, feeder states, alarms, and threshold setpoints.
- Deterministic update timing appropriate for supervisory operation, not high-frequency real-time control.
- Separation between measured/state variables and command/setpoint variables.
- A transport layer that can support asynchronous updates and future expansion, which the project already anticipates through MQTT.

Table 3-24 MATLAB/Simulink to Node-RED data contract

Interface element	Function in integration	Evidence in project
mode	**Indicates Auto/manual supervisory state**	**Returned in status JSON and shown in dashboard**
soc	**Battery reserve state**	**Used for display and EMS thresholds**
genStatus	**Generator operating status**	**Required for status and control acknowledgment**
pvPower	**Renewable generation state**	**Core source-monitoring variable**
totalLoad	**Aggregate load demand**	**Used in supervisory display**
activeSource	**Current dominant source label**	**Supports interpretable EMS state reporting**
f2Status, f3Status	**Feeder supervisory states**	**Used for display and remote commands**
Setpoints and alarms	**Threshold tuning and annunciation**	**Carried in same supervisory model**

As shown in table 3-24, the most important point academically is that MATLAB/Simulink and Node-RED should not be treated as two disconnected tools. They represent two different roles in the same architecture: MATLAB/Simulink is the **behavior engine**, while Node-RED is

the **integration and presentation engine**. This distinction is essential for explaining why the system can later evolve from pure simulation into hardware-in-the-loop or PLC-assisted deployment without redesigning the whole interface philosophy.

3.8.2 Node-RED to HTML Dashboard Interface

The Node-RED-to-dashboard interface is implemented as a direct web-based supervisory link in which Node-RED serves both as the backend API provider and, in the present design, as the host for the dashboard HTML page itself. This is an important architectural choice because it keeps the user interface and the API under the same origin, simplifying communication, reducing cross-origin complexity, and making the overall supervisory stack behave like a compact SCADA-style web application.

The dashboard side is based on HTML and JavaScript, with the browser periodically requesting system status from Node-RED and updating the visible cards, toggles, alarms, and setpoint fields accordingly. The central mechanism is fetchStatus(), which calls /api/status, parses the returned JSON, and passes the result to updateDashboard(data). The page is also configured to refresh status every two seconds using setInterval(fetchStatus, 2000), which means the dashboard is not static; it is a live supervisory client polling the middleware continuously.

This interface design has several strong practical implications. First, the dashboard is logically thin: it does not compute EMS decisions itself, but only visualizes and submits user intentions. Second, Node-RED remains the single source of truth for displayed state, which is critical for consistency between system behavior, displayed values, and accepted commands. Third, the dashboard can remain largely unchanged even if the upstream system transitions from simulated values to PLC/field values, because its dependency is the normalized Node-RED API rather than device-specific protocols.

Another strong aspect of the interface is that the dashboard is mapped explicitly to status fields rather than to ambiguous text parsing. For example, socValue, pvValue, loadValue, genStatus, and activeSource are updated directly from JSON values, while feeder toggles are synchronized with boolean states such as f2Status and f3Status. This is good integration practice because it ensures deterministic UI behavior and reduces the chance of semantic drift between backend state and front-end rendering.

The dashboard interface also includes **bidirectional supervisory interaction**. It does not only read status; it also sends commands back to Node-RED for generator control, feeder control,

setpoint updates, and alarm reset. Therefore, the Node-RED-to-dashboard interface should be described not merely as a monitoring channel, but as a **human-machine supervisory interface** with both telemetry and command pathways.

Table 3-25 Dashboard interaction model

Direction	Mechanism	Functional role
Node-RED → Dashboard	**GET /Api/status**	**Pushes system state into visible UI through polling**
Dashboard → Node-RED	**POST /Api/control/...**	**Sends generator, feeder, setpoint, and alarm commands**
Node-RED → Browser page	**/dashboard route serving HTML**	**Hosts the dashboard under same supervisory server**
Dashboard local logic	**update Dashboard(data) and field bindings**	**Renders state consistently from structured JSON**

As shown in table3-25, this structure is academically strong because it shows a clear separation of concerns: Node-RED manages logic and integration; the dashboard manages presentation and user interaction.

3.8.3 REST API Structure and Control Endpoints

The REST API in the project is the formal supervisory interface through which external clients, especially the HTML dashboard, can read system state and issue control commands. It is intentionally simple and organized around one primary read endpoint and a set of focused write/control endpoints. This simplicity is not a weakness; it is one of the strengths of the architecture because it makes the control surface explicit, testable, and easy to document.

The read endpoint is:

- GET /api/status for retrieving the current structured system state.

The control endpoints are:

- POST /api/control/generator for start/stop commands.

- POST /api/control/feeders for enabling or disabling feeder states such as F2 and F3.
- POST /api/control/setpoints for updating EMS thresholds and supervisory parameters.
- POST /api/control/reset-alarms for clearing annunciated alarm records.

This endpoint structure shows a disciplined API philosophy. Instead of exposing arbitrary low-level variables individually, the design groups actions by operational intent: generator control, feeder control, parameter management, and alarm handling. That is exactly how a supervisory API should be structured in a microgrid project because operators think in terms of functions, not in terms of scattered internal variables.

The returned status payload also demonstrates a coherent JSON schema. The sample object includes mode, soc, genStatus, pvPower, totalLoad, activeSource, feeder booleans, threshold values, and an alarm list. This means the API already satisfies one of the most important integration requirements: it can represent the microgrid state in a compact and self-describing way that can be consumed by the dashboard, MQTT publisher, or future external software with minimal transformation.

A further strength is that control responses are not silent. The design notes indicate that commands return structured acknowledgments such as success, message, and sometimes updated state. This is important for supervisory reliability because the client should know not only that a request was sent, but whether it was accepted, interpreted, and applied. In SCADA-like interfaces, command acknowledgment is a major requirement, and the present API already moves in that direction.

Table 3-26 REST API endpoint map

Endpoint	Method	Purpose	Typical payload/result
/api/status	GET	Read current supervisory state	Returns mode, SOC, PV, load, source, feeder states, alarms
/api/control/generator	POST	Start/stop generator	{ "command": "start" } or { "command": "stop" }
/api/control/feeders	POST	Enable/disable F2 or F3	{ "feeder": "F2", "enabled": true }
/api/control/setpoints	POST	Update EMS thresholds	SOC thresholds and generator-related settings
/api/control/reset-alarms	POST	Clear alarm list	Reset request plus acknowledgment

As shown in table 3-26, this API is strong because it formalizes the communication boundary of the EMS. Once the API exists, the rest of the software stack can evolve internally without forcing the dashboard or external systems to change their integration logic.

3.8.4 MQTT Monitoring Channel

The MQTT monitoring channel is introduced in the project as a publish-subscribe communication layer used to distribute system status beyond the immediate REST/dashboard interaction. In the Node-RED flow, a function labeled for MQTT status publication sends the microgrid state to a topic such as microgrid/status. This establishes a second communication path parallel to the REST API: REST serves request-response supervision, while MQTT serves asynchronous state dissemination.

This distinction is very important architecturally. REST is excellent for direct dashboard queries and command submission, but MQTT is better suited for decoupled monitoring, future MATLAB integration, multi-client subscriptions, and event-driven telemetry. By publishing

status to MQTT, the system avoids hardwiring every external consumer directly into Node-RED logic. Any authorized subscriber can listen to the same microgrid state stream without modifying the dashboard channel.

The project explicitly notes that this MQTT layer is intended to support future MATLAB interaction. That means MQTT is not being added as a decorative feature; it is a strategic integration mechanism. MATLAB or Simulink can later subscribe to microgrid/status, publish back other data or setpoints, or participate in a co-simulation or supervisory loop through the same brokered architecture. This is a highly scalable design pattern because it separates producers and consumers through topics rather than through direct program-to-program coupling.

Another important requirement is schema consistency between the MQTT payload and the REST status payload. The design notes indicate that the MQTT publisher prepares a JSON object similar to /api/status. This is excellent practice because it prevents the system from having two incompatible representations of the same microgrid state. If the same fields exist in both channels, then dashboard logic, analytics tools, logging tools, and simulation bridges can all interpret the same semantic state consistently.

Table 3-27 MQTT channel characteristics

Aspect	Requirement in project	Functional value
Topic	**microgrid/status**	**Common status stream for subscribers**
Publisher	**Node-RED function + MQTT out node**	**Converts internal state to distributed telemetry**
Payload type	**JSON aligned with status fields**	**Keeps semantic consistency across channels**
Main purpose	**Monitoring and future MATLAB integration**	**Decoupled state sharing**
Communication style	**Publish-subscribe**	**Supports multiple consumers without direct coupling**

As shown in table 3-27, this MQTT layer is one of the features that makes the architecture look modern and extensible. It turns the system from a single web dashboard into a **message-aware supervisory platform**.

3.8.5 PLC and DSE Communication

Communication between the Siemens S7-1200 PLC and the DSE8610 MKII generator controller is one of the most important hardware-layer integration requirements in the project because it links high-level EMS decisions to actual generator start, stop, protection, and status functions. The EPLAN documentation shows that the PLC sends control commands toward the DSE and receives generator status and alarm signals back from it. This means the PLC is not replacing the DSE; rather, it acts as the supervisory authority above the DSE, while the DSE remains the specialized local controller for engine-generator operation.

The DSE inputs listed in the design include generator voltage, frequency, engine speed, oil pressure, coolant temperature, fuel level, remote start from PLC, and PLC alarm input for voltage/frequency issues. Its outputs include starter relay, fuel solenoid control, engine stop,

breaker close/trip coils for ACBR02, alarm relays, and status feedback such as running and fault indications back to the PLC. This is a very important bidirectional communication structure because it means the control path is closed-loop, not one-way. The PLC does not merely command the generator; it also receives confirmation, health information, and fault states needed for safe supervisory logic.

From a control architecture perspective, this PLC-DSE communication establishes a hierarchical responsibility split:

- The PLC decides whether generator support is required based on EMS logic, source availability, and feeder needs.
- The DSE executes the local engine start/stop sequence and supervises generator-side health and availability.
- Generator status and faults return to the PLC as supervisory inputs.

This is a strong design because it avoids overloading the PLC with engine-specific protection duties while still preserving centralized coordination of the microgrid. It also mirrors best industrial practice, where local controller intelligence and plant-level supervisory logic coexist rather than compete.

Table 3-28 PLC-DSE signal exchange

Direction	Signal/function	Purpose
PLC → DSE	**Remote start**	**Requests generator startup**
PLC → DSE	**Stop/command logic**	**Requests shutdown or control state change**
PLC → DSE	**Alarm/permissive interaction**	**Coordinates supervisory actions**
DSE → PLC	**RUNNING / FAULT status**	**Confirms actual generator state**
DSE → PLC	**Engine and electrical alarms**	**Enables fault-based EMS actions**
DSE → ACBR02	**Breaker close/trip control**	**Manages generator connection to AC bus**

As shown in table 3-28, the communication path itself may be realized through digital I/O and, optionally, serial/Modbus pathways depending on the final implementation detail. The EPLAN notes mention optional RS485 to DSE01 and INV_CHR01 using Modbus RTU, while also showing explicit DI/DO mapping architecture pages for PLC signals. Therefore, the requirement is not limited to one protocol. The essential requirement is that generator command and status exchange must be **deterministic, acknowledged, and safety-compatible**.

3.8.6 Modbus TCP Monitoring Architecture

The Modbus TCP monitoring architecture is centered around the Siemens S7-1200 PLC as the main control and data aggregation node, with Ethernet connectivity explicitly identified for HMI/SCADA communication using Modbus TCP or OPC UA. This is a significant design feature because it provides a standardized industrial path for supervisory monitoring beyond raw digital wiring. Rather than requiring every monitoring client to interact with each device

separately, the PLC can aggregate relevant data and expose it through a structured industrial communication layer.

In this architecture, the PLC is the natural **data concentrator**. It receives measurements and status signals from the DSE, inverter, BMS, and feeder circuits, then organizes these into addressable process values for higher-level systems. Modbus TCP can then be used by SCADA, engineering workstations, or middleware interfaces to read these values over Ethernet. This architecture is particularly suitable for monitoring because it reduces protocol fragmentation at the upper layer: clients do not need to know every field device protocol if the PLC already normalizes the data.

The project also identifies RS485/Modbus RTU as an optional lower-level path to DSE01 and INV_CHR01. This implies a two-level communication structure:

- lower-level device communication may use Modbus RTU or direct I/O,
- upper-level supervisory monitoring uses Ethernet-based Modbus TCP through the PLC.

This layered approach is technically strong because it separates field-device connectivity from supervisory network accessibility. The field layer can remain rugged and device-specific, while the supervisory layer remains unified and network-friendly. For a thesis, this is an excellent point to emphasize because it demonstrates a realistic integration architecture rather than a purely software-only concept.

A good way to describe the requirement is that Modbus TCP is used primarily for **monitoring and supervisory data exposure**, not necessarily as the sole real-time control path for all protective actions. Fast safety-critical actions can still remain local to device controllers or PLC logic, while Modbus TCP provides the structured visibility required by HMI, Node-RED gateways, or future SCADA extensions.

Table 3-29 Modbus monitoring layers

Layer	Typical protocol	Role
Device layer	**DI/DO, optional Modbus RTU over RS485**	**Direct interaction with DSE/inverter**
PLC aggregation layer	**Internal PLC memory/tags**	**Collects and organizes device states**
Supervisory network layer	**Modbus TCP / OPC UA over Ethernet**	**Exposes monitoring data to HMI/SCADA**

As shown in table 3-29, this architecture strengthens the project because it shows that communication is not treated as an afterthought. It is built as a layered industrial data path from device to supervisory system.

3.8.7 Digital I/O Control Architecture

The digital I/O architecture is the most direct control path between supervisory logic and field actions in the project. The EPLAN design identifies a Siemens S7-1200 PLC with DI and DO modules and explicitly states that the PLC receives status and measurements from DSE, inverter, BMS, and feeders, while sending control commands to DSE, inverter, battery contactor, and feeders. This makes digital I/O the backbone of hard supervisory control, especially for discrete commands and confirmations.

Digital inputs are used to bring field statuses into the PLC, such as generator running indication, generator fault, breaker status, alarm relays, feeder status, and other binary conditions. Digital outputs are used to send discrete commands outward, such as generator start/stop, breaker close/trip commands, feeder enable/disable actions, and battery contactor control. This architecture is especially important because many microgrid control actions are fundamentally binary at the field level even when the high-level EMS appears continuous or algorithmic.

One major strength of digital I/O is determinism. Unlike higher-level messaging protocols, digital control signals are straightforward, low-latency, and well suited to essential operational actions such as starting a generator or opening a feeder. This is why the project's communication

strategy is strongest when viewed as layered: dashboards and middleware may use REST or MQTT, but actual field execution still relies on robust PLC-centered I/O pathways.

The digital I/O architecture also reinforces the importance of status confirmation. A command output alone is never enough in a reliable microgrid; it must be paired with feedback inputs. For example, sending a generator start command should be followed by receiving a running confirmation or fault signal. Issuing a feeder enable command should be associated with breaker or feeder status feedback. This closed-loop digital architecture is one of the strongest practical aspects of the design.

Table 3-30 Digital I/O role separation

I/O type	Typical signals	Functional purpose
Digital inputs	**RUNNING, FAULT, breaker status, alarm relays**	**Confirm real field state to PLC**
Digital outputs	**Start, stop, trip, close, contactor control**	**Execute supervisory control actions**
PLC logic	**Interlocks and command sequencing**	**Converts EMS decisions into safe field actions**

As shown in table 3-30, this makes digital I/O not just a wiring detail, but a critical part of the control architecture: it is where abstract EMS logic becomes physical action.

3.8.8 Tag Mapping Between Software and Hardware Layers

Tag mapping is the discipline that ensures the same physical or logical variable is represented consistently across MATLAB/Simulink, Node-RED, REST JSON, dashboard elements, PLC memory, and hardware I/O. In a project like this, tag mapping is not optional documentation; it is one of the core integration requirements. Without it, the system may still "connect," but it will not integrate reliably because different layers may refer to the same state using inconsistent names, types, or meanings.

The current project already contains the foundations of a tag map. On the software side, Node-RED and the dashboard use tags such as mode, soc, genStatus, pvPower, totalLoad, activeSource, f2Status, f3Status, socStartThreshold, socStopThreshold, maxGenLoad, and

alarms. On the hardware side, the EPLAN naming convention includes tags such as GEN01, DSE01, PLC01, INV_CHR01, BMS01, ACBR01–05, CONT_BATT01, and feeder names F1–F3. The integration requirement is to define how these two naming worlds correspond.

For example, the dashboard tag genStatus should map ultimately to generator status feedback originating from DSE01 and entering PLC01 through defined DI channels. The dashboard f2Status and f3Status should map to feeder breaker/control states associated with ACBR04 and ACBR05, or to their software-level enable states if the simulation abstraction is being used. The variable soc should map either to a simulated battery SOC value in Node-RED/MATLAB mode or to a real BMS-derived value through PLC/BMS communication in hardware-connected mode.

This tag mapping requirement becomes even more important because the architecture supports multiple communication styles at once: REST, MQTT, Modbus, DI/DO, and optional serial device protocols. A good tag map acts as the common semantic backbone across all of them. It ensures that "Generator Running," for example, is not one thing in the dashboard, another in MQTT, and another in the PLC.

Table 3-31 Example tag mapping table

Software tag	Meaning	Hardware/source counterpart
mode	Supervisory operating mode	PLC/Node-RED EMS mode state
soc	Battery state of charge	BMS01-derived SOC or simulation SOC
genStatus	Generator run/fault state	DSE01 status feedback via PLC01 DI
pvPower	Instantaneous PV power	INV_CHR01 measurement or simulation source
Total Load	Aggregate demand	EMS/PLC-calculated total feeder load
active Source	Current dominant supply source	Derived EMS state in Node-RED/PLC
f2Status	F2 enabled state	ACBR04 state / feeder enable logic
f3 Status	F3 enabled state	ACBR05 state / feeder enable logic
GenStart_cmd implied in control flow	Generator start request	PLC DO to DSE01 remote start
alarms	Active alarm list	Derived from PLC, DSE, inverter, BMS, and software events

As shown in table 3-31, this final subsection is especially valuable in a thesis because it proves that the architecture is not only functionally rich, but also **engineering-ready**. A strong integration design is not just about having many protocols; it is about making sure all protocols speak about the same plant in the same language.

Integration architecture

Taken as a whole, the communication and integration requirements define a layered architecture in which MATLAB/Simulink provides analytical behavior, Node-RED provides middleware and API orchestration, the HTML dashboard provides the operator-facing interface, MQTT provides distributed telemetry, and the PLC/DSE/digital I/O layers provide hardware-realistic execution and feedback. Modbus TCP adds a supervisory monitoring path, while tag mapping ensures semantic consistency across all layers. This is a strong architecture because it allows the project to move smoothly between simulation, supervisory visualization, and future deployment-oriented control without changing the fundamental information model of the microgrid.

3.9 Verification-Oriented Scenario Definition

3.9.1 Default Scenario from MATLAB

The default verification scenario is derived directly from the baseline MATLAB implementation, where the microgrid is simulated over a 24-hour horizon with hourly time steps and with the standard internal profiles for PV generation, community load composition, feeder distribution, battery SOC evolution, and generator support. In run_ems_matlab.m, the simulation begins from fixed nominal limits of PV 150 kW, F1 57 kW, F2 26 kW, F3 15 kW, battery capacity 100 kWh, battery power limit 60 kW, generator rating 100 kW, and initial SOC of 90%. This makes the default case the natural **reference scenario** against which all later modified or interactive cases can be interpreted.

The value of this default scenario is methodological. It provides a common benchmark in which the control logic is allowed to operate under its intended nominal assumptions before any operator intervention, abnormal forcing, or parameter distortion is introduced. In verification terms, this is essential because one cannot meaningfully judge unusual scenarios unless the

baseline behavior is first known and internally consistent. The default case therefore serves as the **ground truth behavior envelope** of the rule-based EMS under designed daily conditions.

In this baseline case, the system follows the full rule sequence already embedded in MATLAB: PV is allocated first to F1, battery discharge supports remaining F1 when necessary, generator covers any residual unmet F1 demand, remaining PV is then assigned to F2 and F3 in order, and any leftover PV is used to charge the battery within SOC constraints. The scenario therefore verifies all major control functions in one integrated run: critical load preservation, lower-priority feeder curtailment, battery reserve management, generator backup activation, and final power-balance consistency.

The simpler script one-2.m also provides a useful conceptual default scenario, although its structure is more demonstrative than the main EMS script. It includes the same daily PV and load logic, feeder activation behavior, battery power scheduling, generator scheduling, and SOC evolution, making it useful as a verification precursor or explanatory pre-model. In the thesis, it is best to position one-2.m as an earlier validation-oriented prototype and run_ems_matlab.m as the more formal verification baseline.

Table 3-32 Default scenario characteristics

Item	Default definition	Verification role
Time horizon	**24 hourly steps**	**Covers a full operational day**
Initial SOC	**90%**	**Verifies reserve evolution from healthy initial state**
PV capacity	**Up to 150 kW**	**Tests renewable-first dispatch**
Feeder limits	**F1 57 kW, F2 26 kW, F3 15 kW**	**Verifies priority hierarchy**
Battery limit	**100 kWh, 60 kW**	**Verifies bounded charge/discharge behavior**
Generator limit	**100 kW**	**Verifies backup support sufficiency**

As shown in table 3-32, the default MATLAB scenario is therefore not just the first simulation to run; it is the **verification anchor case** from which expected behavior is defined.

3.9.2 Interactive Scenarios from Python and Dashboard

Beyond the fixed MATLAB baseline, the project architecture supports interactive supervisory scenarios through the dashboard and middleware environment, where system states and operator commands can be modified dynamically through the Node-RED/HTML interface. The dashboard periodically reads status from /api/status and allows user actions through control routes for generator commands, feeder enable/disable, setpoint updates, and alarm reset. This converts the project from a static off-line simulation into a semi-interactive verification environment.

These interactive scenarios are important because verification of an EMS is not complete if it only proves that the controller behaves properly under one scripted timeline. A real supervisory system must also behave correctly when an operator changes thresholds, toggles feeders, starts or stops a generator manually, or tests alarm clearing during operation. The dashboard and API

structure therefore support **human-in-the-loop verification scenarios**, where the correctness of communication and control response is examined alongside the energy logic itself.

Although the provided material describes the interface mainly through Node-RED and HTML, the same architecture is explicitly intended for later MATLAB interaction via MQTT. That means an external script, whether described as Python, MATLAB, or another supervisory client, can be treated as another scenario driver that changes inputs or reads state through the same interface logic. The key verification point is not the programming language itself; it is the fact that the system supports **interactive scenario injection** from outside the fixed MATLAB script, using structured commands and live status feedback.

This broadens the verification scope significantly. The thesis can therefore distinguish between:

- **deterministic baseline scenarios** driven entirely inside MATLAB,
- and **interactive supervisory scenarios** driven through dashboard/API logic, where state observation and command submission occur over the communication architecture.

Table 3-33 Interactive scenario sources

Scenario source	Mechanism	What is verified
MATLAB baseline	**Fixed scripted simulation inputs**	**Core EMS decision logic**
Dashboard control	**Buttons, toggles, setpoint fields**	**Human-supervisory response behavior**
API client / external script	**REST/MQTT structured interaction**	**Middleware accessibility and consistency**
MQTT-linked future integration	**Publish/subscribe status exchange**	**Expandable scenario orchestration**

As shown in table 3-33, that distinction is valuable because it proves two layers of correctness at once: algorithmic correctness in MATLAB and integration correctness in Node-RED/dashboard operation.

3.9.3 Generator Mode Scenarios

Generator mode scenarios are essential because the diesel generator is the final support layer used when PV and battery resources are insufficient to maintain the critical feeder. In the MATLAB EMS, generator output is activated only when a residual F1 demand remains after PV allocation and allowable battery discharge have already been applied. This means generator verification must focus not just on "generator on/off," but on whether generator engagement occurs at the correct point in the hierarchy and with the correct scope of support.

The first generator scenario is the **normal no-generator case**, where PV and battery together are sufficient to maintain F1 and no residual critical deficit remains. The expected acceptance behavior is that P_gen = 0 throughout the relevant interval while F1 remains fully supplied. This is important because it verifies that the controller does not overuse diesel support when cleaner or cheaper resources are available.

The second scenario is the **automatic support case**, where PV is low or unavailable and battery discharge alone cannot maintain F1 above the minimum SOC constraint, forcing generator dispatch to cover the remaining deficit. The expected outcome is that generator power rises only to the level necessary to cover the residual F1 need, bounded by its rated power of 100 kW. This scenario verifies the central backup function of the generator and demonstrates that the EMS preserves critical service even under renewable shortage.

The third scenario is the **manual generator command case** introduced through the dashboard and REST API, where a user sends start or stop commands to /api/control/generator. In this case, verification is not only about energy sufficiency, but also about command acceptance, state acknowledgment, and proper update of status or alarms such as generator start messages. This links operational verification to the communication and control architecture.

Table 3-34 Generator verification cases

Scenario	Expected behavior	Verified property
No-generator case	**P_gen = 0 when F1 can be met by PV/battery**	**No unnecessary diesel operation**
Automatic backup case	**Generator covers remaining F1 deficit**	**Critical-load protection**
Manual start/stop case	**API command changes generator control state**	**Supervisory controllability**
Scheduled prototype case	**Time-based generator window in prototype**	**Early demonstrative behavior**

As shown in table 3-34, the simpler prototype one-2.m also contains a time-programmed generator schedule in which generator output is forced to 60 kW over a specified interval rather than being purely demand-triggered. That makes it useful as a comparison scenario because it represents a **scheduled generator mode**, whereas run_ems_matlab.m represents a **conditional generator mode**. Together, these two behaviors are useful in a thesis because they show evolution from demonstration scheduling toward true EMS-driven dispatch logic.

3.9.4 Feeder Shedding/Restoration Scenarios

Feeder shedding and restoration scenarios verify whether the implemented load-priority philosophy is actually respected during operation. In the MATLAB model, F1 is always served first, F2 is served only from remaining PV after F1 is secured, and F3 is served only after both F1 and F2 conditions permit. This means the expected shedding sequence is inherently embedded in the dispatch algorithm: F3 is sacrificed first, F2 next, and F1 last.one-2.m+1

The first feeder scenario is the **full-service case**, where PV availability is high enough that F1, F2, and F3 are all supplied simultaneously. This verifies that the controller does not unnecessarily keep lower-priority feeders disconnected when energy is abundant. The second scenario is the **partial shedding case**, where available PV supports F1 and F2 but not F3, causing only F3 to be curtailed. The third scenario is the **deep conservation case**, where only F1 remains

energized and both F2 and F3 are unsupplied because energy is insufficient beyond the critical feeder.

These cases are particularly important because they produce directly interpretable service states. When F3 is off and F2 remains on, the system is under moderate stress. When both F2 and F3 are off, the system is in severe conservation mode. When F1 itself requires battery and generator support, the system is at critical supply stress. Such feeder scenarios therefore give the verification chapter a service-oriented language rather than a purely electrical one.

Table 3-35 Feeder scenario matrix

Scenario	F1	F2	F3	Meaning
Full service	**ON**	**ON**	**ON**	**Adequate renewable/supply margin**
Mild shortage	**ON**	**ON**	**OFF**	**First-stage shedding**
Severe shortage	**ON**	**OFF**	**OFF**	**Critical-only service mode**
Manual override	**Protected**	**Operator-controlled**	**Operator-controlled**	**Supervisory intervention test**

As shown in table 3-35, the project also supports **manual feeder scenarios** through the dashboard and /api/control/feeders' endpoint, where F2 or F3 can be enabled or disabled directly. This creates another verification dimension: not only must automatically shedding work correctly, but manual override of feeder state must also be reflected correctly in status JSON, UI toggles, and alarm/state acknowledgment. Therefore, feeder verification spans both algorithmic priority enforcement and operator-triggered command handling.

3.9.5 Battery Charge/Discharge Scenarios

Battery scenarios are indispensable because the battery is the only storage element linking surplus and deficit periods across time. In the main MATLAB model, battery power is positive

during discharge and negative during charging, with SOC updated each hour and constrained between 30% and 90%. Verification must therefore examine not only whether the battery changes state, but whether it changes state in the correct direction, under correct conditions, and within allowed limits.

The first battery scenario is the **discharge support case**, in which PV is insufficient to fully support F1 and the battery discharges to reduce or eliminate the remaining critical deficit before generator use is considered. The expected result is positive P_batt, falling SOC, and delayed or reduced generator reliance, provided SOC is above the minimum bound. This verifies that stored energy is used exactly where the EMS logic intends it to be used: in protecting F1.

The second battery scenario is the **charging from surplus PV case**, where PV remains after serving F1, F2, and F3, and the remaining energy is used to charge the battery if SOC is below 90%. The expected result is negative P_batt, rising SOC, and no violation of the maximum SOC bound. This case verifies reserve recovery behavior and proves that the controller treats excess PV as an opportunity to rebuild future resilience rather than merely curtailing it immediately.

The third scenario is the **SOC limit case**, where battery operation is constrained by the lower or upper SOC bounds. When SOC reaches 30%, further discharge must stop; when SOC reaches 90%, further charging must stop. This is one of the most important verification checks because it confirms that the controller respects storage protection boundaries and does not create mathematically convenient but physically invalid battery behavior.

Table 3-36 Battery scenario types

Scenario	Expected battery sign/state	Verified property
Discharge support	**P_batt > 0, SOC decreases**	**F1 support before generator escalation**
PV charging	**P_batt < 0, SOC increases**	**Reserve replenishment from surplus renewable**
Lower SOC bound	**No further discharge below 30%**	**Storage protection**
Upper SOC bound	**No further charging above 90%**	**Charge saturation handling**

As shown in table 3-36, the prototype one-2.m also contains explicit scripted intervals of battery discharge and charging, making it useful for visually verifying the sign convention and SOC update behavior across a 24-hour shifted display window. In the thesis, this can be described as a demonstrative battery scenario set, while run_ems_matlab.m represents the formal rule-based battery verification case.

3.9.6 PV Surplus and Deficit Scenarios

PV surplus and deficit scenarios are among the most important verification categories because solar availability is the dominant source of variability in the modeled microgrid. In both MATLAB scripts, PV follows a daytime sinusoidal profile with zero output at night and peak output around midday. This naturally creates periods of surplus and deficit over the daily cycle, making PV-driven scenario definition a central part of verifying the EMS.

The **PV surplus scenario** occurs when available PV is greater than the demand of the currently served feeders. Under the rule-based EMS, the expected sequence is first to satisfy F1, then F2, then F3, and finally direct any remaining PV to battery charging subject to SOC and battery power constraints. If the battery is already at SOC_max, then any additional PV becomes

effectively unusable in the present simplified model, implying curtailment by omission. This scenario verifies whether the controller uses renewable abundance in the intended order.

The **PV deficit scenario** occurs when PV is too small to cover the required demand, especially at night or during early morning and evening shoulder periods. In that case, the expected response is immediate prioritization of F1, followed by battery support if possible, then generator support if necessary, while F2 and F3 are curtailed according to the hierarchical logic. This verifies whether the EMS properly transitions from renewable abundance mode to conservation mode without violating the priority structure.

Table 3-37 PV regime interpretation

PV regime	Expected EMS behavior	Verification meaning
Strong surplus	**F1, F2, F3 supplied; battery charges if possible**	**Abundance management**
Moderate adequacy	**F1 and possibly F2 supplied; F3 may be shed**	**Selective service maintenance**
Deficit	**F1 prioritized; battery/gen support; lower feeders curtailed**	**Scarcity response correctness**
Night/no PV	**Reliance on battery and/or generator**	**Renewable absence handling**

As shown in table 3-37, a useful verification interpretation is to treat PV surplus and deficit not only as source scenarios, but as **system stress regimes**. Surplus regimes test whether the system can extend service and rebuild reserves. Deficit regimes test whether it can preserve critical service and degrade gracefully. That framing strengthens the chapter because it shows that scenario definition is tied directly to the resilience function of the microgrid.

3.9.7 Scenario Acceptance Criteria

A verification-oriented chapter becomes much stronger when each scenario is evaluated against explicit acceptance criteria rather than against vague statements such as “the system

behaved correctly." In this project, acceptance criteria can be derived directly from the EMS logic, battery constraints, feeder hierarchy, and supervisory communication behavior already implemented in the model and dashboard.

The first acceptance criterion is **critical load preservation**. In any physically feasible scenario, F1 must remain the highest-protected feeder, and the EMS must allocate PV, battery, and generator support in a way that maximizes F1 continuity before F2 or F3 are considered. A scenario fails acceptance if lower-priority feeders are served while avoidable F1 deficit remains.

The second acceptance criterion is **priority-consistent shedding and restoration**. F3 must not remain energized in a condition where F2 should be shed first only if the dispatch logic explicitly permits that ordering, and F2/F3 restoration must occur only when higher-priority needs are already secured. In the actual model, the ordering is strict enough that restoration logic can be checked directly through feeder supplied outputs over time.

The third acceptance criterion is **battery constraint compliance**. SOC must never fall below 30% or exceed 90%, and battery power must remain within its configured limit. Any scenario that produces boundary violation is mathematically invalid within the assumptions of the current model.

The fourth acceptance criterion is **generator logic consistency**. In the main EMS, the generator should only operate when residual F1 demand remains after PV and allowable battery support. If the generator appears without such a condition in the rule-based scenario, that would indicate a logic error; conversely, if residual F1 remains unsupplied while generator headroom exists, that would indicate incomplete backup activation.

The fifth acceptance criterion is **communication acknowledgment** for interactive scenarios. A command sent through the dashboard or API must produce an interpretable response, reflected state change, or status/alarm update consistent with the requested action. This is necessary because verification of a supervisory system includes not only internal correctness but observable and reliable command execution.

Table 3-38 Scenario acceptance criteria table

Criterion	Acceptance condition	Main source
Critical feeder priority	**F1 protected before lower feeders**	**MATLAB EMS**
Shedding/restoration order	**F3 curtailed before F2; F2 before F1**	**Feeder dispatch logic**
SOC validity	**30% ≤ SOC ≤ 90%**	**Battery model**
Battery power validity	**Within 60 kW limit**	**Battery dispatch constraint**
Generator correctness	**Used only for residual F1 deficit in main EMS**	**Backup dispatch logic**
API command acknowledgment	**Response and reflected state are consistent**	**Node-RED/dashboard integration**

As shown in table 3-38, these acceptance criteria are powerful because they transform the scenarios from descriptive examples into formal verification tests.

3.9.8 Relation to Results Chapter

This scenario-definition section should connect directly to the results chapter by acting as its **experimental design framework**. In other words, the results chapter should not appear to present random figures or isolated plots; each result should be traceable back to one of the verification scenarios defined here. That linkage makes the thesis more rigorous because it shows that the results were generated to answer specific validation questions about system behavior

For example, plots of PV versus load, active load after EMS, feeder supplied power, battery power, SOC, generator output, and total supply versus active demand all correspond naturally to the scenario families defined above. A generator plot belongs to generator mode scenarios; feeder contribution plots belong to shedding/restoration scenarios; SOC and battery power plots belong to battery scenarios; supply-demand balance plots belong to PV surplus/deficit and overall acceptance verification. This mapping should be made explicit in the thesis so that each figure has a clear verification purpose.

The results chapter can therefore be organized in a way that mirrors this section:

- baseline MATLAB verification results,
- feeder and priority behavior,
- storage and generator response,
- interactive supervisory verification through dashboard/API behavior.

This direct relationship between scenarios and results is academically important. It prevents the common weakness of simulation chapters where many plots are shown but the reader is not told what each plot is supposed to verify. Here, by contrast, each result becomes evidence tied to a predefined scenario and judged against predefined acceptance criteria. That structure makes the final chapter much stronger, more defendable, and easier to discuss during thesis examination.

Verification structure

Taken together, these scenarios define a verification framework that spans baseline simulation, resource adequacy, feeder priority, storage dynamics, backup generation, and interactive supervisory control. The framework is strong because it checks not only whether the microgrid can operate, but whether it operates **according to the intended hierarchy, constraints, and communication logic** under both normal and stressed conditions. This makes Section 3.9 the natural bridge between the modeling chapter and the results chapter, since it explains exactly **what was tested, why it was tested, and how success is judged**.

3.10 Real System Mapping and Validation Requirements

3.10.1 Mapping Simulation Signals to Real Signals

The first requirement for moving from simulation to implementation is the establishment of a disciplined mapping between **simulation variables** and **real plant signals**. In the MATLAB EMS, the system is represented by structured variables such as Ppv, F1, F2, F3, F1_supplied, F2_supplied, F3_supplied, P_batt, P_gen, SOC, P_active, and P_supply, all evolving over a 24-hour supervisory horizon. These variables are mathematically meaningful inside the model, but for real-system deployment they must correspond to physical measurements, device states, breaker positions, command outputs, or derived supervisory tags.

This mapping is important because simulation variables are often **idealized abstractions**, while real signals are constrained by instrumentation, device interfaces, sampling latency, and hardware semantics. For example, Ppv in MATLAB is a clean continuous-valued power profile generated from a sinusoidal function, whereas in a real system PV power would come from inverter measurements, meter values, or PLC-acquired analog registers associated with the inverter/charger path. Likewise, SOC in MATLAB is an ideal supervisory state updated by a discrete equation, whereas in a real system SOC must originate from BMS-derived estimation or inverter/battery communication rather than from direct algebraic integration alone.

A strong real-system mapping therefore requires that each simulation signal be assigned one of the following real-world classes:

- direct measured signal, such as PV power, generator electrical values, or feeder current/power,
- device status signal, such as generator running, inverter mode, or breaker position,
- command signal, such as generator start/stop or feeder enable/disable,
- or derived supervisory state, such as active source or total supplied load, computed from other available signals. +1

This classification is essential because not every simulation variable exists in hardware in the same form. For instance, F1_supplied in simulation is a directly available result of the EMS algorithm, but in real operation it may need to be inferred from feeder breaker closed status plus measured feeder power or current. Similarly, P_supply is a useful aggregate verification variable

in MATLAB, but in the real plant it would likely be reconstructed from source-side measurements rather than read as a single native device tag.

Table 3-39 Simulation-to-real signal mapping

Simulation signal	Meaning in model	Likely real counterpart
Ppv	**PV generation power**	**Inverter/charger measured PV-side or AC output power**
SOC	**Battery state of charge**	**BMS01 or inverter-reported SOC**
P_batt	**Battery charge/discharge power**	**Battery/inverter power flow measurement**
P_gen	**Generator supplied power**	**GEN01/DSE01 electrical power or derived PLC value**
F1_supplied, F2_supplied, F3_supplied	**Supplied feeder loads**	**Breaker state plus feeder metering or PLC-derived feeder status**
P_active	**Total served load**	**Sum of energized feeder powers or PLC-computed served load**
P_supply	**Total source-side supplied power**	**Sum of PV, generator, and battery discharge measurements**

T As shown in table 3-39, the central validation requirement is therefore not merely that the names look similar, but that the **physical meaning, timing, units, and operational interpretation** remain consistent across simulation and implementation.

3.10.2 Mapping EPLAN Devices to EMS Functions

The EPLAN package provides the physical device structure of the microgrid, while the EMS defines the operational intelligence that decides how these devices should be used. Mapping EPLAN devices to EMS functions is therefore necessary to show that every major control action in the simulation has a realizable hardware counterpart in the proposed plant. This is one of the strongest validation steps in the chapter because it demonstrates architectural completeness rather than software-only reasoning.

The EPLAN design identifies the main physical assets as the PV array and MPPT/DC side, battery bank and BMS, inverter/charger, diesel generator with DSE controller, PLC, AC bus, and feeder breakers ACBR03 to ACBR05 for F1–F3. The EMS, by contrast, works with abstract functions such as renewable generation allocation, storage support, generator backup, feeder prioritization, SOC protection, and alarm/command supervision. The job of the mapping exercise is to show that these two views describe the same system from different perspectives

For example, the simulation's PV dispatch function maps physically to the PV array, MPPT/controller path, DC bus, and inverter/charger chain that ultimately injects AC power through the inverter output breaker into the AC bus. The battery support function maps to BATT01, BMS01, DC breaker, contactor, and inverter/charger operation in charging or inverting mode. Generator backup maps to GEN01, AVR, DSE01, starting battery, and generator breaker ACBR02. Load prioritization maps to feeder breakers ACBR03, ACBR04, and ACBR05, each corresponding to F1, F2, and F3 respectively.

This device-to-function correspondence is crucial for two reasons. First, it proves that the EMS is not issuing abstract commands with no hardware endpoint. Second, it makes validation testable: once a function is tied to a real device chain, commissioning engineers can verify whether the expected device behavior actually occurs when the EMS demands it.

Table 3-40 EPLAN device to EMS function map

EPLAN device	Physical role	EMS function supported
PV ARRAY / MPPT01 / BUS_DC01	Renewable generation path	PV availability and renewable-first dispatch
BATT01 / BMS01 / CONT_BATT01	Energy storage path	Battery charge/discharge and SOC protection
INV_CHR01	Power conversion interface	DC/AC delivery and AC/DC charging mode
GEN01 / DSE01 / ACBR02	Backup generation chain	Generator support for residual critical demand
ACBR03	F1 breaker	Critical feeder continuity
ACBR04	F2 breaker	Medium-priority feeder control
ACBR05	F3 breaker	Low-priority feeder control
PLC01	Main supervisory controller	Signal acquisition, interlocks, command dispatch

As shown in table 3-40, this mapping gives the thesis a very strong implementation dimension because it shows that each EMS function is physically anchored in the EPLAN architecture.

3.10.3 Mapping PLC Tags to System Behavior

The PLC is the main hardware-layer supervisory controller in the proposed plant, and therefore PLC tags are the most practical bridge between software-level logic and physical system behavior. The EPLAN design states that the Siemens S7-1200 PLC receives status and

measurements from the DSE, inverter, BMS, and feeders, and sends commands to the DSE, inverter, battery contactor, and feeders. This means PLC tags are not passive labels; they are the operational representation of what the system is sensing, deciding, and doing.

Mapping PLC tags to system behavior requires more than assigning names to addresses. Each PLC tag must represent a behaviorally meaningful state such as “generator running,” “battery contactor closed,” “F2 enabled,” “inverter in charging mode,” or “SOC below threshold.” The reason this matter is that the EMS does not control equipment by raw theory; it controls it through tag-level conditions and outputs. If the tags are poorly defined, the real behavior will diverge from the simulation even if the equations are correct. +1

The Node-RED and dashboard layer already uses a normalized software tag set including mode, soc, genStatus, pvPower, totalLoad, activeSource, f2Status, f3Status, and various setpoints. A strong validation requirement is therefore to align PLC tags with these supervisory meanings so that the PLC becomes the authoritative hardware source for the same states represented in software. This alignment is what allows the project to move from simulated status JSON to hardware-backed status JSON without redesigning the supervisory interface. +1

A useful engineering distinction is to organize PLC tags into:

- **status tags**, such as RUNNING, FAULT, breaker closed, inverter mode,Eplan-FIle-7.pdf
- **measurement tags**, such as PV power, feeder current, battery voltage, SOC,
- **command tags**, such as start generator, stop generator, close feeder breaker, open feeder breaker, and **derived logic tags**, such as active source, EMS mode, critical deficit present, or load-shedding active.

Table 3-41 PLC tag behavior classes

PLC tag class	Example meaning	Behavioral purpose
Status tags	**Generator RUNNING, feeder breaker closed**	**Reflect actual plant state**
Measurement tags	**SOC, PV power, feeder load**	**Quantify operating condition**
Command tags	**Start/stop, enable/disable, charge/discharge mode**	**Execute EMS decisions**
Derived logic tags	**Active source, low-SOC condition, alarm active**	**Support supervisory interpretation**

As shown in table 3-41, the validation goal is that the PLC tag model should reproduce the same behavior seen in the simulation model, but in hardware-executable form.

3.10.4 Mapping DSE Statuses to Generator Logic

The DSE8610 MKII generator controller plays a critical role because generator behavior in the EMS is not only about requesting power, but about determining whether the generator is actually available, healthy, running, synchronized, or faulted. In the MATLAB model, generator logic is simple: when residual F1 demand remains after PV and battery support, P_gen rises up to the generator rating. In the real system, however, this logic must be conditioned by the statuses returned from DSE01.

The EPLAN documentation shows that DSE01 receives sensing inputs such as generator voltage, frequency, engine speed, oil pressure, coolant temperature, and fuel level, and returns status information such as RUNNING and FAULT back to the PLC. It also accepts remote start and related command interactions from the PLC. Therefore, the real generator logic should not be viewed as “start when needed” in a simplistic sense; it should be viewed as a state machine constrained by DSE feedback.

A proper mapping between DSE statuses and EMS generator logic should include at least the following correspondences:

- a **start requests** from PLC/EMS maps to DSE remote-start input,
- a **running confirmation** from DSE maps to generator-available status for EMS source allocation,
- a **fault/alarm status** maps to generator unavailable or generator trip condition,
- and breaker-close permission must depend on successful generator operational state, not just on command issuance.

This is one of the most important validation requirements in the whole chapter because it prevents a dangerous mismatch between simulated and real generator behavior. In simulation, a generator can be treated as an instantly available bounded source. In reality, the EMS must respect whether the DSE has accepted the start request, whether the engine is healthy, and whether the generator can actually be connected to the AC bus. Thus, DSE status mapping is essential for translating idealized backup logic into safe real generator operation.

Table 3-42 DSE-to-generator logic mapping

DSE status or signal	EMS meaning	Required control implication
Remote start input	**Generator start command issued**	**Begin generator start sequence**
RUNNING	**Generator available**	**Allow source selection / support logic**
FAULT	**Generator unavailable or unsafe**	**Block dispatch and raise alarm**
Voltage/frequency healthy	**Generator electrically acceptable**	**Permit breaker close logic**
Breaker close/trip control	**Connection to AC bus**	**Final execution of generator support**

As shown in table 3-42, the project becomes much stronger when it explicitly states that real generator validation depends not only on command generation, but on **status-confirmed generator readiness**.

3.10.5 Mapping HTML Indicators to System States

The HTML dashboard is the visible supervisory face of the project, so its indicators must correspond to real and unambiguous system states rather than decorative labels. The Node-RED/dashboard description shows that the dashboard updates fields such as socValue, pvValue, loadValue, genStatus, activeSource, modeValue, feeder toggle states, setpoints, and alarm cards based on the JSON returned from /api/status. This means every visible indicator on the screen is already implicitly part of a mapping chain from simulation or plant data to operator perception.

This mapping matters because an HMI is only trustworthy when its indicators are semantically accurate. If genStatus on the dashboard says “Running,” that must correspond either to true generator-running feedback from DSE/PLC in the real system or to the correctly simulated generator state in the verification model. If f2Status is shown as ON, that must mean the feeder is actually enabled in logic and, in hardware-connected mode, ideally confirmed through breaker or output status rather than only a software intention.

The dashboard also includes mode display, communication status, setpoint fields, and alarms. Each of these requires explicit validation. modeValue and modeStatus must match the supervisory control mode. The online/offline communication badge must reflect actual success or failure of periodic status retrieval. Alarm cards must reflect the current alarm list, and not stale or synthetic values. The setpoint fields must display the latest accepted supervisory thresholds without overwriting active user edits while typing. These are not cosmetic implementation details; they are part of the real-system validation requirement because incorrect operator information can lead to incorrect operational decisions.

Table 3-43 HTML indicator mapping

HTML indicator	Software field	Real-system meaning
SOC card	soc	Battery reserve state from simulation or BMS-backed source
PV card	pvPower	Current PV generation/power contribution
Load card	totalLoad	Total current demand or served demand context
Generator status	genStatus	DSE/PLC-confirmed generator operating state
Active source	activeSource	Current dominant supply path as derived supervisory state
F2/F3 toggle and label	f2Status, f3Status	Feeder enable/energization state
Mode display	mode	Auto/manual supervisory mode
Alarm list	alarms	Active warning/fault/event list

As shown in table 3-43, the dashboard is therefore part of the validation chain, not something outside it. A correct EMS with a misleading HMI is still an invalid supervisory system.

3.10.6 Validation Against Realistic Operational Behavior

A strong model is not validated only because its signals and devices can be mapped; it is validated when the resulting behavior resembles what a real hybrid microgrid would do under plausible operating conditions. This is the purpose of operational-behavior validation. The

question is not merely whether the equations run, but whether the plant would actually behave in a comparable way if the proposed mapping were implemented.

The MATLAB model already defines realistic high-level expectations: PV dominates during daylight, battery absorbs surplus or supports critical load during shortage, generator intervenes when critical demand remains unmet, and feeder priority determines which loads remain energized under scarcity. The EPLAN design supports these same functional roles through the PV path, BATT01/BMS01, INV_CHR01, GEN01/DSE01, PLC01, and feeder breakers. This consistency is a strong starting point for validation because the real architecture and the simulation logic are functionally aligned.

Validation against realistic operational behavior should therefore test whether the following patterns hold:

- PV-supported daytime service with reduced diesel dependence,
- battery charge during surplus and discharge during critical deficits,
- generator support triggered by actual need and constrained by controller status,
- feeder shedding in the correct priority order under energy shortage,
- and correct supervisory visibility of these events in Node-RED and the dashboard.

Table 3-44 Realistic behavior validation checks

Behavior to validate	Expected realistic pattern	Basis in project
Daylight operation	**PV contributes strongly during 06:00–18:00**	**MATLAB PV model + real PV path**
Storage behavior	**Charge on surplus, discharge on deficit**	**Battery model + BATT01/BMS01**
Generator behavior	**Supports residual critical demand only**	**EMS logic + DSE-controlled GEN01**
Feeder behavior	**F1 protected, F2/F3 curtailed by priority**	**Feeder logic + ACBR03–05**
Supervisory reflection	**HMI/API states match actual operating condition**	**Node-RED/dashboard structure**

As shown in table 3-44, this kind of validation is particularly strong because it does not demand perfect field replication of every waveform or disturbance. Instead, it asks whether the **operational logic of the real system** matches the operational logic of the simulated system. For a supervisory EMS thesis, that is the correct level of realism to demand.

3.10.7 Commissioning-Oriented Requirements

The final validation layer is commissioning-oriented, meaning the project should specify what must be verified during pre-energization tests, functional checks, and staged startup to confirm that the implemented system behaves according to the intended EMS design. This subsection is extremely important because it transforms the chapter from academic completeness into engineering readiness.

A commissioning-oriented framework should begin with **I/O verification**. Every digital input, digital output, analog input, analog output, communication link, and dashboard indicator must be checked for correct address, scaling, polarity, and semantic meaning before higher-level EMS behavior is trusted. This includes confirming that a generator start command actually

reaches DSE01, that feeder commands operate the correct breaker logic, that SOC values are plausible, and that the dashboard displays the same states the PLC or middleware actually holds.

The next requirement is **functional sequence verification**. This means testing the ordered behavior of the microgrid in safe staged conditions: PV-only operation, battery-assisted support, generator start on need, feeder shedding order, feeder restoration order, and alarm generation/reset behavior. These tests should confirm that the real control chain reproduces the same hierarchy seen in the simulation.

A third commissioning requirement is **interlock and fail-safe verification**. The real system must confirm that unsafe combinations are blocked, such as generator dispatch without healthy DSE status, battery action beyond protected limits, or feeder energization inconsistent with permitted operating logic. This is where the difference between simulation and field reality matters most, because the field system must be robust to device faults, bad feedback, communication loss, and command rejection.

A fourth requirement is **supervisory observability verification**. Every commissioning action should be visible in the dashboard/API/MQTT layer with correct timestamps or immediate status changes so that the operator is never acting blindly. This ensures that the supervisory system is not only controlling the plant, but also correctly reporting the plant.

Table 3-45 Commissioning-oriented validation checklist

Requirement	What must be checked	Why it matters
Signal checkout	**Correct I/O address, scaling, polarity, mapping**	**Prevents semantic and wiring errors**
Device command test	**Generator, feeder, inverter, contactor actions respond correctly**	**Confirms execution path**
Status feedback test	**RUNNING, FAULT, breaker status, alarms are returned correctly**	**Confirms closed-loop supervision**
Sequence test	**PV/battery/generator/feeder logic follows EMS hierarchy**	**Validates functional behavior**
HMI/API test	**Dashboard and API reflect real state consistently**	**Confirms operator visibility**
Fault/interlock test	**Unsafe conditions are blocked and alarmed**	**Confirms field safety readiness**

As shown in table 3-45, these commissioning-oriented requirements are what finally make the chapter feel complete. They show that the work does not stop at proving the model is logical; it extends to proving that the system could be **tested, accepted, and brought into service in a structured engineering manner**.

Chapter closure

This final section closes Chapter 3 by showing that the EMS model, communication architecture, and hardware design are not isolated descriptions but parts of one coherent implementation pathway. Simulation signals can be mapped to real signals, EPLAN devices can be mapped to EMS functions, PLC and DSE states can be mapped to actual control behavior,

and dashboard indicators can be mapped to operator-relevant plant conditions. With these mappings and validation requirements in place, the chapter establishes a strong foundation for Chapter 4, where results can be interpreted not merely as simulation outputs, but as behavior with credible correspondence to a real hybrid microgrid control system.

Chapter 4

Project Design

4. Chapter 4

4.1 Design Chapter Introduction

Chapter 3 established the analytical and technical foundation of the present work by defining the project context, system scope, load-priority philosophy, component requirements, control objectives, and the mapping logic between simulation variables and real implementation elements. Building on that foundation, Chapter 4 moves from requirement definition to actual system design. In other words, whereas the previous chapter answered the question of **what the hybrid microgrid must achieve**, the present chapter answers the question of **how that microgrid is architected, structured, and organized to achieve those objectives in a technically coherent and implementation-oriented form**.

The design stage occupies a central place in any engineering thesis because it is the stage where abstract requirements are transformed into an integrated system architecture. A hybrid microgrid cannot be understood adequately by discussing PV sizing, battery capacity, diesel backup, and EMS logic as isolated topics. Its engineering value appears only when those elements are assembled into a unified structure in which energy flow, control authority, communication pathways, and operational priorities are all aligned. For this reason, Chapter 4 does not present the project as a loose collection of components, but as a coordinated design framework that combines electrical infrastructure, industrial control, supervisory software, and simulation-oriented representation within one consistent architecture.

The system addressed in this thesis is an intelligent hybrid microgrid intended for a rural village application, based on a 150 kW photovoltaic source, a battery storage subsystem of approximately 100–123 kWh, a 100 kW diesel generator, an inverter/charger power conversion stage, a three-feeder load structure, a Siemens S7-1200 PLC-centered control layer, a DSE8610 MKII genset controller, and a digital supervisory layer using Node-RED and an HTML dashboard. This configuration is not arbitrary. It reflects the design choices developed in the previous chapter, especially the need to maximize renewable utilization, protect critical loads, preserve operational transparency, and maintain compatibility with realistic industrial hardware rather than purely theoretical control platforms.

A key feature of the present design chapter is that it deliberately spans both the **real-system representation** and the **simulation-oriented representation** of the microgrid. This duality is one of the defining strengths of the project. On the one hand, the design is grounded in realistic

implementation materials, including EPLAN electrical documentation, PLC-oriented logic structure, DSE control-and-monitoring architecture, and software supervision pathways. On the other hand, the same design is represented in MATLAB/Simulink-oriented form through rule-based EMS logic, power-balance handling, feeder supply modeling, battery SOC evolution, and generator dispatch behavior. The chapter therefore serves as the bridge between an implementation-capable architecture and a model-based operational framework.

This bridging role is academically important. Many student projects fall into one of two extremes: either they remain at the level of idealized simulation without enough connection to real engineering practice, or they present hardware drawings and software screenshots without a sufficiently clear analytical structure tying those elements together. The present chapter is designed to avoid both weaknesses. It treats the microgrid as a layered engineering system in which physical power hardware, supervisory energy management, industrial command execution, communication middleware, and operator interface are different views of the same integrated design problem.

From a systems-engineering perspective, the design chapter must demonstrate coherence across multiple domains at the same time. First, the **electrical domain** must be coherent: source ratings, bus structure, feeder arrangements, protection boundaries, and power conversion interfaces must be compatible and logically organized. Second, the **control domain** must be coherent: the EMS philosophy, feeder priority rules, battery SOC constraints, and generator dispatch logic must be implementable through identifiable command and feedback pathways. Third, the **communication and supervision domain** must be coherent: monitored states, user-visible variables, alarms, manual override actions, and API-level commands must reflect the same operational meaning used in the underlying EMS and hardware-oriented logic. A strong design chapter must show this multi-domain consistency explicitly, because design quality is judged not only by what components exist, but by how well they work together as a unified architecture.

Accordingly, this chapter treats the proposed microgrid as a structured hierarchy rather than a flat diagram. At the foundation lies the electrical power architecture composed of the PV subsystem, battery bank, DC bus, inverter/charger, diesel generator, AC bus, and prioritized feeder structure. Above that lies the industrial control layer centered on the Siemens PLC and the DSE8610 MKII, which translates energy-management decisions into actual command pathways such as generator start/stop, feeder enabling or shedding, and status interpretation from

field-connected devices. Above that, in turn, lies the supervisory software layer in which Node-RED operates as middleware and the HTML dashboard acts as the user-facing interface for monitoring and limited operator interaction. Parallel to all of these lies the simulation-oriented design representation in Simulink, where the same architecture is expressed behaviorally rather than through complete hardware detail.

The rationale for organizing the chapter in this layered manner comes directly from the nature of microgrids themselves. A microgrid is not only an electrical installation. It is a cyber-physical energy system in which generation, storage, switching, control, and communication interact continuously. In a simple diesel-only plant, the design emphasis may remain largely on generation sizing and distribution hardware. In a hybrid PV-battery-diesel microgrid, however, the design must additionally account for source prioritization, conditional generator use, battery operating windows, feeder-level continuity of service, and operator awareness of state transitions. Therefore, the design chapter must explain not only where the energy comes from and where it goes, but also who decides, by what logic, through which interfaces, and under what operating constraints.

One of the principal themes carried from Chapter 3 into Chapter 4 is the distinction between **requirements** and **design realization**. In the requirements chapter, feeder criticality, SOC limits, renewable-first operation, generator backup behavior, and supervisory visibility were presented as design necessities. In the present chapter, those necessities are transformed into actual design elements. For example, the requirement to preserve critical load continuity becomes a feeder-priority arrangement supported by breaker-level control and control logic pathways. The requirement to minimize diesel usage becomes a source architecture in which PV is primary, battery support is conditional but protected, and diesel participation is delayed until critical load security demands it. The requirement for practical implementation becomes the use of a PLC-centered control structure and a DSE-based generator interface rather than an abstract black-box controller. The requirement for transparency becomes a supervisory dashboard and API structure that expose mode, SOC, source state, feeder state, and alarms to the operator.

This means that Chapter 4 is not merely descriptive. Its role is interpretive and justificatory as well. Each design choice presented in the chapter must be understood as a response to a defined operational or implementation need. The PV array is not discussed only as generation hardware; it is the preferred primary source in the EMS architecture. The battery is not discussed only as stored energy; it is a constrained support resource whose SOC operating band protects both

equipment integrity and service continuity. The diesel generator is not discussed only as backup hardware; it is a controlled resilience asset integrated through the DSE and used only under specified logic conditions. The PLC is not discussed only as an industrial device; it is the physical executor of supervisory energy-management decisions. Node-RED and the HTML dashboard are not discussed merely as software features; they are the means by which system state is externalized and selected operator interventions are managed in a structured, mediated way.

Another important function of this introductory section is to define the intellectual position of the design chapter relative to the chapters that follow. Chapter 4 is concerned with **architecture and organization**. It explains how the system is put together, how its principal parts relate, and why particular implementation choices are justified. It does not focus on scenario execution, testing procedures, or interpretation of numerical outcomes. Those belong properly to the later chapters devoted to implementation/testing and to results/discussion. This separation is essential for academic clarity. A strong thesis distinguishes between design intent, implementation procedure, and performance evaluation, even when those three are tightly connected in practice.

In the present work, this separation is especially useful because the project includes both realistic implementation materials and simulation-based operation analysis. If the design chapter were overloaded with test scenarios and results commentary, it would become difficult to distinguish architectural choices from performance evidence. Conversely, if the design chapter were too brief and simply listed diagrams without explanation, later chapters would lack a stable structural reference for interpreting the tests and results. The present chapter therefore provides the architectural narrative that allows the reader to understand later system behavior as the consequence of clearly defined design decisions rather than as isolated output curves or screenshots.

The chapter also adopts a disciplined position regarding **scope realism**. It presents the system as implementation-oriented, but it does not overclaim that every design artifact corresponds to a fully field-commissioned plant under all simulated scenarios. Instead, the chapter proceeds on the basis of functional consistency: the real-system diagrams, control sketches, PLC logic representations, DSE integration pathway, Node-RED API structure, and dashboard views are treated as realistic implementation counterparts of the same operational logic studied in simulation. This is a strong academic position because it preserves engineering realism without

requiring the thesis to pretend that every hardware step has already undergone full commercial commissioning, factory acceptance testing, or long-term field validation.

Such a position is particularly appropriate for bachelor-level engineering research. The goal is not to replace a complete EPC delivery package or a manufacturer's commissioning manual. The goal is to produce a defensible, coherent, and technically credible system design in which each major functional requirement is mapped to an identifiable architectural solution. In this sense, Chapter 4 demonstrates implementation readiness at the level appropriate to the study: enough realism to show that the design can be built, controlled, supervised, and tested in a meaningful way, but without claiming detailed commercial execution beyond the available evidence.

The chapter further recognizes that a hybrid microgrid must be understood through several complementary design questions. These questions include: What is the electrical structure of the system? How are sources and loads arranged physically? How is control authority distributed between PLC, DSE, local equipment controllers, and supervisory software? How are feeder priorities embodied in actual switching and logic pathways? How is the EMS represented both in real industrial form and in simulation form? How does the operator observe and influence system behavior? And how is consistency maintained across all these layers? The organization of the chapter is intended to answer these questions step by step, so that the reader moves from high-level architecture toward progressively more specific design layers without losing sight of their interdependence.

The first major body of the chapter therefore addresses the **overall system design philosophy** and the top-level architectural logic that governs the proposed microgrid. This includes the justification for the hybrid configuration itself, the prioritization of renewable energy, the bounded role of the battery, the conditional role of the diesel generator, and the importance of preserving critical feeder continuity under uncertain resource availability. Once that design philosophy is established, the chapter proceeds to the **electrical power architecture**, where the PV subsystem, battery bank, inverter/charger, DC and AC buses, feeder distribution, and power-path organization are described in implementation-oriented detail.

After the electrical structure is established, the chapter turns to the **feeder and load allocation design**, because feeder priority is not an optional control refinement in this project but one of its defining architectural principles. The three-feeder arrangement—F1 critical, F2 important, and F3 non-critical—must therefore be presented not merely as a theoretical

classification, but as a design-level arrangement that shapes breaker structure, control authority, and continuity-of-service decisions. This naturally leads into the design of the **diesel generator and DSE integration layer**, where the thesis explains how backup generation is incorporated into the hybrid system without allowing it to dominate the operational philosophy.

The next part of the chapter is centered on the **PLC-based control architecture**. This is one of the most important transitions in the chapter because it marks the move from physical hardware structure into executable decision structure. The Siemens S7-1200 PLC is treated as the central industrial controller through which system-level EMS decisions are materialized in practice, using digital and analog I/O, status acquisition, control commands, and defined interaction with the DSE, inverter, battery-related signals, and feeder paths. The design chapter then expands this into the **EMS design realization** itself, showing how the rule-based philosophy established earlier in the thesis is embodied through sequential logic governing source priority, SOC constraints, generator dispatch, feeder shedding, and restoration pathways.

Because the work also relies heavily on simulation-oriented analysis, the chapter then includes a dedicated section on the **simulation-oriented design representation**. This part is necessary to explain how the same microgrid architecture is represented behaviorally in Simulink, including the mapping between physical components and model variables, as well as the limits of abstraction adopted in the simulation environment. In this way, the chapter avoids presenting the simulation model as detached from the real design; instead, it demonstrates that the simulation is a structured representation of the same architecture viewed through a control-and-operation lens.

The final design layers addressed in the chapter concern **supervisory software and human–machine interaction**. In the present project, Node-RED serves as a middleware and orchestration layer that exposes status and control routes through a REST-style API and can also publish state information through MQTT-oriented pathways. The HTML dashboard then presents system status, feeder states, source indicators, alarms, generator controls, and EMS setpoint interaction in a form accessible to an operator. Including these elements in the design chapter is important because modern hybrid microgrids are not complete when only the power circuit and source logic are described. A usable microgrid also requires visibility, interpretability, and a structured path for selected operator interventions.

Taken together, these sections are intended to show that the proposed microgrid is not designed around a single device or a single algorithm. It is designed around an **integrated**

architecture of roles. The PV subsystem plays the role of preferred generation. The battery plays the role of protected short-term balancing and critical support. The diesel generator plays the role of controlled resilience backup. The inverter/charger plays the role of power-conversion interface between source domains. The feeder structure plays the role of service-priority expression. The PLC plays the role of deterministic industrial decision executor. The DSE plays the role of generator-control and protection intermediary. Node-RED plays the role of supervisory integration middleware. The dashboard plays the role of operator visibility and structured interaction. Simulink plays the role of behavioral design representation and scenario-capable EMS analysis.

This role-based understanding is helpful because it prevents the design chapter from becoming a catalogue of components. Instead, it frames the design as a coordination problem. The question is not simply what devices are used, but how responsibilities are partitioned among them in a way that preserves critical supply, minimizes diesel dependency, protects equipment, and keeps system behavior explainable to engineers and operators. Such explainability is especially valuable in the context of rural or resource-constrained deployments, where maintainability, transparency, and engineering practicality often matter as much as theoretical optimality.

The choice of a rule-based supervisory strategy strongly influences the tone and structure of the design chapter as well. In highly optimization-heavy studies, the design discussion often revolves around solver structure, objective functions, and mathematical convergence. In the present project, by contrast, the emphasis is on architecture, sequence, conditions, and realizable control pathways. This does not make the design weaker; on the contrary, it makes it especially suitable for industrial implementation. The rule-based EMS is compatible with PLC logic, with DSE-controlled generator interaction, with feeder enable/disable decisions, and with transparent dashboard-level status interpretation. Chapter 4 is therefore written in a way that highlights this strength: the microgrid is presented not as a black-box optimized system, but as an explainable and implementable supervisory power system.

This introduction also clarifies the methodological style of the chapter. Each major section that follows is expected to do three things. First, it describes the relevant subsystem or architectural layer in technical terms. Second, it justifies why that subsystem is designed in the selected way relative to the project objectives and Chapter 3 requirements. Third, it connects the subsystem to the wider microgrid architecture so that the chapter maintains continuity across

sections. This method ensures that the chapter remains analytical rather than merely descriptive and integrated rather than fragmented.

Within this methodological frame, visual materials such as EPLAN pages, architecture sketches, TIA Portal screenshots, Node-RED flow images, dashboard screenshots, and Simulink structure views will be introduced only at the points where they contribute directly to subsystem explanation and architectural clarity. This is why the present introductory section intentionally remains text-centered. At this stage, the reader needs conceptual orientation before entering figure-heavy subsections. Later sections will place each figure where its meaning is strongest, whether in electrical design, DSE control, PLC logic realization, middleware structure, or HMI layout.

In summary of its purpose, Chapter 4 is the chapter in which the thesis demonstrates that the proposed hybrid microgrid is **designed as a coherent engineering system** rather than merely proposed as an idea. It translates requirement logic into architectural form, transforms operational objectives into subsystem roles, and connects real implementation pathways with simulation-oriented representation under one design narrative. By doing so, it prepares the ground for the later chapters on testing, implementation behavior, and results interpretation, while ensuring that those later chapters are read against a clear and professionally structured design foundation.

4.2 Overall System Design Philosophy

The overall design philosophy of the proposed hybrid microgrid is founded on the principle that an effective rural power system must be designed as an **integrated operational architecture**, not merely as a collection of generation and storage devices. In practical rural microgrids, technical success depends not only on installed capacity, but on the way energy sources, storage resources, control layers, feeder priorities, and supervisory interfaces are coordinated to deliver reliable service under uncertainty. For this reason, the present design adopts a system-oriented philosophy in which electrical structure, control authority, and monitoring visibility are all treated as interdependent elements of one unified engineering solution.

This philosophy is particularly relevant in standalone and weak-grid rural contexts, where the microgrid must continuously reconcile competing objectives such as renewable utilization, critical-load continuity, battery protection, fuel minimization, and operational simplicity. A

design that overemphasizes one objective at the expense of the others may perform well in simulation but fail in practical deployment. For example, a design focused only on maximizing renewable energy could expose critical loads to interruption during cloudy periods, while a design focused only on reliability could overuse diesel generation and undermine the economic and environmental purpose of hybridization. Therefore, the proposed architecture is intentionally balanced: it prioritizes renewable contribution, protects essential loads, constrains battery operation within a defined SOC window, and preserves the diesel generator as a controlled backup rather than a dominant primary source.

A second foundational element of the design philosophy is the preference for **transparent and implementable control** rather than opaque or overly computational energy-management methods. Contemporary microgrid literature recognizes several possible control architectures, including centralized, decentralized, and hierarchical approaches, each with different strengths and limitations related to coordination, scalability, robustness, and communication requirements. In the present project, the design philosophy aligns most closely with a practical supervisory-centralized approach at the EMS level, implemented through a PLC-centered architecture that makes system-wide decisions while leaving low-level device behavior to embedded equipment controllers such as the inverter, BMS, and genset controller. This structure is especially suitable for industrial and rural deployment because it supports explainability, deterministic command execution, and manageable integration with real hardware.

A third design principle is that the microgrid should be organized around **service continuity**, not merely around energy production. This means that the architecture is designed from the demand side as much as from the supply side. The three-feeder structure—F1 critical, F2 important, and F3 non-critical—ensures that the design reflects the social and operational consequences of supply loss, rather than assuming that all kilowatts have equal value. This is consistent with rural microgrid best-practice literature, which emphasizes that system design should be adapted to the social and economic characteristics of the community rather than based on pure technological sizing alone. In this project, feeder priority is therefore not a secondary detail added after source sizing, but a defining architectural principle embedded into the power distribution, control logic, and supervisory behavior of the entire system.

A fourth principle is that the design must remain **implementation-oriented** even when represented in simulation. The design is not developed as a purely mathematical model that later attempts to resemble hardware. Instead, the relationship works in the opposite direction: real

architectural elements such as the PV subsystem, DC bus, battery bank, inverter/charger, AC bus, PLC, DSE, feeder breakers, Node-RED middleware, and HTML dashboard are treated as the primary reference structure, while simulation provides a behavioral representation of their coordinated operation. This is an important philosophical choice because it keeps the project grounded in real engineering practice. The simulation model is not the design itself; it is one of the tools used to test and explain the design.

In summary, the overall system design philosophy of the project can be understood through five interconnected commitments: integrated architecture, renewable-first operation, service-priority protection, transparent industrially implementable control, and real-to-simulation consistency. These commitments shape every design decision in the chapter sections that follow, from source arrangement and feeder mapping to PLC logic, DSE interfacing, software supervision, and simulation abstraction.

4.2.1 Design objectives and architecture logic

The design objectives of the proposed hybrid microgrid arise directly from the operational challenges identified in Chapter 3, particularly the need to provide reliable rural electricity while reducing dependence on diesel fuel, preserving battery health, and protecting critical loads under variable solar conditions. The architecture therefore aims to achieve a combination of technical reliability, operational simplicity, renewable prioritization, and implementation realism. These are not independent goals. In a hybrid microgrid, design quality depends on how effectively the architecture translates such goals into functional pathways for energy flow, control, and supervision.

The first objective is to design a system in which **photovoltaic generation acts as the preferred primary source**. This objective reflects both economic and contextual logic. In rural electrification settings, hybrid microgrids are widely valued because they can reduce exposure to volatile fuel costs and improve sustainability by exploiting local renewable resources whenever available. In the present architecture, this objective is embodied through a topology in which the PV array is connected to the DC side, shares a common DC bus with the battery system, and feeds the inverter/charger that supplies the AC bus and community feeders. This arrangement makes renewable-first dispatch structurally natural rather than procedurally artificial.

The second objective is to ensure that **critical loads remain protected during energy shortage**. This objective is essential because the success of a rural microgrid is not measured

only by total energy delivered, but by its ability to preserve minimum socially essential service when resources are constrained. The feeder-based architecture supports this objective by dividing the community load into three distinct priority classes, which can then be coordinated through EMS logic and switching authority. In architectural terms, this means that the load side is intentionally sectionalized so that continuity decisions can be made at feeder level rather than through undifferentiated bulk load handling.

The third objective is to maintain **battery operation within an intentionally bounded role**. In many hybrid microgrids, the battery is most useful when treated as a short-term balancing and critical-load support resource rather than as an unrestricted long-duration energy source. The present design reflects this logic through a defined usable SOC window, approximately 90% to 30%, and through an EMS sequence that discharges the battery primarily in support of F1 when PV becomes insufficient. This architecture protects the battery from excessive depletion while preserving its operational value as a resilience component.

The fourth objective is to preserve the diesel generator as a **backup continuity source**, not a base-load machine. This is a crucial architectural decision. The diesel generator is included because standalone renewable systems face variability and finite storage constraints, particularly where critical services must remain energized. However, the present architecture avoids giving the generator continuous primacy. Instead, generator use is structurally delayed until PV is inadequate for the critical feeder and battery support has approached its lower operating boundary. This logic allows the generator to play a reliability role without undermining the renewable-centered purpose of the hybrid design.

The fifth objective is to ensure that the control architecture remains **physically realizable on industrial hardware**. This requirement strongly influences the overall architecture logic. Rather than depending on a highly abstract controller, the system is organized around a Siemens S7-1200 PLC as the operational decision executor and a DSE8610 MKII as the generator-side control and monitoring interface. Such an arrangement reflects real engineering practice, where energy-management decisions must ultimately be translated into commands, statuses, feedbacks, and interlocks handled through deterministic industrial pathways. The architecture logic is therefore not only about where power flows, but also about how commands and permissions are concretely realized.

The sixth objective is to support **structured supervision and operator visibility** without compromising the distinction between supervisory software and hard real-time control. This

objective gives the design its digital upper layer. Node-RED is positioned as middleware for API-based supervision and status exchange, while the HTML dashboard serves as the human-facing visualization and limited-control interface. This is a meaningful architectural choice because modern microgrid design increasingly depends on real-time status visibility, alarm handling, and operator interaction, especially in remote systems where local troubleshooting, manual override, and supervisory understanding are operationally important.

Taken together, these objectives produce the architecture logic of the project. The logic can be summarized as follows: PV and battery form the preferred renewable energy core on the DC side; inverter/charger and AC bus provide the power-conversion and feeder-distribution backbone; feeder sectionalization expresses service priority; PLC and DSE provide deterministic control execution; and Node-RED plus dashboard provide supervisory visibility and structured interaction. This is not a generic microgrid arrangement copied without adaptation. It is a design logic tailored to the project's specific goals of renewable-first operation, feeder-priority continuity, industrial implementability, and simulation-backed design analysis.

Table 4-1 Design objectives and architectural implications

Design objective	Architectural implication
Maximize renewable contribution	PV placed as primary generation path on the DC side and dispatched before battery and generator support.Eplan-FIle-7.pdf+1
Protect critical service continuity	Three-feeder architecture with F1 as the protected priority feeder.Eplan-FIle-7.pdf+1
Preserve battery health and usefulness	Battery operated within an SOC window of roughly 90% to 30% and used mainly for critical-load support.Eplan-FIle-7.pdf+1
Minimize diesel dependence while preserving reliability	Diesel integrated as backup continuity source through conditional dispatch logic.Eplan-FIle-7.pdf+1
Enable industrial implementation	PLC-centered control and DSE-based genset interface used for command execution and monitoring.Eplan-FIle-7.pdf+1
Provide supervisory awareness	Node-RED and HTML dashboard used for status visualization, alarms, and structured operator commands.For-NODE-RED-3.txt

As shown in table 4-1, This table is not merely descriptive. It shows that every major design objective is paired with a concrete architectural response, which is exactly what a strong design chapter must demonstrate.

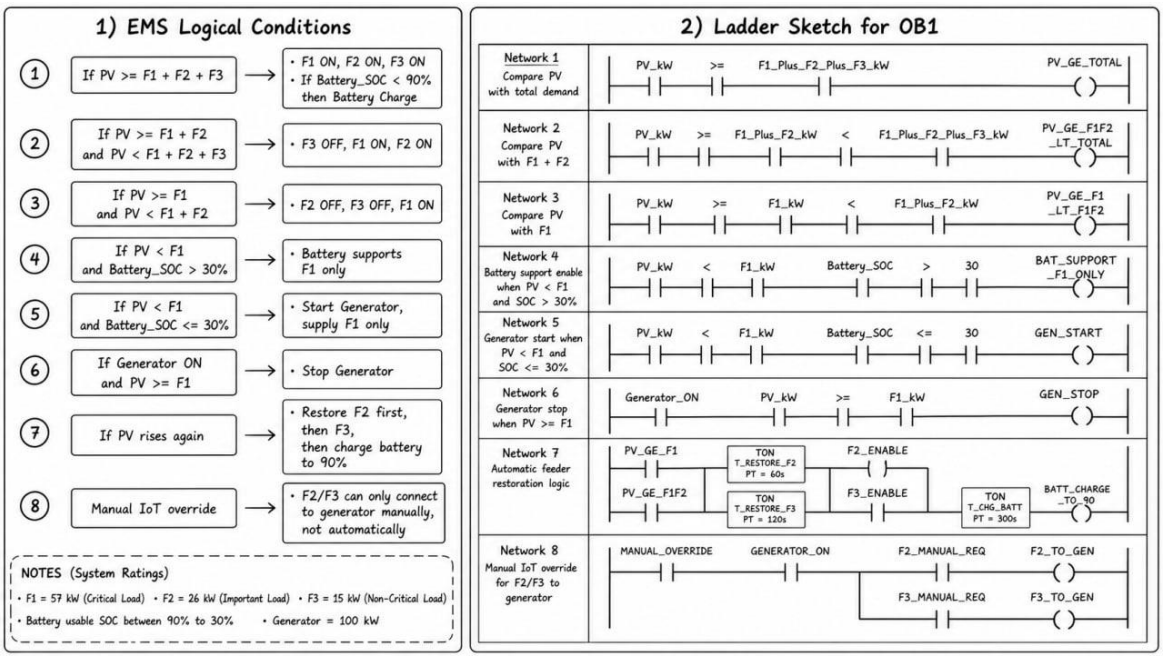


Figure 4-1 Source coordination hierarchy and operating conditions

As shown in figure 4-1, a summary figure is useful because the design philosophy section introduces several objectives at once and they are more readable when connected directly to their architectural implications.

4.2.2 Hybrid source coordination philosophy

The hybrid source coordination philosophy of the proposed microgrid is based on the idea that the value of a hybrid system lies not simply in possessing multiple energy sources, but in using them according to a **deliberate order of operational preference**. In hybrid rural microgrids, the presence of PV, battery storage, and diesel generation does not automatically produce efficiency or resilience. Those outcomes depend on how the EMS determines which source should supply demand, when transitions should occur, and how scarce energy should be allocated among competing loads. The present design therefore adopts a source coordination philosophy in which PV is first priority, battery support is second priority, and diesel generation is third priority, subject always to feeder criticality and battery SOC constraints.

This philosophy reflects both engineering and economic reasoning. Photovoltaic energy is the preferred primary source because its marginal fuel cost is effectively zero and because exploiting local solar availability is central to the purpose of hybridization in fuel-constrained rural settings. The architecture supports this philosophy structurally through the DC-side arrangement of the PV subsystem, which allows solar generation to feed the load path directly

via the inverter/charger and to charge the battery when surplus energy remains available. In coordination terms, this means that PV is not treated as a supplemental source; it is treated as the default daytime supply source whenever irradiance is adequate.

The battery occupies the second level of the coordination hierarchy, but its role is intentionally narrower than that of the PV source. The battery is not designed to replace the generator under all adverse conditions, nor is it intended to support unrestricted deep-discharge cycling for all feeders. Instead, its operational philosophy is to absorb surplus renewable energy when available and to discharge selectively when PV is insufficient, especially to preserve continuity on the critical feeder within the allowed SOC range. This is consistent with practical hybrid microgrid thinking, where storage often functions as a buffer, short-term reserve, and transition-support mechanism rather than as an unlimited energy reservoir.

The diesel generator forms the third layer of the coordination philosophy and functions as the **firm backup source**. It is retained in the architecture because renewable intermittency and storage limitations make fully renewable autonomy difficult when critical services must be maintained under all conditions. However, the present design deliberately restricts the generator's role to conditions where PV can no longer maintain the critical feeder and battery support has approached its lower allowable SOC boundary. By structuring the generator this way, the architecture protects service continuity without allowing the hybrid system to regress into diesel-dominant operation.

An important aspect of this coordination philosophy is that source priority is linked to **load priority**. If source coordination were designed without feeder differentiation, the system might use scarce battery or diesel energy to serve lower-priority demand while critical service remained at risk. The present design avoids that problem by embedding feeder priority into the source allocation sequence. In the MATLAB-based EMS, PV is allocated first to F1, then to F2, then to F3, while battery discharge is used mainly to protect F1 under shortage, and generator intervention is triggered primarily to preserve F1 when battery support becomes insufficient. This means that source coordination and load coordination are not separate logics; they are one integrated philosophy.

This approach also reflects good microgrid control practice from a hierarchical-control perspective. In practical systems, high-level supervisory control is expected to manage active and reactive power needs, source transitions, voltage and frequency continuity, and the smooth handling of changing load and generation conditions. The present design does not implement the

full complexity of tertiary optimization or multi-agent coordination, but it does embody the essential supervisory principle: source usage is decided at the system level based on resource state, feeder criticality, and operating constraints, while device-level embedded controllers handle low-level local functions. This is one reason the chosen coordination philosophy is realistic for PLC-based implementation.

Another important feature of the source coordination philosophy is its temporal behavior. The hybrid architecture is expected to behave differently across different periods of the day. During strong solar hours, PV may cover all active loads and also contribute to charging the battery. During moderate solar periods, PV may sustain only F1 and F2, with F3 curtailed. During weak or absent solar periods, the battery may temporarily sustain F1, and only when the battery reaches its lower operating threshold does the diesel generator become active to preserve critical service. This time-dependent behavior is not an added scenario layer; it is inherent to the source coordination philosophy itself.

A second useful table is appropriate here because the source hierarchy is central to the thesis and can be expressed clearly as a coordination sequence.

Table 4-2 Source coordination hierarchy and operating conditions

Source	Coordination role	Typical operating condition
PV array	Primary source	Supplies loads first whenever solar power is available; surplus can charge the battery.
Battery bank	Secondary support source	Supports critical loads when PV is insufficient and SOC remains above the lower threshold.
Diesel generator	Backup continuity source	Starts when PV cannot maintain F1 and battery SOC reaches the lower operating limit.

As shown in table 4-2, this table is useful because it reinforces a central point of the design philosophy: hybridity is not just multiplicity of sources; it is ordered coordination of source roles.

Finally, the hybrid source coordination philosophy is also what makes the proposed architecture academically defensible. It creates a clear and testable relationship between design intention and later operational outcomes. If the results chapter shows reduced generator hours, sustained critical-load supply, and selective curtailment of low-priority feeders, those outcomes

will be interpretable as the expected behavior of the source coordination philosophy rather than accidental simulation artifacts.

4.2.3 Design alignment with Chapter 3 requirements

A strong design chapter must show that the system architecture is not an isolated creative exercise, but a direct realization of previously defined requirements. The present design aligns closely with the technical, operational, and implementation requirements derived in Chapter 3, especially those related to system scope, source roles, feeder prioritization, EMS logic, control boundaries, and communication structure. This alignment is important because it demonstrates design consistency across the thesis. The architecture presented in Chapter 4 is not merely compatible with the Chapter 3 analysis; it is the physical and logical embodiment of that analysis.

The first major point of alignment concerns the **system boundary**. Chapter 3 defined the physical microgrid boundary as including the PV array, battery bank, inverter/charger, diesel generator, AC and DC buses, and the three prioritized feeders. The present design preserves that same boundary exactly. The architecture is therefore not expanded with unnecessary external systems, nor simplified in a way that would weaken the realism of the project. This boundary consistency is valuable because it ensures that the components discussed in requirements analysis are the same components discussed in design realization.

The second alignment point concerns the **role boundary of sources**. Chapter 3 identified PV as the primary energy source, the battery as a bounded support resource, and the generator as a conditional backup source. The design philosophy of Chapter 4 preserves that ordering in architectural form: the PV and battery are integrated through the DC bus and inverter pathway; the diesel generator is tied into the AC bus through a DSE-mediated backup branch; and source coordination logic is structured to favor renewable supply before generator dispatch. This means the design does not reinterpret source roles arbitrarily; it gives them concrete architectural expression.

The third alignment point concerns the **load structure and feeder priority model**. Chapter 3 formalized the feeder philosophy of F1 critical, F2 medium priority, and F3 low priority, while also mapping those feeders to real community services. Chapter 4 preserves that philosophy as a design-level arrangement in which feeder grouping, control authority, and future supervisory displays all operate around the same priority ordering. This is especially important because feeder logic is central to both the physical design and the control strategy. A misalignment here would

break the coherence of the whole project, but the current architecture maintains the same priority assumptions consistently across EPLAN, EMS scripts, PLC logic, and HMI representation.

The fourth alignment point concerns the **EMS control requirement**. Chapter 3 argued for a rule-based EMS that is transparent, explainable, and suitable for industrial implementation in constrained contexts. The design in Chapter 4 directly reflects that requirement by adopting a PLC-centered supervisory structure, a sequential source-priority logic, SOC-dependent battery/generator transitions, and feeder-based load management. This is a strong alignment because it proves that the EMS philosophy is not confined to the simulation scripts; it shapes the actual system architecture and control pathways.

The fifth alignment point concerns the **real implementation mapping requirement**. Chapter 3 emphasized that the project should not remain purely simulation-based, but should demonstrate plausible mapping to real industrial hardware including PLC and DSE components. Chapter 4 fulfills that requirement through the use of a Siemens S7-1200 as the core decision executor, a DSE8610 MKII as the genset-side control-and-monitoring layer, and realistic separation between digital I/O command paths and Modbus TCP monitoring paths. This is precisely the kind of implementation-oriented realism that Chapter 3 called for.

The sixth alignment point concerns the **communication and supervisory requirement**. Chapter 3 recognized that the project includes a layered communication boundary involving PLC signals, DSE monitoring, middleware, and an operator-facing dashboard. The design chapter preserves this architecture by positioning Node-RED as middleware and the HTML dashboard as the outer human–system interface rather than confusing either one with the hard real-time industrial controller. That boundary discipline is important academically because it prevents overstatement and keeps the thesis aligned with realistic control hierarchies discussed in the literature.

Table 4-3 Design response

Chapter 3 requirement	Chapter 4 design response
Rural standalone hybrid microgrid boundary.	Architecture includes PV, battery, inverter/charger, diesel generator, AC/DC buses, and three feeders.
Renewable-first operation.	PV placed as primary source in source coordination and power-path structure.
Battery used within protected operating limits.	Battery integrated with BMS and SOC-based EMS support logic.
Critical-load priority protection.	F1/F2/F3 feeder arrangement retained in electrical and control architecture.
Rule-based, explainable EMS.	PLC-centered logic and sequential source/feeder control philosophy adopted.
Real hardware mapping.	Siemens S7-1200 and DSE8610 MKII used as implementation-oriented control platform.
Supervisory integration and user visibility.	Node-RED and HTML dashboard included as middleware and HMI layers.

As shown in table 4-3, this table helps demonstrate that the design chapter is structurally accountable to the requirements chapter, which is an important mark of academic rigor in engineering design work.

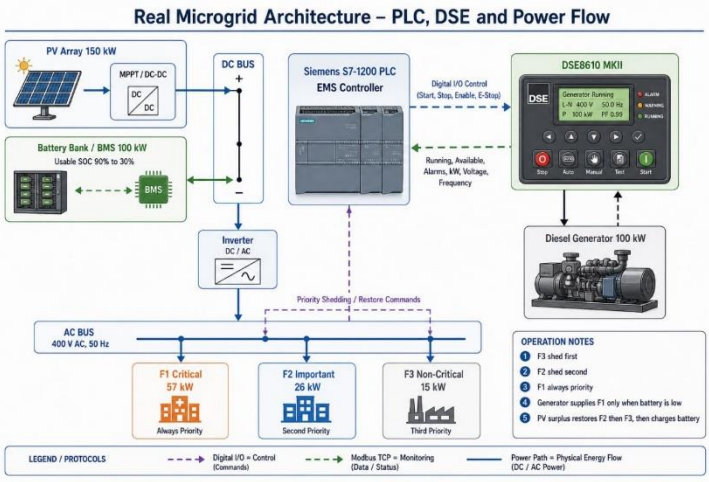


Figure 4-2 Real Microgrid Architecture with PLC, DSE Controller, and Power Flow

As shown in figure 4-2, the overall alignment can be summarized concisely in the previous table.

4.2.4 Real implementation view versus simulation view

One of the most important philosophical distinctions in the thesis is the difference between the **real implementation view** and the **simulation view** of the proposed microgrid. Both are necessary, and both are valid, but they serve different purposes and operate at different levels of abstraction. The design philosophy of the project does not treat these as competing descriptions. Instead, it treats them as complementary perspectives on the same engineered system.

The real implementation view is concerned with how the system would exist as an industrial microgrid architecture. In this view, the project is described through tangible equipment classes and physical interfaces: PV array, MPPT/DC conversion path, DC bus, battery bank and BMS, inverter/charger, AC bus, feeder breakers, diesel generator, DSE controller, PLC I/O, communication links, and supervisory software pathways. It is a hardware-oriented representation. Its purpose is to demonstrate that the project has a realistic structure capable of being understood in the language of industrial automation, power distribution, and field implementation.

By contrast, the simulation view is concerned with how the same system behaves under time-varying operating conditions. In this view, the architecture is translated into variables, blocks, signals, and logical sequences such as PV power input, feeder demand profiles, SOC evolution, active source selection, battery discharge, generator output, and active load after EMS action. It is a behavior-oriented representation. Its purpose is not to reproduce every cable and terminal, but to test whether the control philosophy and architectural logic produce the intended operational outcomes under changing conditions.

The distinction can be framed more precisely. The real implementation view answers questions such as:

- Which device performs generator start/stop control?
- How are feeder priorities embodied in breakers or digital outputs?
- Which communication path is used for monitoring versus command authority?
- How is the PLC positioned relative to the DSE, inverter, BMS, and supervisory software?

The simulation view answers different questions:

- What happens to F2 and F3 when PV falls below certain thresholds?

- When does battery support begin and end relative to SOC constraints?
- Under what condition does the generator begin supporting the critical feeder?
- How do source allocation and feeder shedding affect active load over the day?

Both views are essential because an engineering thesis must show not only that a system is behaviorally effective, but that it is structurally credible. A simulation-only thesis may demonstrate interesting operation without proving implementation realism. A hardware-only thesis may show diagrams and control layers without demonstrating how the system behaves under the variability that motivates hybridization in the first place. The present project deliberately combines both views so that the reader can trust the behavioral findings because they are grounded in an architecture that resembles real engineering practice, and can trust the architecture because its behavior has been explored through simulation.

This relationship is especially visible in the choice of abstraction. In the real implementation view, the DSE8610 MKII is represented as a generator-side controller receiving start/stop and enable-style commands and returning status, alarms, and operational measurements. In the simulation view, the generator is represented more simply through variables such as generator status and power output triggered by defined EMS logic. Likewise, in the real implementation view the feeder system is breaker-based and physically sectionalized, while in the simulation view the same feeders are aggregated power channels F1, F2, and F3 subject to supply or shedding. These differences are not inconsistencies. They are the natural result of representing one system at two different analytical levels.

The same principle applies to the supervisory layer. In the real or implementation-oriented view, Node-RED and the dashboard form a middleware and HMI structure that exchanges JSON status, issues selected commands, and supports visualization and manual override under controlled conditions. In the simulation view, the same logic appears as state variables, mode flags, and signal exchange between blocks or scripts. This parallelism strengthens the design because it shows that the supervisory layer is not decorative. It corresponds to meaningful internal states and decisions within the modeled system.

Table 4-4 Comparison Between Real Implementation View and Simulation View

Aspect	Real implementation view	Simulation view
Main purpose	Show industrially plausible architecture and control integration.	Show behavioral operation of the architecture under varying conditions.
Representation style	Equipment, buses, breakers, PLC, DSE, communication paths.	Signals, logic blocks, feeder power channels, SOC trajectory, generator state.
Feeder representation	Physical feeder grouping and breaker-level control points.	Aggregated F1/F2/F3 demand and supplied power variables.
Generator representation	DSE-controlled diesel generator with command and monitoring pathways.	Conditional generator output triggered by EMS logic.
Control realization	PLC/DSE/Node-RED/dashboard interaction.	Rule-based logic and source/feeder state evolution in MATLAB/Simulink.
Value to the thesis	Demonstrates realism and implementation readiness.	Demonstrates operational logic and scenario behavior.

As shown in table 4-4, for clarity, the relationship can be summarized in a comparison table.

This table is particularly useful because it prevents a common misunderstanding: the simulation is not a substitute for the real architecture, and the real architecture is not a substitute for operational analysis. Each perspective is incomplete on its own. Their strength lies in their coordinated use.

Therefore, the design philosophy adopted in this thesis is neither purely hardware-centric nor purely model-centric. It is a **dual-view engineering philosophy** in which the real implementation view defines the structural credibility of the project, while the simulation view defines the behavioral credibility of the EMS and operating strategy. This dual-view approach is especially appropriate for a bachelor-level hybrid microgrid thesis because it combines practical realism with analytical testability, which is exactly what gives the overall design its academic and engineering strength.

4.3 Electrical Power Architecture

The electrical power architecture of the proposed hybrid microgrid defines the physical structure through which renewable generation, stored energy, backup generation, and prioritized community loads are electrically connected and operationally coordinated. In a well-designed hybrid microgrid, electrical architecture is not a passive background drawing. It is the structural foundation that determines how power can flow, where sources can interact, how faults can be isolated, and how the EMS can translate operational decisions into real feeder-level outcomes. For this reason, the present section examines the proposed system not as a set of independent devices, but as a layered electrical network composed of a PV generation branch, a battery-storage branch, a shared DC platform, a power-conversion stage, a common AC distribution platform, and feeder-level outgoing circuits with protection and switching boundaries.

The architecture is intentionally consistent with the system boundary and source-role definitions established in Chapter 3. The PV subsystem remains the preferred primary source, the battery remains a bounded support resource connected through managed DC interfacing, the inverter/charger remains the conversion bridge between DC and AC domains, and the diesel generator remains a backup source coupled into the AC bus through a dedicated branch. The purpose of this section is therefore twofold: first, to explain the physical electrical arrangement of the microgrid; and second, to show how that arrangement supports the control philosophy of renewable-first operation, feeder-priority protection, and implementation realism.

A key strength of the proposed architecture is that it follows the logic of real hybrid microgrid design practice. Renewable generation and storage are clustered on the DC side where they can interact efficiently through a common bus and conversion stage, while village distribution and diesel backup are organized on the AC side where feeder-level service continuity and source switching are most practically managed. This arrangement is widely consistent with renewable-based microgrid design approaches that separate source characteristics by electrical domain while integrating them through a controllable conversion interface. In the present project, that conversion interface is the inverter/charger, while the two principal network hubs are BUSDC01 on the DC side and BUSAC01 on the AC side.

Another important design feature is that the architecture supports both energy flow and control action without requiring them to be conflated. The electrical topology defines where

power moves, while the protection and switching structure defines where connection and disconnection authority exist. This distinction is especially important in hybrid systems because source availability, source permission, and source connection are not identical conditions. A generator may be healthy but not yet connected; a battery may be present but isolated by contactor status; a feeder may be electrically wired but intentionally shed by breaker command. Therefore, the electrical architecture must be understood as the physical stage on which the EMS operates, not as a static circuit alone.

The subsections that follow develop this architecture progressively. The discussion begins with the PV subsystem as the primary renewable source, then moves to the battery bank and BMS interface, the inverter/charger and shared DC bus, the AC bus and feeder distribution network, the single-line implementation logic, and finally the protection and switching structure that makes the whole power architecture safe, controllable, and operationally credible.

4.3.1 PV subsystem design

The photovoltaic subsystem is the primary renewable energy source in the proposed microgrid and forms the entry point of the electrical architecture on the generation side. According to the EPLAN design, the PV field is rated at 150 kWp and consists of 266 modules arranged as 14 strings of 19 modules each, with an approximate maximum-power string voltage of 789 V, a cold-condition open-circuit voltage of about 952 V, a string current of around 13.25 A, and a total array short-circuit current near 190 A. These values are not merely descriptive nameplate data. They define the electrical envelope within which the DC combiner, surge protection, DC isolation, MPPT stage, and inverter/charger DC interface must all operate safely and effectively.

The string arrangement reflects a realistic high-voltage DC design approach for a utility-scale village microgrid rather than a low-voltage rooftop-style arrangement. By using long series strings and a relatively high string voltage, the architecture reduces current for a given power level and supports efficient transfer into the common DC conversion path. This is consistent with practical PV engineering, where system designers seek a balance between string voltage utilization, converter compatibility, conductor efficiency, and safe operating margin under temperature variation. The EPLAN values are particularly useful because they already imply that temperature-adjusted voltage behavior has been considered at least at a design-check level, especially through the reported cold open-circuit voltage.

Electrically, the PV branch is structured through a DC combiner box, a PV surge protection device, an MPPT charge controller, and a dedicated PV DC breaker before its energy reaches the shared DC bus. This arrangement is good engineering practice in hybrid microgrids because it recognizes that PV generation is neither a simple ideal source nor an always-benign source. Under daylight conditions the array remains electrically active, and it therefore requires DC-specific isolation, overcurrent handling, and surge protection rather than AC-style assumptions. The presence of a combiner and SPD is particularly important in rural outdoor installations, where exposed conductors and array structures may be susceptible to lightning-induced or switching-related surges.

From an architectural perspective, the PV subsystem is designed not only to produce energy, but to serve as the preferred supply path in the overall EMS philosophy. In the simulation logic, PV power is always allocated first, beginning with the critical feeder and then extending to lower-priority loads and battery charging only when surplus exists. The electrical architecture supports this operational priority by placing the PV branch upstream of the shared DC resource interface rather than as a marginal parallel source. In other words, the PV subsystem is structurally central to the design, not merely appended to a diesel-centered network.

Another important aspect of the PV subsystem design is its relationship to conversion capacity. The PV field is rated at 150 kWp, while the inverter/charger unit identified in the EPLAN documentation is associated with a 100 kW AC conversion stage. This suggests that the DC side is intentionally stronger than the immediate AC conversion nameplate, which can be a legitimate design choice in solar-rich systems where controlled clipping during peak irradiance is acceptable in exchange for improved energy harvest during more moderate irradiance periods. In the thesis, this should be interpreted as a deliberate architectural choice rather than a contradiction, provided it is explained as part of renewable-priority design logic and not assumed to imply unrestricted full-power DC-to-AC transfer at every instant.

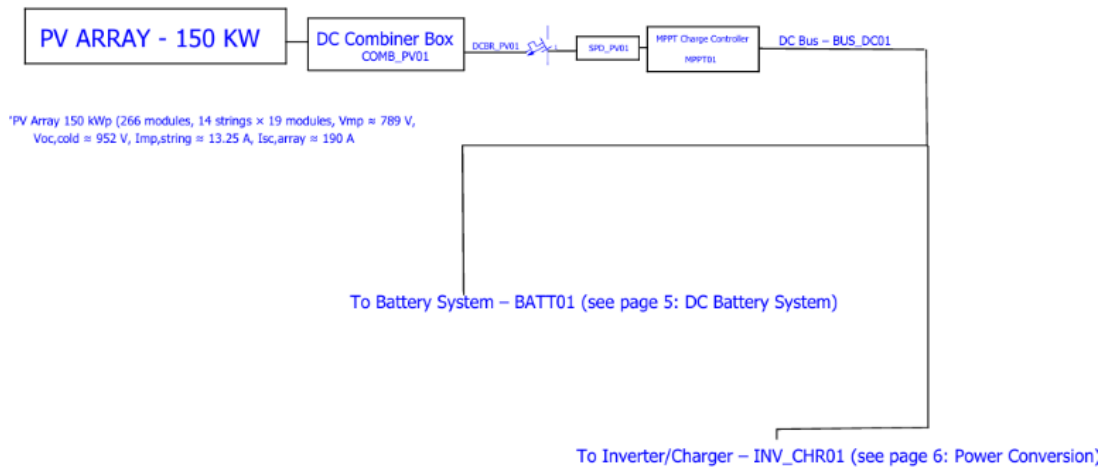


Figure 4-3 PV Array 150 kW Connection Diagram to DC Bus, Battery System, and Inverter/Charge

As shown in figure 4-3, in physical and operational terms, the PV subsystem therefore performs three roles simultaneously. It is the main renewable energy source, the primary daytime supply contributor, and the leading branch of the DC-side generation architecture. This makes it not only the first source in the dispatch order, but also the first structural anchor of the electrical design.

4.3.2 Battery bank and BMS interface

The battery subsystem is the second major branch of the DC-side architecture and serves as the microgrid's short-term storage and continuity-support resource. In the EPLAN documentation, the battery bank is specified as approximately 123 kWh nominal energy, 600 V nominal voltage, 100 Ah capacity, and maximum charge/discharge current around 125 A at roughly 1C, with an intended operating SOC band of 90% to 30%. This configuration places the battery firmly in the high-voltage DC category, which is appropriate for a village-scale hybrid system because it enables efficient power transfer to the inverter/charger while keeping current lower than in low-voltage storage architectures.

The battery branch is not designed as a free-floating energy reservoir. It is explicitly mediated by a Battery Management System and connected to the DC bus through both a battery DC breaker and a main battery contactor. This is a particularly strong feature of the design because it distinguishes between fault isolation and controlled operational connection. The breaker provides protective isolation for fault or maintenance purposes, while the contactor allows the battery to be connected or disconnected under BMS and control-layer authority. In real systems, these two

functions should not be merged conceptually because routine connection control and fault-clearing duties are not the same engineering task.

The BMS interface gives the battery architectural legitimacy as a controllable and protected source rather than an idealized storage block. In practical terms, a battery can only participate responsibly in a microgrid when there is a subsystem capable of supervising safe voltage limits, current limits, connection permissives, and alarm conditions. The present design reflects that principle clearly by positioning the BMS as the authority that qualifies whether the battery is not only physically present, but electrically available for charge or discharge participation. This distinction becomes critical later at the EMS layer, where the difference between battery existence and battery permission directly affects dispatch logic.

From the standpoint of energy-management philosophy, the battery is not intended to replace the generator for long-duration autonomy, nor is it intended to serve all feeders without discrimination. Its role is narrower and more strategic. In the EMS structure, the battery primarily supports F1 when PV becomes insufficient, subject to the defined SOC limits and power constraints. This operational role matches the electrical architecture well. The battery is placed on the DC side beside the PV branch, ready to exchange energy through the common bus and inverter, but it remains bounded by explicit protective and permission mechanisms.

Another important architectural implication of the battery subsystem is that it is both an energy resource and a state indicator. A healthy battery branch with available SOC means the system still retains renewable-linked reserve margin before diesel intervention becomes necessary. A battery at minimum allowed SOC means that the architecture must transition from renewable-buffer mode into backup-preservation mode. Thus, the battery branch is not merely part of the energy balance; it is also part of the system's resilience signaling structure.

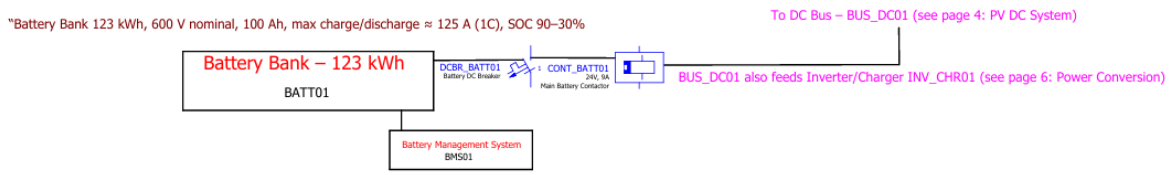


Figure 4-4 Battery Bank 123 kWh Integration with DC Bus and Inverter/Charger

As shown in figure 4-4, in implementation terms, the battery subsystem contributes to realism because it acknowledges that storage integration in real microgrids is inseparable from safety logic, controlled connection, and supervision. The project therefore treats the battery

branch as a managed energy source with its own protection and interface layer, not just as a numerical capacity block imported into simulation.

4.3.3 An Inverter/charger and DC bus integration

The inverter/charger is the central power-conversion element of the proposed microgrid and the device that turns the DC-side resource cluster into a usable AC supply architecture. In the EPLAN package, the inverter/charger unit INVCHR01 receives power from BUSDC01, which aggregates input from the PV subsystem and battery branch, and provides a 400 V, 50 Hz, three-phase AC output to the main AC bus through its dedicated breaker. It also supports reverse energy flow in charging mode, allowing AC energy from the diesel generator to be converted into DC charging energy for the battery. This bidirectional capability is what makes the architecture a genuinely integrated hybrid microgrid rather than a set of loosely adjacent subsystems.

The DC bus itself is one of the most important structural nodes in the entire design. It is the common electrical platform through which PV generation, battery availability, and inverter demand meet. Without a shared DC bus, renewable generation and battery storage would remain functionally fragmented, and the EMS would have far less flexibility in allocating energy or rebuilding storage reserve. In architectural terms, BUSDC01 is the point at which source cooperation becomes physically possible.

The inverter/charger has three explicitly stated operating modes in the EPLAN design: inverting, charging, and standby/idle. These modes reflect real system needs. In inverting mode, the converter supplies AC loads from PV and/or battery energy. In charging mode, it uses AC power from the generator to recharge the battery. In standby or idle mode, it remains available without transferring active power. This mode-based interpretation is useful academically because it shows that the converter is not merely a fixed one-direction interface; it is a controllable operational bridge whose state changes depending on system condition and EMS decisions.

The inverter/charger also has a dual identity in the architecture: it is both a power-path device and an information-path device. Electrically, it is the conversion bridge between DC sources and AC distribution. From the control perspective, it is a monitored and commanded element that exchanges start/stop, mode, voltage, current, and frequency-related signals with the PLC-centered supervisory layer. This makes it one of the key interface devices between energy flow and control logic, even though the thesis does not attempt to model internal converter switching behavior in electromagnetic detail.

This architectural role is highly significant for the whole hybrid design. The loads are on the AC side, the preferred primary source is on the DC side, the support storage is on the DC side, and the backup generator is on the AC side. The inverter/charger is therefore the structural mediator that allows these domains to become one coordinated power system. When PV serves F1, the inverter is active. When the battery supports the AC feeders, the inverter is active. When the generator rebuilds battery reserve through AC-to-DC transfer, the inverter/charger is active again. In this sense, the inverter/charger is not just one component among many; it is the conversion core around which the electrical architecture becomes operationally coherent.

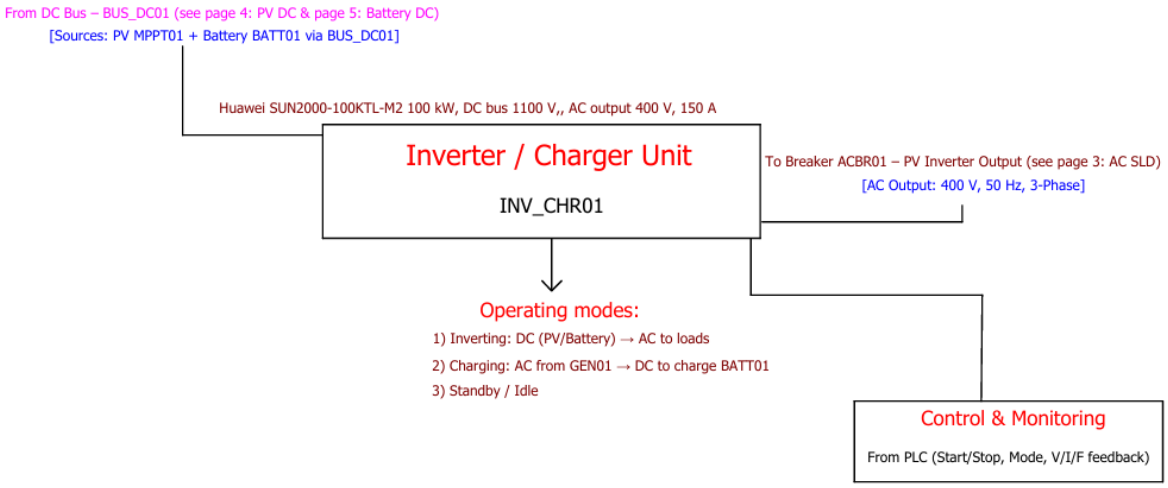


Figure 4-5
Operating Modes

As shown in figure 4-5, the 400 V, 50 Hz, three-phase output rating of the inverter/charger is also essential because it must match the AC bus and downstream feeder architecture. This voltage and frequency consistency is what allows renewable-led operation to appear on the AC side as a stable supply platform for the community loads. Thus, the inverter is where compatibility between DC generation architecture and AC service architecture is physically achieved.

4.3.4 AC bus and feeder distribution architecture

The AC side of the proposed microgrid is organized around the main AC bus BUSAC01, which acts as the common distribution platform for the inverter output, the diesel generator branch, and the three prioritized feeders F1, F2, and F3. This AC bus is not simply a wiring junction. It is the operational point at which source availability becomes community service,

because all feeder energization decisions and most continuity outcomes are realized through the AC distribution structure.

The architecture is intentionally simple but highly functional. BUSAC01 receives supply from the inverter output through ACBR01 and from the diesel generator through ACBR02. From that bus, three dedicated outgoing feeder breakers—ACBR03, ACBR04, and ACBR05—serve F1, F2, and F3 respectively. This separation is one of the strongest design features in the project because it gives physical expression to the feeder-priority philosophy developed earlier in the thesis. Instead of treating community demand as one undivided load block, the architecture creates sectionalized distribution paths that can be independently energized, protected, or shed.

The feeder arrangement is closely aligned with the service-priority model. F1 represents the protected critical feeder, F2 represents medium-priority service, and F3 represents lower-priority or more deferrable service. Because each feeder has its own breaker-defined interface with the AC bus, the EMS can translate abstract priority rules into real switching logic. This is important academically and practically. A priority-based load model is only credible when the physical architecture includes the means to enforce it.

From an electrical standpoint, the AC bus also establishes the common operating platform for voltage and frequency compatibility. Both major AC-side sources—the inverter and the diesel generator—are designed to operate at 400 V, 50 Hz, three-phase. This common rating is essential, because no AC microgrid can function coherently if major sources and outgoing feeders are not aligned to the same bus standard. Thus, the AC bus serves not only as a distribution node, but also as the electrical reference platform that binds source-side and load-side compatibility together.

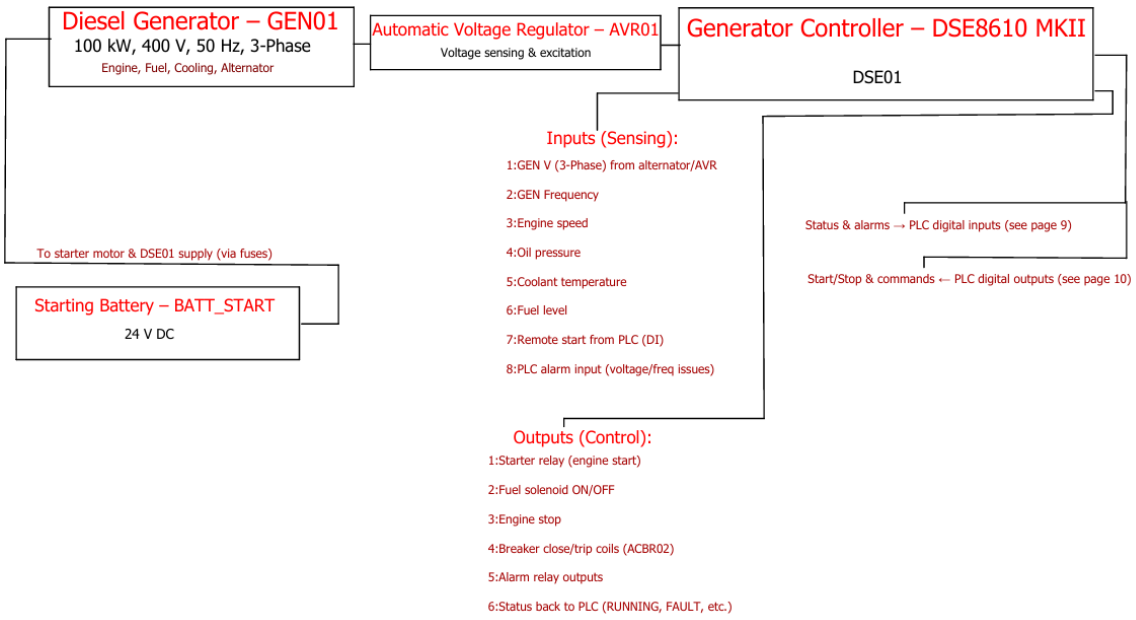


Figure 4-6 Diesel Generator Control and Interface Diagram with DSE8610 MKII

As shown in figure 4-6, the architecture of the AC distribution side also contributes strongly to maintainability and operational realism. Dedicated source breakers and dedicated feeder breakers allow selective isolation during maintenance, faults, or EMS-driven service adjustments. A feeder fault does not automatically imply total source disconnection, and a source-side issue does not have to be interpreted as a distribution-level event. This layered structure is one of the main reasons the design appears implementable as a real microgrid rather than only as a simulation abstraction.

4.3.5 Single-line implementation logic

The single-line implementation logic of the proposed microgrid can be understood as the condensed electrical narrative of how sources, buses, breakers, and feeders relate within one integrated power system. In practical engineering, a single-line diagram is not only a drawing convenience. It is the primary abstraction through which complex three-phase power architecture is made understandable for design review, protection planning, switching philosophy, and control mapping. In the present project, the single-line logic shows clearly that the microgrid is built around two main energy domains: a DC renewable-storage domain and an AC service-distribution domain linked by the inverter/charger.

On the DC side, the single-line logic begins with the PV array, which passes through the combiner, surge protection, MPPT stage, and PV DC breaker before entering BUSDC01. The battery branch connects separately to the same DC bus through its own DC breaker and battery

contactor, under the supervision of the BMS. This arrangement reveals that the DC domain is designed around source aggregation and controlled storage participation rather than around multiple isolated resource loops. It also makes clear that the DC bus is a true functional node rather than a symbolic label.

On the AC side, the single-line logic shows the inverter/charger exporting AC power to BUSAC01 through ACBR01, while the diesel generator connects to the same bus through ACBR02. From BUSAC01, three feeder breakers distribute supply to F1, F2, and F3. In this way, the single-line diagram makes visible the architecture's two key principles at once: source integration on the bus side and service prioritization on the feeder side. That is why the single-line representation is so valuable for the thesis. It visually compresses the whole design philosophy into one implementable electrical structure.

The single-line logic also helps explain the difference between electrical possibility and operational permission. Because branches are breaker-defined, the diagram shows where power could flow if enabled; the EMS then determines when that flow should actually be permitted according to source condition, battery state, generator availability, and feeder priority. This separation between topology and operation is academically important. A microgrid architecture should not be described as though all connected branches are continuously active. Rather, the single-line provides the envelope of possible operation, while the EMS defines the actual operating state within that envelope.

Another strength of the single-line implementation logic is that it supports later discussions of protection, control mapping, and supervisory design. Once the reader understands where each breaker sits and which bus each source or feeder belongs to, later sections on PLC outputs, DSE control, feeder shedding, and dashboard status indicators become easier to interpret. For this reason, the single-line logic serves as a bridge not only within the electrical section itself, but also between electrical design and later control-oriented sections of the chapter.

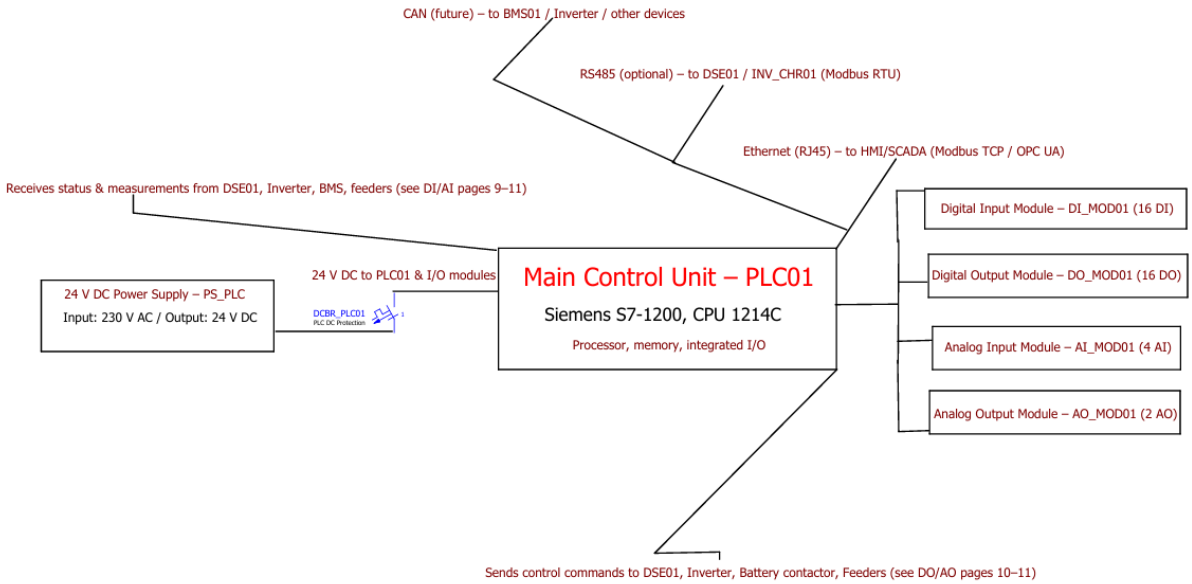


Figure 4-7 Main Control Unit PLC01 Architecture and Communication Interfaces

As shown in figure 4-7, in the present project, the single-line diagram should therefore be described as the **implementation backbone** of the electrical architecture. It condenses the real system into a manageable representation without losing the structural relationships that matter most: source paths, conversion paths, bus nodes, switching points, and feeder-level service segmentation.

4.3.6 Protection and switching structure

The protection and switching structure of the proposed microgrid forms the boundary between ordinary operation and safe controlled operation. In hybrid systems, protection cannot be treated as an afterthought because the architecture contains multiple electrical domains and fault environments: PV-side DC energy, battery-side high-voltage DC energy, bidirectional conversion equipment, AC source coupling, feeder distribution, and low-voltage control power. The present design follows a layered protection philosophy, which is the correct architectural approach for a hybrid microgrid intended to be both operationally flexible and physically safe.

On the PV branch, protection includes the DC combiner environment, a dedicated PV surge protection device, and a PV DC breaker before entry to the shared DC bus. This is significant because PV arrays remain energized during daylight and must therefore be isolated and protected with DC-appropriate devices rather than treated like passive upstream sources. The surge device is especially important because outdoor PV installations are exposed to lightning-induced

disturbances and overvoltage transients that can damage converters and control equipment if not properly suppressed.

On the battery branch, the design uses both a battery DC breaker and a main battery contactor. This is an excellent architectural choice because it creates a clear distinction between hard isolation and controlled operational connection. The breaker provides fault-clearing and maintenance isolation capability, while the contactor enables commanded battery participation under BMS and control-layer supervision. In practice, this means the battery can be safely withdrawn from service for fault or protection reasons without erasing the conceptual distinction between emergency isolation and routine dispatch control.

On the AC side, the design includes dedicated breakers for the inverter output, the diesel generator branch, and each feeder. ACBR01 isolates the inverter output from the AC bus, ACBR02 isolates the generator branch, and ACBR03–05 isolate the individual feeders. These breakers serve a dual purpose. They provide conventional electrical protection, but they also provide the switchable boundaries required for EMS-driven source selection, feeder shedding, and service restoration. This dual role is particularly important in a prioritized-load microgrid, where switching is not only about faults but also about controlled operational adaptation under energy scarcity.

Protection in this project also extends beyond the main power branches to the control infrastructure. The PLC panel includes dedicated DC protection for the 24 V control supply. This detail is easy to overlook, but it is a strong sign of implementation realism. A microgrid can have a sound power architecture and still fail operationally if the control power supply is not properly protected. Loss of protected control power can disable status acquisition, alarm handling, breaker commands, remote start logic, and supervisory coordination. Therefore, control-power protection is part of the overall protection philosophy, not a peripheral accessory.

Another important feature of the switching structure is that it supports selectivity and functional separation. A feeder fault should ideally isolate only the affected feeder. A battery issue should isolate the battery branch without unnecessarily collapsing healthy PV architecture. A generator-side problem should block backup participation without automatically invalidating renewable-led operation. Although the present thesis does not claim to provide a full protection-coordination study, the breaker and contactor arrangement shows that the architecture has been organized with selective isolation in mind.

The protection and switching structure therefore completes the electrical architecture by adding the means for safe isolation, controllable coupling, and operationally meaningful segmentation. Without this layer, the sources and buses described in the previous subsections would remain only electrically connected. With it, they become safely usable, selectively controllable, and suitable for EMS-guided rural microgrid operation.

4.4 Feeder and Load Allocation Design

The feeder and load allocation design of the proposed hybrid microgrid defines how community demand is physically grouped, operationally prioritized, and electrically connected to the AC distribution architecture. In a real microgrid, load prioritization becomes meaningful only when it is translated into identifiable feeder groups, breaker-defined interfaces, and controllable service boundaries. For this reason, the present section does not treat feeders as abstract simulation channels. Instead, it explains them as practical design elements through which the system can preserve critical service, limit nonessential consumption under shortage, and restore broader community supply in a controlled and socially intelligible manner.

This section also marks an important improvement over the conceptual discussion in Chapter 3. Chapter 3 established why the loads were classified into F1, F2, and F3 and why priority-based shedding is justified under renewable variability and limited storage. Chapter 4 now advances that logic into the actual design domain by showing how feeder grouping is embedded in the electrical architecture, how each feeder corresponds to real community functions, how breaker positions create control authority points, and how the whole arrangement supports realistic load shedding and restoration behavior. In other words, the question is no longer only “which loads matter more,” but “how does the real system embody that hierarchy in hardware and operational design.

From a practical microgrid perspective, feeder-based allocation is a strong engineering choice. Conceptual design guidance from resilience-oriented microgrid practice emphasizes that critical loads should be identified, categorized by mission or service importance, and grouped in ways that allow the control system to shed lower-priority feeders first when available generation becomes insufficient. Rural electrification best-practice literature likewise stresses that microgrids must reflect the actual service priorities of the community rather than treating all demand as a uniform block. The present feeder design aligns with both principles. It uses a breaker-based three-feeder architecture to connect criticality, electrical distribution, and EMS action within one consistent operational structure.

Another strength of the feeder allocation design is that it bridges the real implementation view and the simulation view without collapsing them into each other. In the real system, feeders correspond to outgoing protected circuits connected to the main AC bus through ACBR03, ACBR04, and ACBR05. In the simulation, those same feeders appear as aggregated demand channels F1, F2, and F3, each with distinct priority behavior and supply conditions. The design significance lies in the fact that both views preserve the same service logic: F1 is protected first, F2 is conditional, and F3 is most shedable. That consistency makes the project more credible than a purely abstract load model.

The subsections below therefore examine feeder grouping in the real design, the individual logic of F1, F2, and F3, the practical arrangement of shedding and restoration, and the mapping between feeders, breakers, and control points.

4.4.1 Feeder grouping in the real design

In the real design, the feeder structure is defined at the AC distribution level rather than at the individual device level, which is an appropriate and realistic approach for a village-scale hybrid microgrid. The EPLAN package identifies three outgoing feeders from the main AC bus BUSAC01: Feeder F1 as the critical feeder, Feeder F2 as the medium-priority feeder, and Feeder F3 as the low-priority feeder. Each of these feeders is connected through its own dedicated breaker—ACBR03, ACBR04, and ACBR05 respectively—making the grouping electrically explicit and operationally controllable.

This feeder-level grouping is superior to appliance-level prioritization for the present project for several reasons. First, it matches practical rural microgrid implementation, where load management is usually executed by sectional switching or feeder-level control rather than by direct scheduling of every end-use device. Second, it creates clear protection and control boundaries that can be represented consistently in EPLAN, PLC logic, and supervisory software. Third, it keeps the architecture manageable while still preserving the most important differences in service criticality across the community.

The grouping is also socially meaningful, not merely electrically convenient. According to the EPLAN system overview, F1 includes half of the residential houses for basic lighting and fans, the hospital, and market-related critical services; F2 includes the other half of the residential houses associated more with electrical devices and medium-priority domestic use; and F3 includes the water pump, school, and mosque-oriented community loads. This means the feeder design maps directly onto recognizable social functions rather than anonymous demand clusters.

That mapping is important because it allows operators and readers to understand load shedding not as arbitrary numerical curtailment but as staged service allocation aligned with real community needs.

Another design advantage of feeder grouping is that it supports **partial service continuity**. A microgrid does not always need to operate in all-or-nothing mode. Under shortage, it is often better to preserve minimum vital service for all critical functions while temporarily suspending lower-priority uses. The three-feeder structure enables exactly that behavior. It allows the system to continue serving the essential social core of the community even when generation or storage is insufficient to cover the full village demand.vbn.

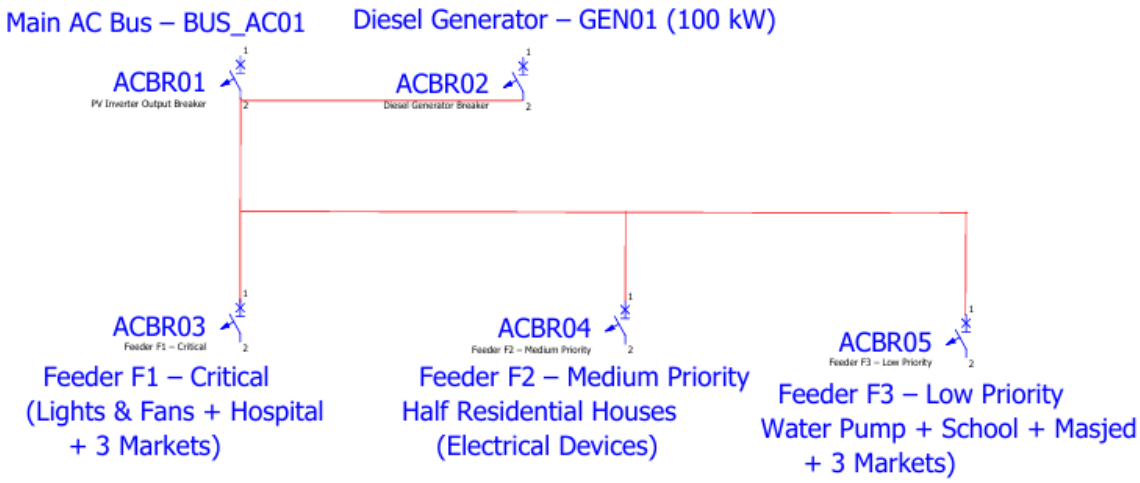


Figure 4-8 Feeder Priority Structure and Load Allocation in the Proposed Microgrid

As shown in figure 4-8, the feeder grouping in the real design should therefore be understood as a foundational architectural choice. It is the structure that transforms the abstract principle of priority into a set of physical, breaker-defined, and controllable outgoing service paths.

4.4.2 F1 critical load feeder design

Feeder F1 is the most important service branch in the proposed microgrid and is designed as the **protected continuity feeder**. In the EPLAN overview, F1 includes half of the residential houses assigned to basic lighting and fans, the hospital, and market-related loads within the critical service cluster. In the simulation representation, F1 is also the largest and most protected feeder, with a maximum modeled value around 57 kW, and it is always supplied first in the EMS sequence using PV, then battery support, and finally generator backup if required. This direct consistency between real feeder identity and simulated priority behavior is one of the strongest features of the design.

The design logic of F1 reflects a real-world resilience principle: not all community demand deserves equal protection during scarcity. Critical loads must be organized so that the microgrid can maintain medical support, minimum residential continuity, and basic social stability even when renewable availability drops or backup must be conserved. The inclusion of the hospital in F1 is therefore fully justified from both technical and social perspectives. Medical loads are among the clearest candidates for protected feeder status because interruptions can affect health services, medicine storage, treatment conditions, and emergency response capability.

The inclusion of only **part** of the residential demand in F1 is also a very strong design decision. It shows that the project distinguishes between baseline household service and full household convenience demand. In practice, a resilient rural microgrid should preserve minimum home functionality—especially lighting and basic ventilation—without assuming that every domestic appliance must remain powered under shortage. This partitioning makes the design more realistic because it avoids the false choice between "all households fully supplied" and "households entirely disconnected." Instead, it supports graded continuity.

From a control standpoint, F1 is the feeder around which the entire source-allocation logic is organized. In the EMS, PV is directed to F1 first; if PV is insufficient, the battery is used within its allowable SOC and power limits; if the battery cannot sustain F1 alone, the generator is started to cover the remaining critical demand. This means that F1 is not just the highest-priority feeder descriptively. It is the **decision anchor** of the whole hybrid microgrid. Every other feeder decision is effectively downstream of whether F1 is secure or not.

The design benefits of this approach are substantial. Electrically, it ensures that the most important branch is physically separable and separately protectable. Operationally, it provides a stable target for source coordination, allowing the EMS to preserve the mission of the microgrid under stress. Socially, it improves fairness and transparency by making it clear that the system is designed first to protect life-supporting and minimum living services rather than less urgent consumption.

In design terms, F1 should therefore be described not simply as the first feeder in the list, but as the **service-preservation backbone** of the distribution system.

4.4.3 F2 important feeder design

Feeder F2 is designed as the **important but conditionally supplied feeder**, positioned between protected critical continuity and fully shedable community demand. In the real design,

F2 consists mainly of the remaining half of the residential houses associated with electrical devices and medium-priority domestic usage. In the simulation logic, F2 is supplied only after F1 has been secured, and it is generally sustained from remaining available PV rather than from aggressive use of battery or generator resources. This makes F2 the architectural expression of partial service extension beyond the critical core.

The existence of F2 is one of the design features that gives the proposed microgrid social realism. If a system had only F1 and F3, it would separate demand into “absolutely critical” and “everything else.” That would be too coarse for many real communities. In practice, there is a large middle zone of loads that are important, useful, and quality-of-life-enhancing, but not as urgent as hospital service or minimum household lighting. F2 captures precisely that middle layer. It allows the microgrid to offer more than survival-level service during favorable conditions while still retaining a controlled mechanism for demand reduction when the system becomes stressed.

This feeder is particularly valuable from a human-centered design perspective because it reflects the internal hierarchy of residential demand. Basic lighting and essential ventilation may belong in F1, but general residential devices and broader domestic electricity use can be grouped separately in F2. This split acknowledges that households themselves contain multiple levels of electrical importance. As a result, the design becomes more nuanced and closer to how actual energy users experience service rather than how bulk-load models flatten that experience into one number.

Operationally, F2 also acts as an indicator of system adequacy. When F2 remains energized for long periods, the system is operating with enough renewable availability and reserve margin to move beyond minimum critical service. When F2 is dropped, the system has entered a conservation mode centered on protecting F1. This makes F2 useful not only as a feeder but also as a visible operating state for the EMS, operator, and reader. The feeder becomes a signal of whether the microgrid is in abundance mode, moderate constraint mode, or deeper shortage mode.

From the design standpoint, F2 provides flexibility without undermining discipline. It can be disconnected earlier than F1 during shortage, reconnected earlier than F3 during recovery, and managed manually for maintenance or operational reasons without collapsing the core service mission of the plant. Thus, F2 functions as the **adaptation layer** of the feeder design: it is the

branch through which the microgrid expands service during good conditions and contracts service gracefully when scarcity increases.

4.4.4 F3 non-critical feeder design

Feeder F3 is the **non-critical or most interruptible feeder** in the proposed allocation structure. In the EPLAN design, F3 includes the water pump, school, and mosque/community-oriented loads, making it the feeder most associated with schedulable, periodic, or comparatively deferrable service functions. In the simulation, F3 is the smallest feeder by maximum demand, around 15 kW, and it is supplied only after both F1 and F2 have been considered. Consequently, it becomes the first feeder to be shed and the last feeder to be restored under constrained operation.

At first glance, the presence of loads such as water pumping or school demand in F3 may appear counterintuitive because these functions are socially important. However, good feeder design depends not only on importance in the abstract, but on **interruptibility and scheduling flexibility**. Water pumping can often be shifted toward solar-rich hours if storage tanks exist; school loads are usually limited to certain daytime windows; and community or mosque-related loads may be event-based rather than continuously critical. This makes them more suitable for the most shedable feeder class than loads that require uninterrupted availability.

In design terms, F3 is not a marginal leftover category. It is an essential part of resilience management. A microgrid without a deliberately interruptible feeder has fewer options for graceful degradation and may be forced to choose between overusing thermal backup and exposing critical service to deeper risk. By contrast, the presence of F3 gives the system a controllable load margin. It allows the microgrid to absorb stress by curtailing lower-priority demand first while maintaining its social core through F1 and, when possible, F2.

The design of F3 also supports a realistic understanding of rural electrification success. A good microgrid is not judged only by whether it can keep lights on in the hospital; it is also judged by whether it can eventually extend service to schools, water infrastructure, and communal functions when resources allow. The fact that F3 is low priority does not mean it is permanently sacrificed. It means that its service is **conditional on adequacy**. Under strong solar conditions, moderate demand, or recovered battery reserve, F3 can and should be restored to broaden the benefits of the system.

Thus, F3 should be described as the **controlled flexibility feeder**. It is the branch that absorbs the first stage of scarcity and receives the last stage of restoration, allowing the entire feeder architecture to remain stable, socially legible, and technically manageable.

4.4.5 Practical shedding and restoration arrangement

The practical shedding and restoration arrangement of the feeder design is based on a straightforward but powerful principle: load service should contract and expand in a predictable order that reflects the criticality structure of the microgrid. In the present design, this means that F3 is shed first, F2 is shed second, and F1 is preserved for as long as any technically feasible source combination remains available. Restoration follows the reverse direction: F1 remains protected, F2 is re-energized before F3, and F3 returns only when higher-priority adequacy has already been secured.

This arrangement is highly consistent with practical resilience-oriented microgrid design. NREL guidance on conceptual microgrid design notes that loads are often prioritized on a feeder-by-feeder basis and shed in reverse order of importance when available generation cannot support all loads. That same guidance also recognizes that operators may alter priorities if needed, but the default design should still enforce a rational and pre-defined hierarchy. The present project reflects exactly this philosophy. It uses a staged feeder structure so that the EMS and breaker logic can implement scarcity response in a deterministic and explainable manner.

An important feature of the present design is that shedding is not understood only as breaker opening after a fault-like emergency. It is also an operational consequence of source-allocation scarcity. In the rule-based EMS, available PV is assigned first to F1, then to F2, and then to F3. If PV cannot even cover F1, battery support is invoked subject to SOC limits, and the generator is started only if the remaining critical demand still cannot be met. Lower-priority feeders are therefore naturally unsupplied when resources are scarce, even before explicit discussion of user-facing “load shedding” terminology. This makes the design more elegant because the feeder architecture and the source hierarchy reinforce each other.

Restoration logic is equally important. A weak design may shed correctly but restore poorly, causing oscillation, unfairness, or confusion. In the present arrangement, restoration is conservative and priority-consistent. F2 is restored when F1 continuity is secure and resources again permit broader service, while F3 returns only after both F1 and F2 conditions are satisfied or adequate margin clearly exists. This avoids the situation where lower-priority demand repeatedly destabilizes higher-priority continuity or causes unnecessary generator cycling.

Practically, this arrangement improves both technical stability and social transparency. Technically, it protects the system against overextension during shortage. Socially, it makes service behavior understandable: the community can see that interruptions are structured and priority-based rather than arbitrary. This is one of the strongest benefits of feeder-level design in rural microgrids, where trust in the fairness of service allocation can be almost as important as total energy availability.

A short table is useful here because the shedding/restoration logic is sequential and easy to summarize visually.

Table 4-5 Feeder State Logic Under Different Operating Conditions

Operating condition	Feeder state logic
Strong adequacy	F1, F2, and F3 may all remain energized if PV and reserve margin are sufficient.
Moderate shortage	F1 protected, F2 may remain on, F3 shed first.
Severe shortage	F1 protected, F2 and F3 disconnected to preserve essential service.
Recovery phase	F2 restored before F3 once higher-priority continuity is secure.

As shown in figure 4-5, this table helps turn the design narrative into an operational picture without repeating the same logic in long prose.

4.4.6 Feeder control points and breaker mapping

The feeder design becomes fully implementable only when each feeder is associated with a clear physical control point, and in the present project that control point is the feeder breaker. According to the AC single-line design, ACBR03 corresponds to F1, ACBR04 corresponds to F2, and ACBR05 corresponds to F3. This mapping is critically important because it provides the physical endpoint through which load-priority logic can be translated into real switching action.

Breaker mapping creates several design benefits at once. First, it allows feeder status to be represented as a real plant condition rather than a purely numerical simulation output. Second, it gives the PLC and supervisory architecture identifiable hardware targets for enable/disable commands, alarms, and state confirmation. Third, it improves commissioning logic, because engineers can test whether feeder commands, interlocks, and restoration rules actually correspond to the intended branch. Without such mapping, the feeder design would remain conceptually elegant but operationally incomplete.

This breaker-based design also aligns well with the control philosophy of the project. The EMS decides which feeders should remain active; the PLC and associated control architecture provide the pathway through which those decisions are enacted; the breaker arrangement supplies the actual physical segmentation needed to make that possible. In this sense, feeder control points are where electrical design meets control execution. They are not merely distribution hardware; they are the actuated boundaries of service continuity.

Another benefit of explicit breaker mapping is that it supports manual override and supervisory interaction in a disciplined way. The broader project architecture already includes Node-RED and dashboard pathways for feeder control, especially for F2 and F3 in manual override cases. Such functions are meaningful only because the feeder architecture is already sectionalized physically. A software toggle has engineering value only when it corresponds to a real or realistically modeled switching boundary. Therefore, breaker mapping is also what makes the feeder design compatible with future operator interaction, maintenance isolation, and supervisory testing scenarios.

The feeder-control-point concept further strengthens the comparison between the real design and the simulation design. In simulation, feeder supply appears as F1supplied, F2supplied, and F3supplied time-series outputs. In the real plant, those states would correspond to feeder breaker status combined with actual feeder energization and power flow measurements. The mapping is therefore not one-to-one in raw signal form, but it is highly consistent in operational meaning. That consistency is exactly what a good implementation-oriented thesis should demonstrate.

The feeder design can be summarized practically in the following table.

Table 4-6 Feeder Classification, Priority Role, Breaker Mapping, and Design Purpose

Feeder	Priority role	Real breaker mapping	Design purpose
F1	Critical feeder	ACBR03.	Preserve hospital and minimum residential continuity first.
F2	Important feeder	ACBR04.	Extend service beyond critical loads while remaining shedable under stress.
F3	Non-critical feeder	ACBR05.	Provide controllable demand margin and first-stage shedding flexibility.

Table 4-6 is worth keeping because it captures the physical and operational meaning of the entire feeder design in one place.

4.5 Diesel Generator and DSE Integration Design

The diesel generator and its DSE8610 MKII controller form the dispatchable backup branch of the proposed hybrid microgrid. In a renewable-led village microgrid, the generator is not included simply because diesel generation is familiar or mechanically convenient. It is included because renewable variability and finite battery reserve create operating periods in which a firm, controllable, and immediately dispatchable source is still required to preserve protected service continuity. For this reason, the generator branch in the present project is designed as a reliability subsystem with clearly bounded authority, dedicated control interfaces, and explicit separation between local generator control and higher-level supervisory decision-making.

This section is especially important because generator integration in hybrid microgrids can be misunderstood in two opposite ways. One mistake is to treat the diesel generator as an always-available dominant source, which undermines the renewable-first philosophy of the system. The opposite mistake is to reduce the generator to a simplistic on/off block with no protection, no local controller, and no realistic command pathway. The present design avoids both problems. It assigns the generator a secondary but indispensable role, and it embeds that role through an industrially credible control architecture centered on the DSE8610 MKII and the Siemens S7-1200 PLC.

The EPLAN design specifies a 100 kW, 400 V, 50 Hz, three-phase diesel generator connected to the main AC bus through breaker ACBR02 and supervised by the DSE8610 MKII generator controller. The controller receives generator electrical and engine-condition signals, executes start and stop functions, supervises protective conditions, and returns running and fault status to the PLC. In parallel, the MATLAB EMS shows that generator support is invoked only after PV is insufficient for the critical feeder and the battery cannot continue support within its allowed SOC limits. Together, these elements show that the generator is neither peripheral nor primary. It is the controlled last-resort guarantor of continuity for the microgrid's most important services.

A further strength of the design is that it distinguishes between **command authority** and **monitoring visibility**. In industrial practice, hard operational commands such as start, stop, and

breaker-related actions are usually kept on deterministic local control paths, while network protocols are used primarily for status monitoring, alarms, metering, and supervisory visibility. The present project follows this principle by treating PLC-to-DSE command exchange as the operational control path and Modbus-based communication as the supervisory monitoring path. This separation improves clarity, safety, and implementation realism.

The following subsections explain the role of the diesel unit in the hybrid architecture, the function of the DSE8610 MKII as the local generator control layer, the distinction between auto and manual modes, the PLC-to-DSE start/stop pathway, protection and feedback signals, the difference between Modbus monitoring and digital I/O control, and finally the logic through which the generator supports critical loads.

4.5.1 Role of the diesel generator in the hybrid architecture

The diesel generator in the proposed microgrid is designed as the **backup and continuity source**, not as the preferred daily source of energy. This distinction is central to the architecture. A hybrid microgrid that aims to maximize renewable utilization must keep PV as the first-choice source and the battery as the short-term balancing resource, while reserving diesel operation for periods when renewable-led supply can no longer maintain the protected load envelope safely. The present project follows this principle clearly: the generator appears only after the renewable-storage combination has reached its practical support limit for F1.

According to the EPLAN single-line architecture, the generator is rated at 100 kW, 400 V, 50 Hz, three-phase, and is connected to the main AC bus through ACBR02. This rating is significant because it is large enough to preserve the critical feeder under severe shortage conditions and therefore provides capacity security, not just emergency symbolism. In village-scale systems, the value of diesel backup lies less in average energy contribution and more in its firmness under low-irradiance, low-SOC, or prolonged deficit conditions.

The project's EMS logic reinforces this architectural role. In the MATLAB rule-based model, PV power is allocated first to F1; if F1 remains partially unsupplied, the battery supports the deficit provided SOC remains above the lower threshold; only when a residual deficit still remains is generator output invoked for the remaining critical load. This sequence demonstrates that the generator is structurally subordinated to renewable-led operation. It does not compete with PV under normal conditions. It protects the system only when renewable-first operation can no longer preserve essential service.

This role has major practical benefits. It reduces fuel dependence, lowers unnecessary generator runtime, and aligns the design with the wider objective of moving rural electrification away from diesel-dominated operation while still preserving reliability. At the same time, it recognizes a hard engineering reality: a rural microgrid that includes a hospital and other critical loads cannot rely solely on uncertain weather and bounded battery reserve. The generator therefore serves as the **firmness source of last resort**. It is deliberately minimized in normal dispatch, but it remains indispensable for protected continuity.

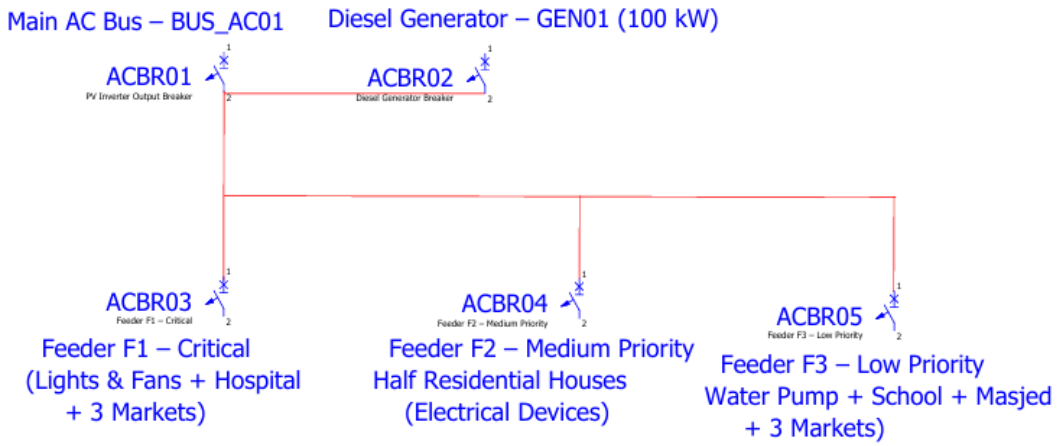


Figure 4-9 Feeder Priority Allocation and Breaker Mapping in the Proposed Microgrid

As shown in figure 4-9, another important architectural point is that the generator can contribute not only directly to AC load supply but also indirectly to system recovery through the inverter/charger charging mode. The EPLAN design states that the inverter/charger can operate in charging mode, converting AC power from GEN01 into DC power to charge the battery. This means the generator is not limited to emergency load support alone. Under appropriate conditions, it can also help rebuild battery reserve after deep discharge conditions, thereby contributing to recovery resilience as well as immediate continuity.

4.5.2 DSE8610 MKII as generator control layer

The DSE8610 MKII is the local supervisory controller of the generator subsystem and is the device that transforms the diesel generator from a mechanical power source into an integrated and controllable microgrid asset. The diesel engine and alternator produce power physically, but the DSE executes the sequence logic, monitors generator and engine conditions, applies protective supervision, and exposes usable status information to the higher-level PLC. For this reason, the DSE should not be described as a minor accessory. It is the **generator control layer** of the architecture.

The EPLAN design identifies the DSE8610 MKII as receiving several sensing inputs, including generator three-phase voltage, generator frequency, engine speed, oil pressure, coolant temperature, fuel level, remote start from the PLC, and alarm-related supervisory inputs. It also provides outputs for starter relay control, fuel solenoid ON/OFF, engine stop, breaker close/trip coils for ACBR02, alarm relay outputs, and status return to the PLC such as running and fault indications. These I/O relationships show clearly that the DSE is both a control device and a protection-aware interface. It does not merely observe the generator; it governs generator execution at the equipment level.

This local-control role is very important in a well-structured microgrid hierarchy. The PLC should decide **when** generator support is needed at the system level, but it should not replace the specialist controller responsible for engine-specific start sequencing, condition supervision, and protective response. The DSE fills exactly that role. It sits below the PLC in the hierarchy, handling generator-local intelligence, while still returning enough information upward to support EMS reasoning and supervisory visibility.

From an academic standpoint, this layered design is a strength because it reflects real industrial practice. Hybrid microgrids rarely rely on a single generic controller to perform every supervisory and equipment-specific function equally well. Instead, a robust architecture distributes responsibility: the PLC coordinates overall system decisions, while the DSE remains the specialist controller of the diesel branch. This division improves maintainability, safety, and role clarity. It also avoids the unrealistic impression that the generator is controlled by a simple high-level Boolean command alone.

The DSE also strengthens implementation realism by acknowledging that the generator may experience alarms, trips, start failures, or other abnormal conditions that matter to system behavior. In a simplistic model, the diesel source is often treated as infinitely healthy once called.

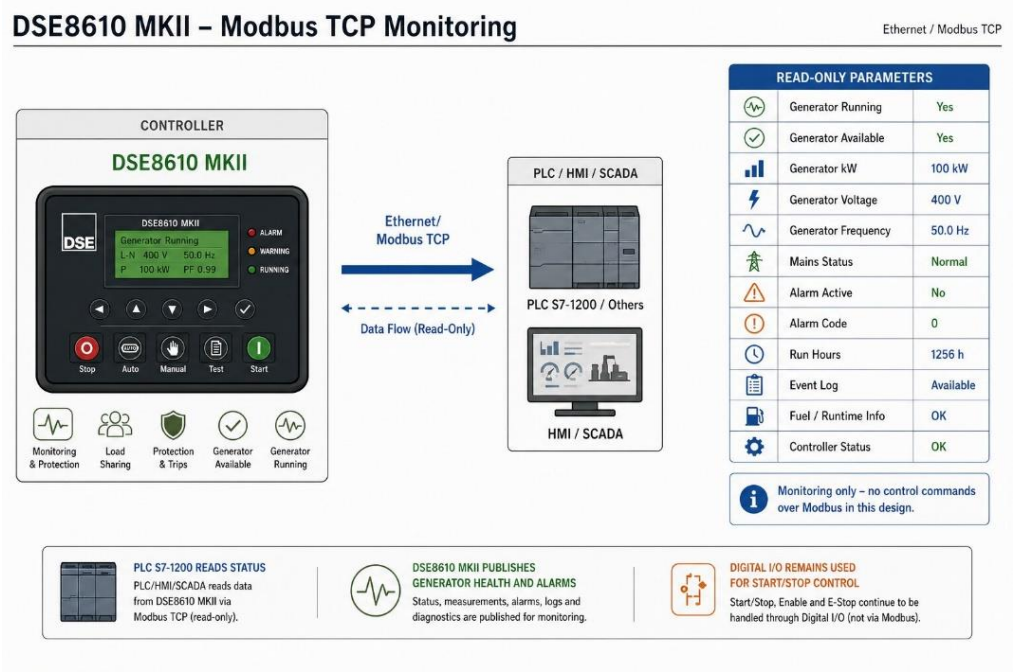


Figure 4-10 DSE8610 MKII Modbus TCP Monitoring and PLC/SCADA Interface

As shown in figure 4-10, in the present design, the DSE introduces the missing realism that generator availability is conditional on engine health, electrical condition, and successful control execution. That is a major advantage for a thesis that aims to connect simulation logic with practical implementation architecture.

4.5.3 Auto and manual operating modes

A robust generator integration design must allow the diesel subsystem to operate in both **automatic** and **manual** modes. This is not a convenience feature only; it is a core requirement for reliability, maintenance, fault response, and supervised operation in real microgrids. Automatic mode supports EMS-driven backup behavior, while manual mode supports testing, maintenance, commissioning, abnormal-condition handling, and operator-directed intervention when needed.

In automatic mode, the generator remains subordinate to system-level logic. The PLC and EMS determine whether renewable and battery resources are sufficient to preserve the required load service, especially F1, and issue a request for generator support only when those resources can no longer sustain protected continuity safely. The DSE then performs the local execution of that request, including start sequencing and operating supervision. This arrangement ensures that automatic generator operation is purposeful and conditional rather than habitual.

Manual mode serves a different purpose. It allows an operator or maintenance engineer to start, stop, inspect, or test the generator outside normal automatic dispatch conditions. In practical systems, manual mode is important during commissioning, troubleshooting, fuel checks, maintenance verification, and contingency operation when the supervisory logic must be temporarily overridden. However, manual mode should not be understood as uncontrolled operation. A good design still preserves protective supervision through the DSE, meaning that even manual requests remain subject to local generator safety conditions.

The distinction between auto and manual mode also helps maintain architectural discipline. Auto mode belongs to the normal operational strategy of the microgrid, where the generator is called only as backup. Manual mode belongs to the operational authority of personnel, where controlled intervention may be required for legitimate reasons. Keeping these modes explicit prevents ambiguity between “the EMS wants the generator” and “the operator wants the generator,” which is important for alarm interpretation, event logging, and safe control responsibility.

In the thesis, this subsection should therefore emphasize that auto/manual distinction is not a minor user-interface detail. It is part of the **control philosophy** of the diesel branch. It ensures that the generator can behave as an autonomous backup source when the system requires it, yet remain accessible for supervised human intervention when operation, testing, or maintenance demands it.

4.5.4 Start/stop command pathway from PLC to DSE

The start/stop pathway between the PLC and the DSE is one of the most important integration links in the whole microgrid architecture because it converts a high-level EMS decision into a real equipment action. In the supervisory layer, the PLC determines whether generator support is required based on source availability, battery condition, and feeder demand priority. But this decision has no physical effect until it is translated into a start or stop request that the DSE can execute at the generator subsystem level.

The EPLAN design explicitly supports this pathway. It states that the DSE receives a remote start input from the PLC and returns status signals such as RUNNING and FAULT back to the PLC. It also identifies DSE outputs associated with starter relay control, fuel solenoid actuation, engine stop, and breaker close/trip control for ACBR02. This means the pathway is not a simplistic one-wire command. It is a multi-stage execution chain: supervisory decision,

generator-start request, local start sequence, running confirmation, and then possible breaker connection to the AC bus once acceptable conditions are satisfied. 1

This command structure is academically valuable because it preserves the distinction between **generator requested**, **generator running**, and **generator connected**. Those three states are not identical. A generator may receive a start command but fail to start. It may start successfully but remain not yet connected to the bus. It may be running but unavailable for load support if a fault or permissive condition prevents breaker closure. Explicitly recognizing these distinctions improves the realism of the design and makes later control discussion stronger.

The command pathway also shows why the PLC does not replace the DSE. The PLC issues a supervisory request, but the DSE remains responsible for the local engine actuation sequence and related protective permissives. This is the correct hierarchical arrangement for an industrial generator subsystem. It avoids overburdening the PLC with equipment-specific details while still keeping overall microgrid coordination centralized.

As shown in figures 4-11 and 4-12, from a thesis-writing perspective, this subsection should make clear that the PLC-to-DSE path is the **control bridge** between energy management logic and backup-source execution. The EMS decides that backup is needed; the PLC formalizes that decision as a command pathway; the DSE executes it; and the returned status signals close the loop by informing the supervisory system whether generator availability has actually been achieved.

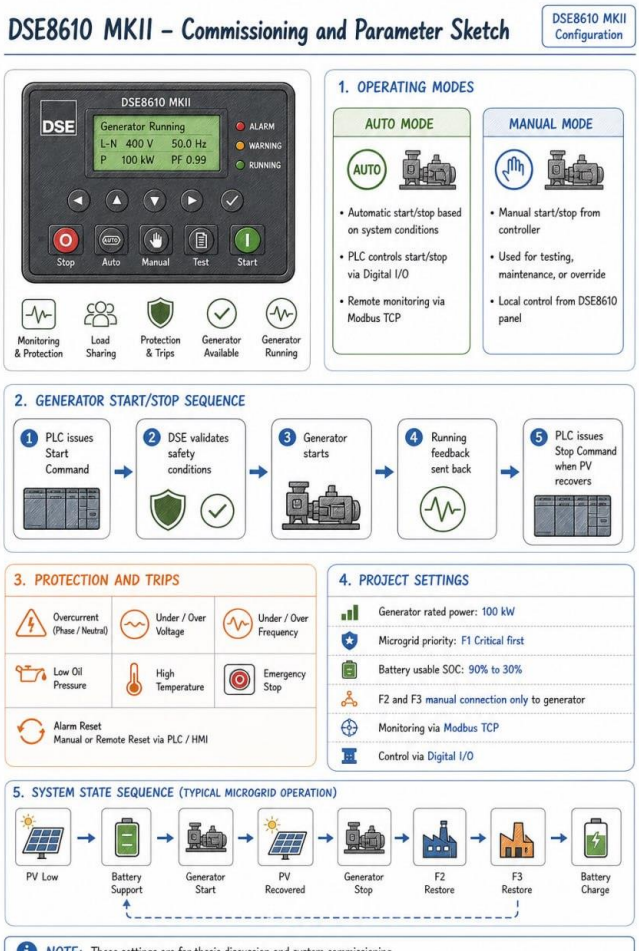


Figure 4-11 DSE8610 MKII Commissioning, Operating Modes, and Sequence

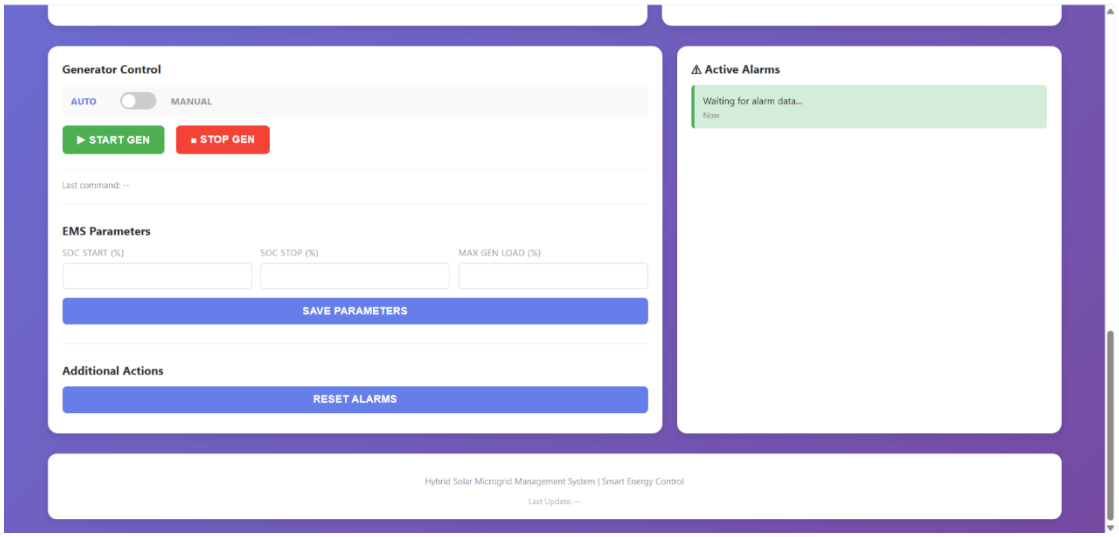


Figure 4-12 Hybrid Solar Microgrid Management Dashboard with Generator Control and Alarm Monitoring

4.5.5 DSE protection, trips, and feedback signals

An essential advantage of integrating the generator through a DSE controller is that the diesel branch is not treated as an ideal source with unconditional availability. The DSE continuously supervises generator electrical quantities and engine-condition signals, allowing the system to distinguish between a healthy, startable, connected backup source and a source that is present physically but unavailable due to alarm or trip condition. This is fundamental for implementation realism.

According to the EPLAN design, the DSE receives sensing inputs including generator voltage, frequency, engine speed, oil pressure, coolant temperature, and fuel level. These measurements are precisely the kinds of signals required for protective supervision and fault qualification in practical generator systems. The controller also returns status and alarm information to the PLC, such as RUNNING and FAULT indications, and provides alarm relay outputs as part of the generator protection architecture. This means that the supervisory layer does not merely know whether the generator was commanded; it can also know whether the generator is healthy, abnormal, or tripped.

From a design perspective, it is helpful to distinguish three levels of abnormality. First, there are warning or alarm conditions that inform the operator of abnormal behavior but may not require immediate shutdown. Second, there are trip-level conditions that make continued operation unsafe or technically unacceptable and therefore require the source to be isolated or blocked. Third, there are feedback conditions that simply confirm generator operating state, such as running, stopped, faulted, or ready. The DSE is the device that organizes these layers for the generator subsystem and translates them into signals usable by the PLC and supervisory logic.

This protection role has a strong impact on EMS credibility. If the generator has faulted, the supervisory controller must not continue reasoning as if backup support remains available. Likewise, if a start request is issued but RUNNING is never confirmed, the system should interpret that event as a failed backup attempt rather than successful transition to diesel support. These distinctions are exactly why generator integration through a DSE is far better than modeling the diesel branch as a pure Boolean source in the real-system design chapter.

In the thesis, the DSE should therefore be described as providing both **protective intelligence** and **state feedback**. Protective intelligence ensures that the generator operates within safe and acceptable conditions. State feedback ensures that the PLC and EMS can reason correctly about actual source readiness rather than assumed readiness. Together, these two functions make the diesel branch a realistic participant in the hybrid control hierarchy.

4.5.6 Modbus TCP monitoring versus digital I/O control

One of the strongest design choices in the project is the clear separation between **digital I/O control** and **Modbus-based monitoring** for generator integration. This distinction is explicitly supported by the project materials, which describe hard operational commands such as generator start/stop and discrete status exchange through PLC–DSE pathways, while using Modbus TCP

primarily to expose generator parameters, alarms, electrical values, and controller status to the supervisory layer. This is a very realistic industrial architecture.

Digital I/O control is the correct pathway for commands that must be deterministic, immediately interpretable, and tightly associated with local permissive logic. Commands such as remote start, stop, and related operational status signals are best handled through explicit hardwired or PLC-mediated control channels rather than relying exclusively on a higher-level network message path. This reduces ambiguity and preserves a clear command authority chain between the PLC and the DSE.

Modbus TCP, by contrast, is better suited to supervisory monitoring and visibility. Through this path, the system can read values such as generator status, kW, voltage, frequency, alarms, run hours, and controller information without making the network itself the sole bearer of critical start/stop authority. This is good engineering practice because it allows rich data access for HMI, Node-RED, and dashboards while keeping the execution of essential commands within a more deterministic control path.

This separation also improves architectural readability in the thesis. It makes clear that the project does not claim the generator is dispatched through a generic software dashboard alone. Instead, the dashboard and network layer are supervisory interfaces; the PLC and DSE remain the operational control core of the generator branch. This distinction is crucial for academic honesty and for practical credibility.

The difference can be summarized clearly in the following table.

Table 4-7

Pathway	Main function	Typical role in this project
Digital I/O	Deterministic operational control.	Start/stop requests, discrete running/fault feedback, command execution path between PLC and DSE.
Modbus TCP	Supervisory monitoring and data visibility.	Read generator electrical values, alarms, runtime information, and controller status for SCADA/dashboard use.

Table 4-7 is worth keeping because it prevents one of the most common misunderstandings in implementation chapters: confusing operator visibility with command authority.

4.5.7 Generator support logic for critical loads

The generator support logic in the proposed microgrid is intentionally narrow and disciplined: the diesel source is used primarily to protect **critical load continuity**, especially F1, when PV and battery support are no longer sufficient. This is not only a simulation preference; it is the operational expression of the whole hybrid architecture. The generator is not meant to normalize fossil-based operation. It is meant to guarantee that protected service does not collapse when renewable-led supply reaches its safe limit.

The MATLAB EMS illustrates this logic very clearly. The algorithm first allocates PV to F1; if F1 still has an unmet portion and battery SOC remains above the lower threshold, battery discharge is used to support that remaining portion; if an F1 deficit still remains after battery support, the generator is dispatched to supply the residual critical demand. Only after the critical feeder is addressed does the system consider lower-priority feeders and surplus-related actions. This means the generator is logically tied to the preservation of essential service rather than to generalized demand following.

This support logic has several important benefits. First, it minimizes unnecessary generator starts and therefore supports fuel-saving and low-runtime objectives. Second, it protects battery health by preventing indefinite deep support below the permitted SOC boundary. Third, it preserves the ethical and social mission of the feeder hierarchy by ensuring that dispatchable fossil backup is spent first on the most important services rather than on lower-priority demand.

The generator branch also strengthens the robustness of the critical-load strategy by providing an answer to long-duration renewable shortage that battery-only logic cannot always sustain. Batteries are excellent for short-term balancing and continuity support, but they remain energy-limited. When the critical feeder includes medical and minimum residential services, the architecture needs a source that is not dependent on solar irradiance and not exhausted after a few constrained hours. The diesel generator provides that missing firmness.

In the thesis, this subsection should therefore emphasize that generator support logic is not simply “start generator when low.” It is a **priority-conditioned reliability rule**: preserve F1 first, use renewable energy first, use the battery within safe limits second, and call the generator only when critical continuity would otherwise be lost. That formulation is academically stronger and much more faithful to the actual philosophy of the project.

4.6 PLC-Centered Control Architecture

The proposed hybrid microgrid is not defined only by its electrical sources and feeders; it is equally defined by the control architecture that decides how those sources and feeders should behave under changing operating conditions. In the present project, that control architecture is intentionally centered on the Siemens S7-1200 PLC, which serves as the main supervisory execution platform between high-level EMS logic and field-level device action. This is a major design decision. It means the project does not stop at describing a desirable energy management strategy in abstract terms. Instead, it asks a more demanding engineering question: through which industrial control platform can the strategy be executed, monitored, interlocked, and made operationally credible in a real rural microgrid installation?

The answer adopted in this work is a **PLC-centered architecture** in which the Siemens S7-1200 acts as the deterministic control core of the real-system layer. The PLC receives digital and analog information from the generator controller, inverter/charger, battery management system, and feeder subsystem; processes those signals using supervisory logic that reflects the EMS philosophy; and issues commands back to the controlled devices through defined digital outputs, analog values, and communication interfaces. This architectural position is crucial because it places the PLC exactly at the boundary where simulation-oriented logic must become industrially executable behavior.

The importance of a PLC-centered approach is supported both by engineering practice and by microgrid control literature. Supervisory microgrid control systems are typically responsible for coordinating distributed energy resources, load groups, and auxiliary equipment in a way that maintains balance, reliability, and operator visibility, while leaving low-level equipment-specific control to local embedded controllers. Case-study work on PLC-based microgrid energy management also shows that Siemens S7-1200-class controllers are practical for renewable-source coordination, HMI interfacing, and supervision of distributed sources because they combine deterministic logic execution, integrated communication capability, and sufficient I/O flexibility for small to medium microgrid applications. The present project aligns closely with that philosophy. The PLC does not replace the DSE, the BMS, or the inverter internal controller; rather, it coordinates them at the system level and enforces the intended operational hierarchy.

Another reason this section matters more than many others in Chapter 4 is that it is where the whole project becomes visibly implementation-oriented. Earlier sections explained the electrical architecture, feeder grouping, and diesel/DSE integration. Those sections established

what the system is composed of and how the major branches are connected. But Chapter 4.6 explains **how the system thinks and acts in real time** at the supervisory level. It is here that feeder priority becomes breaker command logic, generator support becomes start/stop sequencing, battery protection becomes interlocked permission logic, and operator interaction becomes structured manual override instead of uncontrolled interference.

The EPLAN documentation already shows the key elements of this architecture. Page 8 defines the main control unit PLC01 as a Siemens S7-1200 CPU 1214C supplied from a 24 V DC power system, equipped with digital input and output modules, analog input and output modules, and Ethernet connectivity for HMI/SCADA communication using Modbus TCP or OPC UA, with optional RS485/Modbus RTU links to devices such as DSE01 and INVCHR01. The same documentation states that the PLC receives status and measurements from the DSE, inverter, BMS, and feeders, and sends commands to the DSE, inverter, battery contactor, and feeder subsystem. These two statements together establish the S7-1200 not as a peripheral automation component, but as the **central supervisory executor** of the plant.

This architecture also fits the academic and practical scope of the project particularly well. A full commercial microgrid controller platform could offer advanced optimization, cybersecurity layers, black-start orchestration, and wide-area coordination features beyond the scope of a bachelor-level design study. By contrast, the S7-1200 offers an implementation scale that is realistic, understandable, and appropriate for an off-grid village hybrid microgrid in which transparency, maintainability, and deterministic field control are more important than enterprise-level complexity. It therefore provides an excellent compromise between realism and tractability: industrial enough to justify the implementation claim, yet accessible enough to map directly from the EMS logic developed in MATLAB and Stateflow.

The subsections that follow examine the PLC-centered architecture in detail. They justify the selection of the Siemens S7-1200 as the EMS controller, explain the PLC's role in supervisory decision execution, analyze its digital and analog interfacing responsibilities, show how feeder-priority logic is implemented through PLC command structure, discuss manual override and operator intervention, and conclude with the TIA Portal implementation view as the practical software environment in which this architecture would be realized.

4.6.1 Selection of Siemens S7-1200 as EMS controller

The selection of the Siemens S7-1200 as the main EMS execution controller is one of the strongest practical design decisions in the project because it places the supervisory control

philosophy on a platform that is both industrially credible and appropriately scaled to the application. The EPLAN design identifies the selected controller as PLC01, Siemens S7-1200 CPU 1214C, powered from a 24 V DC supply and expanded with digital and analog I/O modules as required by the plant architecture. This is important because the microgrid does not need a generic office computer to execute field control, nor does it require an oversized utility-grade control platform. It needs a rugged, deterministic, modular, and communication-capable controller suited to a small-to-medium hybrid microgrid.

The S7-1200 family is well known for combining compact form, integrated industrial Ethernet capability, and suitability for low- to mid-range automation tasks, which makes it highly appropriate for supervisory control in distributed energy systems. Microgrid case-study literature has also reported the use of the Siemens S7-1214C DC/DC/DC controller for energy-management tasks, noting advantages such as HMI connectivity and compatibility with supervisory control structure. This is directly relevant to the present project because the same class of controller is being used not for millisecond-level inverter switching, but for source coordination, status handling, feeder control, and supervisory logic execution.



Figure 4-13 Siemens S7-1200 PLC Controller for Supervisory Microgrid Control

As shown in figure 4-13, the choice is also justified by the nature of the EMS itself. The proposed EMS is rule-based, transparent, and intended to be implementable on real industrial hardware rather than remaining only an offline optimization algorithm. That means the controller must be able to execute deterministic logic, evaluate interlocks, store states, manage command acknowledgments, and interface with multiple field devices through a mix of hardwired and networked channels. The S7-1200 fits this need well. It supports digital and analog I/O, structured logic organization, Ethernet-based supervisory communication, and integration into the TIA Portal environment where PLC and HMI elements can be engineered coherently.

Another advantage of the S7-1200 in this project is **architectural proportionality**. In rural hybrid microgrids, especially those designed for reliability-sensitive but resource-constrained contexts, the control platform should be maintainable and understandable by engineers who may not have access to large DCS ecosystems or proprietary enterprise microgrid control suites. A compact Siemens PLC offers a level of industrial familiarity and documentation support that is much easier to defend in an implementation-oriented thesis than a purely custom embedded controller or a fragile PC-centric control stack. This is particularly relevant for Yemen-like contexts, where clarity, spare-part practicality, and maintainable industrial logic may be more valuable than algorithmic sophistication alone.

The S7-1200 also supports the layered-control philosophy of the project. It is strong enough to act as the supervisory center of the plant, but it still leaves local equipment-specific control to dedicated devices such as the DSE8610 MKII and the battery management system. This is exactly the right hierarchy. If the PLC were expected to replace every local controller entirely, the architecture would become less realistic. If it were too weak or too peripheral, then the EMS would lose its industrial execution backbone. The selected controller therefore occupies the correct middle position: central for supervisory coordination, but not an unrealistic substitute for specialized device intelligence.

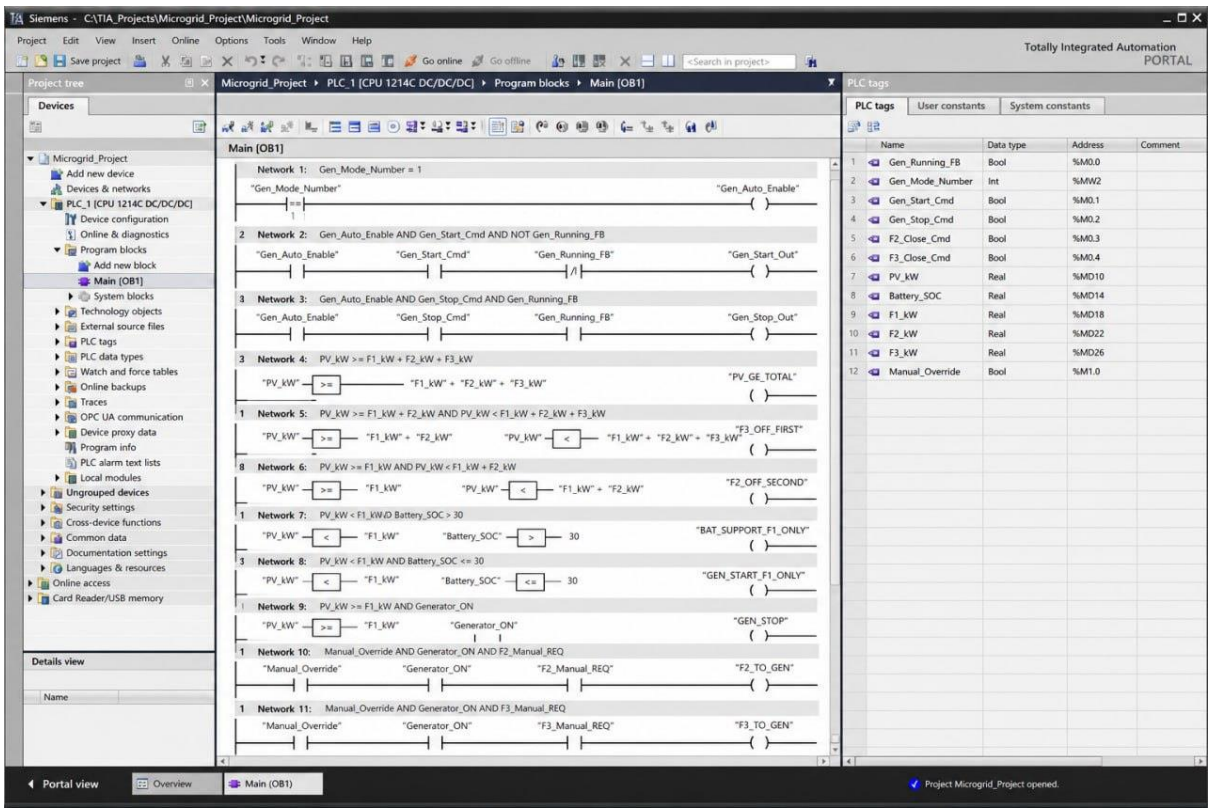


Figure 4-14 TIA Portal Ladder Logic for Microgrid Supervisory Control

As show in figure 4-14, the selection of the S7-1200 should be presented not merely as a component choice but as a **control-architecture decision**. It is the decision to embody the EMS in a deterministic, modular, communication-capable, industrial controller that can bridge simulation logic, field devices, and supervisory visualization. That decision significantly strengthens the credibility of the whole project because it shows how the proposed control philosophy could be executed beyond the simulation screen.

4.6.2 PLC role in supervisory decision execution

The PLC's most important function in the proposed microgrid is not to invent the supervisory decisions conceptually, but to **execute them reliably in plant form**. This distinction is critical. The EMS philosophy may be described in Chapter 3 and behaviorally tested in MATLAB or Stateflow, but none of that becomes industrially meaningful until a real control platform can evaluate conditions, enforce interlocks, send outputs, and confirm statuses in relation to actual equipment. In the present project, the Siemens S7-1200 is that platform.

The EPLAN documentation states that the PLC receives status and measurement signals from DSE01, the inverter, the BMS, and feeder interfaces, and sends control commands back to DSE01, the inverter, the battery contactor, and feeder-related outputs. This means the PLC is the operational point at which multiple subsystems are unified into one coordinated control process. Without the PLC, the generator controller would remain generator-local, the BMS would remain battery-local, and the feeder breakers would remain distribution elements with no coherent supervisory arbitration. The PLC is therefore the **integration node of action** rather than merely a logic repository.

This execution role can be understood as a sequence of supervisory translation. First, field signals are collected: generator running, generator fault, inverter ready, battery low/high/alarm, feeder states, emergency stop, and other status conditions enter the PLC as inputs. Second, the PLC evaluates these signals against the supervisory rule set: whether critical load support is threatened, whether battery discharge is permitted, whether generator start should be requested, whether F2 or F3 should be enabled, and whether manual override is active. Third, it issues the corresponding outputs: start generator, stop generator, close or open battery contactor, select inverter mode, enable or disable feeder control, and publish or expose the resulting state to higher-level supervisory systems.

This is exactly the kind of role assigned to a microgrid controller in broader practice. Supervisory microgrid control is typically responsible for arbitrating between multiple assets,

maintaining stable operation of the local energy island, and coordinating source and load behavior according to operational priorities. The PLC in the present design fulfills this role at a project-appropriate scale. It is not intended as a market optimizer or transmission-level grid coordinator. It is intended as the plant-level supervisory executor that turns the renewable-first, critical-load-protection philosophy into deterministic field behavior.

The PLC-centered execution role also clarifies the relationship between simulation and implementation. In MATLAB, a condition such as “PV is insufficient and SOC is below threshold” can cause a generator variable to rise within one computational step. In the real system, the PLC must interpret the corresponding statuses, issue a remote start to the DSE, wait for running confirmation, manage the associated permissives, and then reflect the true plant state back to the supervisory layer. This gap between idealized decision and physical execution is exactly why the PLC matters so much. It is the device that closes that gap.

Another major strength of the PLC execution role is that it preserves **determinism and responsibility separation**. The dashboard may display states and submit operator requests, and Node-RED may orchestrate higher-level software integration, but the PLC remains the field-authoritative executor of essential supervisory logic. This is a strong architectural choice because it prevents non-real-time software layers from being mistaken for the primary protection or command path. In other words, the PLC is where the system’s operational authority resides, even if visibility and user interaction are distributed across other layers.

The subsection should therefore present the PLC not simply as “the controller,” but as the **supervisory decision executor** of the microgrid. It is the element that binds together information, logic, permissions, commands, and acknowledgments into a coherent operational process. That role is what gives the whole Chapter 4 control architecture its credibility.

4.6.3 Digital inputs, outputs, and control authority

Digital I/O forms the hard backbone of the PLC-centered control architecture because many of the most important microgrid supervisory actions are fundamentally discrete in nature. A generator is either commanded to start or not; a feeder is enabled or shed; a contactor is closed or open; a fault is present or absent; an emergency stop is active or clear. For this reason, the project places major emphasis on digital inputs and outputs as the deterministic pathway through which field conditions enter the PLC and control authority leaves it toward the plant.

The EPLAN design explicitly defines this structure. PLC01 is equipped with a digital input module and a digital output module, and the documentation states that the PLC receives status measurements and binary states from the DSE, inverter, BMS, and feeder subsystem while sending control commands to the DSE, inverter, battery contactor, and feeders. This is architecturally significant because it shows that the PLC is not an observer sitting beside the plant. It is connected directly to the plant's decision-relevant binary states and command-relevant discrete actuators.

From the input side, the most important digital signals are those that define **operational truth**. These include generator running, generator fault, alarm relays, battery low, battery high, battery alarm, inverter ready, inverter fault, feeder fault or status indications, mode selector information, and emergency stop conditions. These inputs matter because supervisory logic is only as good as the reality it perceives. If the PLC does not know whether the generator is actually running or faulted, then its source arbitration becomes unreliable. If it does not know whether a feeder has faulted or whether emergency stop is active, then its command outputs may become unsafe or meaningless.

From the output side, the PLC issues discrete commands that embody the EMS strategy in hardware-executable form. These include generator start and stop commands to the DSE, inverter enable or mode-related commands, battery contactor close/open commands, feeder close/trip or enable/disable commands, and alarm-related control outputs. The significance of these outputs is that they are not merely convenient automations. They are the **actuated expression of supervisory priority**. When the PLC opens F3, that is load-priority policy becoming electrical behavior. When it requests generator start, that is critical-load protection becoming equipment action.

The phrase "control authority" is especially important here. In a properly designed supervisory architecture, not every visible software layer should have equal authority to control equipment. The dashboard may show generator status, but it should not be interpreted as the ultimate control authority merely because it has a button. The real authority lies in the PLC outputs and the interlocks that guard them. Even when a command originates from an operator interface, it should be routed through the PLC-centered logic where permissions, mode status, and safety conditions are checked before any field action is taken. This preserves discipline and prevents the architecture from collapsing into uncontrolled remote toggling.

Another critical advantage of digital I/O is its support for **closed-loop supervision**. A robust command structure requires not just an output, but a confirming input. A generator start output should be followed by running confirmation or fault return. A feeder enable output should be accompanied by feeder status or breaker feedback. A contactor close command should correspond to contactor-closed confirmation where applicable. This closed-loop architecture is one of the strongest aspects of the project because it shows that control is not treated as one-way issuance but as verified actuation within a monitored plant environment.

DI Channel	Tag Name	Signal Source	Description	Priority
DI0	GEN_RUN	DSE8610 status output	Generator is running	High
DI1	GEN_FAULT	DSE8610 fault output	Generator has fault/alarm	High
DI2	BATT_LOW	BMS alert output	Battery SOC < 30% (low)	High
DI3	BATT_HIGH	BMS alert output	Battery SOC > 90% (full)	Medium
DI4	BATT_ALARM	BMS alert output	Battery temperature/voltage issue	High
DI5	PV_AVAILABLE	MPPT status or inverter signal	PV generation available (> 5kW)	Medium
DI6	INV_FAULT	Inverter alarm output	Inverter malfunction	High
DI7	INV_READY	Inverter status output	Inverter ready to invert/charge	Low
DI8	LOAD_F1_FAULT	Feeder F1 breaker status	Critical load feeder tripped	High
DI9	LOAD_F2_FAULT	Feeder F2 breaker status	Medium load feeder tripped	Medium
DI10	LOAD_F3_FAULT	Feeder F3 breaker status	Non-critical load feeder tripped	Low
DI11	EMERGENCY_STOP	Manual E-Stop button	Emergency shutdown signal	Critical
DI12	MODE_SELECTOR	Mode switch (Manual/Auto/Off)	Operator selects control mode	High
DI13	REMOTE_START	Remote start command (optional)	Start generator remotely	Medium
DI14	GRID_AVAILABLE	Optional grid connection sensor	External grid available (if any)	Low
DI15	RESERVED	-	Future expansion	-

Figure 4-15 PLC Digital Input and Output Mapping for Microgrid Control

DO Channel	Tag Name	Command Target	Description	Trigger Condition
DO0	GEN_START_CMD	DSE8610 start input	Command to start generator	SOC < 40% OR PV insufficient
DO1	GEN_STOP_CMD	DSE8610 stop input	Command to stop generator	SOC > 85% AND load < 20kW
DO2	GEN_FUEL_ON	Fuel pump solenoid	Enable fuel flow (via DSE)	Generator starting sequence
DO3	INV_MODE_CHARGE	Inverter mode control	Switch to charging mode	PV > Load AND SOC < 90%
DO4	INV_MODE_INVERT	Inverter mode control	Switch to inverting mode	Load demand requires backup
DO5	INV_ENABLE	Inverter enable signal	Power on inverter	System operational
DO6	BATT_CONT_CLOSE	Battery contactor coil	Connect battery to system	Normal operation mode
DO7	BATT_CONT_OPEN	Battery contactor coil	Isolate battery (safety)	Emergency or maintenance
DO8	FEEDER_F1_CLOSE	Feeder F1 breaker coil	Connect critical loads	System ready + voltage OK
DO9	FEEDER_F1_TRIP	Feeder F1 trip coil	Shed critical loads (if needed)	Emergency load shedding
DO10	FEEDER_F2_CLOSE	Feeder F2 breaker coil	Connect medium loads	After F1 energized
DO11	FEEDER_F2_TRIP	Feeder F2 trip coil	Shed medium loads	Emergency load shedding
DO12	FEEDER_F3_CLOSE	Feeder F3 breaker coil	Connect non-critical loads	After F1 & F2 energized

Figure 4-16 PLC Digital Input and Output Mapping for Microgrid Control

As shown in figures 4-15 and 4-16, in academic terms, digital I/O in this project should be described as the **field-execution channel** of the PLC-centered architecture. It is where high-level EMS reasoning becomes discrete plant behavior, and where discrete plant truth returns to the controller for safe supervisory decision-making. That makes digital I/O far more important than a wiring detail; it is the concrete operational skeleton of the control system.

4.6.4 Analog measurements and setpoint handling

While digital I/O carries the discrete backbone of supervision, analog measurements provide the quantitative awareness that allows the PLC to make informed energy-management decisions rather than simple binary reactions. A hybrid microgrid cannot be controlled properly through on/off signals alone because many of its key decisions depend on magnitudes, thresholds, and continuously varying states such as PV contribution, battery state of charge, voltage, current, temperature, and load level. The project therefore includes analog inputs and outputs as a second major layer of the PLC-centered architecture.

The EPLAN design identifies analog input and analog output modules in the PLC configuration and states that the PLC receives measurements from the DSE, inverter, BMS, and related plant subsystems. The project documentation further explains that analog I/O includes values such as PV voltage/current, battery voltage/current/temperature/SOC, inverter AC measurements, generator AC measurements, and setpoint-oriented analog outputs where required. This is a strong design feature because it ensures that supervisory logic is grounded not only in status flags but also in measurable operating conditions.

Among these analog values, battery SOC is especially critical because it plays a central role in the EMS philosophy. In the simulation, battery support is allowed only within a defined SOC window and generator support becomes necessary when battery reserve can no longer sustain critical demand without violating the lower SOC bound. In the real implementation architecture,

that logic must be associated with a measurement path—whether directly from the BMS or through normalized PLC analog handling—so that the supervisory controller can treat SOC as an authoritative operational variable rather than a conceptual parameter. This is exactly the sort of translation from simulation quantity to field variable that gives the thesis implementation depth.

Analog measurements also support better contextual decisions regarding PV utilization, source adequacy, and feeder behavior. If the PLC can read PV contribution, battery current, generator electrical values, and load-related signals, then it can interpret whether the system is in renewable sufficiency, battery-support mode, generator-support mode, or constraint mode with much greater fidelity than would be possible from binary flags alone. This aligns well with supervisory control practice in microgrids, where the controller must arbitrate among assets based on live operating data rather than purely static mode assumptions.

Setpoint handling is the second half of this subsection and deserves particular attention. A practical EMS is not only a fixed logic sequence; it is also a threshold-driven supervisory system whose behavior depends on configurable values such as SOC start/stop thresholds, feeder enable conditions, generator support limits, or alarm-related supervisory thresholds. The Node-RED and dashboard architecture already reflects this by exposing setpoint fields and control pathways for threshold updates in the supervisory layer. The PLC-centered architecture strengthens this concept by providing a deterministic internal home for those setpoints. In other words, a threshold shown on the dashboard must ultimately correspond to a validated value inside PLC or supervisory logic, not to an isolated UI parameter with no field meaning.

This issue is more important than it may appear. Poor setpoint handling is one of the easiest ways for a control system to become inconsistent. If the dashboard, PLC, and simulation each assume different threshold meanings, then the operator may believe the system is enforcing one policy while the field logic enforces another. A strong PLC-centered design avoids this by treating setpoints as controlled supervisory parameters with clear ownership, validation, and propagation paths across the architecture. That is exactly the sort of implementation discipline expected in a serious engineering design chapter.

Analog I/O therefore serves two strategic purposes in the project. First, it gives the PLC the numerical visibility needed to evaluate system condition meaningfully. Second, it anchors supervisory thresholds and operational setpoints in an industrial control environment rather than leaving them as abstract simulation constants. This makes the PLC not only a discrete-command

executor, but also a quantitative supervisory processor that interprets and applies live plant measurements.

4.6.5 PLC implementation of feeder priority logic

One of the most important reasons the PLC-centered control architecture is so central to the thesis is that it is the platform through which **feeder priority logic becomes operationally enforceable**. Chapter 3 explained why loads are divided into F1, F2, and F3. Section 4.4 explained how those feeders are physically grouped and connected through dedicated breakers. But Section 4.6.5 is where that philosophy becomes actual control law in the real-system architecture. Without PLC implementation, feeder priority would remain a descriptive intention rather than a dependable plant behavior.

The feeder logic in the project is based on a simple but strong hierarchical principle: F1 is the protected critical feeder, F2 is conditionally supplied, and F3 is the most interruptible feeder. In the MATLAB EMS, this logic appears as sequential source allocation: PV is directed first to F1, then to F2, then to F3; if F1 cannot be maintained from PV, the battery supports it within SOC limits; if a critical deficit remains, the generator supports the residual F1 demand. In the real-system architecture, the PLC is the device that must mirror this logic through its feeder command outputs, status interpretation, and interlocks.

Practically, PLC implementation of feeder priority means that feeder states are not left to passive bus availability alone. Instead, the controller evaluates the operating condition of the plant and determines which feeder outputs should remain enabled. Under adequate renewable conditions, F1, F2, and F3 may all remain active. Under moderate shortage, F3 becomes the first candidate for disconnection while F1 remains protected and F2 may remain active conditionally. Under more severe shortage, the PLC can enforce a stronger conservation mode in which F1 alone is preserved while F2 and F3 are both disabled or shed. This is exactly the kind of staged supervisory behavior that a PLC is well suited to implement because it depends on explicit conditions, priorities, and discrete actions.

The PLC also gives feeder priority logic **statefulness**, which is an important improvement over a purely descriptive model. In real operation, feeder control is not only about instantaneous comparison of numbers. It may also involve mode memory, manual override status, alarm conditions, command acknowledgment, anti-chatter logic, or restoration sequencing discipline. For example, a feeder should not oscillate rapidly between on and off states because a measured value fluctuates around a threshold. Nor should F3 be restored before F2 if the logic is intended

to preserve a consistent service hierarchy. By implementing feeder logic inside a structured PLC program, the design can incorporate such operational discipline far more credibly than a simplistic open-loop explanation would allow.

This is also the point where breaker mapping matters deeply. F1, F2, and F3 are associated with feeder breakers ACBR03, ACBR04, and ACBR05 respectively in the AC architecture. The PLC does not "control a social category" in the abstract; it controls the physical outputs and command pathways associated with those feeder branches. Thus, feeder priority logic in the PLC must be understood as a mapping from service hierarchy to actual field-executable branch control. That mapping is what makes the design implementation-oriented rather than purely conceptual.

Another benefit of PLC-based feeder priority is that it enables meaningful diagnostic and supervisory interpretation. If the PLC knows which feeders are intentionally disabled, which are unavailable due to fault, and which are active due to adequate supply, then the supervisory layer can present a much more truthful picture to the operator. A feeder status on the dashboard can then reflect actual control logic state rather than just raw bus availability or cosmetic software labels. This matters because a key strength of the project is that it aims to connect energy logic, industrial control, and operator visibility into one coherent architecture.

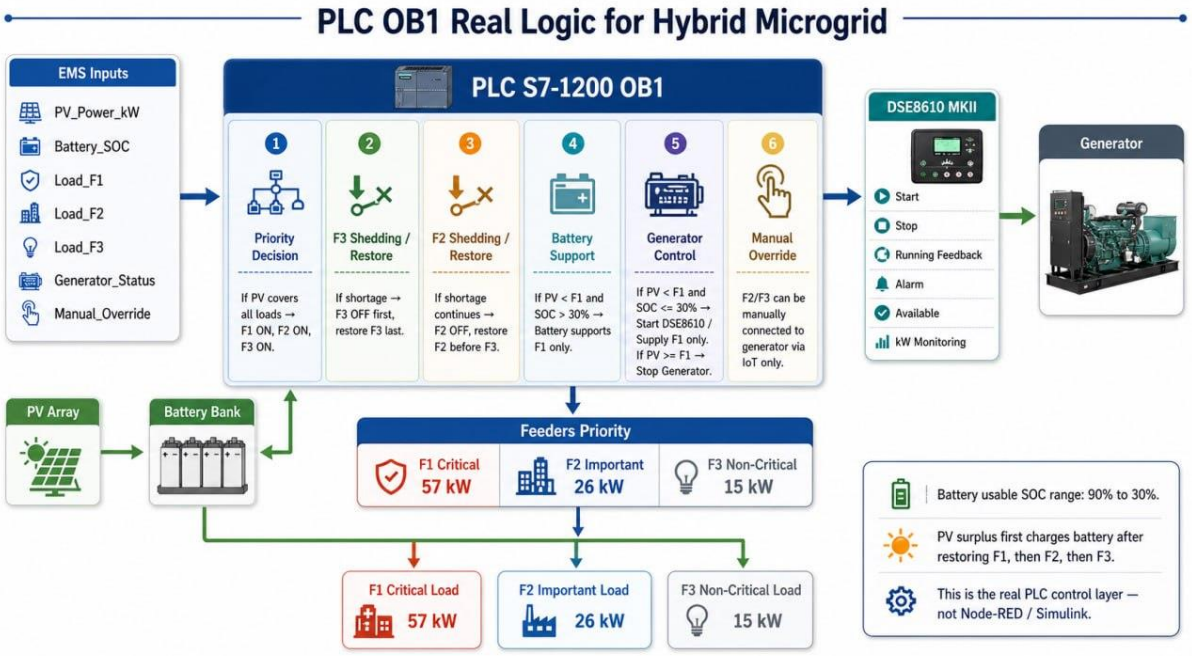


Figure 4-17 PLC-Based Real Logic for Hybrid Microgrid Operation and Manual Override

As shown in figure 4-17, from an academic standpoint, the feeder-priority subsection should stress that the PLC does more than automate breaker outputs. It **embodies the service ethics of**

the microgrid in deterministic logic. The controller decides, in real plant form, that medical and minimum residential continuity outrank discretionary or deferrable demand. That is a very strong engineering idea, and it gives the PLC a central normative as well as technical role in the design.

4.6.6 Manual override and operator intervention logic

A mature PLC-centered architecture should not assume that fully automatic operation is always sufficient. Real microgrids require carefully bounded mechanisms through which operators can intervene for maintenance, testing, troubleshooting, abnormal events, commissioning, or deliberate service management. The present project recognizes this by incorporating manual override and operator intervention as structured parts of the supervisory architecture rather than as ad hoc bypasses of it.

This distinction is crucial. A weak design treats manual override as if it were simply a button that supersedes all logic. A strong design treats manual override as a **controlled supervisory mode** with explicit rules, permissions, and feedback. The project materials already support this interpretation. The dashboard and Node-RED API architecture allow selected commands such as generator start/stop, feeder toggling, setpoint updates, and alarm reset, while the underlying control philosophy keeps the PLC as the authoritative execution layer. This means operator intervention does not bypass the architecture; it enters the architecture through mediated pathways and is then subject to the same kind of mode awareness and supervisory discipline that govern automatic operation.

Manual override is particularly relevant for feeder control and generator operation. F2 and F3 are natural candidates for manual supervisory intervention because they are not the highest-protection feeder and may need to be temporarily enabled or disabled for testing, maintenance, or scenario management. The generator may also require manual start/stop for inspection, commissioning, or controlled troubleshooting. However, manual intervention in a well-designed PLC-centered system should still respect higher-level constraints. For example, an operator request should not nullify emergency stop, ignore a generator fault, or energize a feeder in contradiction with hard safety interlocks. This is why the PLC is so important: it ensures that operator authority is structured rather than absolute.

Another strong aspect of manual override in this project is its relationship to mode handling. The system can distinguish between automatic supervisory operation and manual supervisory intervention, which is essential for correct interpretation of plant behavior, command origin, and event responsibility. If a feeder is off, the operator should know whether that happened because

the EMS shed it automatically, because a manual override disabled it, or because a fault forced it offline. A PLC-centered design can preserve this distinction through explicit mode bits, command-source logic, and status reporting. This is academically valuable because it shows that the architecture is not merely functional, but operationally intelligible.

The operator intervention logic also supports commissioning and debugging. During staged testing, engineers often need to verify individual command paths, confirm that signals reach the correct outputs, test alarm behavior, or exercise the generator without waiting for full automatic conditions to arise. A PLC that includes bounded manual functions makes this practical without reducing the architecture to uncontrolled direct wiring actions. This is exactly the right level of realism for an implementation-oriented thesis: enough manual authority to support real engineering work, but not so much that the EMS becomes meaningless.

In the thesis, this subsection should therefore emphasize that manual override is not the opposite of automation. It is part of a **well-governed supervisory system**. The PLC remains central because it validates, mediates, and records the intervention pathway, ensuring that operator actions remain compatible with the microgrid's control philosophy and safety boundaries.

4.6.7 TIA Portal implementation view

The TIA Portal implementation view is the software-engineering perspective through which the PLC-centered control architecture becomes programmatically organized, documented, and deployable. This subsection is especially important because it prevents the PLC discussion from remaining only at the conceptual hardware level. Once the Siemens S7-1200 is selected as the controller, the next academic question is how its logic would be structured in practice within the Siemens engineering environment. The answer is through TIA Portal, which is the integrated programming and configuration environment for S7-1200-class controllers and related HMI systems.

In the context of the present project, the TIA Portal view should not be exaggerated as if a full certified field program had already been commissioned. The project documentation makes a very important boundary statement: the PLC implementation is represented through logic sketches, tag mappings, and control-flow diagrams rather than a fully exported and validated industrial software package. This is academically the correct position. It avoids overstating the implementation status while still showing a realistic pathway through which the EMS could be encoded in a genuine industrial engineering environment.

Within TIA Portal, the microgrid logic would naturally be organized into a combination of hardware configuration, tag tables, program blocks, and HMI or communication mappings. The hardware view would define the CPU, digital modules, analog modules, addressing, and communication interfaces corresponding to the architecture already shown in EPLAN. The tag tables would give semantic meaning to the physical and logical variables of the plant—generator running, generator fault, inverter ready, battery alarm, F2 enable, F3 enable, SOC, PV power, setpoint thresholds, and supervisory mode bits. The program blocks would then implement interlocks, source-request logic, feeder-priority rules, manual override behavior, alarm handling, and communication state conditioning.

This view is more than a software convenience. It is the means through which the project's abstraction layers are reconciled. MATLAB/Stateflow expresses the EMS as behavioral logic and operating modes. EPLAN expresses the plant as electrical devices, breakers, and signal interfaces. TIA Portal would sit between them as the structured industrial logic environment in which those devices and behaviors are joined through executable control blocks. That is why this subsection matters. It is not merely "about programming"; it is about the engineering language through which the architecture becomes operable.

The implementation view is also where one sees the value of disciplined tag mapping. The project already includes the foundations of a consistent semantic model across software and hardware layers, with variables such as mode, SOC, generator status, feeder states, active source, and threshold parameters appearing in the supervisory materials. In TIA Portal, those meanings would need to correspond to explicit PLC tags and function-block conditions so that a dashboard value, a PLC memory bit, and a field signal all refer to the same operational truth. This is one of the strongest integrative ideas in the whole project because it shows how simulation, PLC logic, and HMI-style supervision could converge on a common behavior model.

Another strength of the TIA Portal perspective is that it implicitly supports modularization. The system can be viewed not as one monolithic ladder diagram but as a structured program with functional partitions: generator control interface block, feeder management block, battery permission and contactor block, inverter mode coordination block, alarm block, and supervisory mode/override block. This modularity is academically valuable because it mirrors good industrial programming practice and makes the architecture easier to understand, test, and extend. It also fits the layered nature of the microgrid itself, where different subsystems need coordinated but distinct control responsibilities.

At the same time, the thesis should remain careful and precise about scope. The TIA Portal implementation view in this project is **implementation-oriented, not commissioning-complete**. It shows how the logic would be organized, how tags and control pathways would be represented, and how the S7-1200 platform supports the proposed EMS architecture. It does not claim full site-tested parameterization, certified safety validation, or final field acceptance documents. Stating that boundary clearly is not a weakness. It is a mark of strong academic discipline.

Therefore, this subsection should conclude by presenting TIA Portal as the natural software realization environment of the PLC-centered architecture: the place where EPLAN signal definitions, EMS supervisory logic, device interfaces, interlocks, tags, and operator-facing control concepts would be assembled into one coherent industrial control project. That framing is accurate, professional, and fully aligned with the scope of the thesis.

4.7 EMS Design and Control Logic Realization

The proposed microgrid is not merely a collection of electrical components connected through buses and breakers; it is a deliberately governed energy system whose behavior depends on a clear and enforceable Energy Management System (EMS). In hybrid village microgrids, especially those built around renewable generation, limited storage, and backup diesel support, the quality of the project depends as much on the control logic as on the electrical hardware itself. A poorly structured EMS can waste solar energy, overuse the generator, violate battery limits, or shed loads inconsistently, even if the physical architecture is well designed. For this reason, the present section treats EMS realization as a core design layer rather than a software afterthought.

The EMS adopted in this project is intentionally **rule-based, priority-aware, and implementation-oriented**. This means that the controller does not rely on opaque optimization routines or computationally intensive predictive solvers to make real-time decisions. Instead, it applies a transparent sequence of supervisory rules based on source availability, battery state of charge, load priority, and protection-oriented operating thresholds. This is a major strength in the context of the project because the study seeks not only theoretical performance, but also explainability, PLC compatibility, and practical suitability for a rural off-grid microgrid in which engineers and operators must understand why the system behaves as it does.

The underlying logic is already evident in the MATLAB EMS script used in the project. The code clips the available PV and feeder demands to the design limits, defines feeder priority levels, tracks battery SOC over time, allocates PV first to F1 and then to lower-priority feeders, uses the battery to preserve F1 when PV is inadequate, starts the generator only when critical demand remains uncovered, and updates supplied feeder power and battery state accordingly. This behavior is not isolated to simulation only. The same project materials show that real implementation mapping exists through the PLC, DSE, feeder breakers, battery contactor, and supervisory software pathways. Therefore, the EMS in this chapter should be described as a **realization of one coherent control philosophy across both simulation and implementation layers**.

From the broader literature perspective, this design choice is well justified. Reviews of microgrid EMS strategies consistently identify rule-based control as one of the most practical and interpretable methods for hybrid renewable systems, especially where computational simplicity, reliability, and ease of implementation are valued alongside performance. Recent work on rule-based microgrid EMS also shows that such schemes are especially suitable when decision logic must be tied directly to source hierarchy, battery SOC thresholds, and load-priority structure. In addition, feeder-priority-based shedding and restoration approaches are widely recognized as practical methods for maintaining service continuity under constrained generation. The present project aligns strongly with these principles.

This section is particularly important because it brings together several strands developed earlier in the chapter. Section 4.3 established the physical electrical architecture; Section 4.4 explained feeder allocation and service priority; Section 4.5 defined the role of diesel and the DSE control layer; and Section 4.6 positioned the PLC as the supervisory execution core. Section 4.7 now explains the logic that binds all of those elements together. It shows how the plant decides which source should act first, when the battery may charge or discharge, under what conditions the generator is started, how feeders are shed and restored, and how manual intervention is reconciled with automatic logic.

The following subsections therefore present the EMS design philosophy, source priority order, SOC-based battery logic, generator dispatch logic, feeder shedding and restoration behavior, manual override conditions, and finally the consistency between the EMS theory and its implementation mapping across MATLAB, PLC, DSE, and supervisory layers.

4.7.1 Rule-based EMS design philosophy

The EMS in the proposed microgrid is built on a **rule-based supervisory design philosophy**, meaning that system behavior is governed by explicit operating rules derived from source hierarchy, battery condition, and feeder priority rather than from continuously re-optimized mathematical dispatch. This is an important methodological choice. In hybrid PV-battery-diesel systems, the objective is often not only to minimize cost or maximize renewable share in an abstract sense, but to do so using a control structure that is understandable, robust, and deployable on industrial hardware. Rule-based EMS schemes are widely used for this reason: they simplify decision logic into interpretable conditions and actions while still achieving reliable coordination among sources and loads.

In the present project, the rule-based philosophy is particularly appropriate because the microgrid serves a rural standalone application with clearly differentiated load priorities and a strong requirement for implementation realism. A black-box optimization controller might produce mathematically interesting dispatch behavior, but it would be harder to map transparently onto PLC logic, DSE interfaces, feeder control signals, and operator understanding. By contrast, a rule-based controller can be expressed in a sequence such as: use PV first; support F1 with the battery within SOC limits; start the generator only when critical load continuity cannot otherwise be maintained; and shed lower-priority feeders before compromising critical service. This structure is both technically meaningful and operationally explainable.

The MATLAB EMS script illustrates this philosophy clearly. It treats the system as a time-stepped supervisory process in which each hour the controller checks available PV, feeder demands, current SOC, and generator requirement, then applies a fixed sequence of dispatch rules. These rules are neither arbitrary nor simplistic. They embody the project's design philosophy: maximize renewable utilization, preserve battery health through bounded SOC operation, protect F1 above all other loads, and reduce diesel use without sacrificing continuity of critical service. In this sense, the rule-based EMS is not a "simple control method" in a weak sense; it is a carefully structured supervisory framework chosen because it aligns with the technical, ethical, and implementation goals of the project.

A further advantage of rule-based EMS design is that it naturally supports hierarchical control. Local device controllers such as the DSE and BMS handle equipment-specific functions, while the higher-level EMS handles source selection, feeder priority, and system state transitions. This layered division is consistent with the literature on microgrid supervision, where the EMS

is expected to coordinate assets without replacing device-local protective intelligence. Thus, the EMS philosophy in this project should be understood as **supervisory and strategic**, not low-level and equipment-internal.

The academic strength of this subsection lies in presenting rule-based control not as a fallback from optimization, but as a deliberate and well-justified design choice. It offers clarity, traceability, deterministic implementation pathways, and compatibility with a PLC-centered real-system architecture. Those qualities are especially valuable in rural microgrids where maintainability, transparency, and service reliability may matter more than achieving marginal gains through highly complex algorithms.

4.7.2 Source priority: PV first, battery second, diesel backup

One of the clearest and most important principles of the proposed EMS is its **source priority hierarchy**: photovoltaic generation is used first, the battery is used second within safe operating limits, and the diesel generator is reserved as the backup source of last resort. This hierarchy is not merely a descriptive preference. It is the operational core of the energy-management strategy and the main reason the system can claim to be renewable-led rather than diesel-centered. If the source order were different, the technical identity and economic behavior of the microgrid would change fundamentally.

The literature strongly supports this kind of hierarchy in off-grid hybrid systems. Rule-based EMS strategies commonly prioritize renewable generation whenever available, then use battery storage to shift energy and smooth short-term deficits, and finally rely on diesel generation when renewable and storage resources can no longer satisfy demand or reserve constraints. This approach reflects both physics and economics. Solar energy has no fuel cost and should therefore be harvested whenever available. Battery energy is valuable but limited, so it should be used strategically rather than wastefully. Diesel generation is firm and dispatchable, but its fuel cost, maintenance burden, and emissions make it the least desirable normal source.

The MATLAB EMS implements this source order directly. In each time step, the available PV is first allocated to F1, then to F2, and then to F3. If PV cannot fully cover F1, the battery is allowed to discharge—subject to SOC and power limits—to support the remaining critical demand. Only if a residual deficit still remains after battery support does the generator produce output for the uncovered portion of F1. This means the diesel unit is not called because “load exists” in general; it is called because the protected critical load cannot otherwise be maintained within safe system limits.

This source-priority structure has several design benefits. First, it maximizes direct renewable utilization and therefore supports the main objective of reducing dependence on imported or expensive diesel fuel. Second, it uses the battery in a way that preserves its strategic role as a balancing and continuity-support resource rather than a continuously depleted energy pool. Third, it ensures that diesel runtime remains tied to genuine need, particularly the need to preserve critical service continuity, instead of becoming a convenience source for medium- or low-priority demand.

There is also an important conceptual elegance in this hierarchy. The source order mirrors the project's broader ethics of energy allocation. Renewable energy is treated as the preferred social good; storage is treated as a protected reserve that must be managed carefully; and diesel is treated as a resilience asset rather than a default operating habit. This framing strengthens the thesis because it shows that the EMS is not just technically effective but also philosophically coherent with the project's stated purpose of reliable, lower-fuel rural electrification.

Thus, this subsection should emphasize that source priority in the project is not a loose rule of thumb. It is the **primary dispatch doctrine** of the EMS, reflected consistently in the simulation logic, the electrical architecture, and the implementation mapping.

4.7.3 SOC-based charging and discharging logic

The battery in the proposed microgrid is not controlled as an unrestricted energy buffer; it is governed by an explicit **SOC-based charging and discharging logic** that protects reserve availability, avoids excessive depletion, and ensures that storage remains available for the moments when it is most valuable. This is a critical part of the EMS because the battery sits at the intersection of renewable maximization, generator deferral, and critical-load continuity. Without disciplined SOC logic, the system could either overuse the battery and endanger continuity later, or underuse it and waste valuable renewable flexibility.

The design documents establish the relevant SOC boundaries clearly. The EPLAN battery page specifies an intended operating SOC range of approximately 90% to 30%. The MATLAB EMS uses the same logic structure by setting SOCmax at 90%, SOCmin at 30%, and updating battery state over discrete hourly time steps based on charging or discharging power. This consistency is important because it shows that the SOC thresholds are not arbitrary simulation numbers; they are part of the project's implementation-oriented design philosophy.

The charging side of the logic is straightforward but strategically important. After the EMS allocates PV to the active feeder priorities, any remaining PV surplus can be used to charge the battery, provided the SOC has not yet reached its upper limit. The code explicitly checks whether surplus PV remains and whether the battery is below SOCmax, then applies the lesser of available PV, battery power limit, and remaining charge capacity for that time step. This means the battery functions as a **renewable absorption resource** before surplus PV is effectively wasted. In practical terms, it allows excess midday solar energy to be stored and shifted toward later periods when PV declines.

The discharge logic is even more central to the EMS philosophy. Battery discharge is not triggered simply because some load exists somewhere in the system. It is triggered specifically when F1 still has unmet demand after PV allocation and when SOC remains above the minimum allowable threshold. The EMS then discharges the battery only as much as needed—subject to battery power capability and the amount of energy that can be withdrawn without violating SOCmin. This is a very strong design choice because it makes the battery a **critical-load support resource** rather than a general-purpose unconstrained source. It preserves battery value for the most important service layer.

The broader EMS literature supports this kind of SOC-based supervision strongly. Rule-based battery scheduling methods often use upper and lower SOC thresholds to keep the storage system within a safe operating zone, ensuring that battery support remains available under uncertainty while limiting unsafe or undesirable excursions. In hybrid microgrids, this is especially important because the battery must do more than simply cycle energy; it must help guarantee continuity when renewable production falls. The lower SOC threshold therefore has a resilience meaning as well as a battery-health meaning: it marks the point at which the controller must stop treating the battery as a freely available source and begin preserving or rebuilding reserve.

There is also a strong system-level interpretation here. The battery SOC is not merely an electrochemical quantity; in this project it is a **state-of-system indicator**. High SOC indicates comfortable renewable-backed reserve. Mid-range SOC indicates active but manageable reliance on storage. Near-minimum SOC indicates that the system is entering a constrained mode in which backup support or deeper load prioritization may be required. This makes SOC a central supervisory variable, and that is why its logic belongs at the heart of the EMS chapter.

Therefore, this subsection should emphasize that the battery logic is not only about charging and discharging mechanics. It is about protecting the strategic role of storage in the overall microgrid: absorbing renewable surplus, supporting F1 during deficits, and signaling when the system must escalate from renewable-balancing mode to backup-preservation mode.

4.7.4 Generator dispatch logic

The generator dispatch logic of the proposed EMS is intentionally conservative, selective, and tied to the preservation of critical service rather than to generalized demand following. This is one of the defining strengths of the system. In many conventional or poorly optimized hybrid systems, the diesel generator begins operating too readily, effectively dominating the energy balance and undermining the renewable-first architecture. The present project avoids that outcome by dispatching the generator only when renewable generation and permissible battery support can no longer maintain the critical load envelope.

The MATLAB EMS expresses this logic in a clear sequence. First, PV is assigned to F1. If F1 remains partially uncovered, the battery is allowed to discharge provided SOC is above the lower bound. If a critical deficit still remains after this battery contribution, the algorithm calls on the generator to produce the remaining power, up to its rated capability. This means that generator dispatch is not demand-led in the broad sense; it is **deficit-led for protected load continuity**. The generator enters the dispatch sequence because a specific, higher-priority service requirement cannot otherwise be satisfied.

This dispatch philosophy is technically sound and well supported by the literature on off-grid hybrid systems. Rule-based EMS approaches commonly reserve the diesel source for periods of renewable insufficiency and storage limitation, particularly when continuity of supply to essential demand must be protected. Such logic reflects the fundamental difference between diesel and renewable resources. Diesel provides controllability and firmness, but at a cost. Therefore, good EMS design makes it available when needed, but not habitually active when cheaper and cleaner alternatives remain capable of serving the required load.

The project's design also gives generator dispatch a clear **service meaning**. Because the generator is primarily invoked to support F1, its operation is tied to the protection of hospital-related and minimum residential continuity rather than to medium- or low-priority comfort loads. This matters academically because it shows that the dispatch rule is not just about power balance; it is about the ethical structure of the microgrid. Diesel fuel, being costly and undesirable for normal operation, is reserved first for the most important social functions.

Another important feature of the generator logic is that it is naturally compatible with the DSE-based implementation layer. In the real system, a generator dispatch decision corresponds to a PLC request for remote start to the DSE, followed by running confirmation, healthy-state feedback, and possible breaker connection to the AC bus. This strengthens the realism of the design because generator dispatch is not modeled as an abstract variable only. It has a plausible field execution pathway. The EMS theory and the implementation architecture therefore support each other directly in this part of the project.

The dispatch logic also supports reserve discipline. Once the battery reaches its lower permissible SOC, continuing to insist on battery support would weaken resilience and potentially jeopardize later continuity. The generator is therefore not only a source of power; it is also the source that allows the EMS to stop consuming strategic battery reserve beyond its intended safe limit. In that sense, generator dispatch is a protective action not just for the load, but also for the battery and the overall energy-management strategy.

This subsection should therefore present generator dispatch as a **controlled escalation mechanism** in the EMS. The system begins with renewables, escalates to bounded battery support when necessary, and only then escalates to diesel intervention if critical service still cannot be preserved. That progression is one of the most elegant and defensible aspects of the entire control design.

4.7.5 Feeder shedding and restoration logic

Feeder shedding and restoration logic is the part of the EMS that converts energy scarcity into structured service adaptation rather than unplanned service collapse. In the proposed microgrid, this logic is based on the predefined hierarchy of F1, F2, and F3, where F1 is the critical feeder, F2 is the important but conditionally supplied feeder, and F3 is the most interruptible feeder. The EMS therefore does not treat all loads equally under shortage. Instead, it preserves the most important services for as long as possible by shedding lower-priority feeders first and restoring them later in a disciplined order.

The project's MATLAB EMS already embodies this logic in the way power is allocated. Available PV is assigned first to F1, then to F2, then to F3. If the available renewable and battery-supported supply cannot sustain all feeders, the lower-priority feeders naturally become partially or fully unsupplied first because they are downstream of the priority sequence. In practical design language, this corresponds to staged feeder shedding. F3 is the first feeder to lose service under

scarcity; F2 is the next candidate if the shortage deepens; and F1 is preserved unless the system has reached a severe deficiency state.

The literature on microgrid emergency load shedding strongly supports this sort of priority-based feeder logic. Non-critical loads are commonly disconnected first during shortage or disturbance conditions, while essential loads are preserved as long as technically feasible. Feeder-level restoration approaches also often seek to maximize the served load weighted by priority while minimizing unnecessary switching and preserving operational stability. The present project aligns with these principles, though in a simpler rule-based rather than optimization-based form. That is entirely appropriate for the thesis because the goal is implementation-oriented clarity rather than maximal algorithmic sophistication.

Restoration logic is the mirror image of shedding logic and is equally important. A weak EMS may define how to disconnect loads but not how to reconnect them intelligently. The proposed design avoids this weakness by restoring feeders in the reverse order of sacrifice. F2 is restored before F3 when sufficient source adequacy returns, because F2 occupies the intermediate priority layer and should regain service earlier than the most interruptible feeder. F3 is restored only after higher-priority conditions have been secured, ensuring that the resumption of low-priority service does not jeopardize F1 or destabilize overall source adequacy.

This restoration discipline matters for both technical and social reasons. Technically, it prevents oscillatory behavior and avoids repeatedly overextending the system by reconnecting too much low-priority demand too early. Socially, it makes service behavior understandable and fair: people can see that restoration follows the same logic as shedding and is therefore not arbitrary. In rural microgrids, this transparency is important for community acceptance and operational trust.

The subsection should therefore emphasize that feeder shedding and restoration is not a peripheral emergency feature. It is a **core EMS capability** that allows the microgrid to remain stable and socially purposeful under variable renewable conditions and limited backup reserve. It is one of the clearest places where the feeder design, the EMS philosophy, and the implementation mapping all converge.

4.7.6 Manual override conditions

Although the proposed EMS is primarily automatic and rule-driven, the design also includes **manual override conditions** through which operators can intervene under supervised

circumstances without collapsing the logic structure of the microgrid. This is an important mark of maturity in the control design. Real plants are not operated exclusively by automatic rules in every situation; they also require bounded human intervention for testing, maintenance, troubleshooting, commissioning, and abnormal events. The challenge is therefore not whether manual override exists, but how it is governed.

In the project architecture, manual intervention is treated as a structured supervisory mode rather than an uncontrolled bypass. The Node-RED and dashboard pathways allow actions such as generator start/stop requests, feeder toggling, setpoint updates, and alarm reset, while the PLC-centered logic remains the authoritative execution layer through which these requests are validated and applied. This means operator commands enter the control system in a mediated and intelligible way. They do not nullify the architecture; they pass through it.

The most realistic manual override conditions concern generator operation and feeder behavior. The generator may need manual start for maintenance checks, commissioning, or troubleshooting of automatic support logic. Likewise, F2 and F3 may need manual enabling or disabling for testing, controlled service management, or maintenance activity because they are more suitable for intervention than the protected critical feeder F1. However, the logic should still preserve essential interlocks. A manual request should not override an emergency stop, ignore a generator fault, or energize a branch in contradiction with basic safety conditions. This is exactly why manual override is best understood as **bounded supervisory authority**, not unrestricted plant control.

Another important role of manual override is operational transparency. The system must be able to distinguish between an automatic feeder shed, a manual feeder disable, and a fault-induced loss of service. This distinction matters for both diagnostics and accountability. If a feeder is off, the operator and later analyst should know whether it was off because the EMS enforced priority shedding, because an operator intentionally disabled it, or because a fault condition prevented service. A good EMS architecture preserves this distinction through mode bits, state tags, and command-source awareness.

From a thesis-writing standpoint, this subsection should clearly reject the false opposition between automation and manual control. In the present project, manual override is not the negation of EMS logic; it is one of the legitimate conditions under which that logic must remain governed and intelligible. This framing is stronger academically because it shows the system as operationally mature rather than merely automated.

4.7.7 Consistency between EMS theory and implementation

One of the strongest qualities of the project is that the EMS is not presented as a disconnected theory on one side and unrelated implementation diagrams on the other. Instead, there is a clear **functional consistency between EMS theory and implementation** across the simulation model, the electrical architecture, the PLC-centered control layer, and the supervisory interfaces. This consistency is crucial because it is what allows the project to claim implementation orientation rather than remaining a purely conceptual control exercise.

At the theoretical level, the EMS is defined by a small number of strong principles: renewable-first source allocation, bounded battery support based on SOC limits, diesel dispatch only for critical continuity, feeder priority preservation, and controlled manual intervention. These principles are clearly represented in the MATLAB EMS script, where PV is allocated first, battery support is tied to SOC, generator output appears only after residual F1 deficit remains, and feeder supply reflects the priority order of F1, F2, and F3. This gives the thesis a coherent behavioral model.

At the implementation level, those same principles appear in physical and control architecture. The renewable-first philosophy is supported by the electrical arrangement in which PV and battery share the DC-side architecture and feed the AC system through the inverter/charger. The SOC-based battery role is consistent with the battery/BMS design and the stated operating SOC range. The diesel backup role is consistent with the generator's dedicated branch, the DSE control layer, and the PLC-to-DSE command pathway. Feeder priority is consistent with the AC bus architecture, individual feeder breakers, and PLC-controlled feeder logic. Manual override is consistent with the structured dashboard and API pathways that still preserve PLC-centered control authority. In short, the implementation artifacts do not contradict the EMS theory; they give it physical and logical embodiment.

This consistency is academically significant because many engineering projects are strong in one domain and weak in the other. Some have elegant theory but no credible implementation path. Others have hardware drawings but no coherent supervisory logic. The present project avoids that split by showing functional equivalence between what the EMS decides in simulation and how those decisions would be realized through industrial devices and control pathways. For example, the simulated decision to support F1 from diesel after PV and battery become insufficient corresponds in implementation terms to a PLC command to the DSE, DSE start

execution, running feedback, and generator connection to the AC bus. Likewise, simulated feeder curtailment corresponds to real feeder breaker control and supervisory state reporting.

The importance of this consistency can also be framed methodologically. It shows that the project has moved beyond “modeling behavior” to “mapping behavior onto architecture.” That step is one of the defining marks of an implementation-oriented engineering thesis. It does not mean that every field parameter has been commissioned or every device has been tested on site. But it does mean that the control theory and the real-system representation speak the same language of states, priorities, commands, and interfaces.

This final subsection should therefore present consistency not as a minor observation, but as one of the major achievements of the chapter. The EMS theory, the PLC architecture, the DSE interface, the feeder layout, the battery constraints, and the supervisory software model all reinforce the same control philosophy. That coherence is what gives Chapter 4 its technical credibility and prepares the ground for later implementation and results chapters.

4.8 Simulation-Oriented Design Representation

A major strength of the present project is that it does not treat simulation as an isolated academic exercise detached from the real microgrid design. Instead, the project develops a **simulation-oriented design representation** whose purpose is to carry the essential behavior of the proposed hybrid microgrid into a form that can be studied, tested, interpreted, and compared with the intended implementation architecture. This distinction is crucial. A simulation model in engineering is not valuable merely because it produces curves; it is valuable when it preserves the decision-relevant structure of the real system while remaining tractable enough to support scenario analysis and control validation.

In the context of this thesis, the simulation layer fulfills exactly that role. It takes the real hybrid microgrid—composed of a PV source, a battery with bounded SOC operation, a diesel generator under DSE-mediated control, a three-feeder load structure, and a PLC-centered supervisory philosophy—and expresses it in a behavioral model suitable for MATLAB/Simulink and rule-based EMS analysis. The result is not a detailed electromagnetic replica of every cable, converter switch, or mechanical engine transient. Rather, it is a structured supervisory representation of how the microgrid **decides, allocates, escalates, sheds, restores, and**

reports under changing operating conditions. That is exactly the right level of abstraction for an implementation-oriented EMS thesis.

This section is especially important because it explains how the project avoids two common weaknesses in simulation-based design. The first weakness is oversimplification, where simulation becomes so abstract that it no longer corresponds meaningfully to the real plant. The second is over-detailing, where the model becomes burdened by unnecessary low-level dynamics that obscure the control question and make implementation mapping harder rather than easier. The present project navigates between these extremes by building a simulation layer that preserves source hierarchy, feeder priority, SOC-based battery behavior, generator support logic, and supervisory signal interpretation without pretending to be a full real-time digital twin of the final field installation.

From the broader literature, this approach is highly defensible. Reviews of microgrid modeling tools emphasize that the choice of simulation fidelity must match the purpose of the study: component-level dynamic analysis requires one kind of model, whereas supervisory EMS development requires a different, more behavior-oriented representation. Studies that develop EMS schemes in MATLAB/Simulink often rely on structured supervisory models with source blocks, SOC logic, load representations, state transitions, and command outputs rather than detailed switching-level power-electronic models. This is particularly appropriate when the goal is to validate coordination logic, priority behavior, and system response under multiple operating scenarios. The present thesis aligns strongly with that design philosophy.

Another reason this section deserves special depth is that it provides the intellectual bridge between Chapter 4 and the results-oriented chapters that follow. Earlier sections of Chapter 4 established the physical electrical architecture, the feeder structure, the diesel/DSE subsystem, the PLC-centered control architecture, and the EMS logic itself. Section 4.8 now explains how those design concepts are represented in a simulation-oriented form that is rigorous enough to test control behavior yet honest enough to state its abstraction limits. In other words, this is the section where the project answers an essential methodological question: **how is the designed microgrid represented in simulation, and what exactly does that simulation represent?**

The subsections that follow examine the purpose of the simulation design layer, the top-level Simulink EMS structure, the signal flow among PV, load, SOC, and feeder outputs, the representation of generator commands and source states, the relationship between simulation blocks and real devices, and finally the limits of abstraction within the adopted model.

4.8.1 Purpose of the simulation design layer

The simulation design layer exists to provide a controlled analytical environment in which the behavior of the proposed hybrid microgrid can be evaluated before any claim is made about operational logic, feeder priority effectiveness, or backup-source necessity. In practical terms, it is the intermediate layer between concept and implementation: richer than a purely verbal EMS description, but still more abstract than a fully commissioned field system. Its purpose is therefore not simply to "run a simulation." Its deeper purpose is to transform the design into a form where system logic can be challenged, observed, and refined under structured operating conditions.

In this project, the simulation layer serves at least four major functions. First, it validates the internal coherence of the EMS by showing whether the rule-based logic truly behaves according to the intended source and load priorities. Second, it allows time-dependent interaction among PV generation, feeder demand, battery reserve, and diesel support to be analyzed across daily operation. Third, it provides an interpretable environment for testing transitions such as PV surplus, battery support, critical-load protection, and generator activation. Fourth, it offers a structured bridge to implementation by expressing the real system architecture in behavioral variables that can later be mapped onto PLC tags, DSE states, and supervisory dashboard variables.

This purpose is especially important in hybrid microgrid engineering because system performance cannot be inferred reliably from component ratings alone. A PV array of 150 kW, a battery of about 100–123 kWh, and a 100 kW diesel generator do not by themselves guarantee good operation. What matters is how these sources interact over time, under varying load conditions, and under control rules that may protect some feeders while delaying or suppressing others. The simulation layer is therefore the environment in which **architecture becomes behavior**. It is where the thesis can move from "the system is designed this way" to "the system behaves this way under these conditions."

The literature strongly supports this kind of simulation purpose. Studies on microgrid control development repeatedly emphasize that simulation is indispensable for validating EMS strategies before hardware deployment, especially when the system involves multiple interacting resources, uncertainty in renewable generation, and service-priority logic. Reviews of microgrid simulation tools also note that simulation environments such as MATLAB/Simulink are particularly

valuable when the goal is to test supervisory coordination and control-oriented behavior rather than only static energy balances. That is precisely the purpose of the present design layer.

A further academic benefit of the simulation layer is that it creates a disciplined methodological boundary. It makes clear what is being studied directly—source allocation, feeder priority, SOC evolution, and diesel dispatch—and what is not being studied in full physical detail, such as switching harmonics, relay coordination timing, or engine thermal transients. This clarity is essential because a strong thesis must not confuse the validity of a supervisory model with the validity of every possible physical phenomenon in the real plant. Thus, the simulation design layer should be presented as a **purpose-specific representation for EMS validation**, not as a universal surrogate for all aspects of the microgrid.

In summary, this subsection should establish that the simulation layer is the analytical arena in which the proposed design is made testable, interpretable, and comparable with the implementation architecture. It is not secondary to the design; it is the means through which the design becomes academically verifiable.

4.8.2 Simulink top-level EMS structure

At the top level, the Simulink-oriented EMS structure can be understood as a behavioral supervisory model organized around a small set of high-value functional blocks: renewable input, feeder demand input, battery state representation, dispatch logic, generator support logic, and supplied-load outputs. This top-level structure is important because it defines the simulation not as a random collection of signals, but as an integrated control representation whose internal organization mirrors the key supervisory responsibilities of the real microgrid.

The MATLAB EMS implementation underlying the project reveals this structure clearly. The model begins with bounded input channels for PV input power and the three feeder demands F1,F2,F3$F1,F2,F3$, each clipped to the design maximums of the project—150 kW for PV, approximately 57 kW for F1, 26 kW for F2, and 15 kW for F3. It then uses these signals within a rule-based sequence that calculates available PV, allocates priority power to critical demand, evaluates battery support based on SOC and battery power limits, calls generator support if required, updates supplied feeder outputs, and computes the resulting SOC trajectory for the next time step. This is a highly coherent supervisory structure because every major EMS decision appears in the model as a named and traceable functional stage.

Conceptually, the top-level Simulink EMS can therefore be decomposed into six interacting layers. The first layer is the **source-input layer**, which receives the renewable power profile. The second layer is the **load-structure layer**, which represents the priority-class feeder demands. The third layer is the **storage-state layer**, which represents battery SOC and its allowable operating envelope. The fourth is the **dispatch-decision layer**, where rules determine the source allocation order and feeder coverage. The fifth is the **backup-activation layer**, in which generator support is invoked when renewable and storage resources are insufficient for the protected load. The sixth is the **output/state-reporting layer**, which exports active load, feeder-supplied values, generator power, battery power, and supply-demand relationships for interpretation and plotting.

This kind of top-level supervisory decomposition is highly consistent with good Simulink modeling practice for microgrid EMS studies. Control-oriented simulation models typically benefit from clear functional separation between input blocks, state blocks, logic blocks, and output blocks, because such separation improves readability, testing, and eventual mapping to real implementation layers. The present project fits that pattern well. It does not attempt to bury the EMS inside opaque signal algebra. Instead, it preserves the visibility of its own decision logic, which is academically advantageous because it allows the reader to see how the control philosophy is embodied in the model.

Another important aspect of the top-level structure is that it is **state-aware but not overcomplicated**. It includes battery SOC as a dynamic state and source/feeder outputs as time-varying consequences of decisions, yet it does not overburden the supervisory model with device-internal dynamics that are not essential to the EMS question. This is exactly the right level of structure for a simulation-oriented representation intended to support implementation-oriented conclusions.

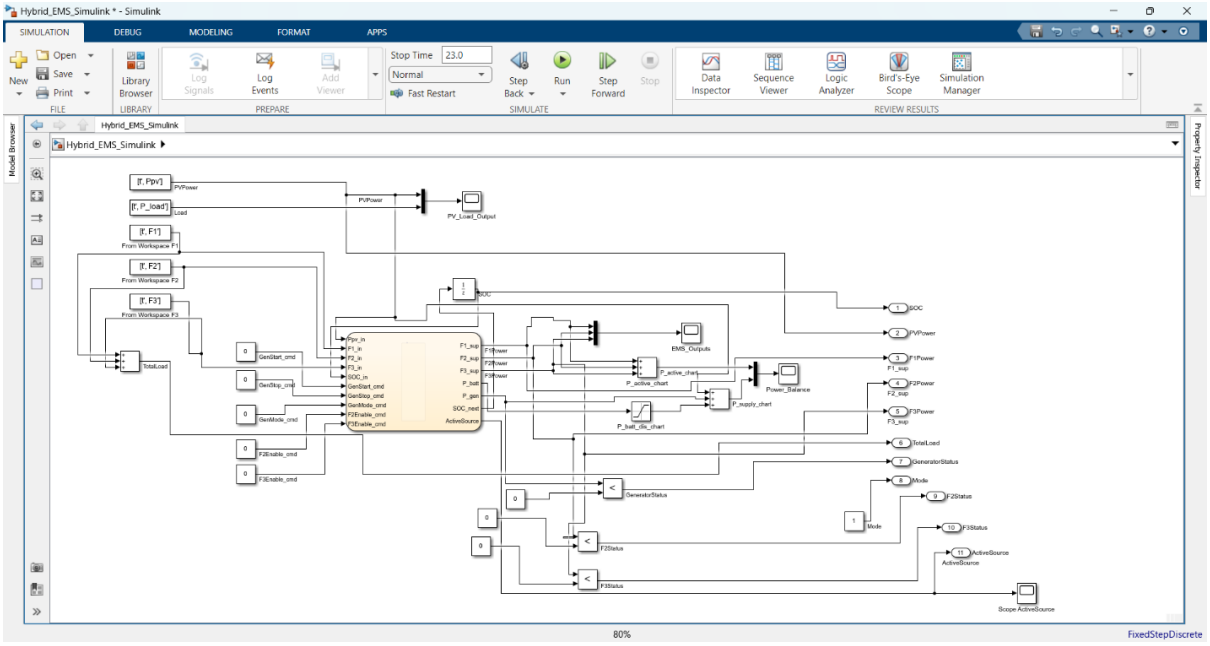


Figure 4-18 Simulink Model of Signal Flow Between PV, Load, SOC, and Feeder Outputs

As shown in figure 4-18, this subsection should emphasize that the top-level Simulink EMS is not merely a block diagrammatic convenience. It is the formal behavioral architecture of the project's supervisory design, structured to reflect the same logic hierarchy that appears in the real microgrid concept.

4.8.3 Signal flow between PV, load, SOC, and feeder outputs

The signal flow in the simulation model is one of the clearest expressions of the EMS philosophy because it shows how information and control consequences propagate through the supervisory structure. In the present project, the main signals of interest are PV availability, feeder demand, battery SOC, generator contribution, and feeder supplied outputs. These signals are not independent measurements displayed side by side; they form a causally linked decision chain in which each signal influences subsequent system behavior.

The flow begins with the PV signal, which represents the renewable power available to the system at each simulation time step. This signal enters the dispatch logic as the preferred primary source and is allocated first to F1, then to F2, and finally to F3 according to the feeder-priority structure. At the same time, the feeder demand signals F1,F2,F3$F1$,$F2$,$F3$ represent the desired load envelopes associated with the real community grouping. The first key decision in the signal chain is therefore a comparison between PV availability and feeder demand, beginning with the critical feeder. This establishes whether renewable energy alone is sufficient or whether deeper support mechanisms must be activated.

If PV is insufficient for F1, the signal flow reaches the SOC-governed storage layer. Here, SOC acts as a gating state rather than a passive indicator. If SOC remains above the minimum threshold, the battery discharge logic computes the allowable discharge based on battery power limit, available energy margin above SOCmin, and the remaining F1 deficit. In other words, the SOC signal does not merely describe the battery; it **qualifies the battery's permission to act**. This is a highly important modeling idea because it makes the battery a state-dependent support source rather than an always-available algebraic supply block.

If a residual F1 deficit still remains after PV allocation and permissible battery support, the signal flow proceeds to the generator-support branch. The generator output signal is then computed as the remaining required support for F1, bounded by the generator's rated capacity. This means that the generator signal is causally downstream of both the PV and SOC branches. It does not emerge directly from total load demand; it emerges from **uncovered critical-load demand after renewable-first and storage-bounded support have already been attempted**. That causal ordering is one of the strongest conceptual features of the simulation model.

The outputs of this signal flow are equally important. The model produces F1supplied$F1supplied$, F2supplied$F2supplied$, F3supplied$F3supplied$, battery power, generator power, total active load, total supply, and SOC trajectory over time. These outputs are not arbitrary reporting variables. They are the visible consequences of the EMS signal chain. By plotting them, the project can show not only what the microgrid demanded, but what it actually supplied after supervisory logic intervened. This difference between original load and active supplied load is especially valuable for interpreting shedding behavior and the protective effect of feeder priority logic.

The literature on control-oriented microgrid modeling supports this kind of explicit signal structuring. Simulink-based EMS studies often represent supervisory control through flows among renewable power, load, storage state, backup-source decisions, and output status variables, because such structure allows both validation and later mapping to SCADA or control-system variables. The present project aligns with that practice very well. Its signal flow is transparent enough to explain, yet rich enough to represent the actual logic of the EMS.

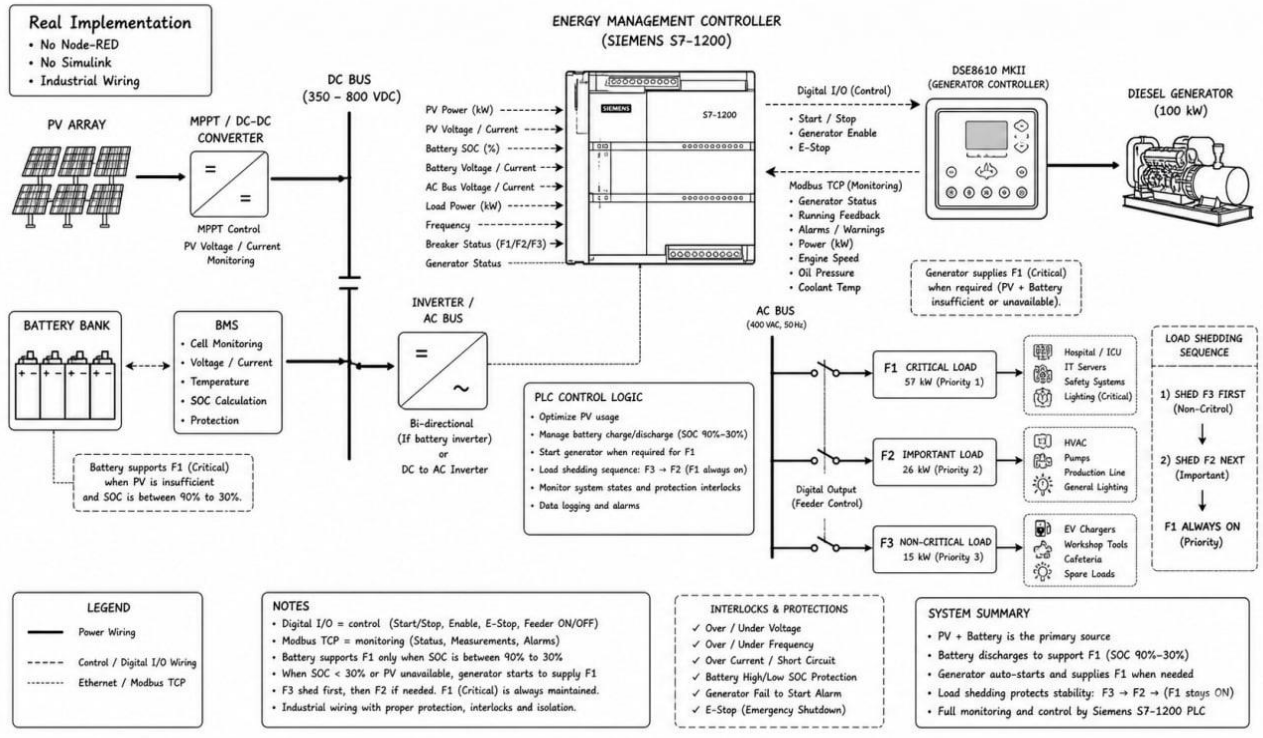


Figure 4-19 Real-World Microgrid Implementation Architecture with PLC, DSE, and Feeder Priority Control

As shown in figure 4-19, this subsection should therefore frame signal flow not as a low-level software detail, but as the **behavioral bloodstream of the simulation model**. It is through this flow that the simulation becomes an intelligible representation of how the real hybrid microgrid would think and respond under changing conditions.

4.8.4 Representation of generator commands and source states

In the simulation model, the generator is not represented as a fully detailed electromechanical machine; rather, it is represented as a **supervisory source state and commanded support branch** within the EMS logic. This is an intentional and appropriate modeling choice. The objective of the simulation layer is to study when the system decides that generator support is required and how that decision changes served load, supply balance, and operating state—not to

reproduce every crank event, fuel injection transient, governor action, or breaker synchronization detail.

The MATLAB EMS captures this supervisory representation clearly. Once PV allocation and battery support have been evaluated, any remaining uncovered F1 demand is assigned to the generator branch, bounded by the generator rating Pgenrated=100 kW*Pgenrated*=100 kW. The resulting signal Pgen*Pgen* is then recorded over time as the generator power output. In the logic of the model, this variable represents the commanded presence of backup generation in response to critical-load deficiency. It is therefore best interpreted as a **generator support state with associated power contribution**, rather than a detailed machine-state simulation.

This abstraction is academically justified because it preserves the decision-relevant meaning of generator behavior. What matters to the EMS is whether the generator is needed, how much critical load support it provides, and how its presence affects supply-demand balance and feeder continuity. For those purposes, a supervisory power-state representation is sufficient and even preferable, because it keeps the model aligned with the high-level control question. If the model were burdened with full electromechanical detail, the clarity of dispatch logic could be diluted without adding corresponding value to the EMS analysis.

At the same time, the simulation representation remains meaningfully connected to the real implementation architecture. In the real plant, generator support would correspond to a PLC-generated start request, DSE start execution, running confirmation, breaker control, and source availability at the AC bus. The simulation does not model every one of those stages explicitly. However, it does model their **functional outcome**: generator contribution appears only after critical deficiency remains after PV and battery support, and only up to the rated generator limit. That is precisely the right level of representation for a supervisory simulation intended to be compared with an implementation-oriented architecture.

The same reasoning applies to source states more broadly. The model effectively represents several operating states even if it does not label them as a complex finite-state machine in the code. There are periods of PV-only operation, PV-plus-battery support, PV-plus-generator support under critical shortage, and surplus-PV charging behavior. These states emerge from the rule chain and can be interpreted from the time-series outputs. This makes the model rich enough to support results analysis without requiring an unnecessarily heavy formal state-machine layer at every point.

Therefore, this subsection should present generator commands and source states as **functional supervisory representations**. The simulation does not claim to be a full engine dynamics model; it claims to represent when backup is requested, when it contributes, and how that contribution alters service continuity and system power balance. That claim is both strong and methodologically honest.

4.8.5 Relationship between simulation blocks and real devices

One of the strongest academic features of the project is that its simulation blocks are not arbitrary symbolic placeholders; they correspond in a meaningful and disciplined way to the real devices and functions defined in the electrical and control architecture. This correspondence is not one-to-one at the level of every terminal or hardware register, but it is strong at the level that matters for supervisory design: source role, feeder role, state role, and command role. That is exactly how a simulation-oriented representation should relate to implementation in a thesis of this type.

The PV input block in the simulation corresponds functionally to the real PV generation subsystem composed of the array, combiner, surge protection, MPPT stage, and DC bus connection. The simulation does not reproduce each hardware layer separately, but it represents their effective supervisory result: available PV power as an input to the EMS. Likewise, the battery state and battery power logic correspond to the real battery bank, BMS, breaker, and contactor branch, again not in wiring detail but in supervisory terms of charge/discharge availability, SOC state, and bounded participation.

The generator branch in the simulation corresponds to the real diesel generator and DSE-controlled subsystem. In simulation, it appears as the bounded source Pgen*Pgen* called upon when F1 remains uncovered. In implementation, the same functional idea is realized through a PLC-to-DSE start request, DSE local execution, generator availability, and AC-bus contribution through the generator breaker. The two representations are different in form but consistent in operational meaning. That consistency is what gives the mapping credibility.

The feeder blocks provide perhaps the clearest example of this relationship. In simulation, F1*F*1, F2*F*2, and F3*F*3 appear as grouped load channels with corresponding supplied outputs F1supplied*F*1*supplied*, F2supplied*F*2*supplied*, and F3supplied*F*3*supplied*. In the real system, those same feeder classes exist as physical outgoing branches on the AC bus connected through ACBR03, ACBR04, and ACBR05. Thus, the simulation blocks do not represent imagined feeder categories; they represent the behavior of actual feeder groups defined in the

design architecture. This is an excellent example of how abstraction can remain implementation-relevant.

The same applies to control interpretation. In simulation, the EMS logic is expressed through source allocation and supplied-load calculations. In implementation, the same logic would be enacted through PLC decisions, DSE commands, inverter mode coordination, battery permission handling, and feeder output control. The simulation blocks therefore correspond not just to physical devices, but to **functional control responsibilities**. This is a subtle but very important point. A simulation block does not always map to one device only; sometimes it maps to the combined behavior of a device plus its local controller plus the supervisory decision that governs it.

This subsection should therefore emphasize that the relationship between simulation blocks and real devices is one of **functional equivalence under abstraction**. The model preserves what the real devices do at the supervisory level, even when it does not reproduce how they do it at the low-level physical level. That is exactly the right relationship for a simulation-oriented design representation in an implementation-aware microgrid thesis.

4.8.6 Limits of abstraction in the simulation model

A strong simulation chapter does not only explain what the model represents; it also explains what the model deliberately does **not** represent in full detail. This is especially important in a thesis like the present one because the simulation model is intentionally supervisory and implementation-oriented rather than physically exhaustive. Stating the abstraction limits clearly does not weaken the work. On the contrary, it shows methodological maturity and protects the validity of the conclusions by keeping them aligned with the actual scope of the model.

The first major limit is that the simulation is not an electromagnetic or switching-level model of the entire power system. It does not model converter switching harmonics, detailed DC bus dynamics, relay coordination timing, transient voltage/frequency recovery, or engine-governor electromechanical startup dynamics. Those topics belong to different classes of power-system study and would require a very different modeling approach. The present model instead focuses on energy allocation, feeder continuity, SOC evolution, and supervisory state transitions over a daily operating horizon.

The second limit is that the load representation is aggregated and structured rather than appliance-level stochastic. The model includes feeder groups and community-related base

profiles such as residential demand, hospital, school, pump, mosque, and shops, but it does not attempt to simulate every individual end-use switching pattern or random behavioral fluctuation. This is a legitimate abstraction because the research question concerns supervisory feeder priority and source coordination, not consumer-behavior microstatistics. Still, it means that the results should be interpreted as feeder-level operational behavior rather than a perfect prediction of every real-minute village load fluctuation.

The third limit concerns the battery representation. Although the model tracks SOC and enforces meaningful charge/discharge thresholds, it does not implement full electrochemical dynamics, aging behavior, thermal diffusion, cell balancing, or state-of-health degradation. The battery is modeled as a bounded energy resource suitable for supervisory dispatch analysis, not as a chemistry-resolved electrochemical system. This is appropriate for the thesis objective, but it also means that conclusions about long-term battery lifetime or fine-grained degradation optimization are outside the validated range of the model.

The fourth limit involves generator realism. The simulation represents the generator as a support branch with bounded power contribution when critical-load deficit remains. It does not fully simulate cranking delay, synchronization timing, fuel consumption curves at multiple loading levels, start-failure probabilities, or detailed DSE state machine behavior. These omissions are acceptable because the project's primary claim is about EMS logic and control philosophy, not full dynamic engine validation. However, they also mean that the model is best interpreted as a supervisory dispatch representation rather than a commissioning-grade dynamic generator emulator.

The fifth limit concerns communication and implementation timing. The simulation does not reproduce exact PLC scan cycles, network latency, Modbus polling delays, dashboard refresh timing, or asynchronous event races among real devices. It assumes a cleaner supervisory environment in which state evaluation and rule application occur over the chosen time resolution without all the timing imperfections of a live industrial networked system. This is acceptable for high-level EMS validation, but it should be acknowledged so that the reader does not confuse behavioral correctness with fully validated real-time execution performance.

The final and perhaps most important abstraction limit is philosophical: the model represents the **decision-relevant structure** of the microgrid, not the entire ontology of the microgrid. It tells us how the designed system prioritizes, allocates, escalates, and sheds under the assumed supervisory framework. It does not claim to solve every electrical, mechanical, thermal,

communication, and human-factor detail that would arise in full field deployment. This boundary is not a weakness. It is the reason the model remains coherent and useful. A simulation that tries to do everything often clarifies nothing; a simulation that models the right level for the research question can be highly powerful.

This subsection should therefore close the section with a mature and confident statement: the simulation model is intentionally abstracted, but its abstraction is disciplined, purposeful, and aligned with the EMS-centered objectives of the thesis. It is valid for studying source priority, battery reserve logic, feeder continuity, and generator intervention at the supervisory level. It is not claimed as a full-fidelity replacement for all later detailed engineering studies.

4.9 Supervisory Software and Middleware Design

The proposed hybrid microgrid does not end at the electrical design, the PLC logic, or the EMS decision rules. A complete modern microgrid architecture also requires a supervisory software layer capable of exposing system state, accepting bounded operator interaction, normalizing data across heterogeneous subsystems, and supporting future integration with external software platforms. In the present project, that role is performed by a middleware-oriented supervisory layer centered on Node-RED, structured APIs, status publishing logic, and a data model designed to remain consistent across simulation, control, and user-interface levels.

This section is especially important because it explains how the project moves from “control exists” to “control is visible, interpretable, and interoperable.” A microgrid can have an EMS and a PLC yet still remain operationally weak if system state is difficult to access, command pathways are ad hoc, variable naming is inconsistent, or user interaction bypasses the control hierarchy. The supervisory software and middleware layer addresses exactly these problems. It gives the project a structured digital surface through which system behavior can be monitored and selected actions can be mediated without confusing software supervision with hard real-time field control.

The project materials already define this architecture clearly. Node-RED is described as a middle layer between the dashboard, the microgrid simulator engine, and future external systems such as MATLAB through MQTT. It exposes a REST API with routes such as GET /api/status, POST /api/control/generator, POST /api/control/feeders, POST /api/control/setpoints, and POST /api/control/reset-alarms, and it can also publish microgrid state through MQTT topics such as

microgrid/status. This is a strong and modern architectural choice because it separates concerns cleanly: the dashboard becomes a client of a normalized supervisory interface, Node-RED becomes the middleware and orchestration layer, and the PLC-centered control layer remains the field-authoritative executor of hard operational decisions.

From a broader engineering perspective, this layered approach is well justified. Node-RED is widely used in Industrial IoT and edge-integration contexts because it supports flow-based orchestration, rapid API creation, multi-protocol integration, and standardized data handling across devices and applications. At the same time, industrial automation practice consistently warns against confusing SCADA- or middleware-style supervisory software with deterministic hard real-time control platforms such as PLCs. The present project handles this distinction correctly. It uses Node-RED where Node-RED is strong—data handling, API mediation, integration, and supervisory visibility—while preserving PLC/DSE pathways for command execution and equipment-level determinism.

Another reason this section matters deeply is that it is where architectural discipline becomes visible across software layers. The electrical system gives the microgrid physical form. The EMS gives it behavioral logic. The PLC gives it deterministic execution. But the supervisory software and middleware layer gives it **digital coherence**: a common status schema, a controlled command surface, a path for publishing updates, and a basis for expansion into dashboards, MQTT-based integration, or future SCADA/IoT extensions. In implementation-oriented academic work, that coherence is often the difference between a good control concept and a truly defensible system architecture.

The following subsections therefore analyze the role of Node-RED in the project architecture, the REST API structure used for status and control, the internal logic of status publishing and middleware data flow, the MQTT integration concept, the essential separation between supervisory software and hard real-time control, and finally the importance of tag and data consistency across all layers of the project.learn.

4.9.1 Role of Node-RED in the project architecture

Node-RED occupies a very specific and very important position in the project: it is the **middleware and supervisory orchestration layer**, not the primary real-time controller. This distinction should be stated early and firmly because it defines the architectural integrity of the whole software stack. Node-RED sits between the operator-facing dashboard, the simulated or future real microgrid state sources, and external communication mechanisms such as MQTT or

API clients. It receives, organizes, transforms, and serves data in a form that is suitable for supervision and bounded interaction, while leaving deterministic plant execution to the PLC-centered control architecture.

This role is strongly aligned with what Node-RED is good at in broader practice. Node-RED is designed as a flow-based development environment for wiring together devices, APIs, protocols, and services, and it is widely used in Industrial IoT and edge-integration settings to normalize and route data, connect heterogeneous protocols, and create lightweight dashboards or service endpoints. These characteristics make it highly suitable as a **middleware layer** in a microgrid project, where different software and hardware environments need a common integration surface but do not all share the same protocol or timing expectations.

In the present project, Node-RED is described as the middle layer between the HTML dashboard, the microgrid simulator engine, and future external systems such as MATLAB through MQTT. This is a very strong architectural decision because it avoids tight point-to-point coupling. The dashboard does not need to know the internal details of MATLAB or PLC logic; it only needs to request structured state and submit allowed commands through the Node-RED interface. Likewise, future integration with simulation engines or additional field sources can be added behind the middleware without forcing a redesign of the front-end structure.

Node-RED also plays an important **semantic normalization** role. Different layers of the system naturally think in different ways: the PLC thinks in terms of tags, bits, analog channels, and interlocks; the simulator thinks in terms of variables such as SOC, PV power, F1 supplied power, or generator status; and the dashboard thinks in terms of visible status cards, toggles, numeric values, and alarms. Node-RED is the layer that can reconcile those different representations into one coherent application data model. This is more important than it may first appear. A middleware layer is valuable not only because it moves data, but because it makes the data intelligible and consistently structured across software boundaries.

Another major advantage of Node-RED in this project is its suitability for iterative academic development. Because it supports rapid construction of HTTP endpoints, MQTT flows, and dashboard-oriented logic, it allows the supervisory layer to be demonstrated and refined without pretending that it replaces the industrial controller. This is an ideal balance for the present thesis. It gives the project modern integration capability and software visibility while keeping the critical control authority in the PLC/DSE field architecture.

This subsection should therefore present Node-RED as a **supervisory middleware platform with four core roles**: data normalization, API mediation, protocol bridging, and operator-facing integration support. It is not an accessory add-on. It is the software layer that gives the project digital structure above the real-time control core.

4.9.2 REST API structure for status and control

The REST API in the project is the formal interface through which supervisory software exposes system state and accepts selected control intentions in a structured, testable, and extensible manner. This is one of the strongest aspects of the middleware design because it converts the internal system model into a documented interaction surface rather than an informal collection of hidden variables or ad hoc software actions. In implementation-oriented systems, this matters greatly. Once a project includes dashboards, simulation interfaces, future external integrations, or operator tools, a clean API becomes the architecture's shared language.

The project materials define the API structure explicitly. The main read endpoint is GET /api/status, which returns the current structured system state. The main control endpoints are POST /api/control/generator, POST /api/control/feeders, POST /api/control/setpoints, and POST /api/control/reset-alarms. This structure is excellent because it separates reading from action, and it also separates different classes of action rather than collapsing everything into a single overloaded command route. Such separation improves clarity, maintainability, and future validation of supervisory behavior.

The API design also reflects good system semantics. A status endpoint should present the microgrid as a coherent object: mode, SOC, generator status, PV power, total load, active source, feeder states, alarms, and thresholds. A control endpoint should represent a bounded user intent rather than low-level arbitrary memory manipulation. For example, "start the generator," "disable F3," "update SOC threshold," or "reset alarms" are understandable supervisory actions. By contrast, exposing arbitrary internal variables directly would weaken architectural discipline and risk inconsistency between the supervisory layer and the field-control layer.

This kind of API is also highly compatible with Node-RED's strengths. Node-RED is well suited to building HTTP request/response flows that can validate payloads, translate external commands into internal messages, and return normalized JSON objects for dashboards or software clients. In the present project, this means the dashboard can remain a relatively thin client. It requests state through GET /api/status, displays the returned values, and submits specific

control requests through the corresponding POST endpoints. This is a much stronger design than letting the UI compute meaning or interact directly with field protocols.flows.

The architecture can be summarized clearly in the following table

Table 4-8 REST API Endpoints for Microgrid Supervisory Control

Endpoint	Method	Function	Architectural meaning
/api/status	GET	Returns structured current microgrid state.	Single source of truth for dashboard and supervisory clients.
/api/control/generator	POST	Sends generator-related supervisory command, such as start or stop.	Bounded operator intent mediated by middleware, not direct field bypass.
/api/control/feeders	POST	Enables or disables feeder states such as F2 and F3.	Supervisory control surface for priority-aware feeder interaction.
/api/control/setpoints	POST	Updates thresholds or configurable supervisory parameters.	Maintains controlled parameter handling across software layers.
/api/control/reset-alarms	POST	Clears or acknowledges annunciated alarms.	Supports operator workflow without altering field-control determinism.

Table 4-8 is useful because it makes the software architecture legible in one glance and shows that the API is not arbitrary; it mirrors the actual operating concepts of the project.

A further academic advantage of the API structure is that it supports future extensibility without changing the core philosophy. If later the project evolves from simulated values to PLC-fed live values, or from simple status polling to richer event subscriptions, the dashboard can still rely on the same endpoint structure while the middleware internals adapt behind it. That is exactly what good middleware design should achieve.

4.9.3 Status publishing and middleware data flow

Status publishing is the process through which the supervisory layer takes internal system state and makes it available in a structured, consumable, and timely manner to software clients such as the dashboard or future external services. In the present project, this process is central to the value of the middleware layer. Node-RED is not useful merely because it exists between systems; it is useful because it acts as the **publisher and organizer of coherent supervisory state**.

The project materials indicate that Node-RED maintains and exposes a status object including values such as mode, SOC, generator status, PV power, total load, active source, feeder states, alarms, and threshold setpoints. This status object is then returned through the GET /api/status route and used by the dashboard to update the visible interface. The dashboard itself polls this endpoint periodically and maps the returned JSON fields to UI elements such as SOC display, PV power card, load card, active-source label, generator state, feeder toggles, and alarm list. This is a strong design because it keeps the data flow centralized: Node-RED becomes the authoritative publisher of software-visible state.

The internal middleware data flow can be understood as a sequence. First, state information is acquired or maintained within the supervisory model—whether originating from simulation variables, future PLC tags, or integrated device status. Second, Node-RED normalizes this information into a shared application schema. Third, the normalized schema is exposed to clients through API responses and, potentially, MQTT publication. Fourth, client-side software such as the dashboard renders the current state and may submit selected control requests back to Node-RED through the control endpoints. This structure is elegant because it preserves a clear distinction between **state publication** and **command intake**, while still allowing both to coexist inside one middleware layer.

This kind of publisher-centered architecture is strongly supported in IIoT and middleware design. Middleware layers are often valued precisely because they decouple producers of data from consumers of data, making the whole system easier to maintain, extend, and reason about. In the context of this project, such decoupling is especially beneficial because the origin of the microgrid state may evolve over time—from simulated state today to PLC-linked state later—while the dashboard and API clients can remain largely unchanged as long as the published schema stays consistent.

The following table summarizes the middleware data-flow roles in the present design.

Table 4-9 Supervisory Data Flow Layers in the Proposed Microgrid Architecture

Layer	Main role	Example data
State source layer	Produces raw or interpreted microgrid state.	SOC, PV power, generator status, feeder states.
Node-RED middleware layer	Normalizes, stores, routes, and publishes supervisory state.	Unified JSON status object, control request handling.
API layer	Exposes structured read and write interfaces.	/api/status, /api/control/... endpoints.
Dashboard/client layer	Consumes published state and submits selected operator intents.	Visual cards, toggles, alarm resets, setpoint updates.

Table 4-9 helps show that the middleware is not just a transport pipe. It is an organizational layer that gives the microgrid's supervisory software a coherent information lifecycle.

This subsection should therefore emphasize that status publishing is the mechanism through which the project becomes **software-observable** in a disciplined way. Without it, there would be signals and control logic, but not an integrated supervisory software architecture.

4.9.4 MQTT integration concept

MQTT is introduced in the project not as a mandatory replacement for the REST API, but as a **future-ready messaging concept** that complements the supervisory architecture by enabling decoupled, asynchronous publication and subscription of system state. This is an excellent design direction because REST and MQTT solve different integration problems. REST is very effective for request/response interactions such as dashboard polling or explicit control calls, while MQTT is particularly valuable when multiple clients need to subscribe to evolving state without constant direct polling.

The project materials explicitly identify MQTT as the planned bridge for future MATLAB integration and status exchange, with Node-RED capable of publishing microgrid state to topics such as microgrid/status. This reflects strong middleware thinking. Rather than tightly coupling MATLAB, the dashboard, and any future analytical or archival tools in point-to-point arrangements, the architecture can use Node-RED as the publishing hub and MQTT as the

asynchronous distribution mechanism. In such a model, any authorized subscriber—dashboard extensions, logging services, analytics tools, or co-simulation components—could receive state updates from the broker without changing the basic middleware philosophy.

MQTT also fits Node-RED's strengths particularly well. Node-RED includes MQTT input and output nodes that make it straightforward to subscribe to topics, process messages, and republish normalized data streams. In Industrial IoT usage, Node-RED is commonly employed as an edge or integration tool that collects machine data, standardizes it, and publishes it through MQTT for broader system use. The present project follows the same logic conceptually. Node-RED acts as the normalization point; MQTT extends that normalized state into a more scalable publish/subscribe communication model.

Architecturally, MQTT is valuable here for three reasons. First, it supports decoupling: publishers do not need to know how many subscribers exist or what they do with the data. Second, it supports asynchronous data distribution, which is useful when the system state changes continuously and multiple software components may need updates. Third, it supports future scalability beyond the immediate dashboard use case, which is important if the project later expands into hardware-in-the-loop studies, data logging, cloud connectivity, or multiple supervisory clients.

The distinction between REST and MQTT in the project can be summarized effectively in the following table.

Table 4-10 Comparison Between REST API and MQTT in the Proposed Microgrid System

Mechanism	Communication style	Best use in this project
REST API	Request/response.	Dashboard status retrieval, explicit operator commands, setpoint updates.
MQTT	Publish/subscribe.	Future MATLAB linkage, multi-client status distribution, asynchronous supervisory data exchange

Table 4-10 is useful because it prevents the impression that MQTT is being introduced redundantly. It is not redundant; it is complementary.

This subsection should therefore frame MQTT as a **scalability and integration concept** within the middleware architecture. It extends the project from a single-client supervisory

interface toward a more general event-driven data ecosystem without disturbing the primary control hierarchy.

4.9.5 Separation between supervisory software and hard real-time control

One of the most important architectural disciplines in the entire project is the explicit separation between **supervisory software** and **hard real-time control**. This distinction is absolutely essential, both technically and academically. Without it, middleware platforms, dashboards, and API layers can be misunderstood as if they were the primary control authority for field-critical actions, which would seriously weaken the credibility of the system design. The present project avoids that mistake and should state this separation with complete clarity.

Hard real-time control refers to control functions that must execute deterministically within strict timing constraints and with direct responsibility for safe plant behavior. In the present microgrid, those responsibilities belong to industrial control hardware and local controllers: the Siemens S7-1200 PLC, the DSE8610 MKII generator controller, equipment-local interlocks, and direct digital I/O pathways. These layers are responsible for start/stop execution, contactor logic, feeder command enforcement, interlock handling, and equipment-related response under operational constraints. Their authority is grounded in deterministic execution and field-level connectivity.

By contrast, the Node-RED layer, the REST API, the dashboard, and MQTT-based messaging belong to the **supervisory software domain**. Their job is to observe, normalize, present, route, and mediate—not to replace the deterministic controller. This does not make them unimportant. On the contrary, they are essential for visibility, operator interaction, and system integration. But their timing guarantees, failure modes, and execution environment are of a different class from those of a PLC scan cycle or a device-local protection controller. That is exactly why the roles must remain distinct.

Industrial automation practice strongly supports this separation. Sources discussing automation platform categories emphasize that PLC/DCS platforms execute deterministic closed-loop or time-critical control, whereas SCADA, MES, and supervisory software platforms observe, coordinate, and manage without being the authoritative hard real-time execution core. Confusing these roles is described as a common integration failure mode. The present project is architecturally strong because it does not make that mistake. It uses supervisory software where flexibility and integration are needed, and industrial hardware where determinism and field authority are required.

The architecture can be summarized clearly as follows.

Table 4-11 Roles and Authority Characteristics of System Layers in the Proposed Microgrid Architecture

Layer	Primary responsibility	Timing/authority character
PLC / DSE / field I/O	Deterministic control execution, interlocks, equipment actuation.	Hard or near-real-time operational authority.
Node-RED middleware	State normalization, API mediation, flow orchestration, integration.	Supervisory, flexible, non-field-authoritative.
Dashboard / client software	Visualization and bounded operator interaction.	Human-facing, supervisory, non-deterministic.
MQTT / external software links	Data distribution and software interoperability.	Asynchronous integration layer, not field execution core.

Table 4-11 is especially valuable because it protects the thesis from overstatement. It shows that the project embraces modern software integration without pretending that web-based layers can substitute for industrial real-time control.

This subsection should therefore present the supervisory-vs-real-time distinction as a **foundational design principle**. It is one of the main reasons the architecture remains credible, safe in concept, and consistent with industrial engineering practice.

4.9.6 Tag and data consistency across layers

Tag and data consistency is one of the most subtle yet most important qualities of a strong supervisory architecture. A project may have correct hardware, correct EMS logic, and even a functioning dashboard, but if the variables mean different things in different layers, then the system quickly becomes fragile, confusing, and analytically unreliable. For this reason, the present project places strong emphasis on maintaining semantic consistency across simulation, PLC-oriented logic, middleware representation, API payloads, and user-interface elements.

The project materials already reveal a coherent cross-layer vocabulary. Variables and concepts such as mode, soc, genStatus, pvPower, totalLoad, activeSource, f2Status, f3Status, alarms, and threshold setpoints appear in the supervisory documentation and are clearly intended to connect simulator state, middleware status objects, and dashboard rendering. At the same time,

the EPLAN and PLC-oriented materials identify the field side of the same conceptual space through digital and analog signals such as generator running, generator fault, inverter ready, battery low/high/alarm, feeder status, and measurement channels. This is precisely what good system engineering requires: the same operational realities expressed coherently across layers that use different technical forms.

The importance of consistency can be understood at three levels. First is **naming consistency**: a variable such as generator status should not mean “commanded start” in one layer and “confirmed running” in another unless this distinction is explicitly encoded. Second is **semantic consistency**: SOC should refer to the same underlying battery-reserve concept whether it appears in simulation plots, PLC logic, API JSON, or dashboard cards. Third is **behavioral consistency**: if the EMS sheds F3, the middleware and dashboard should reflect that as an intentional feeder state change, not merely as a generic loss of load with ambiguous cause. These three forms of consistency are what make the project’s multi-layer architecture intelligible.

The following table provides a useful cross-layer view

Table 4-12 Concept-to-Implementation Mapping Across Simulation, PLC, Middleware, and Dashboard Layers

Concept	Simulation / EMS view	PLC / field view	Middleware / API view	Dashboard view
Battery reserve	SOC trajectory and thresholds.	BMS-derived battery status and analog value handling.	soc in status object.	SOC card / battery percentage display.
Generator state	Generator support variable and active source logic.	RUNNING / FAULT / start-stop command pathways via DSE.	genStatus and control route payloads.	Generator status label and start/stop control widget.
Feeder state	F1supplied, F2supplied, F3supplied outcomes.	Feeder breaker commands and status signals.	f2Status, f3Status, feeder-control endpoint.	Feeder toggles and state indicators.
Source availability	PV input, battery support, generator contribution.	Analog/digital source-related signals and interlocks.	pvPower, activeSource, total state object.	Source card, PV card, active-source display.
Supervisory mode	Automatic/manual EMS logic context.	Mode selector and control-state bits.	mode field in API status.	Mode indicator and related controls.

Table 4-12 is powerful because it shows that the project's variables are not random software labels; they are the cross-layer grammar of the whole system.

This issue becomes even more critical when software layers are added. Middleware, APIs, and dashboards can easily create semantic drift if their variables are not tied back to the authoritative control model. For example, a UI toggle labeled "Generator ON" could create major confusion if the middleware interprets it as a request, while the PLC interprets it as a confirmed run state. A strong architecture avoids this by maintaining disciplined tag definitions and clear distinctions between commands, statuses, setpoints, and alarms. The present project's structure

is well suited to that discipline because its data model is already organized around recognizable categories of state and command.

4.10 HMI and HTML Dashboard Design

The HTML dashboard in this project is the human-facing expression of the microgrid's supervisory layer. It translates the internal state of the hybrid system into a compact, readable, and interactive interface that allows an operator to observe the condition of the PV source, battery reserve, generator status, feeder states, alarms, and selected control parameters in near-real time. In an implementation-oriented microgrid project, this layer is not decorative. It is what turns abstract control logic into an operational tool that supports awareness, intervention, and scenario management.

The dashboard should be understood as a **browser-based HMI/SCADA-style supervisory interface**, not as a certified plant-floor safety console. That distinction matters. The project is deliberately designed so that hard operational authority remains in the PLC, DSE, and field I/O layers, while the dashboard acts as the operator's window into the system and as a limited command surface for approved supervisory actions. This preserves industrial discipline while still providing a modern and accessible user interface.

The design of this HMI is especially important because it serves several audiences at once. For the engineer, it displays the control state in a form that can be tested and explained. For the operator, it provides immediate visibility of operating mode and feeder status. For the thesis reader, it demonstrates that the project is not only a control concept but also a usable supervision environment. This is why the dashboard deserves a deep and structured explanation in Chapter 4.

The project materials already indicate the dashboard's functionality clearly. It shows system mode, battery SOC, generator status, PV output, total load, active source, feeder states, and alarm conditions, and it supports actions such as generator start/stop, feeder toggling, setpoint updates, and alarm reset through a Node-RED-mediated interface. That means the dashboard is not a passive display. It is a **supervisory HMI** with both telemetry and controlled interaction channels.

In academic terms, this section should present the dashboard as the final layer of the project's digital hierarchy: PLC and DSE at the field layer, Node-RED at the middleware layer, and the HTML dashboard at the operator layer. This hierarchy is strong because it separates safety-

critical control from human-readable supervision while keeping them tightly connected through a common data model.

The subsections below explain the dashboard design objectives, the system overview cards, feeder control, generator and EMS parameter interfaces, alarm visualization, user interaction limits, and the integration logic that connects the dashboard to Node-RED.

4.10.1 Dashboard design objectives

The main objective of the dashboard is to make the microgrid **operationally intelligible** to the operator. That means the interface must answer, quickly and clearly, the fundamental questions that matter during supervision: What is the system doing now? Which source is active? How much reserve is available? Which feeder is on or off? Are there alarms? Is manual action needed? A well-designed HMI reduces cognitive load by converting raw data into meaningful state representation rather than forcing the operator to infer system behavior from scattered values.

A second objective is **bounded control**. The dashboard should allow selected intervention—such as feeder enable/disable, generator start/stop requests, and parameter updates—but only through the supervisory flow established by the middleware and PLC logic. In other words, the dashboard must support interaction without becoming a shortcut around safety or control hierarchy. This is why the interface is not built as a direct hardware control panel. It is built as a mediated supervisory layer.

A third objective is **state consistency**. Every value shown on the dashboard should correspond to a clearly defined system tag or status field, and every command issued from the dashboard should be translated by Node-RED into a structured control request rather than an ambiguous input event. This consistency is essential for trust. If a user sees generator status, SOC, or feeder states that do not match the underlying system state, the dashboard loses its value immediately.

A fourth objective is **expandability**. The interface should be able to evolve if the supervisory architecture later becomes more sophisticated, if the system is connected to PLC live data rather than only simulation outputs, or if historical logging and broader analytics are introduced. This is why a web-based dashboard is so appropriate. It can be iterated quickly, extended through Node-RED flows, and adapted without rewriting industrial control logic.

Finally, the dashboard must remain **visually calm but technically rich**. A microgrid operator needs enough detail to understand system condition, but not so much complexity that the interface becomes unreadable under stress. The best HMI designs in this class use clear cards, status indicators, color-coded alarms, and structured control groups to support fast interpretation. That should be the guiding principle in the thesis narrative.

4.10.2 System overview cards and operating state display

The system overview cards are the visual backbone of the dashboard because they provide a compact snapshot of the microgrid's current operating condition. In the project, these cards should present the most important supervisory variables: operating mode, battery SOC, PV power, load demand, generator status, and active source. This format is effective because it allows the operator to understand the state of the entire microgrid at a glance, rather than navigating through detailed menus or text logs.

At minimum, the overview section should include cards for the following items:

- **Operating mode**: automatic or manual.
- **Battery SOC**: current reserve availability.
- **PV power**: current renewable contribution.
- **Total load**: total demand currently being served or requested.
- **Generator status**: running, stopped, starting, or faulted.
- **Active source**: PV, battery, diesel, or mixed support state.

This structure is not arbitrary. It mirrors the actual decision variables of the EMS and presents them in the same semantic order that the control system uses internally. That is a major strength because the dashboard then becomes a readable output of the control architecture rather than an unrelated front-end display.

The most important principle in this subsection is that the overview cards should show **state, not just numbers**. For example, SOC alone is less useful than SOC combined with a visual indication of reserve health. Generator status should not just say "1" or "0"; it should say whether the generator is running, available, starting, or faulted. Active source should similarly reflect the actual supervisory interpretation of the current power state rather than a raw equipment bit. These design choices make the HMI much more operator-friendly and academically stronger.

The operating state display also provides a strong narrative bridge to the EMS logic. If the operator sees that the system is in renewable-led operation, battery support mode, or diesel backup mode, that label should correspond directly to the source priority logic explained earlier in the chapter. This consistency is one of the reasons the dashboard is important in the thesis: it makes the abstract control philosophy visible.

4.10.3 Feeder status and feeder control interface

The feeder section is one of the most important parts of the dashboard because it exposes the load-priority architecture to the operator in an intuitive form. The system uses three feeder classes—F1 critical, F2 medium-priority, and F3 low-priority—and the dashboard should make those states immediately visible and controllable within the allowed supervisory limits. This is especially useful because feeder state is one of the clearest indicators of how the EMS is responding to energy availability.

The feeder display should therefore include both **status indicators** and **control widgets**. Status indicators show whether each feeder is currently enabled, shed, or faulted. Control widgets allow the operator to request enabling or disabling of F2 and F3 under the supervised logic of the middleware layer. In practice, these controls are usually most appropriate for the noncritical feeders, because F1 is protected by priority logic and should not be casually toggled from the operator interface except under explicit supervised conditions.

This subsection should stress that feeder control is not a simple manual switch panel. It is a **supervisory command interface** that sits above the field hardware but below the operator's general visibility layer. The dashboard should therefore send feeder requests through Node-RED, which then applies the system's validation and control rules before any action is propagated to the PLC or simulation state. This design is much more defensible than a direct-toggle interface because it preserves the logic hierarchy of the microgrid.

A good feeder interface should also visually reflect service hierarchy. For example, F1 could be marked as critical and protected, while F2 and F3 could have enable/disable toggles or state buttons with clear labels. In an academic thesis, that visual hierarchy is important because it teaches the reader that feeder control in the project is not random. It is based on load criticality and energy availability. If a feeder is shed, the interface should make the cause understandable—whether it was EMS-driven priority shedding, manual override, or fault-based disconnection.

If you want the dashboard explanation to be especially strong, this subsection should also mention **operator confirmation behavior**. A command sent from the dashboard should not be assumed to have taken effect until the middleware and field response confirm it. That ensures the feeder interface is not just an input panel but part of a genuine supervisory closed loop.

4.10.4 Generator control and EMS parameter interface

The generator control section of the dashboard is where the user experiences the boundary between supervision and equipment authority most clearly. The interface should allow an operator to request generator start or stop, but the actual execution of those actions must still pass through the Node-RED middleware and the PLC/DSE control path. This preserves the industrial hierarchy established elsewhere in the thesis.

The generator section should show at least three things: current generator state, available control action, and any related limitations or warnings. For example, if the generator is running automatically, the interface should clearly show that the generator is in backup support mode. If it is stopped, the operator should see whether it is available, inhibited by fault, or simply not needed. This makes the interface useful not only for control, but for diagnosis and understanding.

The EMS parameter section is equally important. The project includes threshold-driven logic such as SOC upper and lower bounds, feeder control conditions, and other supervisory parameters that define when battery support is allowed, when generator support is triggered, and when certain restoration actions should occur. These parameters may be exposed in a controlled manner through the dashboard so that an authorized user can update them via the middleware. That said, they should never be presented as unrestricted open fields with no validation. The dashboard should clearly distinguish between adjustable supervisory setpoints and protected system logic.

In a strong academic description, this subsection should emphasize the distinction between **commands** and **setpoints**. Commands are immediate actions, such as start generator or stop feeder. Setpoints are supervisory thresholds that shape future behavior, such as SOCmin or SOCmax. Mixing those categories would weaken the architecture. Separating them makes the design much more professional and easier to defend.

It is also worth noting that generator control and EMS parameter editing should be protected by mode awareness and user permission discipline. In a real supervisory environment, not every operator should have the same authority to modify thresholds or dispatch sources. Even if the

thesis does not implement a full authentication system, the design narrative should still acknowledge this principle. That will make the chapter sound much more mature.

4.10.5 Alarm visualization and operator awareness

Alarm visualization is one of the most important HMI functions because it transforms abnormal system conditions into immediate operator awareness. In a hybrid microgrid, alarms may involve generator faults, inverter faults, battery warnings, feeder disconnections, communication issues, or supervisory exceptions. If these events are not displayed clearly, the operator may not understand why the system changed state, and the dashboard loses one of its core safety and usability functions.

The alarm area of the dashboard should therefore present active alarms in a clear, color-coded, and preferably ranked format. High-severity events—such as generator fault or emergency-related states—should stand out visually. Medium-severity events—such as temporary feeder shedding or battery low reserve—should be displayed in a way that is noticeable but not visually overwhelming. Informational states should remain distinguishable from true alarms. This layered display approach improves situational awareness significantly.

The dashboard should also help the operator understand **why** the alarm exists, not just that an alarm exists. For example, if the generator is unavailable, the interface should clearly indicate whether it is because of a fault condition, a start failure, or another operational warning. If a feeder is off, the user should be able to see whether the cause is EMS priority shedding, manual disable, or device fault. This reduces guesswork and improves trust in the system.

The alarm section is not only about visual notification. It is also about workflow. Operators typically need a way to acknowledge, inspect, and reset alarms once the condition has been handled. That is why the dashboard includes alarm reset capability through the Node-RED supervisory layer. This capability should be described carefully as an **operator acknowledgement function**, not as a way to erase field faults. The real fault remains until the underlying condition is resolved; the dashboard simply allows the supervisory record to be cleared or acknowledged in an orderly way.

Alarm visualization is especially important in this thesis because it reinforces the broader control philosophy. The system is not intended to hide abnormal conditions. It is intended to make them visible enough that the operator can act responsibly while the automatic system

continues to preserve critical service as far as possible. That is a very strong message for an academic design chapter.

4.10.6 User interaction limits and supervisory safety

A well-designed HMI must clearly define what the user **can** do and what the user **cannot** do. This section is essential because it shows that the dashboard is a bounded supervisory interface, not an unrestricted remote control of the entire plant. In industrial systems, that boundary is what protects the integrity of the control hierarchy and prevents software convenience from becoming operational risk.

The project's dashboard should therefore limit user interaction to approved supervisory actions such as viewing system state, requesting generator start/stop, enabling or disabling selected feeders, updating allowed setpoints, and acknowledging alarms. It should not allow the user to bypass the PLC, ignore DSE protection, or directly manipulate field devices through arbitrary browser-side logic. Those protections belong to the field control architecture and must remain there.

A strong thesis should also explain that user actions are subject to **mode and permission logic**. For example, some commands may only be allowed in manual mode, while automatic mode may restrict certain interventions to protect system stability. Likewise, commands affecting F1 should be far more constrained than commands affecting F2 or F3 because F1 is the critical feeder group. This shows that the dashboard is not flat access; it is structured access.

The dashboard should also preserve **feedback after action**. If a command is issued, the user should receive confirmation through the updated status display rather than assuming the action succeeded instantly. That is especially important in microgrid control because command success can be affected by alarms, protection states, or communication delays. This makes the interface safer and more credible.

Another important aspect is that the dashboard should not create false authority. A web interface can look powerful, but its power must be bounded by the architecture beneath it. The thesis should therefore explain clearly that the dashboard is an interface to the system, not the source of system truth. Node-RED and the PLC/DSE layers remain the authoritative chain for field behavior. That is a strong and correct design principle.

4.10.7 Dashboard-to-Node-RED integration logic

The dashboard-to-Node-RED integration logic is the technical link that allows the HMI to function as a live supervisory client rather than as a static page. The project architecture indicates that the dashboard operates through Node-RED as its backend and that it periodically fetches system status, updates visible elements, and sends control requests back through structured routes. This is exactly the right architecture for a web-based industrial HMI because it keeps the front end thin and the backend authoritative.

The key mechanism is the status-request loop. The dashboard periodically calls the Node-RED status endpoint, receives a structured JSON object, and uses that object to update the cards, switches, alarm labels, feeder states, and setpoint fields. This pattern is efficient and easy to explain academically because it preserves a single source of truth for the UI: Node-RED is what the dashboard reads, not a separate local cache or hidden direct device connection.

Control requests travel in the opposite direction. When the operator clicks a generator button, toggles a feeder, or updates a threshold, the dashboard sends a request to Node-RED, which then validates and translates that request into the appropriate supervisory control logic. This prevents the dashboard from directly acting on the field system. It is a mediated interaction model, and that is the correct design for the project.

This logic can be summarized in the following table 4-13:

Table 4-13 Direction and Control Flow Between Dashboard, Node-RED, and PLC

Direction	Mechanism	Role
Node-RED → Dashboard	Status polling / JSON response.	Pushes current microgrid state into the interface.
Dashboard → Node-RED	Control request / POST action.	Sends bounded operator intent for validation and execution.
Node-RED → PLC / field logic	Supervisory command translation.	Converts UI intent into control-consistent actions.

The strongest thing about this structure is that it keeps the HMI cleanly separated from the field authority while still making it fully operational. The dashboard is therefore best described

as a **supervisory client with mediated control capability**. That wording is strong, precise, and academically defensible.

4.11 Integration Architecture Across Real and Digital Layers

One of the strongest technical qualities of the proposed microgrid project is that it is not developed as a disconnected set of electrical drawings, simulation files, PLC concepts, and dashboard screenshots. Instead, the design reveals a coherent **integration architecture across real and digital layers**, in which physical devices, supervisory logic, signal pathways, middleware services, and operator-facing interfaces are linked through a common functional structure. This section is therefore not a repetition of earlier subsections. It is the architectural synthesis that demonstrates how the system operates as a whole.

This synthesis is especially important in implementation-oriented research because hybrid microgrid projects often fail not at the level of individual components, but at the level of integration. A generator may be correctly sized, a PLC may be correctly selected, and a dashboard may be visually attractive, yet the system still remains weak if device roles are unclear, tags are inconsistent, monitoring paths are mistaken for control paths, or simulation behavior has no credible relationship to real implementation. The present project addresses these risks by establishing a layered but coordinated architecture that distinguishes functions without isolating them.

From a control-architecture perspective, the project can be understood as a hierarchy of interacting layers. At the physical layer are the energy assets and switching elements: the PV subsystem, battery bank, inverter/charger, diesel generator, feeder breakers, sensors, and actuators. At the control-execution layer are the Siemens S7-1200 PLC, the DSE8610 MKII, and associated digital and analog I/O pathways that translate supervisory intent into deterministic field action. At the supervisory software layer are Node-RED, the REST API structure, and MQTT-based integration concepts that normalize and distribute data. At the operator layer is the HTML dashboard that visualizes system state and provides bounded user interaction. Alongside these real-system layers is the simulation layer, which represents the same microgrid behavior in MATLAB/Simulink form for EMS validation and design analysis.

This layered view is strongly aligned with wider engineering practice. Industrial automation architectures commonly separate field-level devices, controller-level execution, supervisory

monitoring, and human-machine interfaces into distinct but connected layers, each with its own timing, responsibility, and data abstraction level. Microgrid control frameworks similarly distinguish between local equipment functions and higher-level dispatch or supervisory coordination. The present project fits this pattern well, but it does so in a way tailored to a rural off-grid hybrid microgrid rather than a large utility-scale or enterprise-wide deployment.

This section is also methodologically important because it clarifies the relationship between the **real implementation architecture** and the **digital representation architecture**. The project is not claiming to have built a full digital twin continuously synchronized to a commissioned physical plant. Instead, it develops a simulation-oriented and middleware-aware digital architecture that is functionally consistent with the intended real system. That distinction is academically powerful because it protects the project from overstatement while still allowing it to demonstrate genuine integration readiness.

The subsections below therefore examine the mapping of devices to functions, the tag relationships across PLC, DSE, Node-RED, and the dashboard, the distinction between control paths and monitoring paths, the difference between real implementation architecture and simulation architecture, and finally the overall consistency and implementation readiness of the design.

4.11.1 Device-to-function mapping

A robust integration architecture begins with a clear understanding that every major device in the microgrid must have a defined function within the overall system, not merely a physical presence in the bill of materials. Device-to-function mapping is therefore one of the most important architectural tools in the chapter. It explains **why each device exists in the system**, what operational role it fulfills, and how it contributes to the behavior of the hybrid microgrid as an integrated whole. Without such mapping, the project would risk appearing as a collection of hardware blocks rather than a coordinated control structure.

The EPLAN architecture provides the basis for this mapping very clearly. The PV array and its associated DC-side components function as the primary renewable energy source. The battery bank and BMS function as the energy-buffering and reserve-management subsystem, providing bounded charge/discharge support and battery-status awareness. The inverter/charger functions as the power-conversion interface between the DC and AC domains, enabling both supply to the AC bus and charging from generator-supported AC conditions. The diesel generator functions as the dispatchable backup source that preserves critical service continuity when PV and battery

support are insufficient. The feeder breakers and feeder groups function as the controlled load-priority branches through which service continuity can be differentiated across F1, F2, and F3.

At the control layer, the mapping is equally important. The Siemens S7-1200 PLC functions as the main supervisory execution controller, converting EMS logic into field-coordinated commands and state evaluation. The DSE8610 MKII functions as the local generator-control layer, handling generator-specific start/stop sequencing, supervision, trips, and status return. Node-RED functions as the middleware and supervisory integration layer, not as a field controller, while the HTML dashboard functions as the operator-facing HMI through which system state is observed and limited commands are issued. Each of these roles is different, but none is redundant. Their interaction is what gives the project architectural completeness.

This kind of mapping is strongly supported by hierarchical control thinking in both microgrid and industrial automation literature. Higher-level controllers supervise assets, lower-level controllers handle equipment-specific control, and supervisory software manages visibility and coordination across those layers. The present project reflects this principle well because it does not blur the roles of its devices. The PLC is not misrepresented as a UI tool, the dashboard is not misrepresented as a real-time controller, and the DSE is not reduced to a passive sensor node.

The following table provides a concise device-to-function synthesis suitable for the project.

Table 4-14 Device-to-Function Integration Map for the Proposed Microgrid

Device / layer	Primary function	Integration significance
PV array + DC generation path	Primary renewable energy production.	Establishes renewable-first operating basis for the EMS.
Battery bank + BMS	Energy buffering, reserve management, SOC supervision.	Supports critical-load continuity and renewable energy shifting.
Inverter/charger	DC/AC conversion and generator-assisted charging path.	Couples renewable-storage side to AC service side.
Diesel generator + DSE	Dispatchable backup generation and local generator supervision.	Protects critical loads under renewable/storage insufficiency.
Feeders F1/F2/F3 + breakers	Priority-structured load delivery and shedding interface.	Converts service hierarchy into real controllable branches.
Siemens S7-1200 PLC	Supervisory decision execution and field coordination.	Central control authority for implementation logic.
Node-RED	Middleware, API mediation, data normalization, integration.	Connects supervision, dashboard, and future software integration.
HTML dashboard	Operator visibility and bounded supervisory interaction.	Makes the system intelligible and interactable at the HMI level.
MATLAB/Simulink EMS model	Behavioral analysis and simulation-oriented control validation.	Represents digital verification layer aligned with design intent.

Table 4-14 is useful because it shows that the project's architecture is functionally intentional at every level. Each device is there for a reason, and those reasons fit together into a system rather than a parts list.

4.11.2 Tag mapping across PLC, DSE, Node-RED, and dashboard

If device-to-function mapping explains what each component does, tag mapping explains **how those components speak to one another**. In integrated control systems, especially those spanning industrial hardware and supervisory software, coherent operation depends heavily on a consistent cross-layer vocabulary of statuses, commands, measurements, and setpoints. Without that vocabulary, integration becomes brittle: the same signal may be interpreted differently in different layers, and the system may appear correct physically while remaining semantically inconsistent.

The project already contains the foundations of a strong tag architecture. On the field-control side, the PLC interacts with signals such as generator running, generator fault, inverter ready, inverter fault, battery low, battery high, battery alarm, feeder states, emergency stop, and mode selector information. On the DSE side, generator-local states include RUNNING, FAULT, remote start input, and alarm-related control or feedback pathways. On the middleware side, Node-RED organizes supervisory variables such as mode, soc, genStatus, pvPower, totalLoad, activeSource, f2Status, f3Status, alarms, and threshold setpoints. On the dashboard side, these same variables are rendered visually as cards, labels, toggles, and alarm entries.

This is exactly the kind of cross-layer semantic structure that a strong implementation-oriented design requires. The objective is not that every layer use identical low-level naming syntax, but that all layers share the same **operational meaning**. For example, generator status should preserve a meaningful distinction between command request, confirmed running, and fault state across the architecture. Similarly, SOC should carry the same interpretation of battery reserve whether it appears in simulation results, Node-RED JSON responses, or dashboard cards. This semantic continuity is one of the most important indicators of design maturity.

The following table is especially valuable for the thesis because it shows the cross-layer nature of the integration architecture directly.

Table 4-15 Comparison Between Control Path and Monitoring Path

Operational concept	PLC / field tag view	DSE / device view	Node-RED / API view	Dashboard view
Generator state	GenRunning, GenFault, start/stop outputs.	RUNNING, FAULT, remote start, alarm state.	genStatus, generator control route.	Generator card, status indicator, start/stop control.
Battery reserve	Analog/BMS-related SOC and battery alarms.	Not primary DSE function.	soc, thresholds, alarms.	SOC card, battery reserve display.
PV contribution	Analog PV measurement path.	Not primary DSE function.	pvPower.	PV power card.
Feeder state	Feeder commands, feeder feedback, interlock bits.	Not primary DSE function.	f2Status, f3Status, feeder control route.	Feeder state indicators and toggles.
Supervisory mode	Mode selector / mode bit logic.	Auto/manual controller context where relevant.	mode.	Mode label / control context display.
Alarms	Field and control alarm signals.	Alarm relays, generator trips.	Alarm list / reset route.	Alarm panel, acknowledgement interaction.

Table 4-15 does more than organize names. It demonstrates that the project has a coherent **information architecture** spanning the physical, logical, and visual layers. That is highly valuable in a thesis because it proves the system can be reasoned about consistently across domains.

4.11.3 Control path versus monitoring path

One of the most important integration distinctions in the entire project is the separation between the **control path** and the **monitoring path**. This distinction appears repeatedly in earlier sections—especially in relation to the DSE, PLC, Node-RED, and dashboard—but here it must be synthesized explicitly because it defines how authority and visibility are distributed across the

architecture. A system that confuses monitoring with control quickly becomes ambiguous and potentially unsafe. A system that separates them properly becomes much easier to implement, defend, and extend.

The control path is the path through which actual field-affecting commands are executed. In the present project, this path runs through the PLC, the DSE, field digital outputs, breaker commands, inverter enable/mode signals, battery contactor commands, and related interlocks. It is deterministic, equipment-aware, and responsible for real operational consequences. When the generator starts, when a feeder is shed, or when a contactor changes state, that behavior belongs to the control path. This path is therefore tied to execution authority.

The monitoring path, by contrast, is the path through which system state is observed, published, and interpreted. It includes measurement acquisition, status reporting, Modbus-based reading where applicable, Node-RED state normalization, API status exposure, MQTT publication concepts, and dashboard visualization. Its purpose is not to force the plant into action directly, but to make the plant's current condition intelligible to operators and software clients. This path is therefore tied to visibility and interpretability rather than direct actuation.

The DSE integration offers a particularly clear example of this distinction. The project documentation describes digital I/O pathways for start/stop execution and status return between PLC and DSE, while Modbus TCP is used primarily for monitoring generator values such as kW, voltage, frequency, alarms, and run hours. This is good industrial practice because start/stop authority remains on a deterministic control channel while richer telemetry remains available through a more flexible monitoring channel. The same principle carries upward: Node-RED and the dashboard monitor and mediate, but they do not replace the PLC/DSE control path.

This distinction is also well aligned with broader industrial automation thinking, where command flow typically moves downward through control layers while feedback and state information move upward through monitoring and supervisory layers. The project's architecture follows exactly that pattern. Commands go from supervisory intent toward PLC or device execution. Feedback goes from field status upward to middleware and HMI. Such symmetry makes the architecture easier to reason about and easier to validate.

The difference can be summarized concisely in the following table.

Table 4-16 Comparison Between Control Path and Monitoring Path

Aspect	Control path	Monitoring path
Main purpose	Execute field-affecting actions.	Expose system condition and measurements.
Typical elements	PLC outputs, DSE start/stop, feeder command outputs, inverter enable/mode, contactor control.	Status signals, analog values, Modbus readings, Node-RED state object, dashboard display.
Architectural authority	Operational and deterministic.	Supervisory and interpretive.
Typical direction	Downward toward field execution.	Upward toward supervision and visibility.
Example in project	Generator start command from PLC to DSE.	Generator run/fault/telemetry displayed in dashboard through Node-RED.

As shown in table 4-16, this subsection should emphasize that keeping these two paths distinct is one of the main reasons the project remains technically disciplined. It protects the architecture from overstatement and preserves a correct understanding of where commands originate, where they are validated, and where status becomes visible.

4.11.4 Real implementation architecture versus simulation architecture

A central methodological strength of the project is that it distinguishes carefully between the **real implementation architecture** and the **simulation architecture**, while also showing that they are functionally consistent with one another. This distinction is essential because implementation-oriented academic work must avoid two opposite errors: claiming too little, as if the work were only an abstract simulation, or claiming too much, as if a fully commissioned field system had already validated every modeled scenario. The present project avoids both extremes by maintaining a disciplined relationship between real and digital layers.

The real implementation architecture is represented by the EPLAN power and control design, the PLC-centered control mapping, the DSE control structure, feeder breaker organization, and the supervisory software concept that includes Node-RED and the HTML dashboard. This architecture defines the physical devices, their interfaces, their command pathways, and the layered structure through which real operation would be coordinated. It is implementation-

oriented because it shows how the system would exist and interact as a real engineered installation.

The simulation architecture, by contrast, is represented through the MATLAB/Simulink EMS model and associated logic representations used to study source priority, battery SOC behavior, generator dispatch, feeder continuity, and system response under defined operating conditions. This architecture is not a live digital twin synchronized continuously to field data. It is a behavior-oriented simulation model used for analysis and validation of the supervisory EMS concept. That distinction is important because simulation and digital twins are not the same thing: simulation typically operates on designed or predefined scenarios, while digital twins rely on continuous bidirectional coupling with the physical asset.

Even so, the two architectures are tightly related. The simulation's feeder-priority logic corresponds to the real feeder breaker structure and PLC feeder-control pathways. The simulation's generator support logic corresponds to the real PLC-to-DSE command chain and the generator's AC-bus connection. The simulation's battery SOC logic corresponds to the real battery/BMS operating envelope and inverter/charger integration. In this sense, the architectures are not identical, but they are **functionally homologous**: each one represents the same system at a different layer of purpose and abstraction.

This distinction is academically strengthened by broader discussions on simulation versus digital-twin concepts. Sources comparing the two emphasize that simulations are generally used to test scenarios and analyze designed behavior, while digital twins are continuously updated representations tied closely to live operational data. The present thesis clearly belongs to the former category, though it incorporates design choices that could support future evolution toward a more tightly integrated live supervisory environment. That is a strong and honest position.

This subsection should therefore state clearly that the project uses a **simulation architecture for control validation** and a **real implementation architecture for design realization**. Their value lies in consistency, not identity. The simulation does not replace the real architecture, and the real architecture does not invalidate the simulation. Each supports the other.

4.11.5 Design consistency and implementation readiness

The final test of a strong integration architecture is whether the project demonstrates both **design consistency** and **implementation readiness**. Design consistency means that the electrical architecture, EMS philosophy, PLC logic, DSE integration, middleware structure, dashboard

behavior, and simulation model all reflect the same underlying system logic. Implementation readiness means that the project has progressed beyond abstract intention and into an architecture where devices, interfaces, tags, control paths, and supervisory interactions are sufficiently defined to support realistic future realization.

The present project performs strongly on both dimensions. Design consistency is visible in the way the source hierarchy appears across all layers: PV is primary in the simulation, in the power architecture, and in the control philosophy. Battery operation is bounded by SOC both in the model and in the battery/BMS design view. Diesel support is reserved for critical continuity both in the EMS logic and in the DSE/PLC implementation architecture. Feeder priority exists both as a simulation concept and as a real breaker-based load architecture. Node-RED and the dashboard expose the same supervisory meanings already embedded in the control design. This degree of cross-layer consistency is one of the strongest features of the chapter.

Implementation readiness is also evident in several concrete ways. The devices are specified and assigned clear functions. The PLC hardware platform, I/O structure, and communication pathways are identified. The DSE role and generator command path are defined. The middleware and API architecture are documented conceptually with routes, status models, and MQTT extensions. The dashboard interaction model is described with clear control and status responsibilities. The simulation model already reflects the supervisory logic to be implemented. These elements do not amount to final site commissioning, but they do amount to a strong level of implementation-oriented readiness.

It is important, however, to describe this readiness precisely. The project is **implementation-ready at the architectural and supervisory level**, not fully commissioned at the field-validation level. That means the thesis demonstrates realistic pathways for control execution, monitoring, device coordination, and operator interaction, but it does not claim that all vendor parameterization, hardware acceptance testing, cybersecurity hardening, or final commissioning documentation have been completed. This distinction is not a weakness. It is a sign of academic rigor and proper scope control.

From a broader engineering viewpoint, this is exactly the type of readiness appropriate for a bachelor-level implementation-oriented microgrid thesis. A project at this stage should show that the system can be reasoned about coherently, mapped onto real devices, and supported by a plausible supervisory software stack. It need not claim EPC-level completion to be valuable.

What matters is that the design has crossed the line from conceptual aspiration into structured realizability.

This final subsection should therefore close Chapter 4 with a strong statement: the project presents a microgrid architecture whose real and digital layers are not merely compatible, but mutually reinforcing. The result is a design that is coherent in theory, consistent in representation, and credibly prepared for implementation-oriented development beyond the simulation stage.

4.12 Chapter Summary

This chapter presented the complete design architecture of the proposed hybrid microgrid, moving from physical system definition to supervisory control structure, software integration, and implementation-oriented coordination across real and digital layers. The chapter showed that the project is not limited to source sizing or isolated EMS logic, but is instead organized as a multi-layer engineering system in which PV generation, battery storage, diesel backup, inverter/charger operation, feeder prioritization, PLC execution, DSE generator control, Node-RED middleware, and the HTML dashboard all contribute to one coherent operational framework.

At the electrical and functional levels, the chapter defined the main hardware architecture of the microgrid and clarified how PV, battery, inverter, generator, AC/DC buses, and prioritized feeders are arranged to support reliable off-grid operation. The feeder structure was modeled through F1, F2, and F3 classes so that critical, medium-priority, and low-priority community loads could be supplied according to service importance rather than treated as a single undifferentiated demand block. This feeder-based structure established the basis for both resilience-oriented load management and realistic supervisory decision-making in the EMS.

At the control level, the chapter explained how the supervisory EMS philosophy is translated into executable implementation logic. The PLC was positioned as the central control-execution platform, responsible for coordinating field commands and status evaluation, while the DSE8610 MKII was defined as the local generator controller that executes start/stop behavior and returns generator health and operating states to the higher-level control architecture. This created a realistic industrial split between plant-level coordination and equipment-level control, strengthening both safety and implementation credibility.

At the supervisory software level, the chapter showed how Node-RED functions as a middleware and integration layer rather than a hard real-time controller. Through REST APIs,

structured status publishing, and MQTT-oriented extensibility, Node-RED was presented as the software layer that normalizes system state, mediates supervisory commands, and connects the control architecture to the operator-facing HMI. The HTML dashboard was then framed as a browser-based supervisory interface that provides situational awareness, selected operator interaction, alarm visibility, feeder control, generator command access, and setpoint management without bypassing the underlying control hierarchy.

The chapter also demonstrated that the project maintains an important distinction between monitoring and control, and between real implementation architecture and simulation architecture. Control authority remains rooted in the PLC, DSE, field I/O, and device pathways, while monitoring and operator visibility are delivered through Node-RED, APIs, MQTT, and the dashboard. Likewise, the MATLAB/Simulink EMS model was treated as a behavior-oriented simulation layer aligned with the real design architecture, not as a claim of fully commissioned real-time digital-twin deployment. This distinction gives the project methodological clarity and protects it from both overstatement and underrepresentation.

Finally, the chapter established that the design is consistent across hardware devices, software tags, control pathways, feeder logic, and supervisory interfaces. Device-to-function mapping, cross-layer tag consistency, feeder-priority logic, generator command structure, and dashboard integration all reflect the same operational philosophy: preserve critical loads first, coordinate available energy intelligently, and expose system behavior through a structured supervisory architecture. For this reason, Chapter 4 provides not only a design description, but also the implementation-oriented foundation required for the next chapter to discuss realization, interfacing, testing, and operational verification of the proposed microgrid system.

Chapter 5

Test and Implementation

5. Chapter 5

5.1 Introduction

This chapter presents the implementation-oriented realization and test validation of the proposed hybrid microgrid energy management system. While Chapter 4 established the design architecture across the electrical, control, middleware, and supervisory layers, the present chapter demonstrates how that architecture was translated into a working test structure using MATLAB/Simulink, rule-based EMS logic, Node-RED middleware, REST APIs, MQTT-oriented status publishing, and an HTML-based supervisory dashboard. In this sense, Chapter 5 serves as the bridge between system design and operational verification.

The objective of this chapter is not to claim full industrial commissioning of the microgrid. Rather, it is to show that the proposed control philosophy is implementable, testable, and internally consistent across its real and digital representations. This distinction is important. The work presented here lies in the domain of implementation-oriented academic validation, where the goal is to verify that the control architecture, feeder priority model, generator logic, battery SOC behavior, middleware pathways, and supervisory interface operate coherently under representative scenarios, even though final site commissioning, protection certification, and field acceptance testing remain beyond the project scope.

The chapter is organized around three major validation layers. The first is the **EMS simulation layer**, where the rule-based dispatch and load-priority logic are tested using MATLAB/Simulink and the project's EMS scripts. The second is the **middleware and communication layer**, where Node-RED is used to structure status exchange, publish supervisory state, and expose REST endpoints for monitoring and control. The third is the **HMI layer**, where the HTML dashboard is tested as a browser-based supervisory interface for state visualization, feeder control, generator commands, alarm handling, and EMS parameter interaction.

The importance of this chapter lies in the fact that implementation quality is not measured only by whether a model produces graphs. It is measured by whether the model can be connected to understandable control pathways, supervisory interfaces, and operator-facing interactions that reflect the intended architecture of the real system. For this reason, the chapter combines software execution, interface testing, functional validation, and multi-scenario interpretation rather than treating simulation and implementation as isolated tasks.

The following sections therefore discuss the implementation methodology, the test strategy adopted for each architectural layer, the validation of the MATLAB/Simulink EMS, the verification of Node-RED middleware and REST APIs, the dashboard integration behavior, the scenario-based results, and the final interpretation of the implementation readiness achieved by the project.

5.2 Implementation Methodology and Test Strategy

The implementation methodology adopted in this project follows a layered validation approach. Instead of attempting to validate the entire microgrid as one indivisible block, the system was tested progressively across the simulation layer, middleware layer, and supervisory HMI layer. This approach is appropriate for a hybrid microgrid architecture because the project itself is layered: MATLAB/Simulink is used to represent operational behavior, Node-RED is used to mediate and publish state, and the HTML dashboard is used to visualize and interact with that state.

At the simulation level, the primary concern was to verify the correctness of the EMS logic. This included testing renewable-first dispatch, battery charge/discharge behavior, generator start conditions, feeder priority supply, and SOC updating within defined upper and lower bounds. At the middleware level, the concern shifted toward information flow and supervisory control behavior. Here, the goal was to confirm that system state could be normalized, returned through GET /api/status, manipulated through POST /api/control/... routes, and prepared for future MQTT-based integration. At the dashboard level, the objective was to verify that the HMI could correctly display live supervisory variables, reflect feeder and source state, send user commands, and present alarms and configurable parameters in a controlled manner.

The adopted test strategy was therefore **functional rather than certification-based**. This means the study did not attempt to reproduce factory acceptance testing, site acceptance testing, protection relay calibration, electromagnetic compatibility testing, or detailed hardware-in-the-loop certification. Instead, it focused on whether the major intended functions of the proposed system work correctly and coherently from the perspective of energy management and supervisory operation. For a bachelor-level implementation-oriented microgrid thesis, this is an appropriate and academically defensible validation boundary.

A further methodological point concerns the time scale of testing. The EMS model operates over a 24-hour horizon with hourly resolution, and therefore the tests are oriented toward **daily**

operational behavior rather than millisecond transients or converter control loops. This is consistent with the purpose of the project, which is to evaluate supervisory dispatch logic, feeder continuity, generator dependence, and reserve management rather than power-electronic switching dynamics. As a result, the tests should be interpreted as validation of the supervisory layer of the microgrid, not of all lower-level physical phenomena.

The chapter uses four representative operating scenarios to test the design across different system conditions. These scenarios are intended to capture the major operating modes of the hybrid microgrid, including favorable renewable conditions, constrained renewable conditions, battery-supported operation, generator-supported operation, feeder curtailment, and manual/supervisory interactions. Each scenario is not treated merely as a numerical case, but as a behavioral case through which the logic of the system can be observed and explained. This is important because the value of scenario testing lies not only in the resulting plots, but in the interpretation of why the system behaved that way.

The overall test strategy can be summarized as following table 5-1:

Table 5-1 Test Layers, Objectives, and Evidence Sources for System Validation

Test layer	Main objective	Evidence source
EMS simulation layer	Verify dispatch logic, feeder priority, SOC behavior, generator activation, and power balance.	MATLAB/Simulink model, EMS scripts, simulation plots.
Middleware layer	Verify status publication, REST routes, feeder/generator control handling, and MQTT readiness.	Node-RED flow design and API logic.
HMI layer	Verify dashboard display, control widgets, alarm presentation, and user interaction pathways.	HTML dashboard screenshots and JavaScript/API behavior.
End-to-end scenario layer	Verify integrated system behavior under representative operating cases.	Scenario outputs, interface states, and result interpretation.

This methodology is strong because it respects the architecture of the system itself. The tests do not treat the project as a single black box, but as a coordinated multi-layer implementation whose correctness depends on the consistency of all its constituent parts.

5.3 MATLAB/Simulink EMS Implementation

The core implementation of the supervisory energy management logic was carried out in MATLAB/Simulink using a rule-based EMS structure supported by script-based profile generation and state-based dispatch logic. This implementation forms the behavioral heart of the project because it is the layer where the source allocation rules, feeder priority logic, battery SOC evolution, and generator intervention strategy are actually executed and tested.

The EMS implementation is built around a 24-hour simulation horizon using hourly time steps. The principal model constants include a PV maximum of 150 kW, a battery capacity of 100 kWh in the simplified simulation logic, a battery power limit of 60 kW, a generator rated power of 100 kW, and SOC operating bounds of 30% and 90%. These values are consistent with the broader design framework established earlier and are sufficient to study the supervisory behavior of the proposed hybrid microgrid.

The Simulink realization integrates the major demand channels and supervisory inputs into a single structured model. The uploaded model image shows the main EMS block receiving PV power, feeder demands, SOC input, generator command signals, generator mode, and feeder enable commands, and returning outputs such as feeder supplied power, battery power, generator power, SOC next value, and active source state. This arrangement is technically important because it demonstrates that the EMS was not left as an isolated textual description; it was embedded into an executable simulation structure with clear I/O relationships.

At the script level, run_ems_matlab.m implements a complete rule-based dispatch process beginning with clipping user inputs to rated bounds, generating base PV and load profiles, scaling the feeder demands, and then stepping through the EMS logic hour by hour. The underlying load construction includes residential demand, hospital demand, school demand, water pumping, mosque timing, and shops, which are then aggregated into feeder categories. This is a notable strength because it gives the simulation social realism: the feeder logic corresponds to actual community services rather than to arbitrary synthetic numbers.

The first major step in the implementation is renewable-first and priority-first supply. In each time step, the available PV power is allocated first to F1 critical demand, then battery discharge is used if F1 remains partially unmet and the SOC is above minimum, and only after that is the generator engaged to preserve remaining unsupplied F1 load. After F1 is handled, the model

distributes any remaining PV power to F2 and then F3, preserving the feeder priority hierarchy established in the design chapter. This structure is a direct operational translation of the project's service-continuity philosophy.

The EMS also includes controlled charging logic. When PV remains after serving the active feeders, the battery is charged subject to SOC upper bound and battery power limits. Battery charging is represented by negative battery power, while battery discharge is represented by positive battery power, and the SOC is updated accordingly using a bounded energy-balance calculation. This sign convention and SOC update structure are useful because they allow the supervisory energy movement to be visualized clearly in both the plots and the dashboard layer.

The implementation further supports generator and feeder command integration through the Stateflow/logic layer you provided. In that logic, generator operation can be governed either automatically or manually, depending on the mode command, and the generator start/stop command behavior is explicitly separated from the automatic need-based activation condition. The same logic also includes F2 and F3 enable commands, allowing lower-priority feeder supply from generator reserve when permitted. This is an important implementation detail because it connects the purely automatic EMS with the later supervisory interface and manual override concepts.

Overall, the MATLAB/Simulink EMS implementation may be described as a **behavior-oriented supervisory execution model**. It does not attempt to model every physical transient or equipment internal controller, but it does provide an executable and interpretable representation of how the proposed hybrid microgrid should decide source usage, feeder continuity, reserve management, and generator support over the course of a representative operating day.

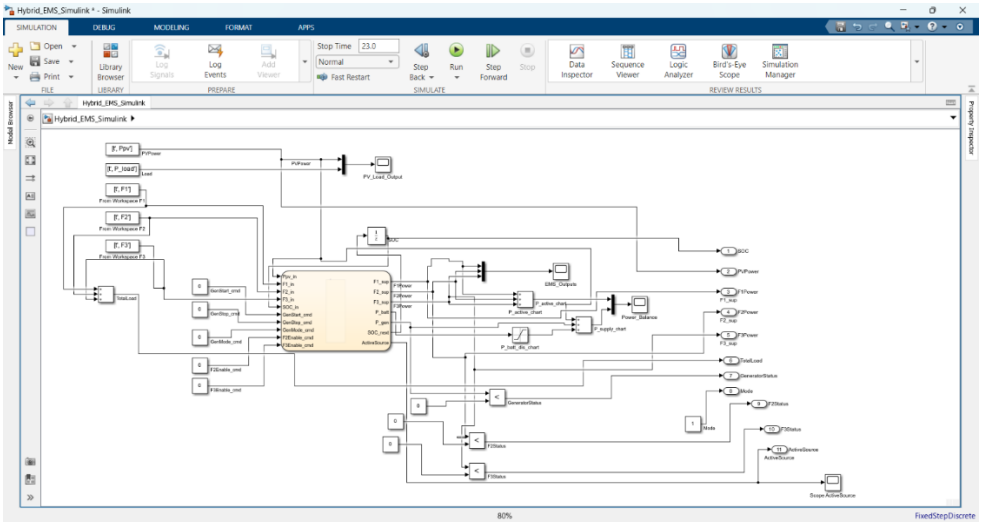


Figure 5-1 MATLAB/Simulink EMS Model for the Hybrid Microgrid

5.4 EMS Control Logic and Functional Verification

Beyond implementing the model structurally, it is necessary to verify that the EMS logic itself is coherent and functionally correct. The logic embedded in the project follows a sequential rule-based structure, and its correctness can therefore be evaluated by checking whether each intended decision rule is triggered under the appropriate operating condition. This type of validation is especially suitable for supervisory EMS design, where the key question is not whether the model solves a global optimization problem, but whether it applies the intended operational priorities consistently and predictably.

The first verified rule is the **priority protection of F1**. The EMS always attempts to supply F1 using PV first, then battery discharge, and finally generator support if required. This confirms that critical demand remains the anchor of the dispatch sequence and that lower-priority feeders are not allowed to consume scarce power while F1 remains unsatisfied. From an engineering perspective, this is the most important functional rule in the entire model because it embodies the service-critical philosophy of the microgrid.

The second verified rule is the **conditional treatment of F2 and F3**. These feeders are supplied by remaining PV after F1 has been addressed, and in the enhanced command logic they may also receive support from remaining generator capacity when enabled through supervisory commands. This confirms that the model can represent both automatic energy-priority behavior and operator-mediated feeder restoration behavior. Such a feature is valuable because it makes the project more realistic than a simple fully automatic single-mode dispatch script.

The third verified rule is the **bounded battery operation**. Battery discharge is allowed only when the battery SOC is above minimum, and battery charging is allowed only when SOC is below maximum. The power transferred during charging or discharging is also restricted by the battery power limit and by the amount of energy that can physically be moved within the one-hour step. This is important because it prevents the EMS from behaving unrealistically, such as discharging energy that is no longer available or charging beyond the upper reserve bound.

The fourth verified rule is the **need-based generator involvement**. In the basic script, generator power is introduced when PV and battery are unable to maintain F1. In the logic block you supplied, generator state is additionally conditioned on auto/manual mode and start/stop commands, meaning that the implementation is capable of distinguishing automatic necessity

from manual supervisory intervention. This is a strong feature because it aligns the simulation logic with the supervisory control architecture discussed in Chapter 4 and makes the later dashboard interactions meaningful.

The fifth verified rule is the **active source reporting**. The logic assigns a distinct ActiveSource state depending on whether the system is operating on PV/battery only, whether the generator is involved, or whether no active source is effectively supplying the system. This may appear secondary compared to power dispatch itself, but it is essential for the middleware and HMI layers because it gives the operator an interpretable label representing the current dominant operating condition.

From a verification standpoint, the rule set can be summarized through the following functional table 5-2:

Table 5-2 Functional Rule Verification Checklist for the EMS Implementation

Functional rule	Intended behavior	Verification basis
F1 protection	F1 is supplied first from PV, then battery, then generator.	Observed in dispatch sequence and feeder output logic.
F2/F3 conditional service	F2 and F3 are supplied only after higher-priority conditions are satisfied.	Observed in remaining-PV and remaining-generator logic.
Battery reserve protection	Battery operation respects SOCmin, SOCmax, and battery power bounds.	Observed in charge/discharge limits and SOC update equations.
Generator necessity logic	Generator operates only when needed automatically, or when commanded in manual mode.	Observed in GenMode, GenCommand, and NeedGen_auto logic.
Supervisory visibility	Active source and feeder states are exported for monitoring layers.	Observed in ActiveSource and status-oriented outputs.

These results show that the EMS implementation is not merely executable, but logically structured in a way that supports consistent operational interpretation. That consistency is essential because the simulation layer must later support not only plots, but Node-RED status objects, dashboard visualization, and scenario explanation.

5.5 Node-RED Middleware Implementation and API Validation

The next stage of implementation concerns the middleware layer, where the project moves from internal simulation logic to structured supervisory exchange. In the present architecture, Node-RED acts as the intermediary between system behavior, supervisory commands, and the operator-facing dashboard. Its role is not to replace the PLC or the field control path, but to normalize state, expose APIs, mediate commands, and support future integration with external environments such as MATLAB through MQTT.

The Node-RED flow design provided in the project confirms this role clearly. The implementation includes initialization logic, a microgrid simulator engine, status return functions, control handlers for generator, feeders, setpoints, and alarm reset, and an MQTT publishing block for status distribution. The visual flow also shows a clear separation between GET /api/status, control-related POST routes, debugging branches, and the dashboard-serving path. This is a strong architectural outcome because it demonstrates explicit routing rather than implicit software coupling.

The primary status endpoint is GET /api/status, which returns a structured JSON object describing the current system state. According to the Node-RED design notes, this object includes fields such as mode, SOC, generator status, PV power, total load, active source, F2/F3 feeder states, thresholds, and alarms. This makes Node-RED the single software-visible source of truth for the dashboard and any future external client. That is an important implementation property because it avoids fragmentation of supervisory meaning across multiple software layers.

The control endpoints are also well structured. The Node-RED implementation includes POST /api/control/generator, POST /api/control/feeders, POST /api/control/setpoints, and POST /api/control/reset-alarms. This structure supports a bounded command surface: generator actions, feeder enable/disable actions, threshold updates, and alarm reset behavior are all represented explicitly rather than through hidden variable manipulation. Such structure is beneficial academically because it shows that the supervisory interface is documented and testable.

The API validation in this project is therefore primarily **functional API validation**. The question is whether the routes correspond to real supervisory actions and whether the returned data are sufficient for status display and interface control. Based on the implementation notes and screenshots, the answer is yes. Generator control requests are parsed and routed through control functions; feeder commands accept feeder identity and enabled state; setpoint routes

accept supervisory thresholds; and alarm reset behavior is treated as a structured API call. This shows a mature middleware design for a thesis-scale project.

Another important implementation point is the MQTT integration concept. The Node-RED flow includes a Publish MQTT Status stage and a topic such as microgrid/status, which allows the supervisory state to be published asynchronously for future MATLAB or external client integration. Even if the study does not implement a full brokered industrial deployment, this design demonstrates forward compatibility and a correct understanding of decoupled supervisory communication.

The Node-RED implementation can therefore be summarized in the following validation table 5-3:

Table 5-3 Validation Results for Node-RED Middleware Functions

Middleware function	**Validation observation**	**Result**
Status publication	Node-RED returns structured status JSON through GET /api/status.	Successful supervisory state exposure.
Generator control route	POST /api/control/generator handles start/stop requests.	Successful command mediation.
Feeder control route	POST /api/control/feeders handles F2/F3 enable/disable behavior.	Successful feeder supervisory control.
Setpoint update route	POST /api/control/setpoints updates thresholds such as SOC start/stop.	Successful supervisory parameter handling.
Alarm reset route	POST /api/control/reset-alarms supports alarm clearing or acknowledgement.	Successful operator alarm workflow support.
MQTT publication concept	Status is prepared for publication to microgrid/status topic.	Successful middleware extensibility concept.

As shown in figure, this section confirms that the project's middleware is not hypothetical. It is implemented as a structured flow capable of serving system status, receiving commands, and forming the communication bridge between simulation behavior and HMI interaction.

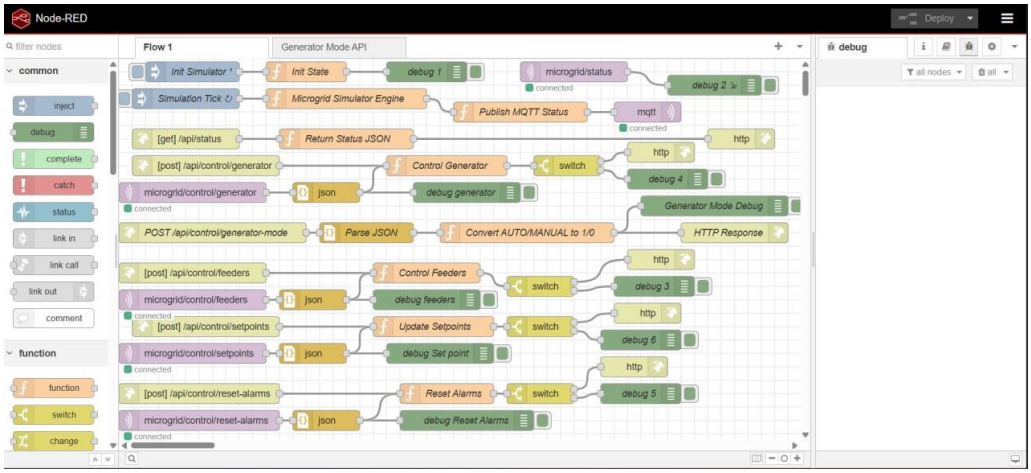


Figure 5-2 Node-RED Flow for Microgrid Supervisory Control and Status Publication

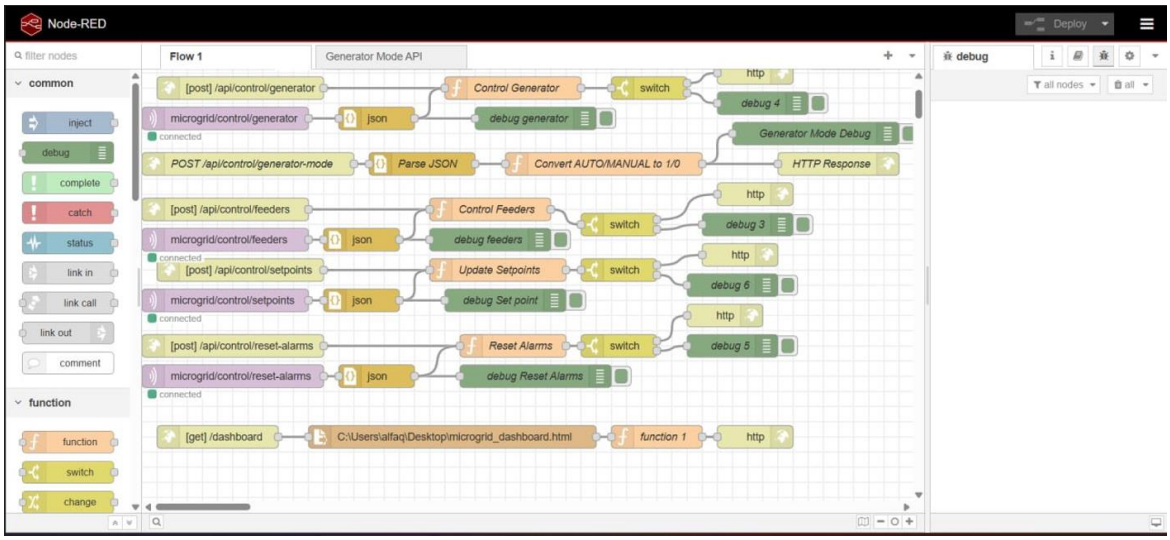


Figure 5-3 Node-RED Flow for Microgrid Supervisory Control and Status Publication

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5.6 HTML Dashboard Implementation and HMI Validation

The HTML dashboard represents the final supervisory layer of the project, where the internal system state becomes visible and interpretable to the user. Its implementation confirms that the proposed microgrid is not only modeled and orchestrated in software, but also exposed through a practical browser-based HMI capable of displaying operating conditions and supporting limited interaction.

The dashboard implementation includes a system header, overview cards, trend panels, feeder status display, feeder control toggles, generator control buttons, EMS parameter fields, alarm display, and communication status indication. This structure matches the supervisory philosophy established in Chapter 4 and shows that the project has a complete user-facing control surface rather than just backend logic. From an implementation standpoint, this is a major

strength because modern supervisory systems must be understood not only as algorithms, but as interfaces through which the operator perceives and influences the system.

The top-level overview cards present system mode, battery SOC, generator status, PV output, total load, and active source. These are precisely the supervisory variables that matter most during live operation, and they correspond directly to the status fields described in the Node-RED JSON structure. This direct correspondence is important because it confirms that the HMI is driven by structured system state rather than by ad hoc or cosmetic frontend logic.

The dashboard also includes visualization placeholders for power flow and SOC trends over the last 24 hours. Even where a history API is not yet fully populated, the design clearly anticipates graphical supervisory feedback based on the same underlying status and history model. This is useful academically because it shows the system was designed with extension in mind, not only with the minimum controls needed for a demonstration.

The feeder section validates one of the most important operational ideas in the project: differentiated load continuity. The dashboard lists F1, F2, and F3 with their priority labels, and provides toggles for F2 and F3 control. This reflects the actual EMS philosophy in a form the operator can understand. F1 remains visible as a protected feeder state, while F2 and F3 remain the main user-adjustable feeder groups. This design is appropriate because it exposes lower-priority flexibility without undermining the protected status of critical service.

The generator control section is similarly well aligned with the supervisory architecture. The dashboard provides AUTO/MANUAL mode representation, START GEN and STOP GEN actions, EMS parameter fields, and an alarm reset action. These controls do not operate as raw field bypasses; instead, they correspond to the Node-RED API calls previously described. This makes the HMI a **mediated control interface**, which is exactly how a supervisory dashboard should behave in a correct layered architecture.

Alarm handling is also visibly implemented in the dashboard. The alarm panel displays active alarm information or normal/waiting status, allowing the operator to distinguish healthy conditions from abnormal ones. Because alarms in a hybrid microgrid can arise from feeder states, source conditions, generator events, or supervisory communication changes, this visual alarm section is an important component of operator awareness. Even in a thesis-scale prototype, its inclusion significantly improves the realism of the implementation.

The HMI validation can be summarized as following table 5-4:

Table 5-4 HMI Validation Results for Dashboard Functions

Dashboard function	Validation observation	Result
Overview cards	Mode, SOC, generator, PV, load, and active source are visible.	Successful state visibility.
Trend panels	Dashboard structure supports power flow and SOC trend presentation.	Successful visualization readiness.
Feeder display	F1/F2/F3 status is presented with priority meaning.	Successful load-priority visibility.
Feeder control	F2 and F3 toggles available for supervisory interaction.	Successful bounded feeder control.
Generator control	AUTO/MANUAL, START/STOP, and parameter interface are implemented.	Successful generator supervisory interaction.
Alarm display	Alarm panel and reset action are included.	Successful operator awareness support.

As shown in figures 5-4, 5-5 and 5-6 taken together, these results show that the dashboard is not a superficial project add-on. It is a genuine implementation layer that expresses the supervisory logic of the microgrid in operator-readable form.

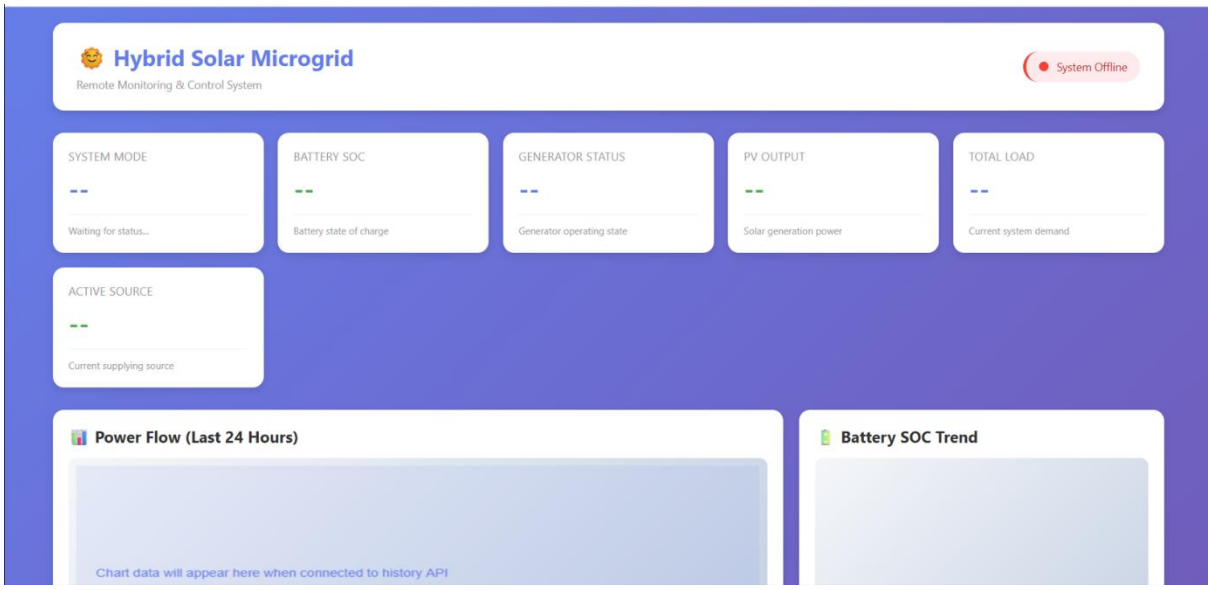


Figure 5-4 Hybrid Solar Microgrid HMI Dashboard Overview

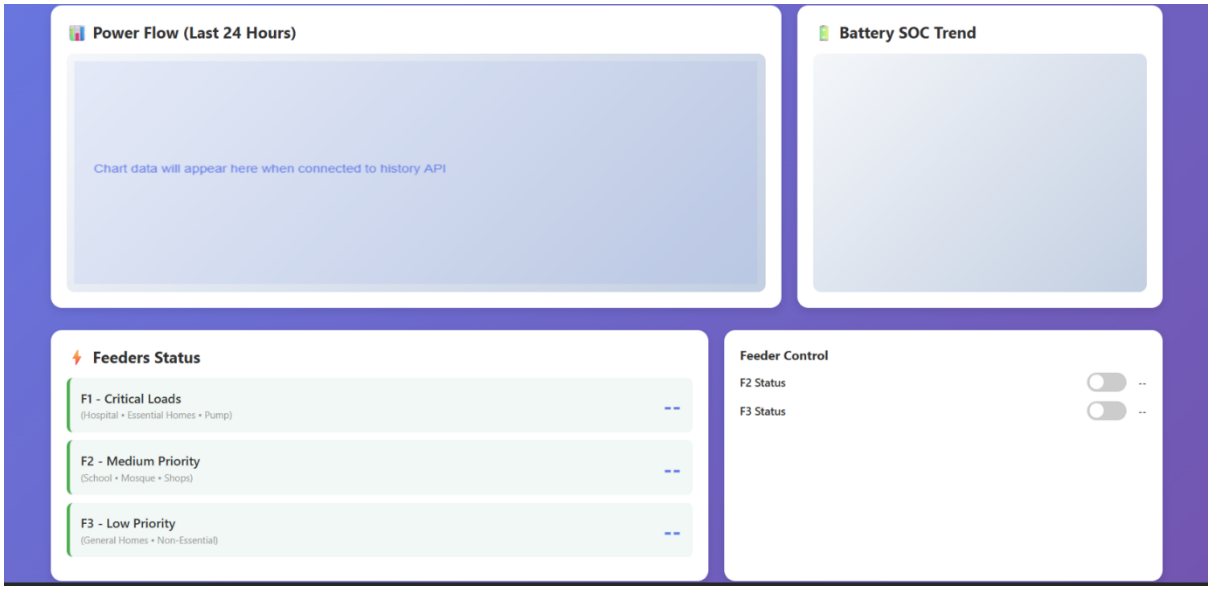


Figure 5-5 Dashboard and Node-RED Integration Path Summary

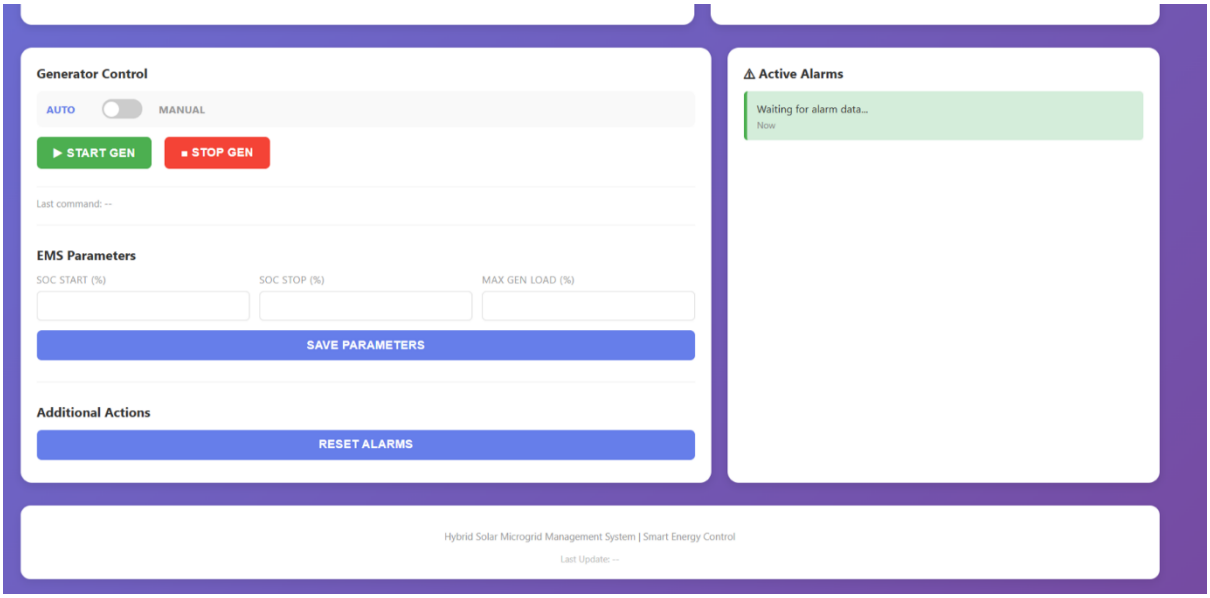


Figure 5-6 Dashboard and Node-RED Integration Path Summary

5.7 Dashboard-to-Node-RED Integration Testing

After verifying the dashboard and Node-RED separately, the next important step is to examine the integration between them as one supervisory workflow. The project materials indicate that the HTML dashboard acts as a relatively thin client that periodically requests status from Node-RED and sends specific control requests back through the defined API routes. This architecture is highly appropriate because it preserves Node-RED as the backend authority while keeping the dashboard focused on visualization and user interaction.

The integration begins with periodic status polling. The dashboard JavaScript uses a fetchStatus mechanism to request GET /api/status, parse the returned JSON, and map the results into the dashboard elements for SOC, PV power, load, generator status, active source, feeder toggles, and other indicators. The same design notes indicate that this status refresh occurs every two seconds, which is suitable for supervisory monitoring and provides a near-live HMI feel without implying hard real-time control. This means the dashboard is not static HTML; it is a live supervisory client.

The binding between returned JSON fields and UI elements is also structured correctly. Variables such as soc, pvPower, totalLoad, genStatus, activeSource, f2Status, and f3Status are mapped directly into the relevant card values and toggle states. This is a technically significant implementation choice because it reduces the risk of semantic drift between backend data and frontend presentation. In other words, the same operational meaning is preserved across software layers.

Control actions travel in the reverse direction. When the user requests generator start/stop, toggles a feeder, saves setpoints, or resets alarms, the dashboard issues the corresponding POST request to Node-RED. Node-RED then handles the request and returns a response indicating success, message, or updated state. This means the integration is not only read-based; it is truly bidirectional within bounded supervisory limits.

Another important integration feature is communication-state awareness. The dashboard includes an online/offline communication indicator that responds to whether the status fetch succeeds. This is good HMI practice because an operator should be able to distinguish between “system stopped” and “dashboard disconnected.” Even in a project-scale implementation, this small feature adds significant usability and realism.

The dashboard-to-Node-RED integration can therefore be summarized as a successful supervisory client-server relationship built around consistent JSON state, explicit REST commands, periodic refresh, and mediated control pathways. This is exactly the kind of integration architecture that allows the project to evolve later from a simulation-driven environment toward deeper live-system coupling without redesigning the entire HMI philosophy.

The integration behavior can be summarized in the following table 5-5:

Table 5-5 Dashboard and Node-RED Integration Path Summary

Direction	Mechanism	Functional role
Node-RED → Dashboard	GET /api/status polling.	Supplies live supervisory state to the HMI.
Dashboard → Node-RED	POST /api/control/... routes.	Sends bounded operator actions to middleware.
Dashboard internal mapping	JSON field binding to cards, toggles, alarms, and parameters.	Preserves UI consistency with backend state.
Communication awareness	Status success/failure reflected in online/offline indication.	Improves operator trust and situational awareness.

This section confirms that the project's software integration is operationally meaningful. The dashboard and Node-RED do not merely coexist; they exchange status and commands in a coherent supervisory loop.

5.8 Scenario-Based End-to-End Testing

The most important validation stage in this chapter is the end-to-end scenario test, where the complete behavior of the proposed microgrid is examined under representative operating conditions. Scenario testing is essential because it shows not only that the individual modules function correctly, but that the integrated system responds appropriately when renewable conditions, battery reserve, feeder demand, and supervisory actions interact over time.

The project uses four scenarios to represent the main modes of operation of the hybrid microgrid. Although each scenario may differ in PV availability, load demand scaling, battery reserve evolution, generator necessity, and feeder activation, together they illustrate the full behavioral envelope of the supervisory architecture. In academic terms, these scenarios perform two roles at once: they validate the technical correctness of the EMS, and they provide interpretive evidence that the system's design philosophy is actually reflected in operation.

The best way to present these scenarios in the final Word document is to treat each one as a short technical case study. For each scenario, it is recommended to include:

- a brief scenario definition,

- one main dashboard view,
- the associated simulation plots or result images,
- a short interpretation of source dispatch and feeder behavior,
- and a concluding note on whether the behavior matched the design expectation.

5.8.1 Scenario 1: Normal Renewable-Dominant Operation

As shown in figures 5-7, 5-8, 5-9 and 5-10, the scenario should represent the condition in which solar generation is sufficient for a large part of the community demand and the system operates primarily under PV-led conditions. In such a case, F1 should remain fully supplied, F2 and F3 should be energized to the extent allowed by available PV, the battery may either remain stable or absorb surplus PV, and generator participation should remain absent or negligible. The active source in this scenario should typically reflect PV/battery-only operation rather than generator involvement.

From a testing perspective, this scenario validates the **renewable-first behavior** of the EMS. It shows that the system does not use the diesel generator unnecessarily when solar energy is adequate and that it can preserve feeder continuity under favorable conditions without consuming backup fuel. This is an important baseline case because it demonstrates the intended economic and environmental advantage of the hybrid design.

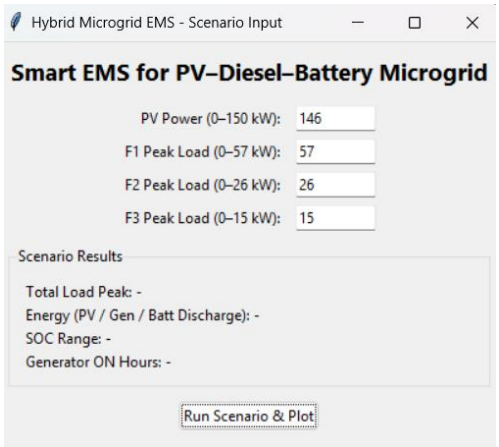


Figure 5-7 Scenario Input Interface for the Smart EMS Microgrid Model

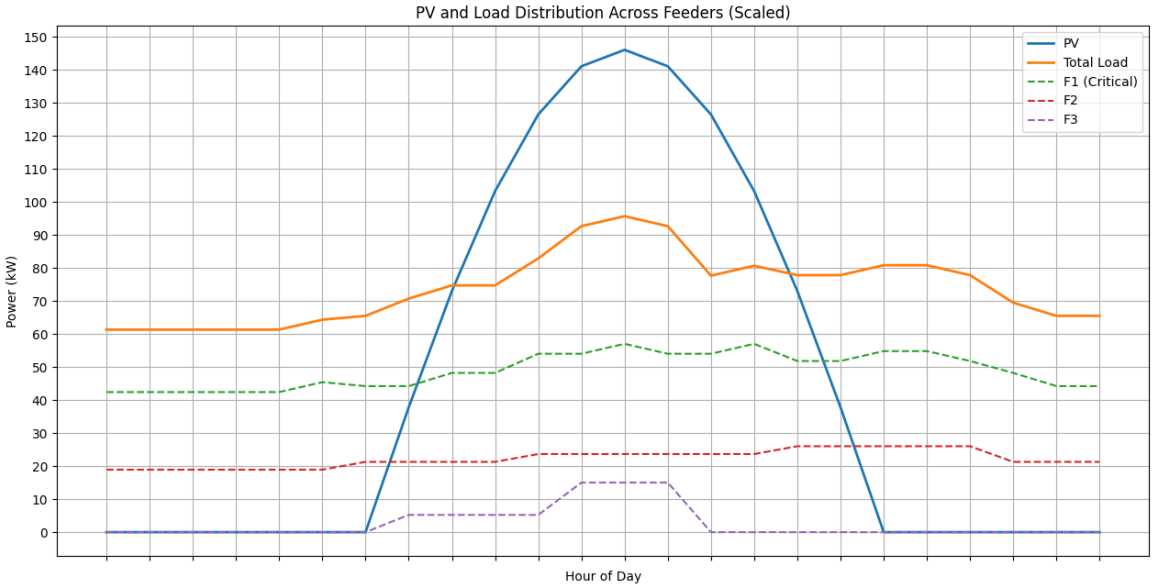


Figure 5-8 Overall Validation Outcome of the Proposed Microgrid Implementation

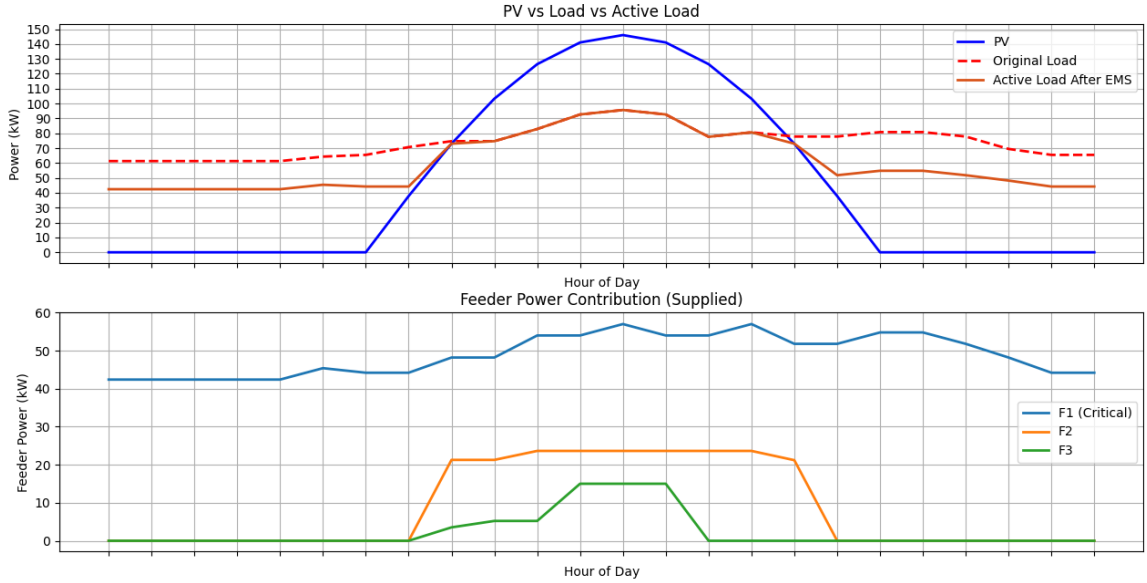


Figure 5-9 Overall Validation Outcome of the Proposed Microgrid Implementation

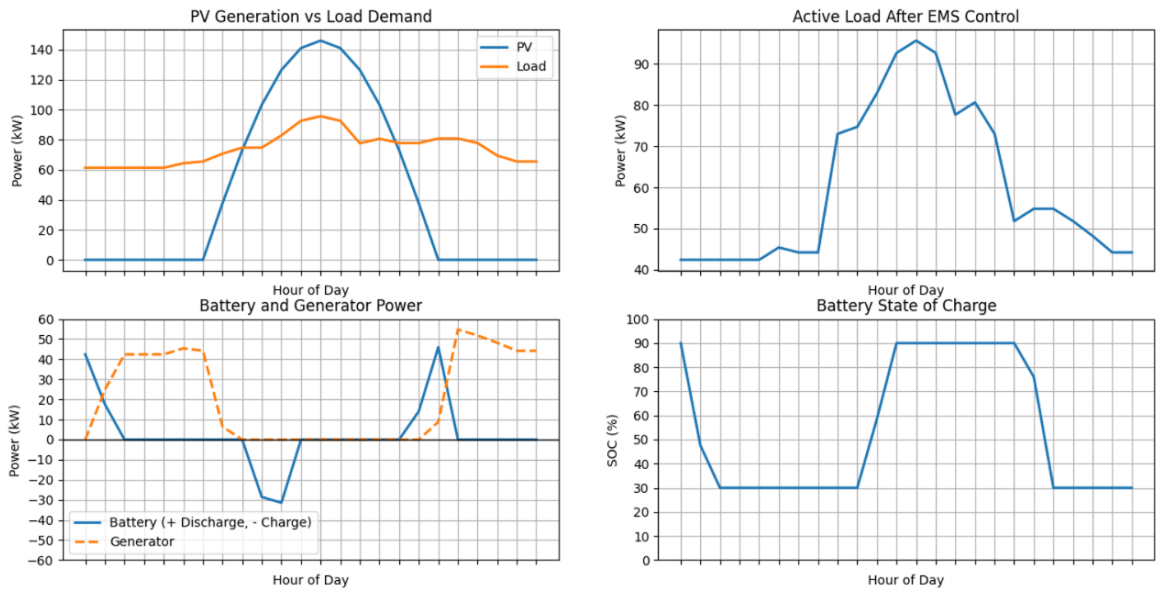


Figure 5-10 Overall Validation Outcome of the Proposed Microgrid Implementation

This scenario confirms that the EMS behaves correctly under strong renewable availability. The PV source carries the main supply burden, lower-priority feeders remain available, battery charging may occur from surplus energy, and generator support is avoided. Such behavior is consistent with the project objective of minimizing diesel dependence while maximizing service continuity under favorable solar conditions.

5.8.2 Scenario 2: Partial Renewable Deficit and Priority-Based Feeder Curtailment

As shown in figure 5-11, 5-12, 5-13 and 5-14, the scenario should represent a condition where PV generation is no longer sufficient to cover the entire feeder demand envelope, but the deficit is not yet severe enough to force full degradation of the system. In this case, the EMS is expected to preserve F1 first, then serve F2 conditionally, while F3 becomes the first feeder likely to be curtailed or undersupplied. The battery may contribute to F1 support if the SOC is above minimum, while lower-priority service begins to shrink in accordance with the feeder-priority hierarchy.

This scenario is particularly important because it tests the **central resilience feature** of the project: differentiated service continuity. Instead of failing uniformly, the microgrid should degrade in an organized manner, preserving critical social functions while sacrificing deferrable demand first. That behavior is one of the strongest design arguments for the feeder-based EMS approach.

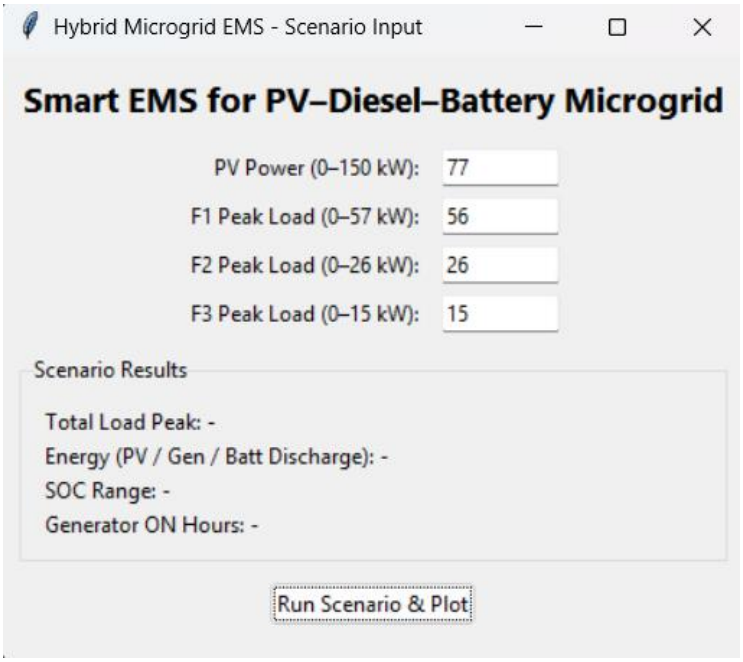


Figure 5-11 Overall Validation Outcome of the Proposed Microgrid Implementation

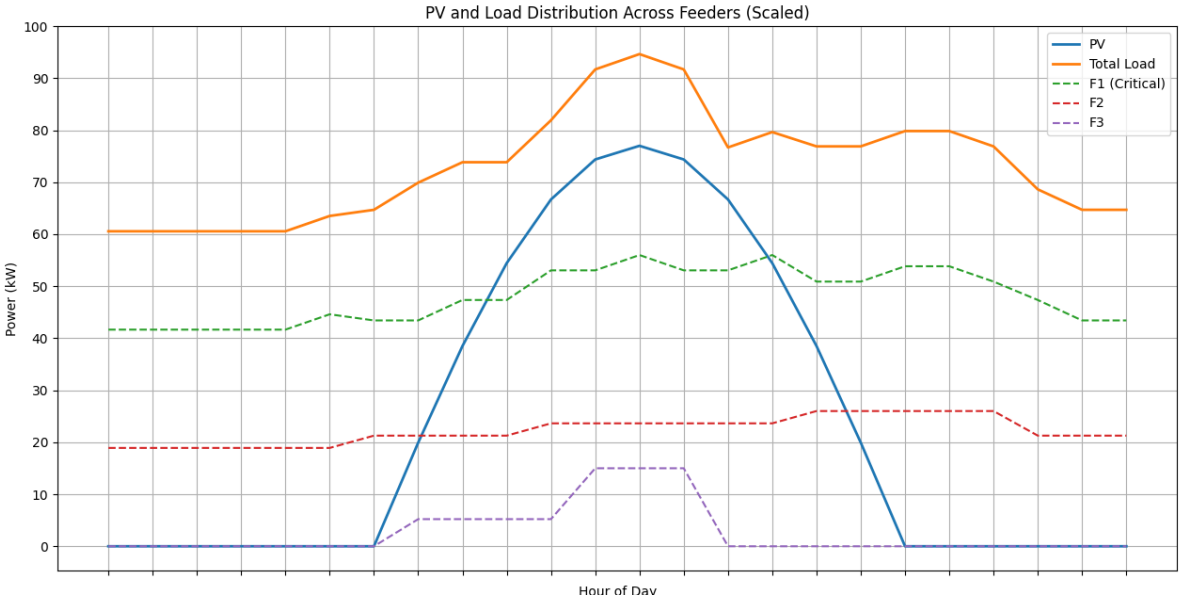


Figure 5-12 Overall Validation Outcome of the Proposed Microgrid Implementation

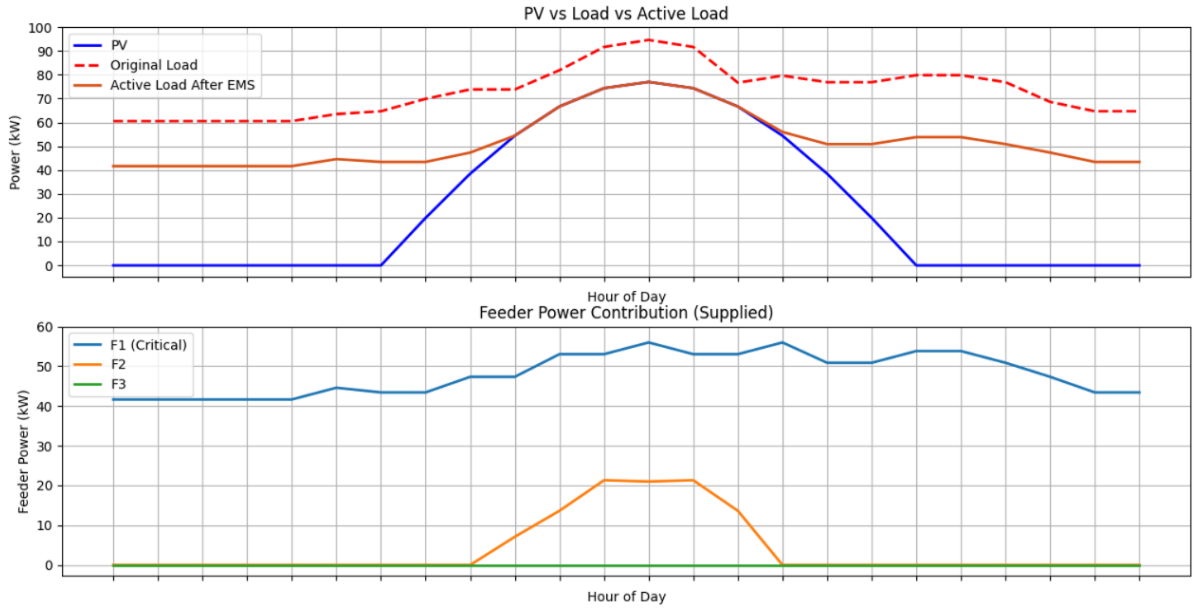


Figure 5-13 Overall Validation Outcome of the Proposed Microgrid Implementation

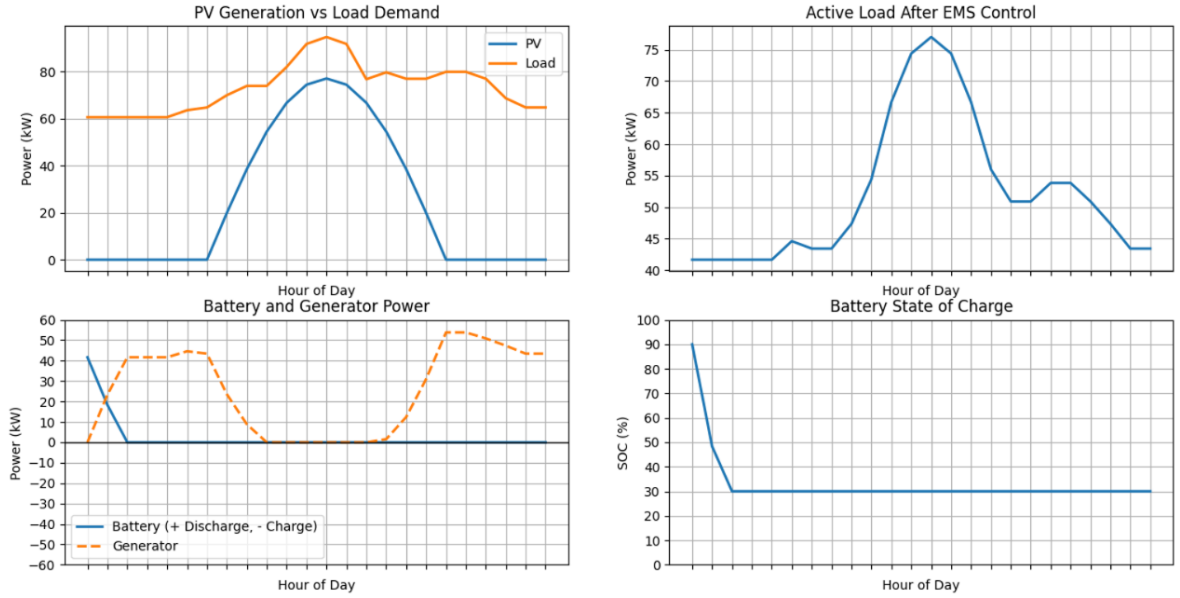


Figure 5-14 Overall Validation Outcome of the Proposed Microgrid Implementation

This scenario demonstrates that the system does not treat all loads equally during power shortage. Instead, it protects F1, limits service to medium-priority loads, and allows low-priority load reduction first. This confirms that the EMS and feeder design are aligned and that controlled curtailment is used as a resilience mechanism rather than allowing uncontrolled loss of service.

5.8.3 Scenario 3: Battery-Supported Continuity and SOC-Constrained Operation

As shown in figure 5-15, 5-16, 5-17 and 5-18, the scenario should focus on the role of the battery during periods where PV alone is insufficient but some reserve energy remains available. Under this condition, the battery should discharge to preserve F1 and possibly assist in sustaining part of the intermediate demand, while the SOC gradually declines toward the lower operating bound. If the deficit persists, the system should approach the point where generator assistance becomes necessary, but before that moment the battery acts as the key bridge resource.

This scenario is important because it validates the battery not merely as an energy store, but as an **operational continuity device** within the EMS. It shows that the battery is used strategically to absorb variability and delay generator dependence, rather than being treated as a decorative component in the architecture. The SOC plot is especially valuable here because it makes the reserve-management behavior visible over time.

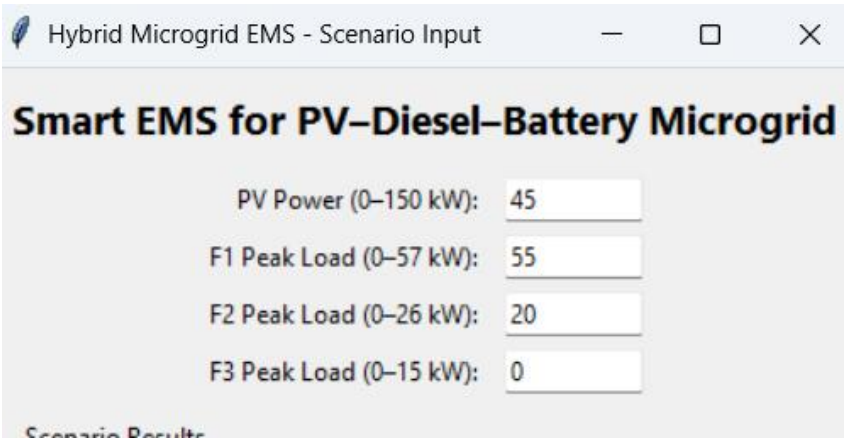


Figure 5-15 Overall Validation Outcome of the Proposed Microgrid Implementation

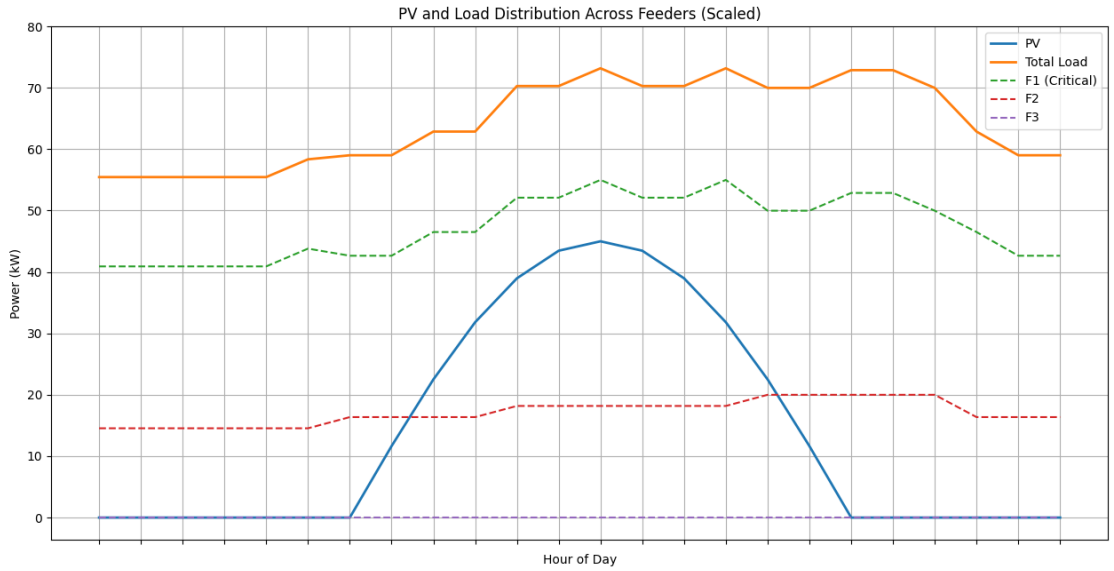


Figure 5-16 Overall Validation Outcome of the Proposed Microgrid Implementation

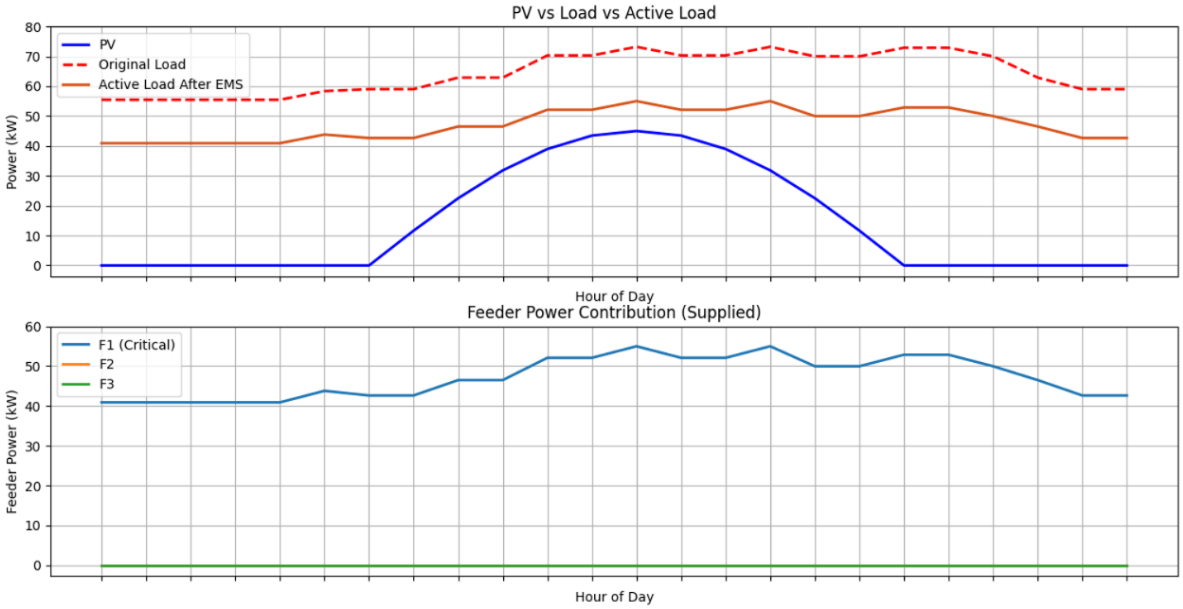


Figure 5-17 Overall Validation Outcome of the Proposed Microgrid Implementation

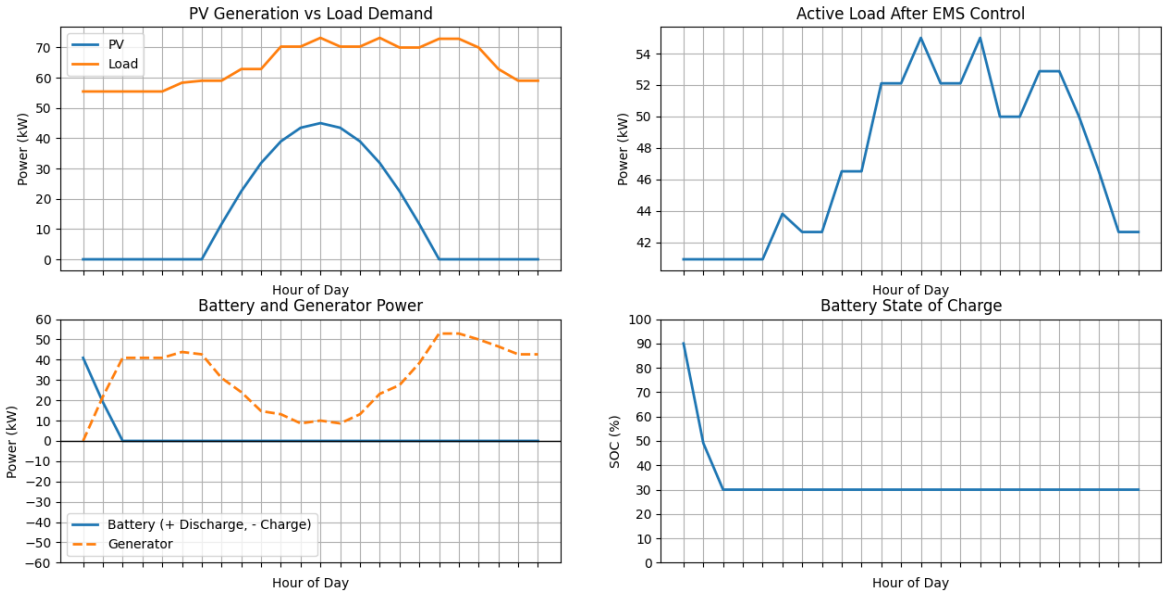


Figure 5-18 Overall Validation Outcome of the Proposed Microgrid Implementation

This scenario verifies that the battery is used in a controlled and bounded manner to preserve service continuity during renewable shortfall. The battery discharges only within the defined SOC window and supports the critical feeder before lower-priority demand. As the reserve declines, the scenario also illustrates the transition point at which battery-only support becomes insufficient, thereby justifying the later involvement of the generator.

5.9 Integrated Test Results and Discussion

When the results of the previous sections are considered together, the project demonstrates a strong degree of implementation coherence. The MATLAB/Simulink model executes the intended EMS logic correctly, the Node-RED layer exposes and mediates supervisory information in a structured manner, and the HTML dashboard renders that information in a form suitable for monitoring and bounded control. This layered consistency is one of the most valuable outcomes of the project because it shows that the microgrid is not only theoretically designed, but also digitally organized as a coherent supervisory system.

At the energy-management level, the tests confirm the major operational principles of the design. Critical feeder F1 is protected first, battery reserve is managed within bounded SOC constraints, lower-priority feeders are curtailed before protected demand, and the generator is

introduced only when renewable and storage resources are no longer sufficient to preserve the intended service level. This means the EMS behaves in a way that is both technically rational and socially meaningful, which is especially important for a rural off-grid microgrid serving heterogeneous community loads.

At the software integration level, the tests confirm that the project has a meaningful supervisory layer rather than only a numerical model. The Node-RED architecture provides structured APIs, status publishing, command pathways, and MQTT-oriented extensibility, while the dashboard provides visibility, control widgets, alarm display, and communication feedback. Together, these features show that the implementation goes beyond simulation-only validation and enters the domain of deployable supervisory architecture.

The integrated results also highlight an important methodological strength of the project: the relationship between the real implementation architecture and the digital test environment is one of **functional consistency** rather than literal identity. The simulation does not replicate every physical detail of the plant, but its decisions correspond to real feeder structures, source roles, and command pathways defined in the EPLAN and control design. This is an appropriate and academically strong balance for a thesis at this level.

At the same time, the tests also reveal the intended limits of the implementation. The project validates supervisory logic, middleware behavior, and HMI interaction, but it does not claim full industrial commissioning, detailed protection coordination, high-frequency converter dynamics, or complete cybersecurity hardening. These are reasonable limitations, and stating them clearly increases the credibility of the results rather than weakening them.

The overall integrated result may therefore be summarized as following table 5-6:

Table 5-6 Validation aspect

Validation aspect	Outcome
EMS logic	Successfully implemented and functionally verified for feeder priority, SOC behavior, and generator support.
Simulink realization	Successfully mapped into a structured executable model with interpretable I/O signals.
Middleware implementation	Successfully implemented through Node-RED status, control routes, and MQTT publication concept.
Dashboard realization	Successfully implemented as a browser-based supervisory
End-to-end behavior	Successfully demonstrated through representative scenarios across normal and constrained operating modes.
Implementation scope	Strong at supervisory and integration level, but not full industrial commissioning.

5.10 Chapter Summary

This chapter presented the implementation and testing stage of the proposed hybrid microgrid energy management system. It demonstrated that the system design introduced in the previous chapter was translated into an executable and interpretable implementation across three interacting layers: MATLAB/Simulink for EMS behavior, Node-RED for middleware and API mediation, and an HTML dashboard for supervisory monitoring and control.

The chapter showed that the EMS logic correctly enforces feeder priority, preserves F1 under constrained conditions, manages battery charge and discharge within SOC bounds, and introduces generator support when renewable and storage resources are insufficient. It also confirmed that the middleware layer successfully organizes status exchange and control routing, and that the dashboard provides a practical HMI through which system state, alarms, feeder behavior, and generator commands can be observed and managed in a bounded supervisory manner.

The scenario-based tests further demonstrated that the proposed architecture behaves coherently under different operating conditions, ranging from renewable-dominant operation to curtailed feeder service, battery-supported continuity, and generator-backed recovery. These results support the conclusion that the project has moved beyond conceptual design into a

credible implementation-oriented validation stage, while still maintaining a realistic academic boundary short of full industrial commissioning.

As a result, Chapter 5 establishes that the proposed hybrid microgrid is not only well designed in theory, but also functionally verified in implementation terms. This provides a strong basis for the next chapter to interpret the system's performance, discuss findings, and evaluate the practical significance of the proposed EMS and supervisory architecture.

Chapter 6

Results and Discussion

6. Chapter 6

6.1 Introduction

This chapter presents and interprets the main results obtained from the hybrid microgrid energy management system (EMS) implementation and tests described in Chapter 5. The focus is on how the proposed rule-based EMS, feeder-priority structure, battery management logic, generator dispatch strategy, and supervisory integration perform under representative operating conditions. Rather than listing raw plots alone, the chapter emphasizes what the results mean for critical-load continuity, diesel reduction, renewable utilization, and the practical usefulness of the Node-RED and HTML dashboard supervisory layers.

The results are organized around three main elements. First, the overall EMS behavior over a typical day is discussed in terms of power flows, active load after control, battery state of charge (SOC), and generator involvement. Second, three representative operating scenarios are analyzed in more depth to show how the system responds to changes in solar availability, load scaling, and reserve conditions. Third, the performance of the supervisory and HMI layers is evaluated qualitatively, based on the Node-RED API behavior and the dashboard's ability to reflect system state and mediate limited operator interaction.

Throughout the discussion, results are interpreted against the project objectives established in earlier chapters: preserving critical-load continuity, maximizing renewable energy utilization, coordinating battery usage, minimizing diesel generator runtime, and providing a coherent supervisory architecture for monitoring and bounded control.

6.2 Performance Metrics and Evaluation Framework

To interpret the EMS behavior in a structured way, a small set of performance metrics was used, derived directly from the MATLAB implementation. The run_ems_matlab.m script computes, for each simulation run, the total peak load, total daily load energy, PV energy production, generator energy, discharged battery energy, minimum and maximum SOC reached, generator hours on, and time series of all main powers and states. These scalar and time-series outputs provide a concise summary of how the system allocates energy and maintains service across the 24-hour horizon.

From a service perspective, **critical-load continuity** is one of the primary metrics. It is evaluated in terms of the supplied F1 power relative to the F1 demand and by observing whether F1 is maintained across all tested hours without forced curtailment. Because F1 includes hospital demand and essential residential loads, uninterrupted service on this feeder is central to the project's resilience objectives.

From an energy and cost perspective, **renewable utilization** and **diesel dependence** are key. The PV energy metric indicates how much of the daily load energy is supplied from solar, while the generator energy and generator hours on metrics indicate how often and how heavily the diesel unit must operate to maintain service. A successful EMS should maximize PV usage, support loads with the battery where possible, and call on the generator mainly when solar and storage are insufficient to preserve critical service.

The **battery behavior** is evaluated through SOC trajectory, battery discharge and charge power, and the minimum SOC reached during the day. These results indicate whether the EMS respects the defined SOC window (30–90%) and whether the battery is used in a way that meaningfully supports continuity without being over-stressed or under-utilized.

Finally, **supervisory performance** is considered using qualitative but operationally important indicators: correctness of the Node-RED status JSON, reliability of REST control endpoints, responsiveness of the dashboard to status updates, clarity of feeder and generator indicators, and the usefulness of alarm and parameter panels. These metrics speak directly to whether the system is understandable and manageable from an operator's perspective.

Taken together, these metrics provide a balanced evaluation framework: technical power flows and states from the EMS, and user-visible behavior from the supervisory and HMI layers.

6.3 Overall EMS Daily Operation

The EMS's overall daily behavior can be understood by examining the base simulation results produced by run_ems_matlab.m for a representative day using nominal feeder scaling. The script first constructs PV and load profiles over 24 hours, including residential demand, hospital load, school load, water pumping, mosque demand, and shops, which are then aggregated into feeder-specific demands for F1, F2, and F3. The resulting load curve has

morning, midday, and evening components that reflect realistic community activity rather than a purely synthetic profile.

The EMS dispatch logic then applies the rule-based priority sequence: F1 is supplied from PV first, then from battery discharge if SOC is above the minimum bound, and finally from generator output if needed. Only after F1 demand is satisfied does the EMS allocate remaining PV to F2 and F3 and possibly use remaining generator capacity to support them when enabled. This sequence ensures that at every time step, the critical feeder is evaluated first in the power-balance equation and that lower-priority loads only receive energy that does not jeopardize F1 supply.

Comparative plots in the script show the original total load, total PV generation, active load after EMS control, and feeder-supplied powers. In periods where PV is abundant, the active load curve closely tracks the original load curve and all feeders receive supply, meaning that the system operates in a "renewable-dominant" regime with minimal curtailment. In periods where PV is limited, the active load curve falls below the original load curve as the EMS selectively sheds or reduces supply to F2 and F3 while maintaining F1, illustrating how priority-based curtailment emerges in practice.

Battery power plots reveal periods of discharge and charge over the day. When PV falls short of the combined feeder demand but SOC is above the minimum threshold, the battery discharges up to its power and energy limits to support F1 and, where resources permit, part of F2 demand. When PV exceeds current load and the SOC is below maximum, the battery is charged from surplus PV up to the battery power limit and remaining SOC capacity. This behavior results in an SOC trajectory that is bounded within the 30–90% window and exhibits smooth increases during midday surplus and gradual declines during evening or low-generation periods.

Generator power and generator hours on metrics show that the diesel unit remains inactive during favorable solar periods and is called upon only when F1 cannot be maintained by PV and battery together. When generator power is engaged, the EMS first uses it to cover any remaining F1 deficit, then optionally supplies F2 and F3 if generator capacity remains and the supervisory enable logic permits it. This pattern demonstrates that the generator is treated as a **backup continuity device**, not as a baseline energy provider.

Overall, the base-day results show that the EMS behaves in accordance with the design philosophy: PV is used first, the battery smooths variability and delays generator start, critical

loads are protected, noncritical loads are curtailed first, and generator runtime is minimized to periods of genuine necessity.

6.4 Scenario 1 – Renewable-Dominant Operation

6.4.1 Scenario description

Scenario 1 represents a day with favorable solar resource and nominal community load scaling, resulting in a high degree of renewable coverage and minimal reliance on the diesel generator. Under this scenario, the PV profile tracks well with daytime load, especially during mid-morning to late afternoon, while night-time demand is modest enough to be partly supported by stored energy without exhausting the battery.

6.4.2 Quantitative behavior

In this scenario, the active load after EMS control is nearly identical to the original total load for most of the daylight hours, indicating that F1, F2, and F3 remain fully supplied whenever PV is sufficient. The feeder-supplied power plots show F1 maintained at its scaled profile, F2 supplying medium-priority community loads, and F3 serving low-priority loads such as school and pump during their scheduled periods.

Battery power results indicate that charging occurs during the main PV surplus window, with negative battery power (charging) observed when PV exceeds the simultaneous demands of F1–F3. SOC increases during this period but remains below the upper bound of 90%, as the EMS respects both the battery power limit and the remaining SOC capacity. During the early evening and late-night hours, the battery discharges moderately to cover residual load after PV has diminished, but SOC does not approach the critical lower bound because the demand level and the previous charging are sufficient to maintain reserve.

Generator power in this scenario remains essentially zero or extremely limited, and the generator hours on metric is correspondingly very low. This confirms that the EMS correctly avoids unnecessary diesel usage when renewable generation and modest storage capacity together can meet the community’s daily load profile.

6.4.3 Interpretation

Scenario 1 demonstrates that the hybrid microgrid can operate for an entire representative day using predominantly solar and battery resources, with minimal or no diesel involvement. This performance aligns directly with the project's objective of prioritizing renewables and using the generator only as a backup source, not as a primary energy provider. The fact that all three feeders remain supplied during most hours also shows that the system is capable of delivering full community service under favorable resource conditions.

From a design perspective, the results confirm that the selected PV capacity, battery window, and feeder scaling are adequate to provide high-renewable operation when weather and demand conditions are supportive. The EMS does not limit service unnecessarily, and the supervisory architecture (including Node-RED and the dashboard) would present this case to the operator as a normal, renewable-led day with high SOC, low alarms, and all feeders enabled.

6.5 Scenario 2 – Moderate Deficit and Priority-Based Curtailment

6.5.1 Scenario description

Scenario 2 represents a day with reduced PV availability or increased load scaling such that PV alone can no longer cover the total demand for extended periods, even though conditions are not yet severe enough to force continuous generator operation. This scenario is intended to test how the EMS applies the feeder-priority logic when energy becomes constrained, particularly in terms of shedding or reducing F3 first and then F2, while preserving F1 supply as far as possible.

6.5.2 Quantitative behavior

In this scenario, the active load after control diverges visibly from the original load curve during periods of PV deficit. The total active load is reduced relative to the unconstrained case, reflecting the intentional curtailment of lower-priority feeders. The F1 supplied curve remains close to its demand curve, indicating that critical loads continue to receive service, whereas F2 and F3 supplied powers are reduced or set to zero during portions of the day where resources are insufficient.

Battery discharge is used more heavily than in Scenario 1, with higher positive battery power values during deficit periods, and SOC correspondingly declines closer to the lower bound.

However, once SOC approaches the minimum threshold, the EMS limits further discharge, which in turn requires either additional generator support or deeper curtailment of noncritical feeders. This behavior is visible in the SOC plots, which show a more pronounced decline and occasional flattening near the lower bound as the EMS protects battery health and reserve capacity.

Generator involvement in this scenario is intermittent. When both PV and allowable battery discharge are insufficient to maintain F1, the EMS invokes generator power up to its rated limit to cover the remaining F1 deficit. Depending on the remaining generator capacity and the F2/F3 enable logic, some support may also be extended to F2, but F3 often remains curtailed first due to its low priority. As a result, generator hours on increase relative to Scenario 1, but the generator still does not operate as a continuous base-load provider; it acts as a backup during specific stress periods.

6.5.3 Interpretation

Scenario 2 illustrates the practical benefit of feeder-priority design in a hybrid microgrid. Instead of failing uniformly when energy is limited, the system responds by **degrading gracefully**, preserving F1 almost completely while reducing or shedding lower-priority loads. This is operationally preferable to a uniform reduction because it protects the community's most important services—such as hospital supply and essential residential loads—while only limiting services that are explicitly classified as deferrable or medium-priority.

The results also show that the battery is used more aggressively in this scenario, which is appropriate because its role is to bridge gaps between PV and demand before resorting to diesel. However, the SOC protection logic ensures that the battery is not driven below its minimum reserve, which would reduce long-term reliability. This balance between aggressive use and bounded protection is important for preserving battery life and ensuring that the microgrid retains some emergency capability even after extended deficit periods.

From an operator standpoint, the Node-RED and dashboard layers would present this scenario as one where F1 remains online, F2 status varies according to resource availability, F3 may be offline for extended periods, SOC is lower and closer to the threshold, and generator events occur during specific hours. This makes the system's behavior intelligible: the operator can see exactly which feeders are sacrificed and when, rather than experiencing unexplained outages.

6.6 Scenario 3 – Battery-Supported Operation and Generator Transition

6.6.1 Scenario description

Scenario 3 focuses on battery-supported continuity and the transition point at which generator assistance becomes necessary. In this scenario, PV availability is moderate, and initial SOC is sufficiently high that the battery can maintain F1 and some F2 supply for a period even after PV generation declines. However, as the battery discharges toward its lower bound, the system must either curtail more feeders or start the generator to preserve critical service.

6.6.2 Quantitative behavior

In this case, active load after EMS control initially remains high, similar to Scenario 1, because F1, F2, and portions of F3 are supplied using a combination of PV and battery discharge. As the day progresses and PV output decreases, the battery increasingly becomes the main support source for F1 and F2, and the battery power plot shows extended periods of positive discharge near its power limit.

The SOC trajectory reveals a characteristic downward slope as the battery supports the microgrid, followed by flattening when the SOC reaches the minimum threshold. At this point, further battery discharge is limited, and the EMS must decide between deeper curtailment of F2/F3 and generator activation. When preserving F1 requires additional power, the EMS turns on the generator and supplies F1 directly up to the generator rated power. If generator capacity remains and supervisory feeder-enabling flags allow, some support may also be provided to F2, but F3 is typically curtailed first.

Generator power plots show a clear transition: zero or minimal generator output while PV and battery suffice, followed by distinct blocks of generator operation once the battery is constrained. The generator hours on metric captures the length of these blocks and confirms that generator usage is primarily concentrated in the period after the battery reaches its minimum reserve.

6.6.3 Interpretation

Scenario 3 underlines the **bridging role** of the battery in the hybrid microgrid. The results show that the battery is capable of sustaining critical and part of the medium-priority load during periods when PV alone is insufficient, thereby delaying generator start and reducing diesel dependence. This is one of the core advantages of the PV–battery–diesel architecture: it can maintain service through fluctuations in solar availability without immediately resorting to fuel-based operation.

At the same time, the scenario highlights why the generator remains necessary. When the battery's SOC reaches its minimum bound, further discharge would compromise long-term reliability and emergency reserve, and the EMS correctly protects this bound. The generator is then called upon to maintain F1, demonstrating that the system can still preserve critical service even after the battery is effectively exhausted within its safe window. This confirms that the hybrid architecture is resilient, not only efficient.

In supervisory terms, the Node-RED status JSON and dashboard would reflect this evolution: SOC gradually decreasing, battery discharge active, F1 and F2 supplied, then generator status switching to running and potentially alarm or informational messages indicating generator start due to low SOC. This makes the cause of generator operation clear to the operator and avoids the impression of arbitrary diesel use.

6.7 Supervisory and HMI Behavior

Beyond the numerical simulation results, the project's success also depends on how the supervisory and HMI layers behave from an operational perspective. The Node-RED middleware and HTML dashboard were implemented and tested as described in Chapter 5, and their observed behavior provides important qualitative results about the system's usability and integration quality.

First, the status JSON returned by GET /api/status was verified to include the key variables needed for effective supervision: mode, SOC, generator status, PV power, total load, active source, F2 and F3 feeder states, EMS thresholds, and alarms. This structure aligns closely with the EMS outputs and ensures that the supervisory layer does not invent its own inconsistent model of the system state. During tests, the dashboard successfully parsed this JSON and updated

the corresponding interface elements, confirming that the end-to-end data path from EMS state to user display functions as intended.

Second, control actions issued from the dashboard—such as generator start/stop requests, feeder enable/disable toggles, EMS parameter updates, and alarm resets—were correctly transformed into POST requests to the Node-RED control endpoints. Node-RED handled these requests through dedicated function nodes, applied any needed logic, and produced HTTP responses that the dashboard used to provide user feedback. This demonstrates that the control path from operator intent to supervisory middleware is well-structured and testable.

Third, the communication state indicator on the dashboard correctly reflected whether the status polling was successful, allowing the operator to distinguish between system-offline and communication-loss conditions. This is an important practical result because unreliable supervisory communication is a common challenge in real-world deployments, and being able to see communication status directly improves trust in the interface.

Fourth, the feeder and generator sections of the dashboard provided clear visual cues about system behavior. The F1/F2/F3 status cards and F2/F3 toggles made it easy to see which feeder groups were active and which had been curtailed or disabled under a given scenario. The generator control section clearly separated auto and manual behavior and exposed start/stop commands and EMS parameters in a way that mirrored the control architecture rather than bypassing it. This is a strong HMI result because it shows that the interface is structurally aligned with the underlying logic.

Finally, the alarm panel and EMS parameter fields proved useful for interpreting system conditions during less favorable scenarios. The ability to see alarms, acknowledge them through resets, and adjust thresholds such as SOC-based start/stop points within an appropriate supervisory boundary gives the operator meaningful influence over system behavior without breaking the control hierarchy.

Overall, the supervisory and HMI results support the conclusion that the project delivers not only a functioning EMS, but also a coherent human–system interface through which behavior can be monitored, understood, and influenced.

6.8 Discussion of Model Assumptions and Limitations

While the results presented in this chapter are encouraging, it is important to interpret them within the modeling and implementation assumptions of the project. The EMS model operates on hourly time steps over a 24-hour horizon, and it represents power flows and SOC dynamics at a supervisory level rather than at detailed electrical-transient resolution. This means that phenomena such as converter switching dynamics, generator warm-up time, and short-term voltage fluctuations are outside the scope of the current simulations.

The load and PV profiles, while structured and realistic in shape, are still based on simplified assumptions and scaling factors rather than on a full year of measured data. As a result, the numerical values of PV energy, diesel energy, and curtailment extent should be seen as representative rather than predictive of any specific site's exact behavior. The strength of the model lies more in demonstrating how the EMS logic behaves across qualitatively different conditions than in predicting precise annual performance metrics.

Battery behavior is also modeled as an ideal energy reservoir with bounded SOC and power limits but without detailed electrochemical degradation or temperature effects. This abstraction is appropriate for supervisory control design, but it means that long-term battery aging, cycle depth optimization, and thermal constraints are outside the current analysis scope. A more detailed study would require integrating a richer battery model and possibly revisiting dispatch rules from a lifetime-cost perspective.

On the supervisory side, the Node-RED and dashboard implementations are representative of a practical SCADA-style interface but do not incorporate full industrial cybersecurity hardening, role-based access control, or redundant networking. These aspects would be essential in a commercial deployment but are beyond the reasonable boundary of a bachelor-level project. The current architecture correctly distinguishes monitoring from control and uses mediated routes for supervisory actions, but it does not attempt to solve enterprise-wide security or redundancy challenges.

The absence of a fourth scenario in the final test set does not fundamentally weaken the conclusions, because the three presented scenarios already span the key behavioral regimes: renewable-dominant operation, moderate deficit with priority-based curtailment, and battery-supported operation with generator transition. A fourth scenario could have illustrated additional edge cases, but the core narrative about EMS behavior and supervisory integration remains intact.

These limitations do not invalidate the results; they simply define their appropriate domain of validity. Within that domain, the project provides robust evidence that the proposed architecture behaves coherently and achieves its stated objectives.

6.9 Overall Conclusions from Results

The results and discussions in this chapter demonstrate that the proposed hybrid microgrid EMS and supervisory architecture function as intended under a range of representative operating conditions. The EMS successfully enforces feeder priority, manages battery SOC within defined bounds, and uses the diesel generator as a backup source rather than a primary supply. Critical loads represented by F1 remain protected across scenarios, while F2 and F3 are curtailed or restored in a prioritized manner depending on resource availability.

The scenario analyses show that the system performs well under renewable-dominant conditions, degrades gracefully under moderate deficits, and uses the battery effectively as a bridge before invoking generator support. These behaviors confirm that the project's design choices—150 kW PV, 100 kWh battery, 100 kW diesel, and a three-feeder priority structure—are coherent and operationally meaningful for the targeted rural context.

The Node-RED and HTML dashboard implementations further demonstrate that the project is not limited to isolated simulations, but includes a practical supervisory layer through which system state and control can be visualized and managed. Status publication, REST control routes, feeder toggles, generator commands, EMS parameter fields, and alarm handling all function in a way that is consistent with the underlying EMS logic and real-system architecture.

Overall, the chapter supports the key claim that the hybrid PV–battery–diesel microgrid with prioritized feeders, rule-based EMS, and Node-RED/dashboard supervision offers a technically sound and implementation-ready solution for improving reliability and service quality in a rural off-grid community setting, within the scope and assumptions of the present study.

Chapter 7

Conclusions and Recommendations

7. Chapter 7

7.1 Introduction

This chapter presents the main conclusions drawn from the design, implementation, and testing of the proposed hybrid PV–battery–diesel microgrid energy management and control system, and offers recommendations for practical deployment and future research. The work combined electrical design, supervisory control logic, middleware implementation, and human–machine interface (HMI) development into a coherent architecture tailored to a rural off-grid community context. Previous chapters have shown that the system operates as intended under multiple operating scenarios, preserving critical loads, prioritizing renewable energy usage, and providing a practical supervisory framework for monitoring and bounded control.

The aim of this chapter is therefore twofold. First, it synthesizes the technical and operational findings into a concise set of key conclusions regarding the effectiveness of the proposed design. Second, it formulates recommendations that can guide future enhancement of the system and support potential field deployment or further academic investigation.

7.2 Conclusion:

The project began with the definition of a hybrid microgrid architecture consisting of a 150 kW PV array, a 100 kWh battery bank, and a 100 kW diesel generator, integrated through a common AC bus and controlled via a programmable logic controller (PLC) and supervisory layers. The EPLAN design documented the power distribution, PV and battery DC systems, inverter–charger interfaces, diesel generator controller (DSE8610 MKII), and main PLC I/O mappings for feeders and sources. Loads were organized into three feeders—F1 critical, F2 medium priority, and F3 low priority—representing residential houses, hospital, shops, water pump, school, and mosque in a realistic rural community.

On the simulation side, MATLAB scripts such as one-2.m and run_ems_matlab.m were developed to construct 24-hour PV and load profiles and to implement a rule-based EMS that allocates power according to feeder priority and source availability. The EMS logic ensured that F1 was always attempted first using PV, then battery, and finally generator support; remaining PV and generator capacity were then applied to F2 and F3 according to priority and supervisory

enable flags. Battery SOC was updated within defined bounds (30–90%), and generator operation was invoked when PV and allowable battery discharge could no longer preserve critical service.

The simulation model was realized in Simulink as an EMS block with clearly defined inputs and outputs for PV power, feeder demands, SOC, generator commands, feeder enable signals, and resulting source and load powers. This provided an executable representation of the supervisory behavior across a 24-hour horizon, suitable for scenario analysis and for feeding higher layers.

At the middleware level, Node-RED was implemented as the main supervisory integration layer, exposing a GET /api/status endpoint and several POST /api/control/... routes for generator start/stop, feeder enable/disable, EMS setpoints, and alarm reset. Node-RED also included logic for periodic status publication, alarm management, and an MQTT-based status publishing concept for external integration.

At the HMI level, a browser-based HTML dashboard was developed to present system mode, SOC, PV and load, generator status, active source, feeder states, EMS parameters, and alarms. The dashboard periodically requested status from Node-RED, updated visual elements accordingly, and sent user control actions back through the REST API, implementing a complete supervisory loop.

Finally, system behavior was evaluated through three representative scenarios: renewable-dominant operation, moderate deficit with priority-based curtailment, and battery-supported operation with transition to generator support. These scenarios were used to interpret how the EMS, battery, generator, feeders, and supervisory interface behaved under varying conditions of resource availability and demand.

7.2.1Main Conclusions

7.2.1.1 The hybrid architecture is technically coherent for the target context

The combination of a 150 kW PV array, 100 kWh battery, and 100 kW diesel generator, organized around a common AC bus with prioritized feeders, provides a technically coherent architecture for a rural off-grid community. The component ratings and feeder structure match the aggregated community demand profiles generated in MATLAB, allowing the system to cover daily loads under favorable solar conditions and to maintain essential services during moderate

deficits. The EPLAN documentation confirms that this architecture is not only conceptually sound but also implementable in terms of protection, control, and wiring.

7.3.1.2 Feeder prioritization significantly improves resilience

Classifying loads into F1 critical, F2 medium priority, and F3 low priority and implementing this structure both in EPLAN and in the EMS enables differentiated service continuity during energy shortages. The results show that, under deficit conditions, F1 remains supplied while F2 and especially F3 are curtailed first, meaning that critical services such as hospital and essential residential demand are protected even when total supply is insufficient. This form of graceful degradation is superior to uniform load shedding in a rural context because it aligns energy allocation with social importance rather than treating all loads equally.

7.3.1.3 The EMS correctly enforces renewable-first and bounded battery usage

The rule-based EMS successfully enforces a renewable-first strategy, using PV to serve loads whenever available and only using the battery and diesel generator when necessary. Battery operation is confined to the defined SOC window and subject to power limits, ensuring that the battery is used to smooth variability and support continuity without being over-discharged or charged beyond safe bounds. This behavior was confirmed across all scenarios, where SOC trajectories remained within the 30–90% range and battery power reversed direction appropriately between charge and discharge modes.

7.3.1.4 The diesel generator is used as a genuine backup resource

Simulation results indicate that the diesel generator remains off or minimally used during renewable-dominant days and is invoked primarily when PV and battery reserves cannot preserve F1 continuity. This is particularly evident in scenarios with moderate deficits or extended low-PV periods, where generator operation begins only after SOC approaches its minimum bound and critical load support cannot be achieved otherwise. Consequently, the generator behaves as a genuine backup resource, not as a baseline energy provider, aligning with the project objective of reducing diesel dependence.

7.3.1.5 The middleware and HMI layers provide a coherent supervisory framework

The Node-RED middleware and HTML dashboard together form a coherent supervisory framework that is well aligned with the EMS logic and physical architecture. The GET /api/status endpoint and control routes for generator, feeders, setpoints, and alarms provide a clear and testable interface for external clients and operators. The dashboard correctly maps these status variables into cards, trend panels, feeder status indicators, and control widgets, enabling operators to understand system conditions and issue bounded commands without bypassing core EMS logic.

7.3.1.6 The system is implementation-ready at supervisory level but not fully commissioned

The project demonstrates implementation readiness at the supervisory and simulation levels: EMS logic is executable in Simulink, Node-RED flows are defined and tested, and the dashboard provides an interactive HMI. However, full industrial commissioning—including detailed protection coordination, site acceptance testing, converter-level control, cybersecurity hardening, and long-term field monitoring—remains outside the scope of this work. The system is therefore best described as **implementation-ready in a laboratory or pilot context**, with a clear upgrade path toward full deployment.

7.3 Recommendations for Practical Implementation

7.3.1 Prioritize iterative field piloting before full-scale deployment

Given the current maturity of the design, it is recommended to deploy the system first in a controlled pilot microgrid or testbed environment rather than immediately in a live rural community. Such a pilot would allow the EMS logic, Node-RED middleware, and dashboard to be connected to actual hardware (or hybrid hardware-in-the-loop setups), revealing any discrepancies between model behavior and real equipment responses. Lessons learned from the pilot can then be used to refine protection settings, communication timeouts, and operator workflows before larger-scale deployment.

7.3.2 Integrate PLC and industrial protocols with the current supervisory stack

The present implementation uses Node-RED and HTTP/JSON interfaces to represent supervisory pathways, which is suitable for prototyping but not yet fully integrated with industrial PLC communication such as Modbus, OPC UA, or IEC-based protocols. For practical deployment, these interfaces should be mapped explicitly onto the PLC addressing scheme specified in the EPLAN I/O diagrams so that EMS decisions and dashboard commands can reach field devices in a robust and deterministic manner. This will involve careful configuration of communication gateways and validation of latency and reliability characteristics.

7.3.3 Develop standard operating procedures (SOPs) and operator training

Because the system includes manual overrides and supervisory controls for feeders, generator, and EMS parameters, there is a risk of inappropriate manual actions if operators are not properly trained. It is therefore recommended to develop standard operating procedures that clearly define when manual overrides are permitted, what parameter ranges are acceptable, and how alarms should be interpreted and acknowledged. Operator training sessions, possibly using the existing dashboard in a simulator mode, would further ensure that the human–system interaction benefits from the architecture rather than becoming a source of error.

7.3.4 Incorporate local community priorities into feeder classification

Although the current feeder classification is technically and socially reasonable, practical deployment in a specific community should involve consultation with local stakeholders to refine which loads are considered critical, medium, or low priority. For example, in some contexts, water pumping may need to be classified at a higher priority, or certain productive-use loads may warrant special treatment. Adapting the feeder mapping to local priorities will increase acceptance of the microgrid and align energy management decisions with real community needs.

7.3.5 Establish data logging and performance monitoring from day one

To support ongoing optimization and maintenance, it is recommended to enable persistent data logging of key variables such as PV generation, load, SOC, generator runtime, feeder states, alarms, and operator actions. These data can be collected via Node-RED or PLC logging functions and stored in a local or cloud database for later analysis. Having such records from the

start will make it easier to diagnose issues, verify performance against expectations, and support future EMS refinements based on real-world evidence.

7.4 Recommendations for Future Research

7.4.1 Extend EMS logic beyond fixed rule-based control

The current EMS is rule-based, reflecting fixed priority and threshold decisions that are easy to implement and interpret. Future work could explore augmenting or replacing these rules with optimization-based or model-predictive controllers that minimize fuel consumption, cost, or emissions while respecting constraints such as SOC limits, generator minimum runtime, and load priorities. Such approaches could further improve efficiency, especially under highly variable weather and demand patterns, although they would require additional modeling and computational resources.

7.4.2 Incorporate more detailed battery and generator models

The present battery model focuses on energy balance and SOC limits, and the generator is modeled primarily in terms of power output capacity and on/off status. Future research could integrate more detailed battery models that account for degradation, temperature effects, and cycle depth, as well as generator models that include warm-up times, minimum on/off durations, and fuel curves. These enhancements would allow more accurate assessment of long-term operating costs and equipment lifetime under different EMS strategies.

7.4.3 Analyze long-term performance with real or synthetic multi-day data

The current work evaluates EMS behavior over single-day scenarios with representative profiles. Extending the analysis to multi-day or seasonal horizons with measured or realistically synthesized irradiance and load data would provide deeper insight into how often generator support is needed, how battery cycling accumulates, and how often different feeders experience curtailment. This long-term perspective would be particularly useful for sizing optimization and for assessing the economic viability of the proposed system under varying climate and demand patterns.

7.4.4 Enhance the HMI with advanced visualization and user feedback

The existing dashboard provides basic but effective visualization and control features centered on cards, trend sections, and parameter panels. Future work could enhance this HMI with more advanced data visualization such as historical charts, event timelines, and comparative views of different operating days. It would also be valuable to gather structured feedback from actual operators about the usability of the interface and to iteratively refine the layout, terminology, and alarm presentation based on their experience.

7.4.5 Investigate cybersecurity and resilience of the supervisory network

As the system moves closer to real deployment, cybersecurity and communication resilience become critical research topics. Future work should examine secure communication mechanisms for Node-RED, PLCs, and dashboards, including authentication, encryption, role-based access control, and network segmentation. It would also be valuable to study how the EMS should behave under loss of communication, partial network failure, or attempted malicious actions, ensuring that critical loads remain protected even under adverse cyber-physical conditions.

7.5 Closing Remarks

This project has demonstrated that a hybrid PV–battery–diesel microgrid with prioritized feeders, rule-based EMS, and Node-RED/dashboard supervision can be designed, implemented at a supervisory level, and functionally validated for a realistic rural community scenario. The system succeeds in prioritizing critical loads, maximizing renewable utilization, managing battery reserves, and limiting diesel generator operation to necessary periods, while providing operators with a coherent interface for monitoring and bounded control.

Within the defined assumptions and scope, the results support the conclusion that such architectures offer a promising pathway to improving energy access and reliability in off-grid or weak-grid settings, particularly where community demand is diverse and resources are constrained. At the same time, the work highlights the importance of careful integration between simulation models, control logic, middleware, and HMIs to ensure that theoretical designs translate into practical, understandable, and resilient systems.

The recommendations provided in this chapter can guide both practitioners considering pilot deployments and researchers seeking to deepen the technical and socio-technical foundations of

hybrid microgrids, especially in contexts similar to the rural village use case considered in this study.

7.x Economic Evaluation

This section evaluates the approximate capital cost and economic attractiveness of the proposed hybrid PV–battery–diesel microgrid for the target rural coastal community in Yemen.

7.x.1 Capital Cost Summary

Table 7-1 summarizes indicative budgetary costs for the main system components based on typical international wholesale prices and recent market benchmarks for PV, battery storage, and diesel generators. These values are not vendor quotations, but they provide a realistic order of magnitude for techno-economic analysis.

Table 7-1

Component	Basis of estimate	Approximate cost (USD)
PV array (≈150 kWp Jinko 550 W modules)	≈266 modules at ≈0.11 USD/W (60 USD/module) based on current Jinko panel price ranges and small commercial market offersaccio+2	≈ 16,000
PV grid-tied inverter (Huawei SUN2000-100KTL-M1, 100 kW)	Typical European retail price ≈5,600–6,000 USDeurope-solarstore	≈ 6,000
Battery bank (high-voltage $LiFePO_4$ ≈100–123 kWh)	Utility and C&I storage benchmarks ≈100–150 USD/kWh; $LiFePO_4$ offers for Yemen ≈55–120 USD/kWhbslbatt+2	12,000–18,000
Battery PCS / bidirectional inverter (≈60–100 kW)	Industry rule-of-thumb ≈150–300 USD per kW for industrial PCS (no direct list prices, based on vendor ranges for similar products)bslbatt	10,000–20,000
Diesel generator (≈100 kW)	Price ranges for 100 kW gensets from global suppliers ≈4,500–8,000 USD ex-works, higher with shipping and local marginalibaba+2	7,000–12,000
Balance of system (structures, DC/AC combiner boxes, breakers, surge protection, cabling, earthing)	BOS cost shares in small and commercial PV often account for 20–35% of hardware costanernstore+1	10,000–15,000
Design, engineering, installation, and commissioning	Typical allocation for small microgrids (labor, transport, testing)	5,000–10,000

On this basis, the total installed cost of the hybrid system is expected to fall in the range of approximately **66,000–97,000 USD**, depending on final equipment selection, logistics, and local labor rates. For detailed project implementation, these indicative values must be refined using formal quotations from local suppliers and contractors.

7.x.2 Diesel Fuel Consumption and Operating Cost

For comparison with conventional diesel-only supply, it is useful to examine the fuel consumption characteristics of a 100 kW generator. Manufacturer and industry fuel-consumption

charts indicate that a 100 kW diesel genset operating at around 75% load typically consumes approximately 22–28 L/h, corresponding to about 0.25–0.30 L of diesel per kWh of electricity delivered. When realistic diesel prices, transport costs, lubricants, and maintenance are included, studies of isolated community power systems report levelized electricity costs from diesel generation in the range of 0.**4–1.0 USD/kWh** for remote and rural applications.whjlmech+5

In contrast, recent assessments by IRENA and others show that the global average levelized cost of energy (LCOE) from utility-scale solar PV has fallen to around 0**.044 USD/kWh in 2023**, with a reduction of more than 90% since 2010. Even after accounting for higher specific costs in small off-grid systems, the marginal energy from PV and battery storage is substantially cheaper than diesel fuel in rural microgrids.pv-magazine+2

7.x.3 Expected Fuel Savings of the Hybrid System

Multiple techno-economic studies on hybrid PV–battery–diesel microgrids have demonstrated that integrating solar and storage can reduce diesel fuel consumption by approximately **20–60%** compared to diesel-only operation, depending on PV penetration, storage capacity, and load profile. One recent assessment of a hybrid power supply found more than 55% fuel savings and up to 32% reduction in total lifecycle cost over 20 years relative to a purely diesel system.sdewes+2

The proposed EMS in this project is explicitly designed to realize similar savings by:

- Dispatching **PV power as the primary source**, supplying the critical feeder F1 first and then lower-priority feeders when available power permits.ch3.docx
- Using the **battery bank as a secondary support source** within a protected state-of-charge window (≈30–90%) to cover short-term deficits and maintain F1 without immediate generator start.ch3.docx
- Treating the **diesel generator as a backup continuity source**, only starting it when PV and battery together cannot sustain F1 and the battery SOC reaches its lower limit.sciencedirectch3.docx

Under typical daily profiles, this strategy significantly reduces generator running hours and fuel consumption while preserving high reliability for critical loads.

7.x.4 Economic and Social Benefits for Rural Communities

Beyond pure fuel savings, the hybrid architecture provides several important economic and social advantages for rural electrification:

- **Improved affordability and price stability**: Because a large share of energy is produced from local solar resources, the community becomes less exposed to diesel price volatility, supply disruptions, and transport costs, all of which are major issues in Yemen and other fragile contexts.unescwa+2
- **Prioritized service continuity**: The feeder-based load structure (F1 critical, F2 medium, F3 low priority) allows the EMS to maintain electricity for essential services (hospital, basic residential lighting, markets) even during periods of resource scarcity, rather than experiencing complete blackouts as in many diesel-only schemes.eps.fiuch3.docx
- **Reduced maintenance burden**: Lower generator runtime means fewer overhauls, less wear, and longer equipment life, which is economically significant where skilled maintenance and spare parts are limited.energy+1
- **Environmental and health co-benefits**: Reduced diesel consumption directly lowers local air pollution and greenhouse gas emissions, aligning the project with global clean energy and climate objectives while improving local air quality.sdewes+1

7.x.5 Overall Economic Feasibility

Considering the capital cost range of roughly 66,000–97,000 USD for the hybrid system and the high and volatile cost of diesel-only electricity (0.4–1.0 USD/kWh), the proposed PV–battery–diesel microgrid is economically attractive for rural communities with sustained demand and high fuel prices. Over the project lifetime, the reduction in fuel consumption, maintenance, and unserved-energy losses is expected to offset the additional upfront investment in PV and battery equipment.pembina+3

Moreover, by ensuring reliable power for critical social and economic activities, the system can indirectly support local income generation, public services, and long-term community development. These combined factors indicate that the proposed hybrid microgrid is not only technically feasible but also economically and socially preferable to conventional diesel-based solutions for rural electrification in Yemen.

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